

Pathway Analysis Report

This report contains the pathway analysis results for the submitted sample ". Analysis was performed against Reactome version 95 on 19/12/2025. The web link to these results is:

<https://reactome.org/PathwayBrowser/#/ANALYSIS=MjAyNTEyMTkwODUyMDlfNTMzOQ%3D%3D>

Please keep in mind that analysis results are temporarily stored on our server. The storage period depends on usage of the service but is at least 7 days. As a result, please note that this URL is only valid for a limited time period and it might have expired.

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1. Introduction

Reactome is a curated database of pathways and reactions in human biology. Reactions can be considered as pathway 'steps'. Reactome defines a 'reaction' as any event in biology that changes the state of a biological molecule. Binding, activation, translocation, degradation and classical biochemical events involving a catalyst are all reactions. Information in the database is authored by expert biologists, entered and maintained by Reactome's team of curators and editorial staff. Reactome content frequently cross-references other resources e.g. NCBI, Ensembl, UniProt, KEGG (Gene and Compound), ChEBI, PubMed and GO. Orthologous reactions inferred from annotation for Homo sapiens are available for 14 non-human species including mouse, rat, chicken, puffer fish, worm, fly and yeast. Pathways are represented by simple diagrams following an SBGN-like format.

Reactome's annotated data describe reactions possible if all annotated proteins and small molecules were present and active simultaneously in a cell. By overlaying an experimental dataset on these annotations, a user can perform a pathway over-representation analysis. By overlaying quantitative expression data or time series, a user can visualize the extent of change in affected pathways and its progression. A binomial test is used to calculate the probability shown for each result, and the p-values are corrected for the multiple testing (Benjamini-Hochberg procedure) that arises from evaluating the submitted list of identifiers against every pathway.

To learn more about our Pathway Analysis, please have a look at our relevant publications:

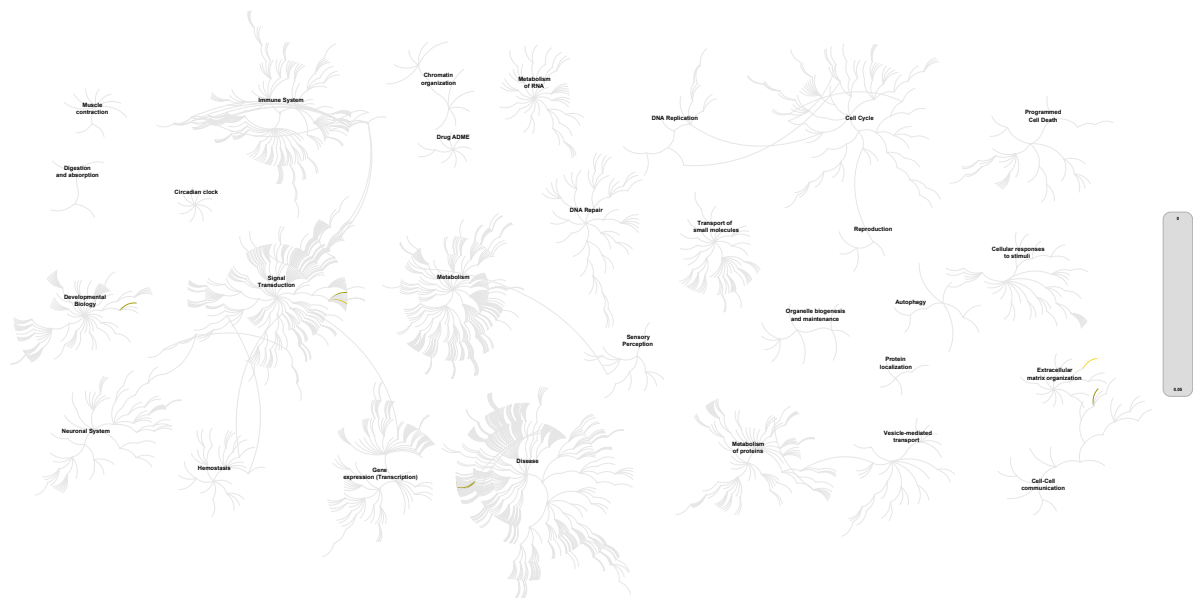
Fabregat A, Sidiropoulos K, Garapati P, Gillespie M, Hausmann K, Haw R, ... D'Eustachio P (2016). The reactome pathway knowledgebase. *Nucleic Acids Research*, 44(D1), D481–D487. <https://doi.org/10.1093/nar/gkv1351>. 

Fabregat A, Sidiropoulos K, Viteri G, Forner O, Marin-Garcia P, Arnau V, ... Hermjakob H (2017). Reactome pathway analysis: a high-performance in-memory approach. *BMC Bioinformatics*, 18. 

2. Properties

- This is an **overrepresentation** analysis: A statistical (hypergeometric distribution) test that determines whether certain Reactome pathways are over-represented (enriched) in the submitted data. It answers the question 'Does my list contain more proteins for pathway X than would be expected by chance?' This test produces a probability score, which is corrected for false discovery rate using the Benjamini-Hochberg method. [↗](#)
- 4014 out of 6405 identifiers in the sample were found in Reactome, where 2588 pathways were hit by at least one of them.
- All non-human identifiers have been converted to their human equivalent. [↗](#)
- IntAct interactors were included to increase the analysis background. This greatly increases the size of Reactome pathways, which maximises the chances of matching your submitted identifiers to the expanded pathway, but will include interactors that have not undergone manual curation by Reactome and may include interactors that have no biological significance, or unexplained relevance.
- This report is filtered to show only results for species 'Homo sapiens' and resource 'all resources'.
- The unique ID for this analysis (token) is MjAyNTEyMTkwODUyMDIhNTMzOQ%3D%3D. This ID is valid for at least 7 days in Reactome's server. Use it to access Reactome services with your data.

3. Genome-wide overview



This figure shows a genome-wide overview of the results of your pathway analysis. Reactome pathways are arranged in a hierarchy. The center of each of the circular "bursts" is the root of one top-level pathway, for example "DNA Repair". Each step away from the center represents the next level lower in the pathway hierarchy. The color code denotes over-representation of that pathway in your input dataset. Light grey signifies pathways which are not significantly over-represented.

4. Most significant pathways

The following table shows the 25 most relevant pathways sorted by p-value.

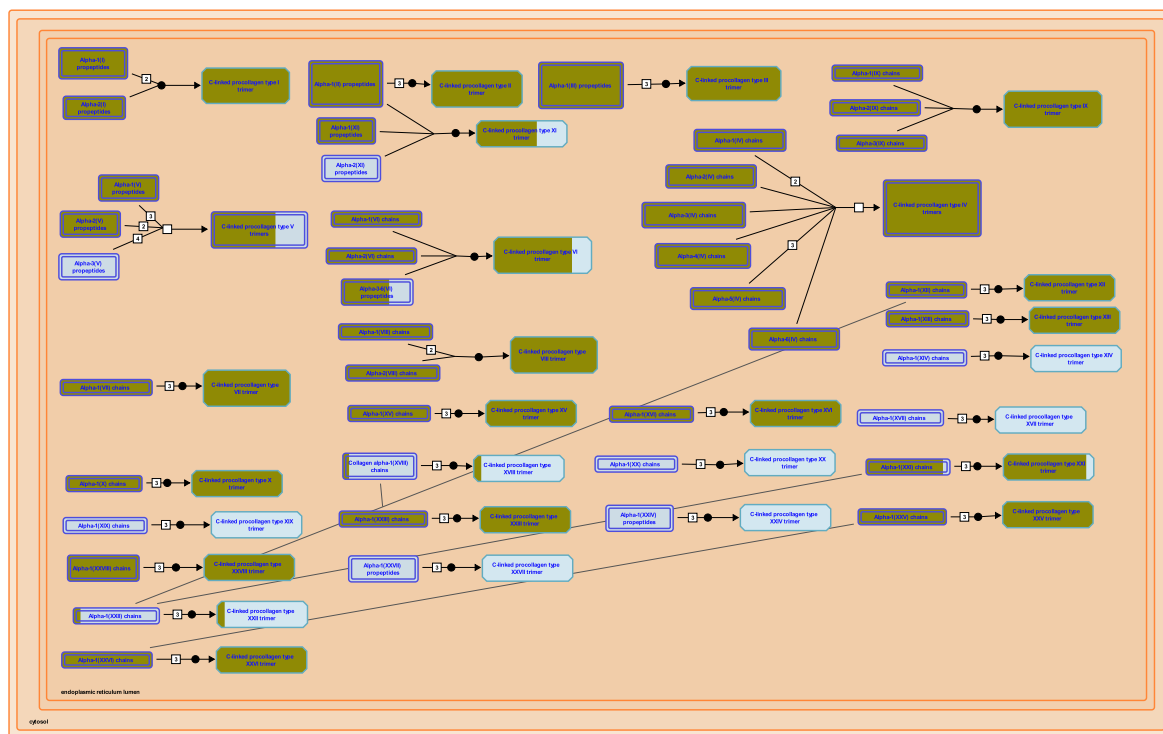
Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
Collagen chain trimerization	33 / 44	0.002	0.002	1	22 / 28	0.002
RHOG GTPase cycle	47 / 78	0.003	0.015	1	6 / 6	3.78e-04
Developmental Lineage of Pancreatic Ductal Cells	44 / 77	0.003	0.038	1	2 / 2	1.26e-04
Defective B4GALT7 causes EDS, progeroid type	15 / 21	8.46e-04	0.041	1	1 / 1	6.30e-05
Defective B3GALT6 causes EDSP2 and SEMDJL1	15 / 21	8.46e-04	0.041	1	1 / 1	6.30e-05
Regulation of CDH11 mRNA translation by microRNAs	11 / 14	5.64e-04	0.042	1	4 / 4	2.52e-04
CDC42 GTPase cycle	83 / 159	0.006	0.043	1	6 / 6	3.78e-04
Defective EXT1 causes exostoses 1, TRPS2 and CHDS	12 / 16	6.44e-04	0.047	1	3 / 3	1.89e-04
Defective EXT2 causes exostoses 2	12 / 16	6.44e-04	0.047	1	3 / 3	1.89e-04
RHOJ GTPase cycle	34 / 59	0.002	0.056	1	3 / 3	1.89e-04
Defective B3GAT3 causes JDSSDHD	15 / 22	8.86e-04	0.057	1	1 / 1	6.30e-05
Regulation of signaling by NODAL	9 / 12	4.83e-04	0.078	1	3 / 3	1.89e-04
RHOA GTPase cycle	78 / 154	0.006	0.08	1	6 / 6	3.78e-04
Post-transcriptional silencing by small RNAs	6 / 7	2.82e-04	0.084	1	3 / 3	1.89e-04
Leading Strand Synthesis	11 / 16	6.44e-04	0.088	1	4 / 4	2.52e-04
Polymerase switching	11 / 16	6.44e-04	0.088	1	4 / 4	2.52e-04
cGMP effects	12 / 18	7.25e-04	0.092	1	4 / 4	2.52e-04
RUNX3 regulates RUNX1-mediated transcription	4 / 4	1.61e-04	0.095	1	2 / 2	1.26e-04
Crosslinking of collagen fibrils	15 / 24	9.67e-04	0.099	1	13 / 13	8.19e-04
RHOQ GTPase cycle	34 / 63	0.003	0.107	1	3 / 5	3.15e-04
Anchoring fibril formation	10 / 15	6.04e-04	0.116	1	4 / 4	2.52e-04
RAC1 GTPase cycle	93 / 191	0.008	0.119	1	5 / 6	3.78e-04
HS-GAG biosynthesis	24 / 43	0.002	0.12	1	13 / 14	8.82e-04
Gap-filling DNA repair synthesis and ligation in GG-NER	16 / 27	0.001	0.126	1	2 / 2	1.26e-04
Diseases associated with glycosaminoglycan metabolism	30 / 56	0.002	0.131	1	21 / 23	0.001

* False Discovery Rate

5. Pathways details

For every pathway of the most significant pathways, we present its diagram, as well as a short summary, its bibliography and the list of inputs found in it.

1. Collagen chain trimerization (R-HSA-8948216)



The C-propeptides of collagen propeptide chains are essential for the association of three peptide chains into a trimeric but non-helical procollagen. This initial binding event determines the composition of the trimer, brings the individual chains into the correct register and initiates formation of the triple helix at the C-terminus, which then proceeds towards the N-terminus in a zipper-like fashion (Engel & Prockop 1991). Most early refolding studies were performed with collagen type III, which contains a disulfide linkage at the C-terminus of its triple helix (Bächinger et al. 1978, Bruckner et al. 1978) that acts as a permanent linker even after removal of the non-collagenous domains.

Mutations within the C-propeptides further suggest that they are crucial for the correct interaction of the three polypeptide chains and for subsequent correct folding (refs. in Boudko et al. 2011).

References

Byers PH, Click EM, Harper E & Bornstein P (1975). Interchain disulfide bonds in procollagen are located in a large nontriple-helical COOH-terminal domain. Proc Natl Acad Sci U S A, 72, 3009-13

Bächinger HP, Brückner P, Timpl R & Engel J (1978). The role of cis-trans isomerization of peptide bonds in the coil leads to and comes from triple helix conversion of collagen. *Eur J Biochem*, 90, 605-13. [🔗](#)

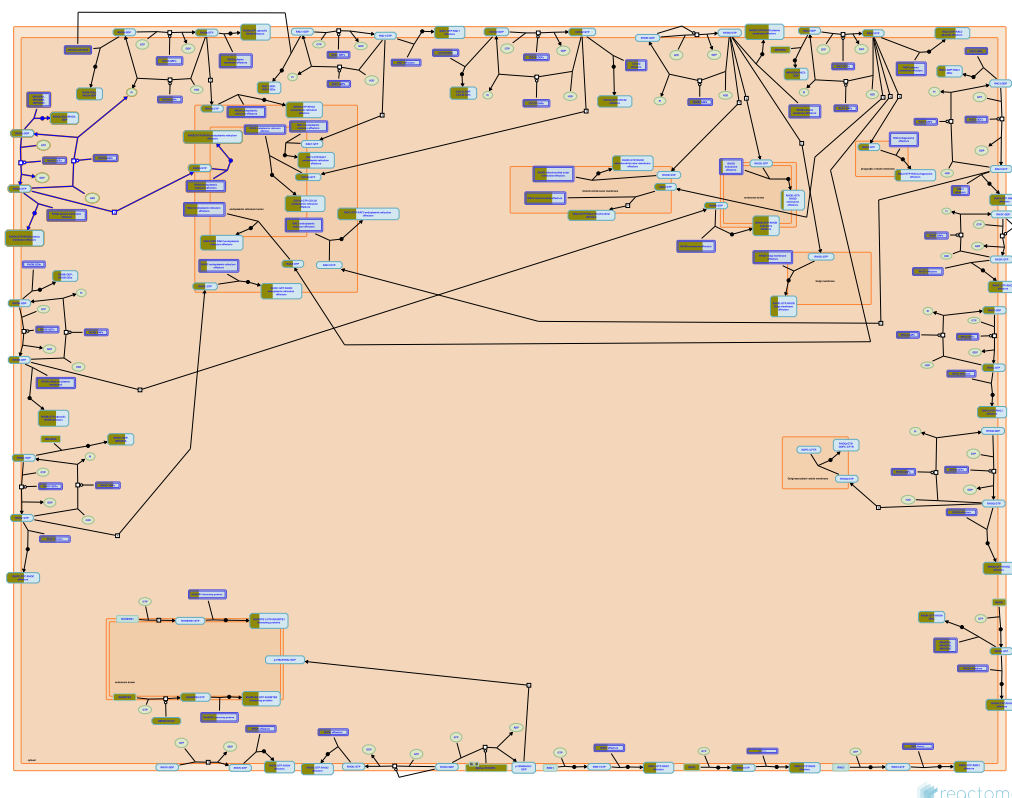
Edit history

Date	Action	Author
2012-04-11	Authored	Jupe S
2012-05-24	Reviewed	Canty-Laird EG
2016-11-03	Edited	Jupe S
2016-11-11	Created	Jupe S
2025-11-15	Modified	Weiser JD

31 submitted entities found in this pathway, mapping to 36 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
COL10A1	Q03692	COL11A1	P12107	COL12A1	Q99715
COL13A1	Q5TAT6	COL15A1	P39059, Q5TAT6	COL16A1	Q07092
COL1A1	P02452	COL1A2	P08123	COL21A1	Q96P44
COL23A1	Q86Y22	COL25A1	Q9BXS0	COL26A1	Q86Y22, Q96A83
COL28A1	Q2UY09	COL2A1	P02458	COL3A1	P02461
COL4A1	P02462, Q14055	COL4A2	P08572	COL4A3	Q01955
COL4A4	P53420	COL4A5	P29400	COL4A6	Q14031
COL5A1	P20908	COL5A2	P05997	COL6A1	P12109
COL6A2	P12110	COL6A3	A8TX70, P12111	COL7A1	P29400, Q02388
COL8A1	P27658	COL8A2	P25067	COL9A1	P20849
COL9A3	Q14050				

2. RHOG GTPase cycle (R-HSA-9013408)



This pathway catalogues RHOG guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and RHOG effectors. RHOG is a RAC-related RHO GTPase, ~70% identical to RAC1 (Vincent et al. 1992, de Curtis 2008). RHOG is broadly expressed in different tissue types. It regulates the cytoskeleton, acting either upstream of or in parallel to RAC1. RHOG regulates cell polarity, adhesion, migration and invasion, contributing to the formation of lamellipodia and invadopodia (Gauthier-Rouvière et al. 1998, Al-Koussa et al. 2020). The ortholog of RHOG is required for neuronal development in *C. elegans* (de Curtis 2008). RHOG is involved in VEGF signaling and angiogenesis (El Baba et al. 2020). RHOG cooperates with RAC1 and CDC42 in malignant cell transformation (Roux et al. 1997) and may contribute to invasiveness of cancer cells (Al-Koussa et al. 2020).

References

- de Curtis I (2008). Functions of Rac GTPases during neuronal development. *Dev. Neurosci.*, 30, 47-58. [🔗](#)
- El Baba N, Farran M, Khalil EA, Jaafar L, Fakhoury I & El-Sibai M (2020). The Role of Rho GTPases in VEGF Signaling in Cancer Cells. *Anal Cell Pathol (Amst)*, 2020, 2097214. [🔗](#)
- Al-Koussa H, Atat OE, Jaafar L, Tashjian H & El-Sibai M (2020). The Role of Rho GTPases in Motility and Invasion of Glioblastoma Cells. *Anal Cell Pathol (Amst)*, 2020, 9274016. [🔗](#)
- Vincent S, Jeanteur P & Fort P (1992). Growth-regulated expression of rhoG, a new member of the ras homolog gene family. *Mol Cell Biol*, 12, 3138-48. [🔗](#)
- Gauthier-Rouvière C, Vignal E, Mérianne M, Roux P, Montcourier P & Fort P (1998). RhoG GTPase controls a pathway that independently activates Rac1 and Cdc42Hs. *Mol Biol Cell*, 9, 1379-94. [🔗](#)

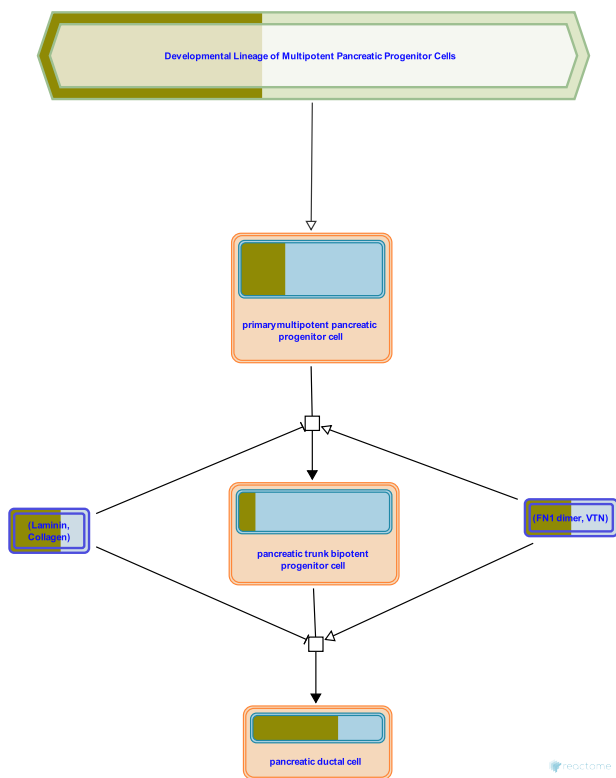
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Date	Action	Author
2017-07-25	Created	Orlic-Milacic M
2020-07-14	Authored	Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

49 submitted entities found in this pathway, mapping to 49 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ANKLE2	Q86XL3	ARFGAP1	Q9NP61	ARFGAP3	Q9NP61
ARHGAP1	Q07960	ARHGAP10	Q5T5U3	ARHGAP21	Q5T5U3
ARHGAP35	Q9NRY4	ARHGDIA	P52565	ARHGDIB	P52566
ARHGEF16	Q5VV41	ARHGEF26	Q96DR7	ARHGEF5	Q12774
CAV1	Q03135	CDC42	P60953	CDC42EP1	Q00587
CYFIP1	Q7L576	DEPDC1B	Q8WUY9	DIAPH3	Q9NSV4
DOCK2	Q92608	DOCK3	Q8IZD9	DOCK4	Q8N1I0
DOCK5	Q9H7D0	DSG2	Q14126	ELMO2	Q96JJ3
EMD	P50402	ERBIN	Q96RT1	ESYT1	Q9BSJ8
GARRE1	O15063	HSPE1	P61604	IQGAP2	Q13576
ITGB1	P05556	LAMTOR1	Q6IAA8	LETM1	O95202
MCF2	P10911	MPP7	Q5T2T1	NDUFA5	Q16718
OPHN1	O60890	PAK4	O96013	RAB7A	P51149
RHOG	P84095	SHMT2	P34897	STBD1	O95210
STX5	Q13190	TRIO	O75962	VAMP3	Q15836
VANGL1	Q8TAA9	VAV1	P15498	VAV3	Q9UKW4
YKT6	O15498				

3. Developmental Lineage of Pancreatic Ductal Cells (R-HSA-9925563)



Ductal epithelial cells line the pancreatic ducts in a single layer, changing the shape as ducts increase in size from cuboidal, through low columnar, to high columnar epithelium (Liggitt and Dintzis, “Pancreas”, pp. 241-250). Ductal epithelial cells may have cilia or microvilli on their apical surface (Liggitt and Dintzis, “Pancreas”, pp. 241-250). Besides ductal epithelial cells, main pancreatic ducts sometimes include goblet cells (Liggitt and Dintzis, “Pancreas”, pp. 241-250).

Ductal epithelial cells develop from bipotent pancreatic trunk progenitor cells, which also give rise to endocrine progenitors and are derived from the primary multipotent pancreatic progenitor cells (reviewed in Shih et al. 2013, Jennings et al. 2015).

References

Liggitt D & Dintzis SM (2018). *Pancreas, "Pancreas", in "Comparative Anatomy and Histology. A Mouse, Rat, and Human Atlas.", edited by Treuting, Dintzis and Montine. , pp. 241-250.*

Shih HP, Wang A & Sander M (2013). Pancreas organogenesis: from lineage determination to morphogenesis. *Annu Rev Cell Dev Biol*, 29, 81-105. [🔗](#)

Jennings RE, Berry AA, Strutt JP, Gerrard DT & Hanley NA (2015). Human pancreas development. *Development*, 142, 3126-37. [🔗](#)

Edit history

Date	Action	Author
2024-10-17	Authored	Orlic-Milacic M
2024-10-17	Created	Orlic-Milacic M
2025-02-21	Reviewed	Li NT
2025-02-25	Edited	Orlic-Milacic M

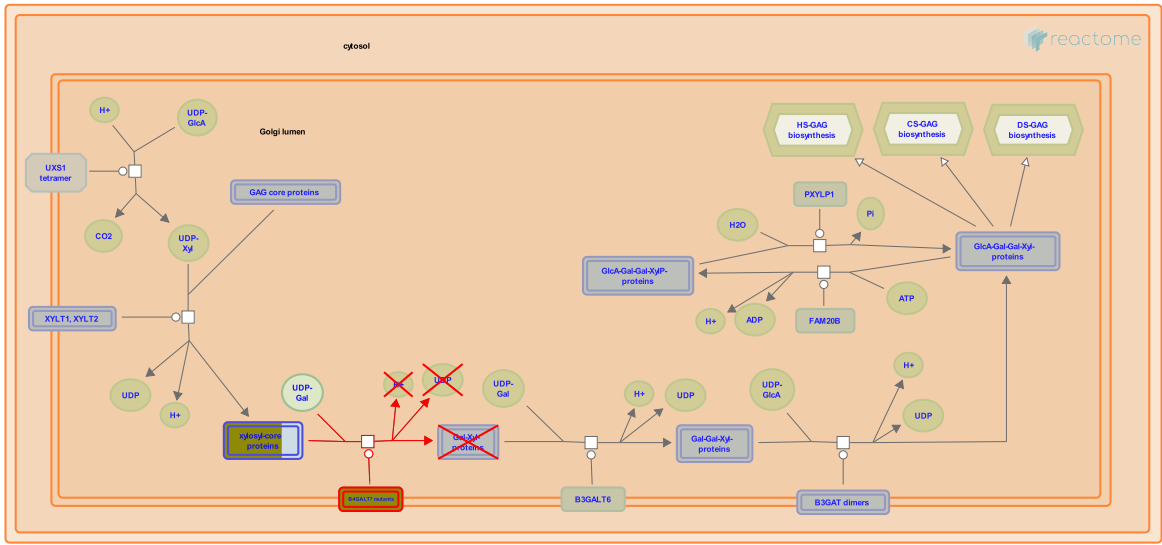
Date	Action	Author
2025-03-05	Modified	Weiser JD

32 submitted entities found in this pathway, mapping to 46 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ANXA1	P04083	ANXA2	P07355	ANXA3	P12429
CA2	P00918	CD74	P04233	CLDN10	P78369
COL11A1	P12107	COL1A1	P02452	COL1A2	P08123
COL2A1	P02458	COL3A1	P02461	COL5A1	P20908
COL5A2	P05997	FN1	P02751	KRT14	P08727
KRT19	P08727	KRT7	P08729	LAMA1	P25391
LAMA2	P24043	LAMA4	Q16363	LAMA5	O15230
LAMB1	P07942	LAMB2	P11047, P55268	LAMC1	P11047
LAMC3	Q9Y6N6	ONECUT1	Q9UBC0	PROM1	O43490
SLC4A4	Q9Y6R1	SOX9	P48436	TFPI2	P48307

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
ANXA1	ENST00000257497	ANXA2	ENST00000451270	ANXA3	ENST00000264908
ANXA4	ENST00000394295	CA2	ENST00000285379	CD74	ENST00000009530
CLDN10	ENST00000299339	GATA6	ENST00000269216	KRT19	ENST00000361566
KRT7	ENST00000331817	ONECUT1	ENST00000305901	PROM1	ENST00000447510
SLC4A4	ENST00000264485	SOX9	ENST00000245479	TFPI2	ENST00000222543

4. Defective B4GALT7 causes EDS, progeroid type (R-HSA-3560783)



Diseases: Ehlers-Danlos syndrome.

Ehlers–Danlos syndrome (EDS) is a group of inherited connective tissue disorders, caused by a defect in the synthesis of collagen types I or III. Abnormal collagen renders connective tissues more elastic. The severity of the mutation can vary from mild to life-threatening. There is no cure and treatment is supportive, including close monitoring of the digestive, excretory and particularly the cardiovascular systems. Defective B4GALT7, a galactosyltransferase important in proteoglycan synthesis, causes the progeroid variant of EDS (MIM:130070). Features include an aged appearance, developmental delay, short stature, generalized osteopenia, defective wound healing, hypermobile joints, hypotonic muscles, and loose but elastic skin (Okajima et al. 1999).

References

Okajima T, Fukumoto S, Furukawa K & Urano T (1999). Molecular basis for the progeroid variant of Ehlers-Danlos syndrome. Identification and characterization of two mutations in galactosyltransferase I gene. J Biol Chem, 274, 28841-4. [🔗](#)

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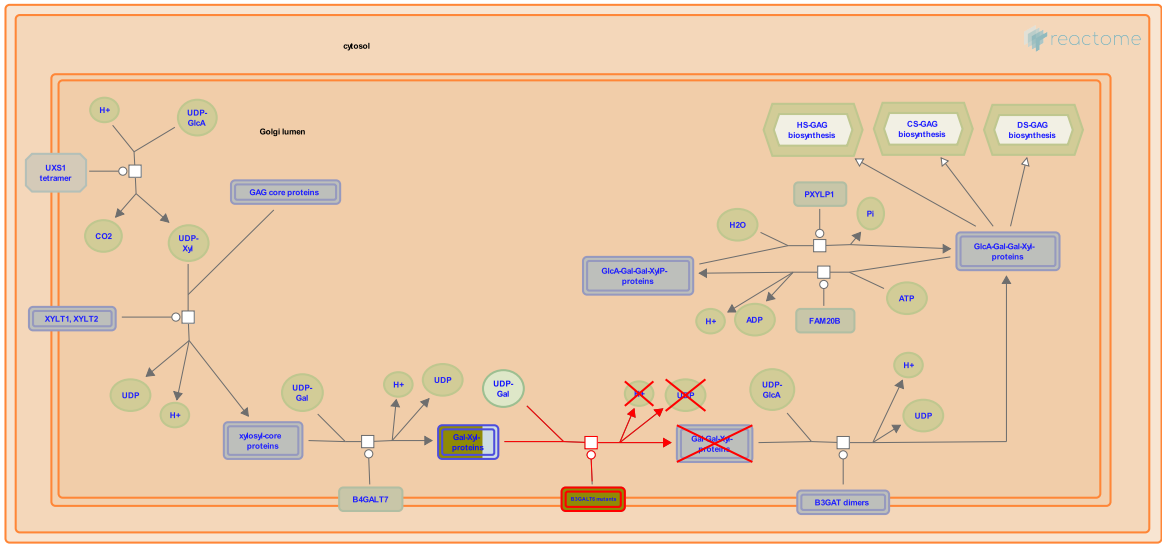
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2013-05-21	Edited	Jassal B
2013-05-21	Authored	Jassal B
2013-05-21	Created	Jassal B
2014-07-09	Reviewed	Spillmann D
2025-06-02	Modified	Stephan R

15 submitted entities found in this pathway, mapping to 15 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
B4GALT7	Q9UBV7	BCAN	Q96GW7	BGN	P21810
CSPG4	Q6UVK1	DCN	P07585	GPC1	P35052
GPC2	Q8N158	GPC3	P51654	GPC4	O75487

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
GPC5	P78333	GPC6	Q9Y625	HSPG2	P98160
SDC1	P18827	SDC2	P34741	SDC4	P31431

5. Defective B3GALT6 causes EDSP2 and SEMDJL1 (R-HSA-4420332)



Diseases: Ehlers-Danlos syndrome, spondyloepimetaphyseal dysplasia.

The biosynthesis of dermatan sulfate/chondroitin sulfate and heparin/heparan sulfate glycosaminoglycans (GAGs) starts with the formation of a tetrasaccharide linker sequence attached to the core protein. Beta-1,3-galactosyltransferase 6 (B3GALT6) is one of the critical enzymes involved in the formation of this linker sequence. Defects in B3GALT6 causes Ehlers-Danlos syndrome progeroid type 2 (EDSP2; MIM:615349), a severe disorder resulting in a broad spectrum of skeletal, connective tissue and wound healing problems. Defects in B3GALT6 can also cause spondyloepimetaphyseal dysplasia with joint laxity type 1 (SEMDJL1; MIM:271640), characterised by spinal deformity and lax joints, especially of the hands and respiratory compromise resulting in early death (Nakajima et al. 2013, Malfait et al. 2013).

References

Malfait F, Kariminejad A, Van Damme T, Gauche C, Syx D, Merhi-Soussi F, ... De Paepe A (2013). Defective Initiation of Glycosaminoglycan Synthesis due to B3GALT6 Mutations Causes a Pleiotropic Ehlers-Danlos Syndrome-like Connective Tissue Disorder. *Am. J. Hum. Genet.* [\[link\]](#)

Nakajima M, Mizumoto S, Miyake N, Kogawa R, Iida A, Ito H, ... Ikegawa S (2013). Mutations in B3GALT6, which Encodes a Glycosaminoglycan Linker Region Enzyme, Cause a Spectrum of Skeletal and Connective Tissue Disorders. *Am. J. Hum. Genet.* [\[link\]](#)

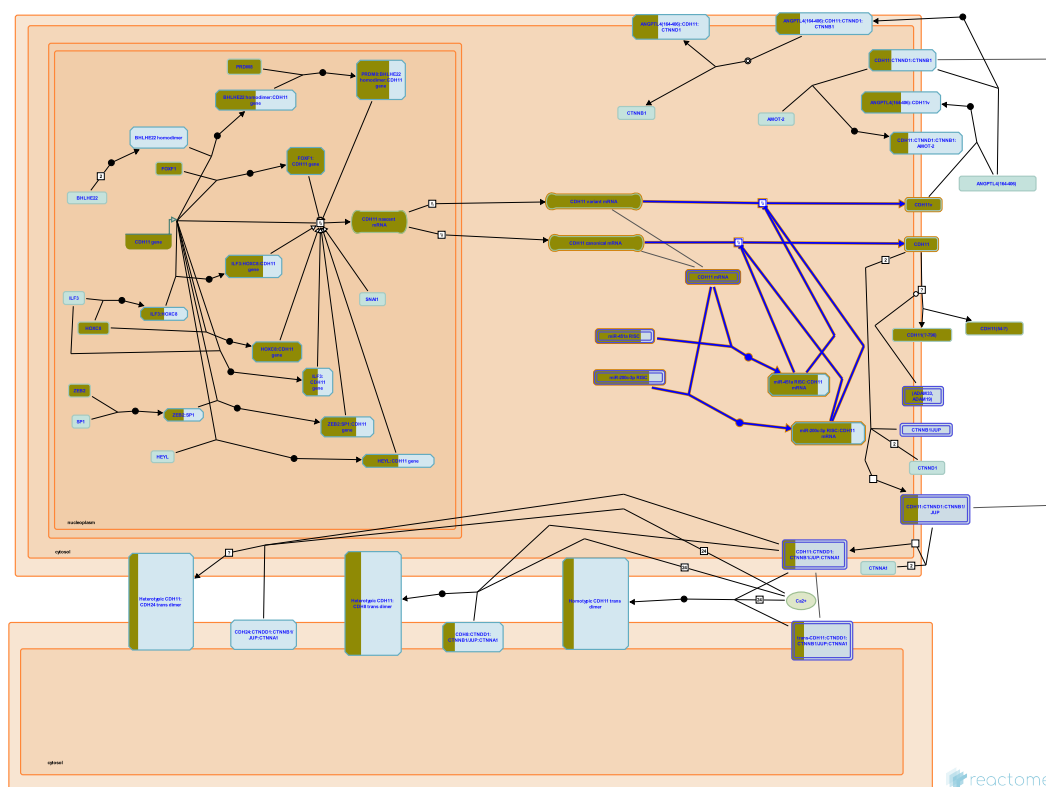
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2013-09-03	Edited	Jassal B
2013-09-03	Authored	Jassal B
2013-09-03	Created	Jassal B
2014-07-09	Reviewed	Spillmann D
2025-06-02	Modified	Stephan R

15 submitted entities found in this pathway, mapping to 15 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
B3GALT6	Q96L58	BCAN	Q96GW7	BGN	P21810
CSPG4	Q6UVK1	DCN	P07585	GPC1	P35052
GPC2	Q8N158	GPC3	P51654	GPC4	O75487
GPC5	P78333	GPC6	Q9Y625	HSPG2	P98160
SDC1	P18827	SDC2	P34741	SDC4	P31431

6. Regulation of CDH11 mRNA translation by microRNAs ([R-HSA-9759811](#))



Several microRNAs are implicated in posttranscriptional regulation of CDH11 gene expression. The diagram only shows microRNAs reported to downregulate CDH11 mRNA and/or protein levels by at least 2 studies, with at least one study providing human evidence, and at least one study providing mechanistic evidence in the form of luciferase reporter assay and/or formation of the microRNA-mRNA complex. For a more detailed description of criteria for microRNA annotation please refer to Huntely et al. 2016.

The following CDH11-targeting microRNAs are strongly supported by experimental evidence and directly shown in the diagram:

miR-200c-3p (Luo et al. 2013; Van der Goten et al. 2014)

miR-451a (Yamada et al. 2018; Wang et al. 2020; Wang et al. 2021)

The following microRNAs probably target CDH11, but additional evidence is needed to directly represent them in the diagram:

miR-103-2-5p (Gao et al. 2020)

miR-127-3p (Dong et al. 2021)

miR-148-5p (Kawagoe et al. 2020)

References

Wang T, Zhou Q & Shang Y (2020). MiRNA-451a inhibits airway remodeling by targeting Cadherin 11 in an allergic asthma model of neonatal mice. *Int Immunopharmacol*, 83, 106440. [🔗](#)

Yamada Y, Arai T, Sugawara S, Okato A, Kato M, Kojima S, ... Seki N (2018). Impact of novel oncogenic pathways regulated by antitumor miR-451a in renal cell carcinoma. *Cancer Sci*, 109, 1239-1253. [🔗](#)

Wang T, Zhou Q & Shang Y (2021). Downregulation of miRNA-451a Promotes the Differentiation of CD4+ T Cells towards Th2 Cells by Upregulating ETS1 in Childhood Asthma. *J Innate Immun*, 13, 38-48. [🔗](#)

Van der Goten J, Vanhove W, Lemaire K, Van Lommel L, Machiels K, Wollants WJ, ... Arijs I (2014). Integrated miRNA and mRNA expression profiling in inflamed colon of patients with ulcerative colitis. *PLoS One*, 9, e116117. [🔗](#)

Luo D, Wilson JM, Harvel N, Liu J, Pei L, Huang S, ... Shi H (2013). A systematic evaluation of miRNA:mRNA interactions involved in the migration and invasion of breast cancer cells. *J Transl Med*, 11, 57. [🔗](#)

Edit history

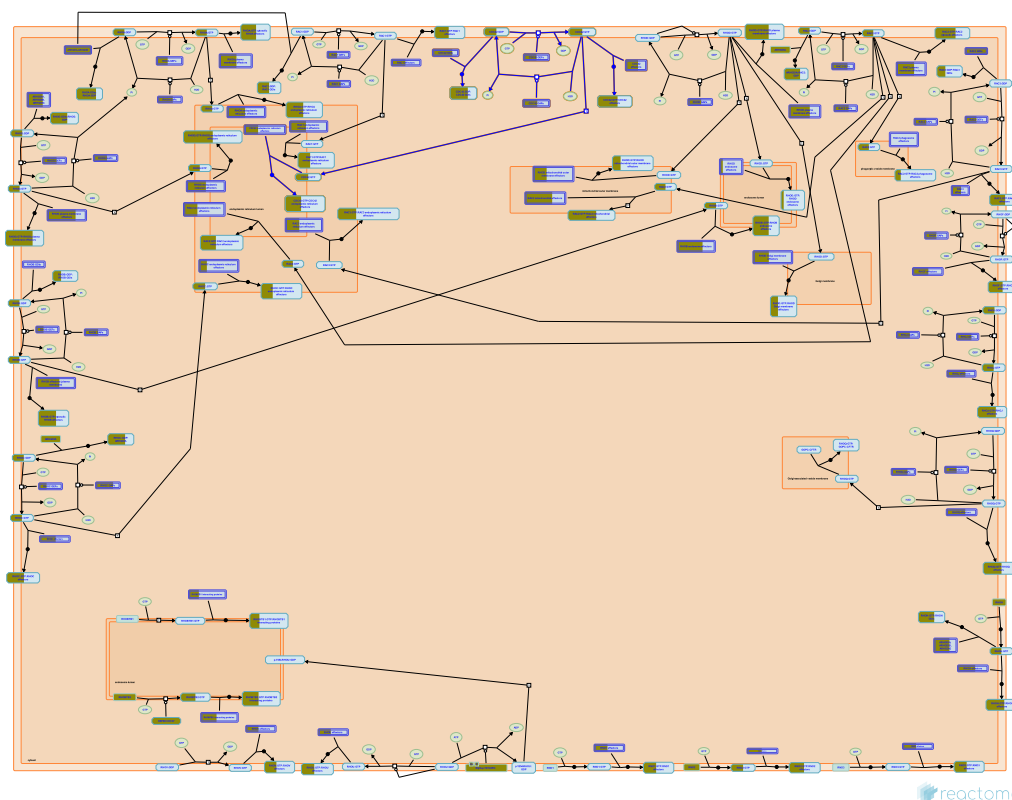
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2021-12-13	Created	Orlic-Milacic M
2022-06-02	Authored	Orlic-Milacic M
2023-04-18	Reviewed	Brasch J
2023-04-19	Edited	Orlic-Milacic M
2023-05-17	Modified	Matthews L

6 submitted entities found in this pathway, mapping to 11 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
AGO1	Q9H9G7, Q9UKV8, Q9UL18	CDH11	P55287, P55287-2	MOV10	Q9HCE1
TNRC6A	Q8NDV7	TNRC6B	Q9UPQ9	TNRC6C	Q9HCJ0

Input	Ensembl Id
CDH11	ENST00000268603, ENST00000394156

7. CDC42 GTPase cycle (R-HSA-9013148)



This pathway catalogues CDC42 guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and CDC42 effectors. CDC42 is one of the three best characterized RHO GTPases, the other two being RHOA and RAC1. By regulating the cytoskeleton, CDC42 regulates cell polarity across different species, from yeast to humans (Pichaud et al. 2019, Woods and Lew 2019). CDC42 is an essential regulator of polarized morphogenesis in epithelial cells, where it coordinates formation of the apical membrane and lumen formation, as well as junction maturation (Pichaud et al. 2019). CDC42 plays a role in cell-to-cell adhesion and cell cycle regulation (Xiao et al. 2018). CDC42 takes part in the regulation of membrane trafficking. Dysfunction of several CDC42-specific GEFs has been shown to impair intracellular trafficking (Egorov and Polishchuk 2017). CDC42 participates in insulin synthesis and secretion and contributes to the pathogenesis of insulin resistance and diabetic nephropathy (Huang et al. 2019). CDC42 is often dysregulated in cancer because a number of GEFs and GEF activators that act upstream of RAC1 and CDC42 are known oncogenes (Aguilar et al. 2017; Maldonado et al. 2018; Zhang et al. 2019; Maldonado et al. 2020). CDC4 promotes cancer cell proliferation, survival, invasion, migration and metastasis (Xiao et al. 2018), especially under hyperglycemia (Huang et al. 2019).

References

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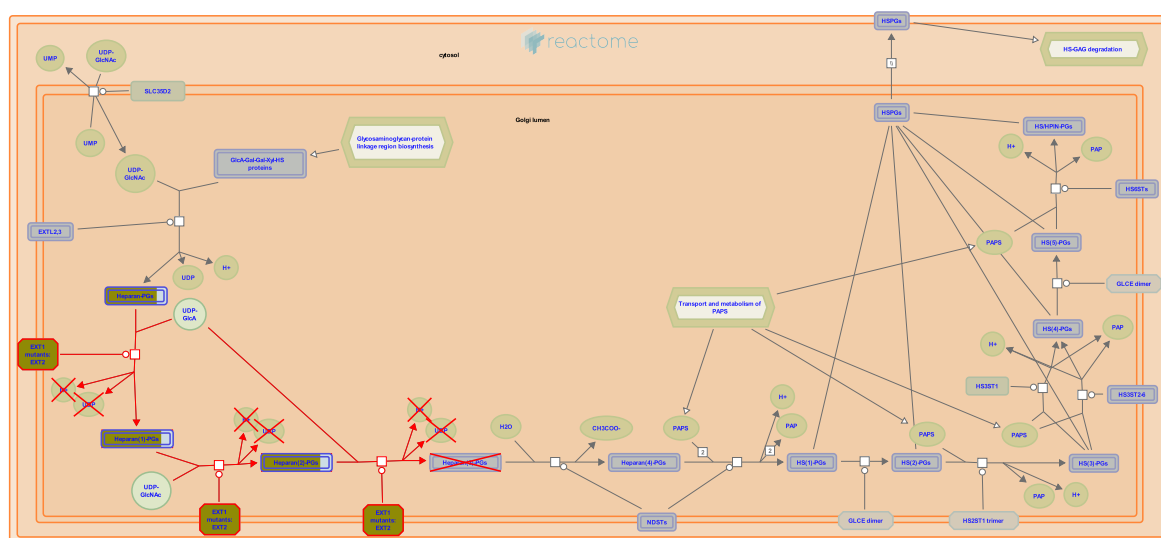
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2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M
2021-05-17	Reviewed	Das ME
2021-06-10	Revised	Orlic-Milacic M
2021-06-10	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

84 submitted entities found in this pathway, mapping to 88 Reactome entities

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ABR	Q12979	ARAP2	Q8WZ64	ARAP3	Q8WWN8
ARFGAP1	Q9NP61	ARFGAP2	Q8N6H7	ARFGAP3	Q9NP61
ARHGAP1	Q07960	ARHGAP10	A1A4S6, Q5T5U3	ARHGAP20	Q9P2F6
ARHGAP21	Q5T5U3	ARHGAP22	Q7Z5H3	ARHGAP24	Q8N264
ARHGAP26	Q9UNA1	ARHGAP30	Q7Z6I6	ARHGAP31	Q2M1Z3
ARHGAP35	Q9NRY4	ARHGAP42	A6NI28	ARHGAP44	Q17R89
ARHGAP9	Q9BRR9	ARHGDIA	P52565	ARHGDIB	P52566
ARHGEF12	Q9NZN5	ARHGEF15	O94989	ARHGEF16	Q5VV41
ARHGEF25	Q86VW2, Q96PX9	ARHGEF26	Q96DR7	ARHGEF4	Q9NR80
ARHGEF5	Q12774	ARHGEF9	O43307	BAIAP2	Q9UQB8
CAV1	Q03135	CDC42	P60953	CDC42BPB	Q9Y5S2
CDC42EP1	Q00587	CDC42EP2	O14613	CDC42EP3	Q9UKI2
CDC42EP4	Q9H3Q1	DAAM1	Q9Y4D1	DEF6	Q9H4E7
DEPDC1B	Q8WUY9	DIAPH3	Q9NSV4	DLC1	Q96QB1
DNMBP	Q6XZF7	DOCK10	Q96BY6	DOCK8	Q8NF50
FAM13B	Q9NYF5	FARP1	Q9Y4F1	FGD4	Q96M96
FHOD3	Q8IVF7	FNBP1	Q96RU3	FNBP1L	Q5T0N5
GIT2	Q14161	GNA13	Q14344	IQGAP2	Q13576
IQGAP3	Q86VI3	LAMTOR1	Q6IAA8	MCF2	P10911
MYO9B	Q13459	OPHN1	O60890	PAK1	Q13153
PAK4	O96013	PAK5	Q9NQU5, Q9P286	PAK6	Q9NQU5
PLEKHG1	Q9ULL1	PLEKHG4B	Q96PX9	PREX2	Q70Z35
RAB7A	P51149	RACGAP1	Q9H0H5	RASGRF2	O14827
SH3PXD2A	Q5TCZ1	SRGAP1	Q7Z6B7	STARD13	Q9Y3M8
STEAP3	Q658P3	STOM	P27105	SYDE1	Q6ZW31
TRIO	O75962, Q96PX9	VAMP3	Q15836	VANGL1	Q8TAA9
VAV3	Q9UKW4	WDR81	Q562E7	WIPF1	O43516

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
WIPF2	Q8TF74	WIPF3	A6NGB9	YKT6	O15498

8. Defective EXT1 causes exostoses 1, TRPS2 and CHDS ([R-HSA-3656253](#))



Diseases: hereditary multiple exostoses.

Heparan sulfate (HS) is involved in regulating various body functions during development, homeostasis and pathology including blood clotting, angiogenesis and metastasis of cancer cells. Exostosin 1 and 2 (EXT1 and 2) glycosyltransferases are required to form HS. They are able to transfer N-acetylglucosamine (GlcNAc) and glucuronate (GlcA) to HS during its synthesis. The functional form of these enzymes appears to be a complex of the two located on the Golgi membrane. Defects in either EXT1 or EXT2 can cause hereditary multiple exostoses 1 (Petersen 1989) and 2 (McGaughran et al. 1995) respectively (MIM:133700 and MIM:133701), autosomal dominant disorders characterized by multiple projections of bone capped by cartilage resulting in deformed legs, forearms and hands. Trichorhinophalangeal syndrome, type II (TRPS2 aka Langer-Giedion syndrome, LGS) is a disorder that combines the clinical features of trichorhinophalangeal syndrome type I (TRPS1, MIM:190350) and multiple exostoses type I, caused by mutations in the TRPS1 and EXT1 genes, respectively (Langer et al. 1984, Ludecke et al. 1995). Defects in EXT1 may also be responsible for chondrosarcoma (CHDS; MIM:215300) (Schajowicz & Bessone 1967, Hecht et al. 1995).

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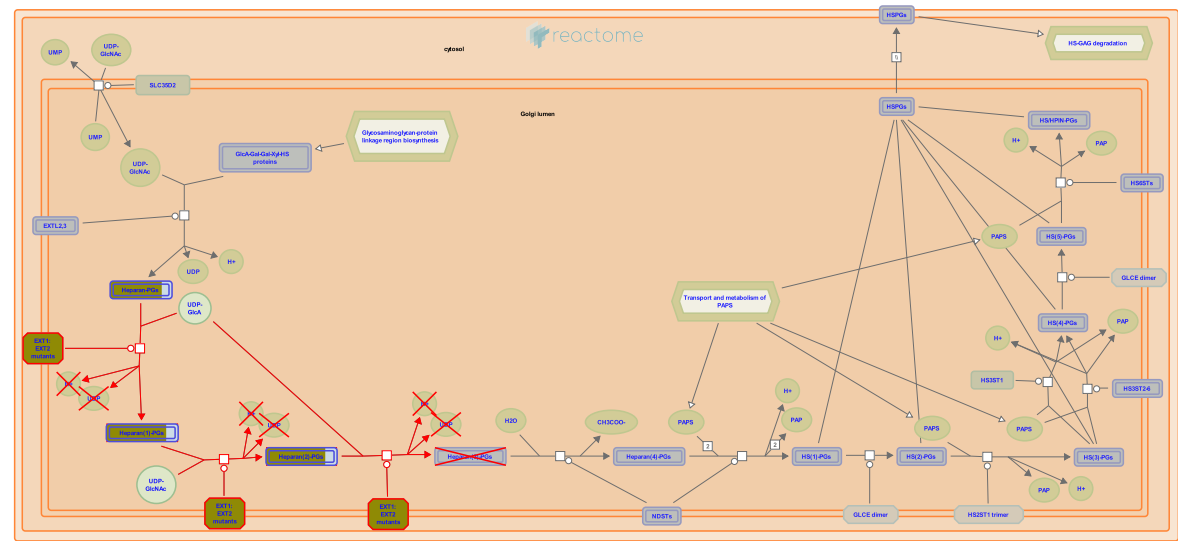
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Date	Action	Author
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2013-05-31	Authored	Jassal B
2013-05-31	Created	Jassal B
2014-07-09	Reviewed	Spillmann D
2025-05-30	Reviewed	Matthews L
2025-06-02	Modified	Stephan R

12 submitted entities found in this pathway, mapping to 12 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
EXT1	Q16394	EXT2	Q93063	GPC1	P35052
GPC2	Q8N158	GPC3	P51654	GPC4	O75487
GPC5	P78333	GPC6	Q9Y625	HSPG2	P98160
SDC1	P18827	SDC2	P34741	SDC4	P31431

9. Defective EXT2 causes exostoses 2 (R-HSA-3656237)



Diseases: hereditary multiple exostoses.

Heparan sulfate (HS) is involved in regulating various body functions during development, homeostasis and pathology including blood clotting, angiogenesis and metastasis of cancer cells. Exostosin 1 and 2 (EXT1 and 2) glycosyltransferases are required to form HS. They are able to transfer N-acetylglucosamine (GlcNAc) and glucuronate (GlcA) to HS during its synthesis. The functional form of these enzymes appears to be a complex of the two located on the Golgi membrane. Defects in either EXT1 or EXT2 can cause hereditary multiple exostoses 1 (Petersen 1989) and 2 (McGaughran et al. 1995) respectively (MIM:133700 and MIM:133701), autosomal dominant disorders characterised by multiple projections of bone capped by cartilage resulting in deformed legs, forearms and hands.

References

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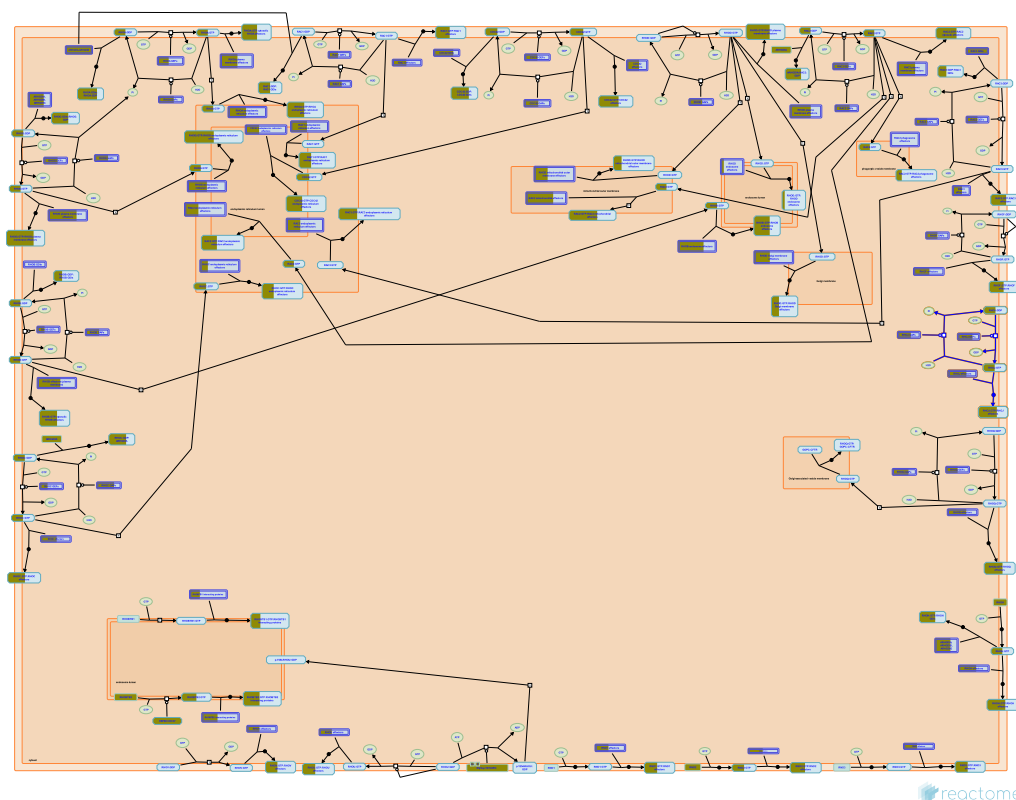
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2013-05-31	Created	Jassal B
2014-07-09	Reviewed	Spillmann D
2025-05-30	Reviewed	Matthews L
2025-06-02	Modified	Stephan R

12 submitted entities found in this pathway, mapping to 12 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
EXT1	Q16394	EXT2	Q93063	GPC1	P35052

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
GPC2	Q8N158	GPC3	P51654	GPC4	O75487
GPC5	P78333	GPC6	Q9Y625	HSPG2	P98160
SDC1	P18827	SDC2	P34741	SDC4	P31431

10. RHOJ GTPase cycle (R-HSA-9013409)



This pathway catalogues RHOJ (also known as TCL or RHOI) guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), and RHOJ effectors. No GDP dissociation inhibitors (GDIs) have been shown to interact with RHOJ. RHOJ, together with RHOQ (TC10), belongs to the CDC42 subfamily of RHO GTPases and shares 85% and 78% amino acid similarity with RHOQ and CDC42, respectively (Vignal et al. 2000). RHOJ regulates the cytoskeleton, including formation of lamellipodia and actin filaments (Vignal et al. 2000, Abe et al. 2003, Aspenstrom et al. 2004, Shi et al. 2016). RHOJ is highly expressed in endothelial cells, regulating their motility and, consequently, vascular morphogenesis (Leszczynska et al. 2011; Shi et al. 2016). As a part of the VEGF signaling cascade, RHOJ promotes retinal angiogenesis (Fukushima et al. 2013). RHOJ plays a role in cancer cell migration and cancer-related angiogenesis (Shi et al. 2016). RHOJ is involved in adipocyte differentiation (Nishizuka 2003; Shi et al. 2016).

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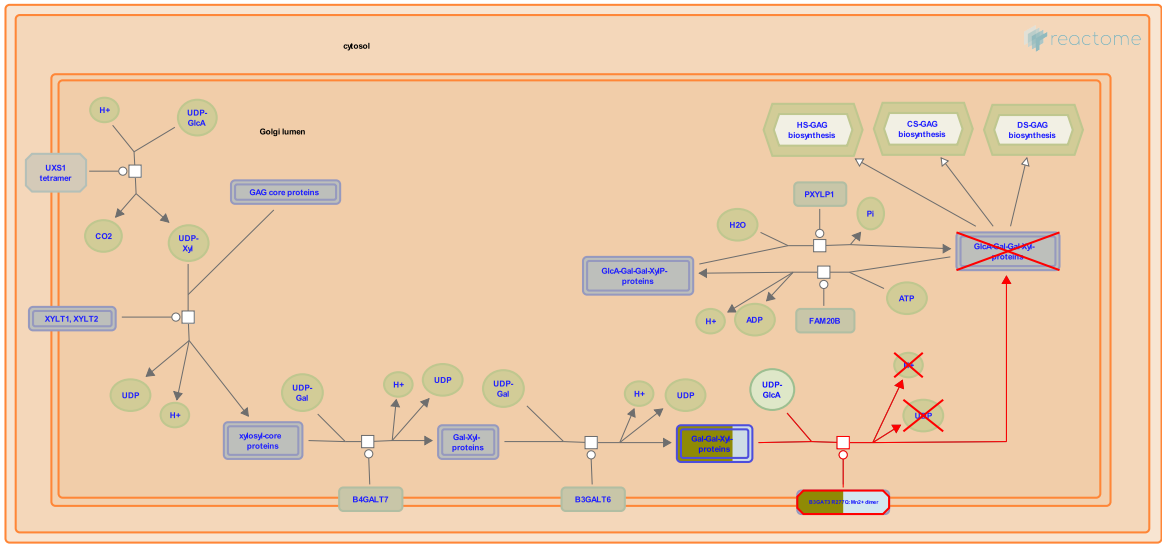
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2021-02-25	Edited	Orlic-Milacic M
2021-05-17	Reviewed	Das ME
2021-06-10	Revised	Orlic-Milacic M
2021-06-10	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

35 submitted entities found in this pathway, mapping to 35 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ARHGAP1	Q07960	ARHGAP10	Q5T5U3	ARHGAP21	Q5T5U3
ARHGAP26	Q9UNA1	ARHGAP35	Q9NRY4	CAV1	Q03135
CDC42	P60953	CDC42BPB	Q9Y5S2	CDC42EP1	Q00587
DEPDC1B	Q8WUY9	DIAPH3	Q9NSV4	DOCK8	Q8NF50
FHOD3	Q8IVF7	FNBP1	Q96RU3	FNBP1L	Q5T0N5
GIT2	Q14161	GJA1	P17302	IQGAP3	Q86VI3
LAMTOR1	Q6IAA8	MPP7	Q5T2T1	OPHN1	O60890
PAK1	Q13153	PAK4	O96013	RAB7A	P51149
RHOJ	Q9H4E5	SLC1A5	Q15758	SLC4A7	Q9Y6M7
STEAP3	Q658P3	STOM	P27105	SYDE1	Q6ZW31
TRIO	O75962	VAMP3	Q15836	VANGL1	Q8TAA9
WIPF2	Q8TF74	WWP2	O00308		

11. Defective B3GAT3 causes JDSSDHD (R-HSA-3560801)



Diseases: congenital heart defect, Larsen syndrome.

Galactosylgalactosylxylosylprotein 3-beta-glucuronosyltransferases1, 2 and 3 (B3GAT1-3) are involved in forming the linker tetrasaccharide present in heparan sulfate and chondroitin sulfate. Defects in B3GAT3 cause multiple joint dislocations, short stature, craniofacial dysmorphism, and congenital heart defects (JDSSDHD; MIM:245600). This is an autosomal recessive disease characterized by dysmorphic facies, elbow, hip and knee dislocations, clubfeet, short stature and cardiovascular defects (Steel & Kohl 1972, Bonaventure et al. 1992, Baasanjav et al. 2011). JDSSDHD has phenotypic similarities to Larsen syndrome (Larsen et al. 1950).

References

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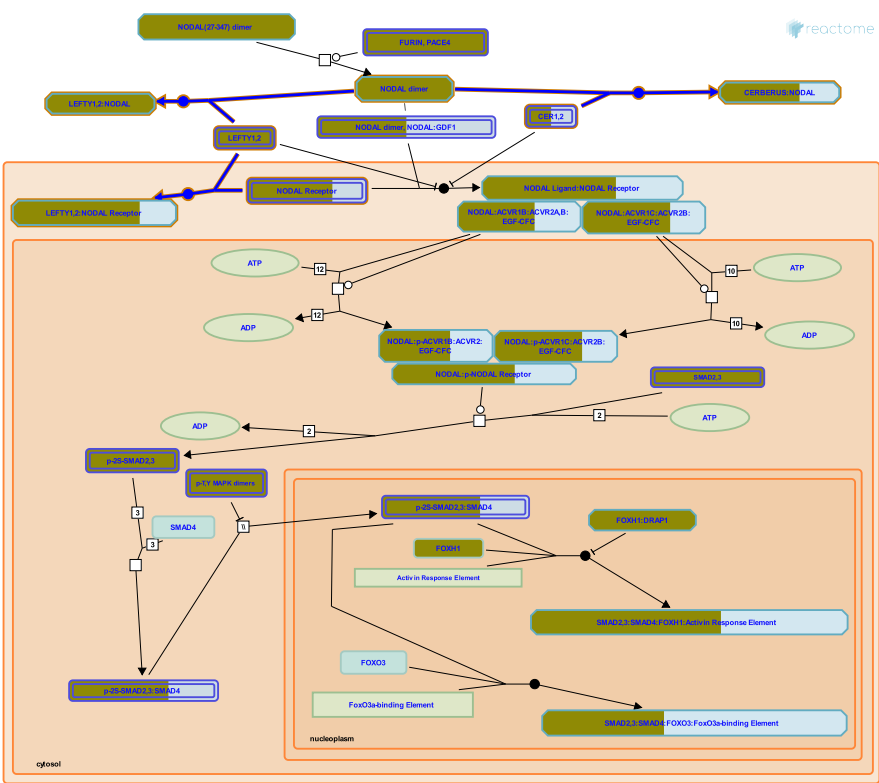
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2013-05-21	Created	Jassal B
2014-07-09	Reviewed	Spillmann D
2025-06-02	Modified	Stephan R

15 submitted entities found in this pathway, mapping to 15 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
B3GAT3	O94766	BCAN	Q96GW7	BGN	P21810
CSPG4	Q6UVK1	DCN	P07585	GPC1	P35052
GPC2	Q8N158	GPC3	P51654	GPC4	O75487
GPC5	P78333	GPC6	Q9Y625	HSPG2	P98160
SDC1	P18827	SDC2	P34741	SDC4	P31431

12. Regulation of signaling by NODAL (R-HSA-1433617)



Cellular compartments: extracellular region, plasma membrane.

Mature NODAL can form heterodimers with LEFTY1, LEFTY2, or CERBERUS. The heterodimers do not activate the NODAL receptor. LEFTY1 and LEFTY2 also bind CRIPTO and CRYPTIC coreceptors and prevent them from interacting with other components of the NODAL receptor. By these mechanisms LEFTY1, LEFTY2, and CERBERUS negatively regulate NODAL signaling (reviewed in Shen 2007, Schier 2009).

References

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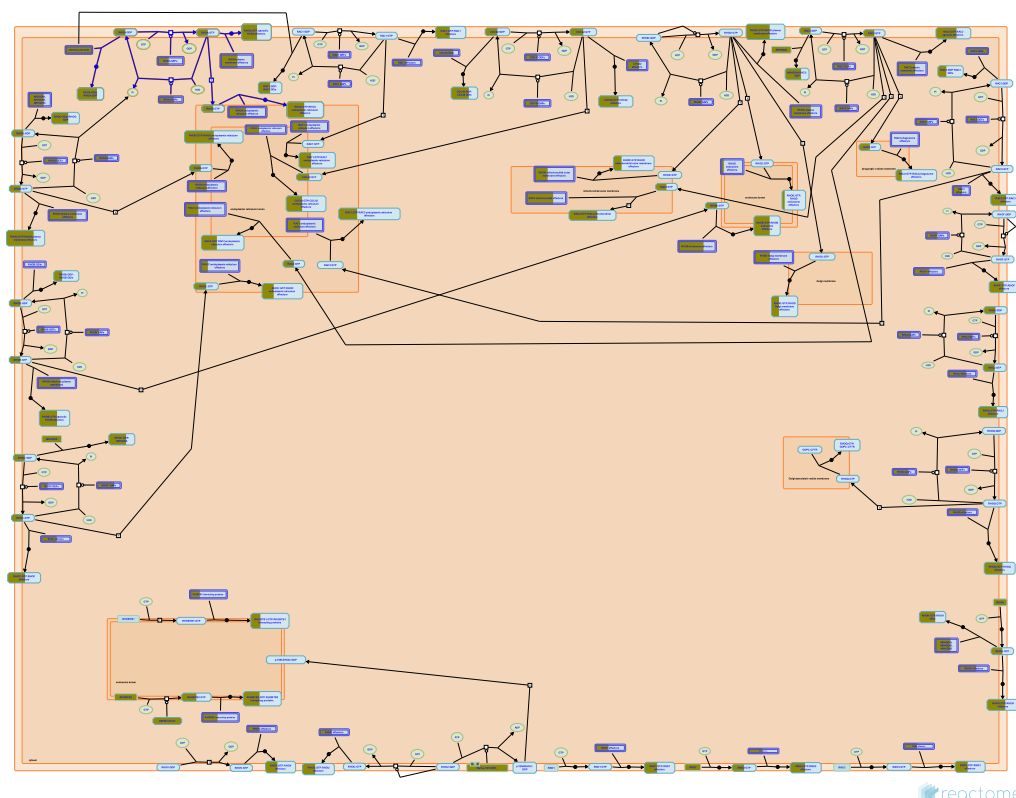
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2011-07-11	Authored	May B
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2011-08-18	Reviewed	May B
2023-10-12	Modified	Weiser JD

9 submitted entities found in this pathway, mapping to 10 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ACVR1C	Q8NER5	ACVR2A	P27037	ACVR2B	Q13705
CER1	O95813	CRIPTO	P13385	CRIPTO3	P51864
LEFTY1	O00292, O75610	LEFTY2	O00292	NODAL	Q96S42

13. RHOA GTPase cycle (R-HSA-8980692)



This pathway catalogues RHOA guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and RHOA effectors. RHOA is one of the three best characterized RHO GTPases, the other two being RAC1 and CDC42, and is the founding member of the RHO GTPase family (Zhou and Zheng 2013). RHOA regulates the cytoskeleton and cell contractility (Lessey et al. 2012), thus playing a role in a number of cellular functions, such as adhesion, migration, survival, division, vesicle trafficking and gene expression (Zhou and Zheng 2013). RHOA-regulated processes are involved in mechanotransduction (Lessey et al. 2012; Marjoram et al. 2014), neuronal development (Zhou and Zheng 2013; Fujita and Yamashita 2014; Hu and Selzer 2017), immune system development (Zhou and Zheng 2013; Ricker et al. 2016; Bros et al. 2019), and cardiovascular regulation (Zhou and Zheng 2013; Cai et al. 2016; Shimokawa et al. 2016). RHOA mutations are frequently found in cancer (Kataoka and Ogawa 2016). Toxins of numerous pathogens target RHOA to hijack the host cytoskeleton (Jamilloux et al. 2018).

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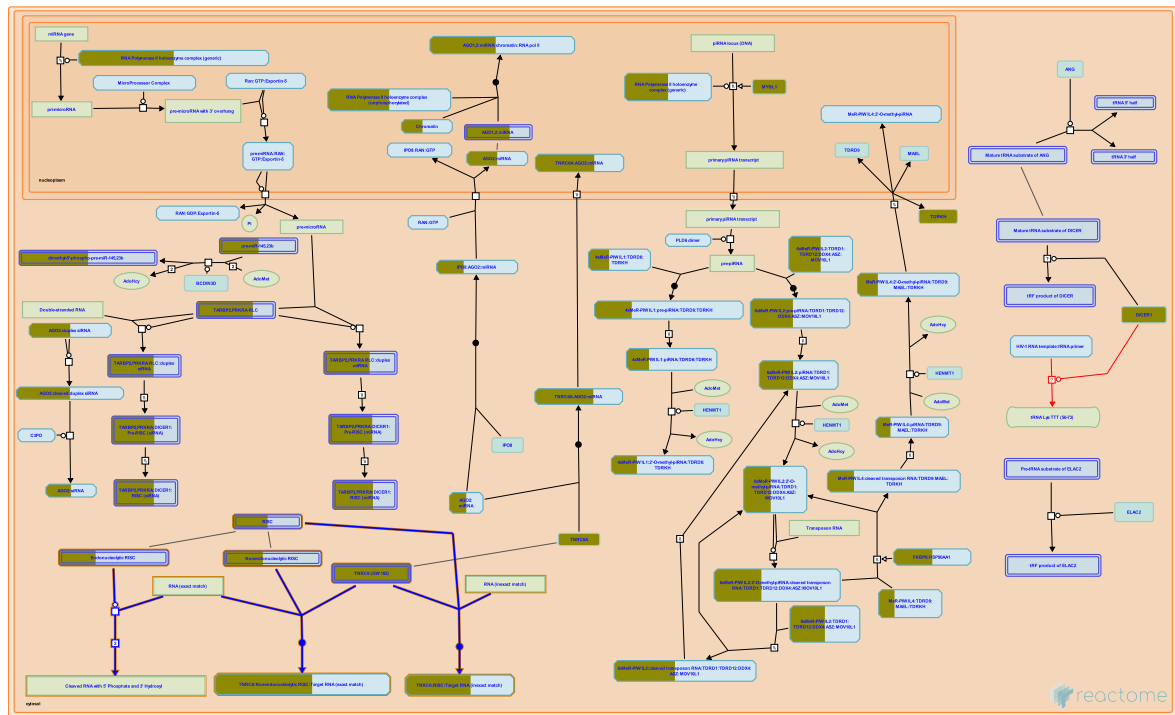
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2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M

78 submitted entities found in this pathway, mapping to 80 Reactome entities

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ARHGAP10	A1A4S6, Q5T5U3	ARHGAP18	Q8N392	ARHGAP19	Q14CB8
ARHGAP20	Q9P2F6	ARHGAP21	Q5T5U3	ARHGAP22	Q7Z5H3
ARHGAP23	Q9P227	ARHGAP24	Q8N264	ARHGAP26	Q9UNA1
ARHGAP28	Q9P2N2	ARHGAP30	Q7Z6I6	ARHGAP31	Q2M1Z3
ARHGAP35	Q9NRY4	ARHGAP42	A6NI28	ARHGAP44	Q17R89
ARHGAP8	P85298	ARHGAP9	Q9BRR9	ARHGDIA	P52565
ARHGDIB	P52566	ARHGEF12	Q9NZN5	ARHGEF15	O94989
ARHGEF17	Q96PE2	ARHGEF2	Q92974	ARHGEF25	Q86VW2
ARHGEF4	Q9NR80	ARHGEF5	Q12774	ATP6AP1	Q15904
BCAP31	P51572	CAV1	Q03135	CAVIN1	Q6NZI2
CIT	O14578, O14578-3	DAAM1	Q9Y4D1	DDR GK1	Q96HY6
DEF6	Q9H4E7	DEPDC1B	Q8WUY9	DIAPH3	Q9NSV4
DLC1	Q96QB1	DOCK2	Q92608	ERBIN	Q96RT1
FAM13A	O94988	FARP1	Q9Y4F1	FHOD3	Q8IVF7
FLOT1	O75955	HMOX2	P30519	IQGAP3	Q86VI3
MCF2	P10911	MID1	Q9NQW6	MYO9B	Q13459
OBSCN	Q5VST9	OPHN1	O60890	PAK1	Q16512
PREX2	Q70Z35	RACGAP1	Q9H0H5	RASGRF2	O14827
RHOA	P61586	RHPN1	Q8TCX5	RHPN2	Q8IUC4
SRGAP1	Q7Z6B7	STARD13	Q9Y3M8	STBD1	O95210
STK10	O94804	STOM	P27105	STX5	Q13190
TEX2	Q8IWB9	TJP2	Q9UDY2	TMEM87B	Q8NBN3
TRIO	O75962	VAMP3	Q15836	VANGL1	Q8TAA9
VAV1	P15498	VAV3	Q9UKW4	YKT6	O15498

14. Post-transcriptional silencing by small RNAs (R-HSA-426496)



Cellular compartments: cytosol.

Small RNAs act with components of the RNA-induced silencing complex (RISC) to post-transcriptionally repress expression of mRNAs (reviewed in Nowotny and Yang 2009, Chua et al. 2009). Two mechanisms exist: 1) cleavage of target RNAs by complexes containing Argonaute2 (AGO2, EIF2C2) and a guide RNA that exactly matches the target mRNA and 2) inhibition of translation of target RNAs by complexes containing AGO2 and an inexact matching guide RNA or by complexes containing a nonendonucleolytic Argonaute (AGO1 (EIF2C1), AGO3 (EIF2C3), or AGO4 (EIF2C4)) and a guide RNA of exact or inexact match. Small interfering RNAs (siRNAs) and microRNAs (miRNAs) can serve as guide RNAs in both types of mechanism.

RNAi also appears to direct chromatin modifications that cause transcriptional gene silencing (reviewed in Verdel et al. 2009).

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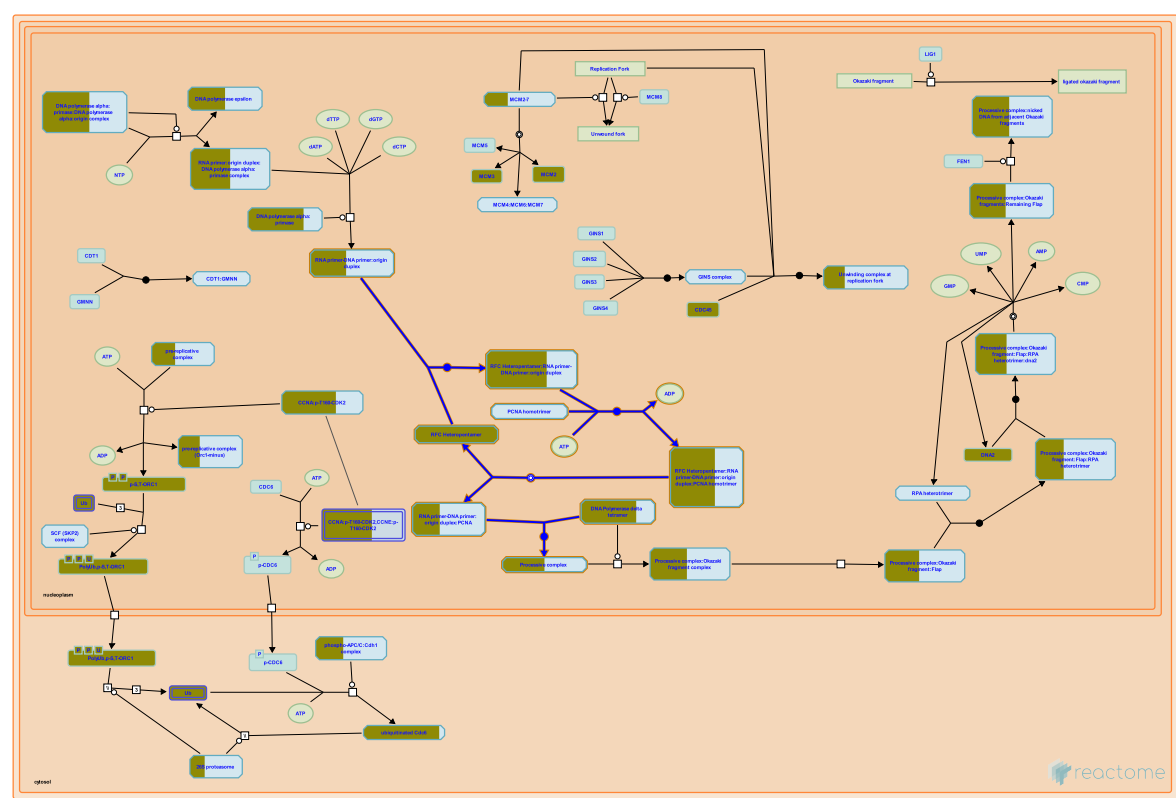
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2009-06-10	Authored	May B
2009-06-17	Created	May B
2012-02-11	Reviewed	Tomari Y
2025-11-15	Modified	Weiser JD

4 submitted entities found in this pathway, mapping to 6 Reactome entities

Input	UniProt Id	Input	UniProt Id
AGO1	Q9H9G7, Q9UKV8, Q9UL18	TNRC6A	Q8NDV7
TNRC6B	Q9UPQ9	TNRC6C	Q9HCJ0

15. Leading Strand Synthesis (R-HSA-69109)



Cellular compartments: nucleoplasm.

The processive complex is responsible for synthesizing at least 5-10 kb of DNA in a continuous manner during leading strand synthesis. The incorporation of nucleotides by pol delta is quite accurate. However, incorporation of an incorrect nucleotide does occur occasionally. Misincorporated nucleotides are removed by the 3' to 5' exonucleolytic proofreading capability of pol delta.

References

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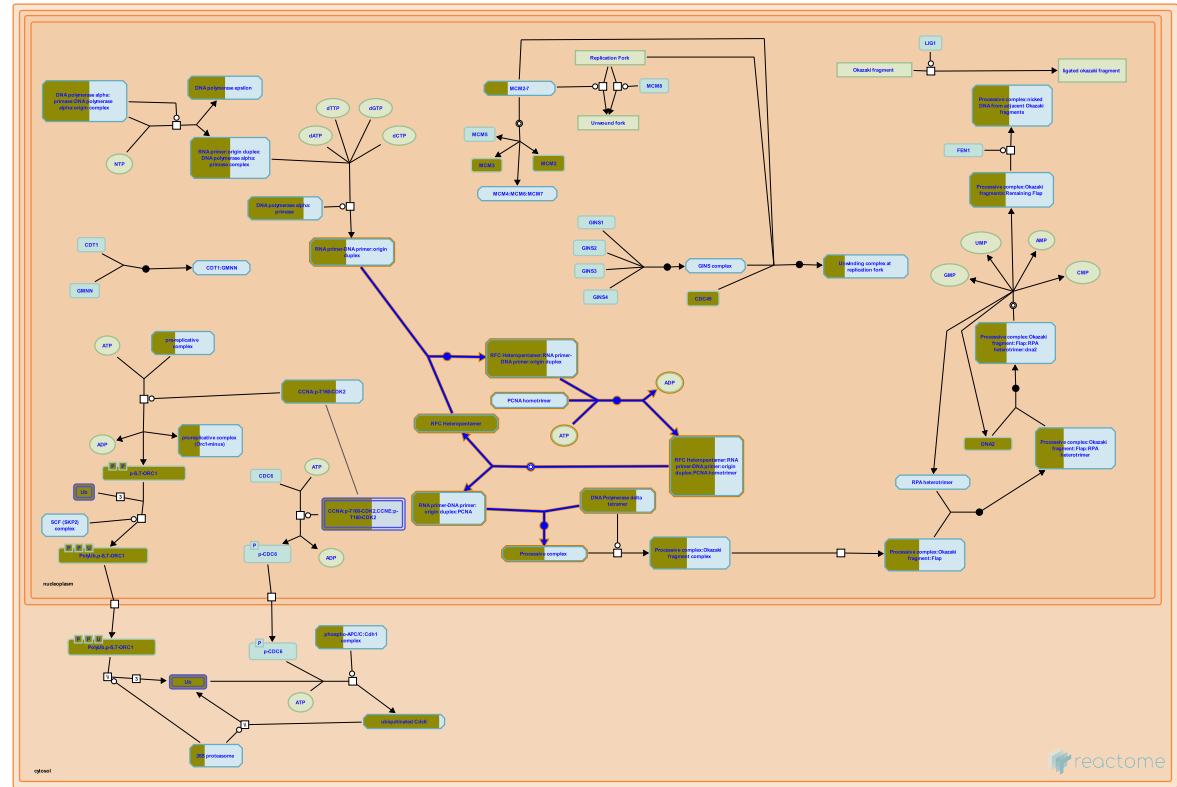
Date	Action	Author
2025-11-15	Modified	Weiser JD

10 submitted entities found in this pathway, mapping to 12 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CDC27	Q15054	POLA1	P09884	POLA2	Q14181
POLD2	P49005	POLD4	Q9HCU8	PRIM1	P49642

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
RFC1	P35251	RFC3	P40937, P40938	RFC4	P35249, P35250
SPP1	P49642				

16. Polymerase switching (R-HSA-69091)



Cellular compartments: nucleoplasm.

After the primers are synthesized, Replication Factor C binds to the 3'-end of the initiator DNA to trigger polymerase switching. The non-processive nature of pol alpha catalytic activity and the tight binding of Replication Factor C to the primer-template junction presumably lead to the turnover of the pol alpha:primase complex. After the Pol alpha-primase complex is displaced from the primer, the proliferating cell nuclear antigen (PCNA) binds to form a "sliding clamp" structure. Replication Factor C then dissociates, and DNA polymerase delta binds and catalyzes the processive synthesis of DNA.

References

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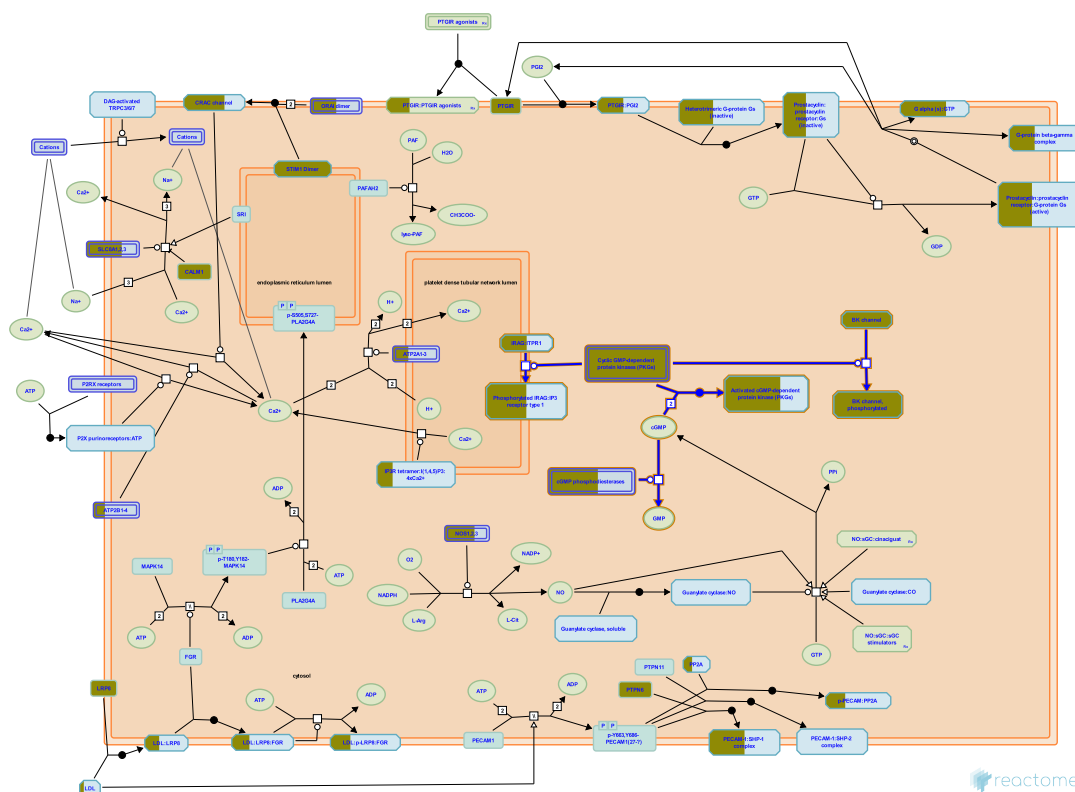
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Date	Action	Author
2025-11-15	Modified	Weiser JD

10 submitted entities found in this pathway, mapping to 12 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CDC27	Q15054	POLA1	P09884	POLA2	Q14181
POLD2	P49005	POLD4	Q9HCU8	PRIM1	P49642
RFC1	P35251	RFC3	P40937, P40938	RFC4	P35249, P35250
SPP1	P49642				

17. cGMP effects (R-HSA-418457)



Cellular compartments: cytosol.

Cyclic guanosine monophosphate (cGMP) is an important secondary messenger synthesized by guanylate cyclases. cGMP has effects on phosphodiesterases (PDE), ion-gated channels, and the cGMP-dependent protein kinases (cGK, Protein Kinase G or PKG). It is involved in regulation of several physiological functions including vasodilation, platelet aggregation and neurotransmission. Elevation of intracellular cGMP activates PKG (Haslam et al. 1999) which regulates several intracellular molecules and pathways including the vasodilator-stimulated phosphoprotein (VASP) (Halbrugge et al. 1990) and the ERK pathway (Hood and Granger 1998, Li et al. 2001). cGMP mediates nitric oxide (NO)-induced vascular smooth muscle relaxation (Furchgott and Vanhoutte 1989). Phosphodiesterase 5 (PDE5) hydrolyzes cGMP; the PDE5 inhibitor sildenafil (Viagra) increases intracellular cGMP and thereby can be used as a treatment for erectile dysfunction (Corbin and Francis 1999). The role of the cGMP and PKG in platelet activation was controversial as increases in platelet cGMP levels were observed in response to both platelet agonists (thrombin, ADP or collagen) and inhibitors (NO donors such as sodium nitroprusside), but it is currently accepted that PKG inhibits platelet activation (Haslam et al. 1999). Consistent with this, nitric oxide (NO) donors that inhibit platelet activation enhance intracellular cGMP (Haslam et al. 1999). cGMP also plays an important stimulatory role in GPIb-IX-mediated platelet activation. Platelet responses to cGMP have been proposed to be biphasic, consisting of an early stimulatory response that promotes platelet activation followed by a delayed platelet inhibition that serves to limit the size of platelet aggregates (Li et al 2003).

References

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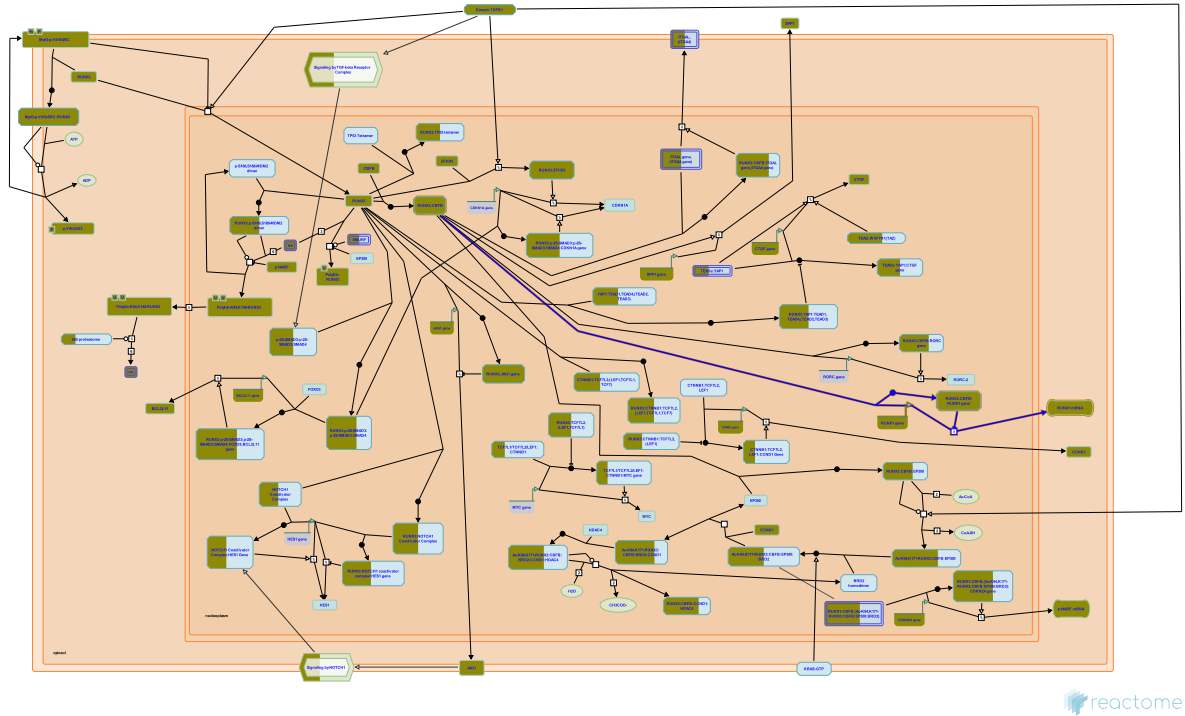
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Date	Action	Author
2009-04-22	Created	Jupe S
2009-06-03	Authored	Akkerman JW
2010-06-07	Edited	Jupe S
2010-06-07	Reviewed	Kunapuli SP
2025-11-15	Modified	Weiser JD

12 submitted entities found in this pathway, mapping to 14 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
IRAG1	Q9Y6F6	KCNMA1	Q12791	KCNMB1	Q16558
KCNMB2	Q9NPA1, Q9Y691	KCNMB3	Q9NPA1, Q9Y691	KCNMB4	Q86W47
PDE1C	Q01064	PDE2A	O00408	PDE5A	O76074
PDE9A	O76083	PRKG1	Q13976-1	PRKG2	Q13237

18. RUNX3 regulates RUNX1-mediated transcription (R-HSA-8951911)



RUNX3 binds to Runx response elements in the distal (P1) promoter of the RUNX1 gene, repressing RUNX1 transcription (Spender et al. 2005).

References

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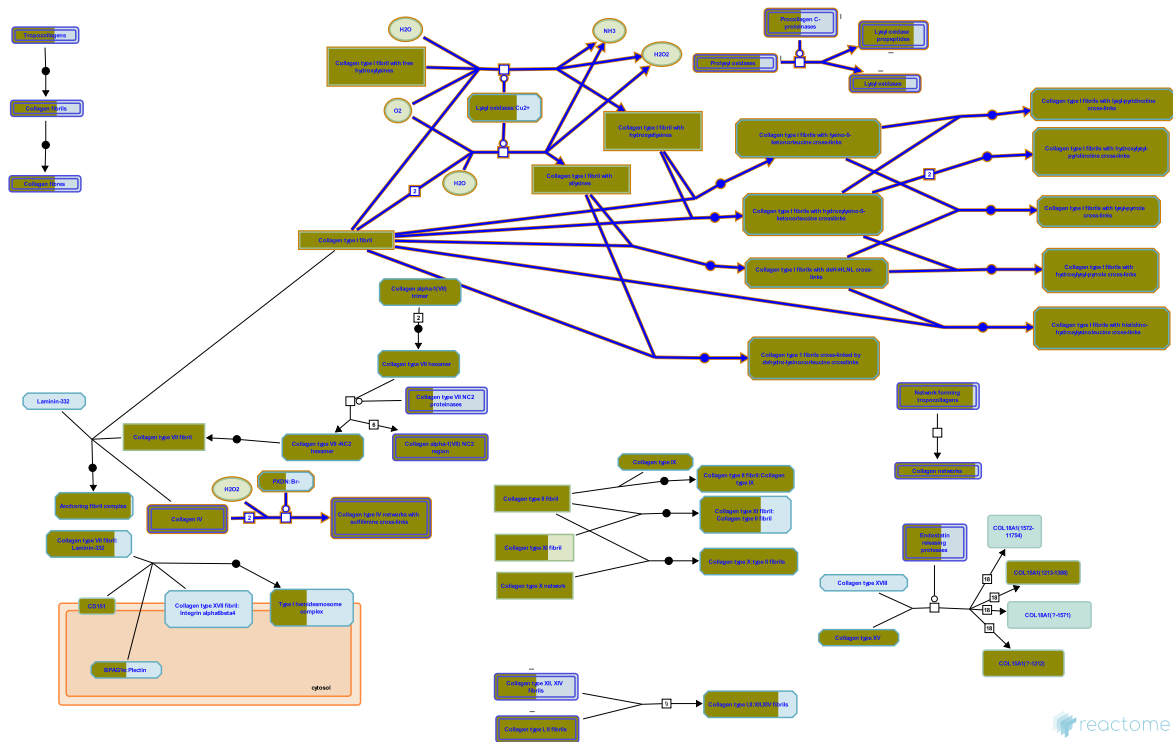
Date	Action	Author
2016-12-12	Created	Orlic-Milacic M
2016-12-13	Authored	Orlic-Milacic M
2017-01-31	Edited	Orlic-Milacic M
2017-01-31	Reviewed	Ito Y, Chuang LS
2023-03-08	Modified	Matthews L

3 submitted entities found in this pathway, mapping to 4 Reactome entities

Input	UniProt Id	Input	UniProt Id
CBFB	Q13951	RUNX3	Q13761

Input	Ensembl Id
RUNX1	ENSG00000159216, ENST00000344691

19. Crosslinking of collagen fibrils (R-HSA-2243919)



After removal of the N- and C-procollagen propeptides, fibrillar collagen molecules aggregate into microfibrillar arrays, stabilized by covalent intermolecular cross-links. These depend on the oxidative deamination of specific lysine or hydroxylysine residues in the telopeptide region by lysyl oxidase (LOX) with the subsequent spontaneous formation of covalent intermolecular cross-links (Pinnell & Martin 1968, Siegel et al. 1970, 1974, Maki 2009, Nishioka et al. 2012). Hydroxylysine is formed intracellularly by lysine hydroxylases (LH). There are different forms of LH responsible for hydroxylation of helical and telopeptide lysines (Royce & Barnes 1985, Knott et al. 1997, Takaluoma et al. 2007, Myllyla 2007). The chemistry of the cross-links formed depends on whether lysines or hydroxylysines are present in the telopeptides (Barnes et al. 1974), which depends on the proportion of collagen lysines post-translationally converted to hydroxylysine by LH. The lysine pathway predominates in adult skin, cornea and sclera while the hydroxylysine pathway occurs primarily in bone, cartilage, ligament, tendons, embryonic skin and most connective tissues (Eyre 1987, Eyre & Wu 2005, Eyre et al. 2008). Oxidative deamination of lysine or hydroxylysine residues by LOX generates the allysine and hydroxyallysine aldehydes respectively. These can spontaneously react with either another aldehyde to form an aldol condensation product (intramolecular cross-link), or with an unmodified lysine or hydroxylysine residue to form intermolecular cross-links.

The pathway of cross-linking is regulated primarily by the hydroxylation pattern of telopeptide and triple-helix domain lysine residues. When lysine residues are the source of aldehydes formed by lysyl oxidase the allysine cross-linking pathway leads to the formation of aldimine cross-links (Eyre & Wu 2005). These are stable at physiological conditions but readily cleaved at acid pH or elevated temperature. When hydroxylysine residues are the source of aldehydes formed by lysyl oxidase the hydroxyallysine cross-linking pathway leads to the formation of more stable ketoimine cross-links.

Telopeptide lysine residues can be converted by LOX to allysine, which can react with a helical hydroxylysine residue forming the lysine aldehyde aldimine cross-link dehydro hydroxylysino norleucine (deHHLNL) (Bailey & Peach 1968, Eyre et al. 2008). If the telopeptide residue is hydroxylysine, the hydroxyallysine formed by LOX can react with a helical hydroxylysine forming the Schiff base, which spontaneously undergoes an Amadori rearrangement resulting in the ketoimine cross link hydroxylysino 5 ketonorleucine (HLKNL). This stable cross-link is formed in tissues where telopeptide residues are predominantly hydroxylated, such as foetal bone and cartilage, accounting for the relative insolubility of collagen from these tissues (Bailey et al. 1998). In bone, telopeptide hydroxyallysines can react with the epsilon-amino group of a helical lysine (Robins & Bailey 1975). The resulting Schiff base undergoes Amadori rearrangement to form lysino-hydroxynorleucine (LHNL). An alternative mechanism of maturation of ketoimine cross-links has been reported in cartilage leading to the formation of arginoline (Eyre et al. 2010).

These divalent crosslinks greatly diminish as connective tissues mature, due to further spontaneous reactions (Bailey & Shimokomaki 1971, Robins & Bailey 1973) with neighbouring peptides that result in tri- and tetrafunctional cross-links. In mature tissues collagen cross-links are predominantly trivalent. The most common are pyridinoline or 3-hydroxypyridinium cross-links, namely hydroxylysyl-pyridinoline (HL-Pyr) and lysyl-pyridinoline (L-Pyr) cross-links (Eyre 1987, Ogawa et al. 1982, Fujimoto et al. 1978). HL-Pyr is formed from three hydroxylysine residues, HLKNL plus a further hydroxyallysine. It predominates in highly hydroxylated collagens such as type II collagen in cartilage. L-Pyr is formed from two hydroxylysines and a lysine, LKNL plus a further hydroxyallysine, found mostly in calcified tissues (Bailey et al. 1998). Trivalent collagen cross-links can also form as pyrroles, either Lysyl-Pyrrole (L-Pyrrole) or hydroxylysyl-pyrrole (HL-Pyrrole), respectively formed when LKNL or HLKNL react with allysine (Scott et al. 1981, Kuypers et al. 1992). A further three-way crosslink can form when DeH-HLNL reacts with histidine to form histidino-hydroxylysino norleucine (HHL), found in skin and cornea (Yamauchi et al. 1987, 1996). This can react with an additional lysine to form the tetrafunctional cross-link histidinohydroxymerodesmosine (Reiser et al. 1992, Yamauchi et al. 1996).

Another mechanism which could be involved in the cross-linking of collagen IV networks is the sulfilimine bond (Vanacore et al. 2009), catalyzed by peroxidasin, an enzyme found in basement membrane (Bhave 2012).

To improve clarity inter-chain cross-linking is represented here for Collagen type I only. Although the formation of each type of cross-link is represented here as an independent event, the partial and random nature of lysine hydroxylation and subsequent lysyl oxidation means that any combination of these cross-linking events could occur within the same collagen fibril .

References

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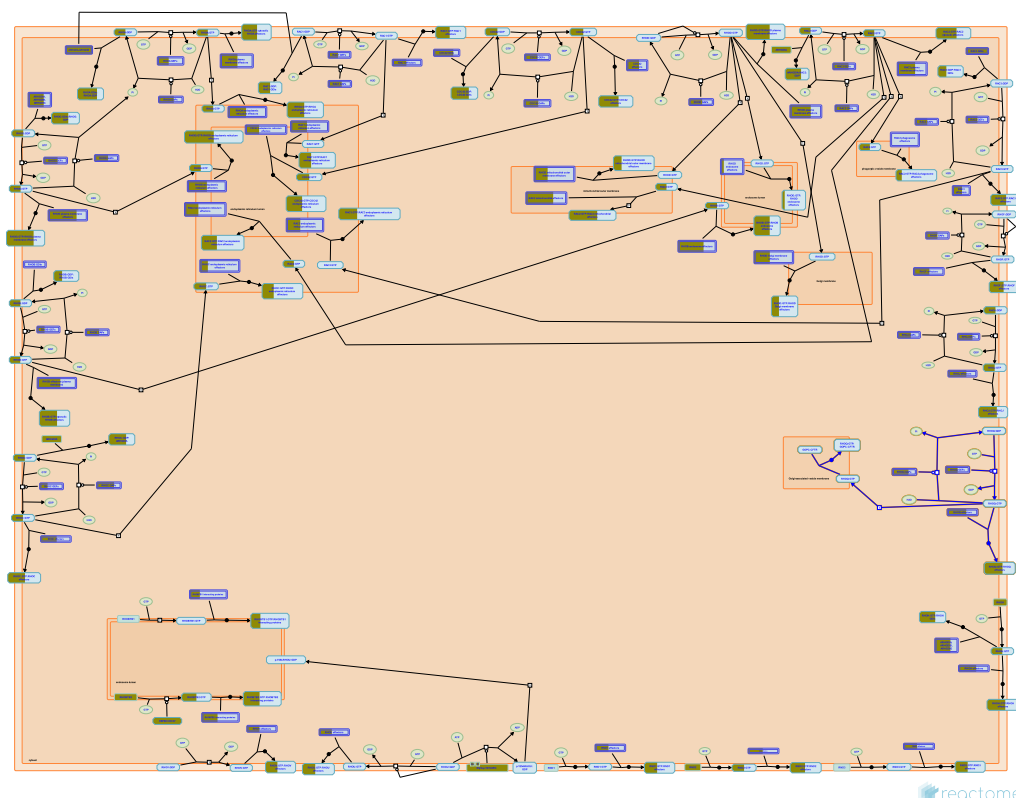
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2012-04-30	Authored	Jupe S
2012-05-09	Created	Jupe S
2012-10-08	Reviewed	Kalamajski S, Raleigh S

Date	Action	Author
2012-11-12	Edited	Jupe S
2012-11-19	Reviewed	Ricard-Blum S
2025-11-15	Modified	Weiser JD

16 submitted entities found in this pathway, mapping to 17 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
BMP1	P13497	COL1A1	P02452	COL1A2	P08123
COL4A1	P02462	COL4A2	P08572	COL4A3	Q01955
COL4A4	P53420	COL4A5	P29400	COL4A6	Q14031
COL7A1	P29400	LOX	P28300	LOXL1	P58215, Q08397
LOXL2	Q9Y4K0	LOXL3	P58215	PCOLCE	Q15113
PXDN	Q92626				

20. RHOQ GTPase cycle (R-HSA-9013406)



This pathway catalogues RHOQ (also known as TC10) guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), and RHOQ effectors. No GDP dissociation inhibitors (GDIs) have been shown to interact with RHOQ. Two GDIs, ARHGDIA (also known as Rho-GDI alpha) (Michaelson et al. 2001; Murphy et al. 2001) and ARHGDIB (also known as D4-GDI) (Zhang et al. 2009), were specifically shown not to interact with RHOQ. RHOQ, together with RHOJ (TCL), belongs to the CDC42 subfamily of RHO GTPases, with the three family members sharing 65-85% of homology (Murphy et al. 1999, Vignal et al. 2000, Donnelly et al. 2014; Shi et al. 2016). RHOQ is highly activated on exocytosing vesicles and recycling endosomes. RHOQ GTPase activity is necessary for fusion of vesicles with the plasma membrane (Kawase et al. 2006, Donnelly et al. 2014). RHOQ is required for insulin-stimulated glucose uptake (Kanzaki and Pessin 2003; Saltiel and Pessin 2003) and is involved in insulin signaling in adipocytes (Satoh 2014). While some studies reported RHOQ involvement in neurite outgrowth (Abe et al. 2003, Gonzalez-Billault et al. 2012) and regulation of membrane addition during axon formation (Dupraz et al. 2009, Gonzalez-Billault et al. 2012), other studies failed to identify RHOQ as a regulator of neurite outgrowth (Murphy et al. 1999, Murphy et al. 2001).

References

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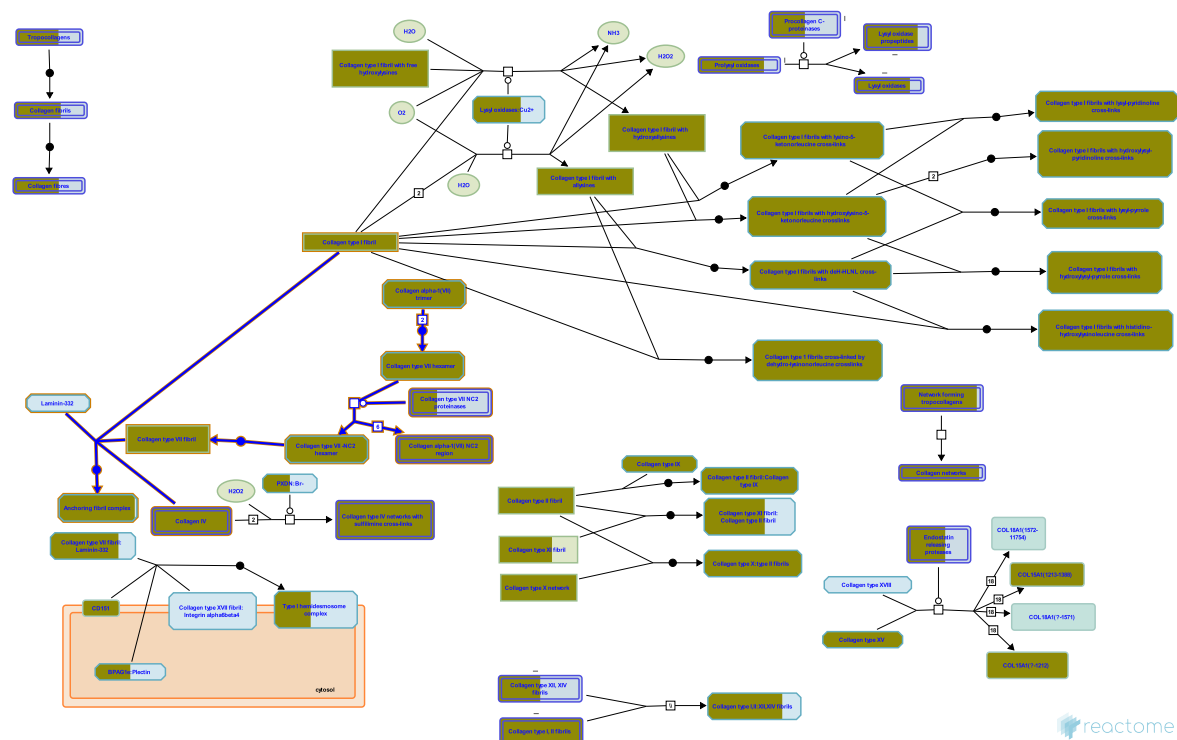
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Date	Action	Author
2017-07-25	Created	Orlic-Milacic M
2020-07-14	Authored	Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M
2021-05-17	Reviewed	Das ME
2021-06-10	Revised	Orlic-Milacic M
2021-06-10	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

35 submitted entities found in this pathway, mapping to 35 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ARHGAP1	Q07960	ARHGAP10	Q5T5U3	ARHGAP21	Q5T5U3
ARHGAP26	Q9UNA1	ARHGAP35	Q9NRY4	ARHGEF9	O43307
CAV1	Q03135	CDC42	P60953	CDC42BPB	Q9Y5S2
CDC42EP1	Q00587	CDC42EP2	O14613	CDC42EP3	Q9UKI2
CDC42EP4	Q9H3Q1	DEPDC1B	Q8WUY9	DIAPH3	Q9NSV4
DLC1	Q96QB1	FNBP1	Q96RU3	GIT2	Q14161
GJA1	P17302	IQGAP3	Q86VI3	LAMTOR1	Q6IAA8
MPP7	Q5T2T1	OBSCN	Q5VST9	OPHN1	O60890
PAK1	Q13153	PAK4	O96013	RAB7A	P51149
SLC1A5	Q15758	SLC4A7	Q9Y6M7	STEAP3	Q658P3
STOM	P27105	SYDE1	Q6ZW31	VAMP3	Q15836
VANGL1	Q8TAA9	WWP2	O00308		

21. Anchoring fibril formation (R-HSA-2214320)



Cellular compartments: extracellular region.

Collagen VII forms anchoring fibrils, composed of antiparallel dimers that connect the dermis to the epidermis (Bruckner-Tuderman 2009, Has & Kern 2010). During fibrillogenesis, the nascent type VII procollagen molecules dimerize in an antiparallel manner. The C-propeptide is then removed by Bone morphogenetic protein 1 (Rattenholl et al. 2002) and the processed antiparallel dimers laterally aggregate (Villone et al. 2008, Gordon & Hahn 2010).

References

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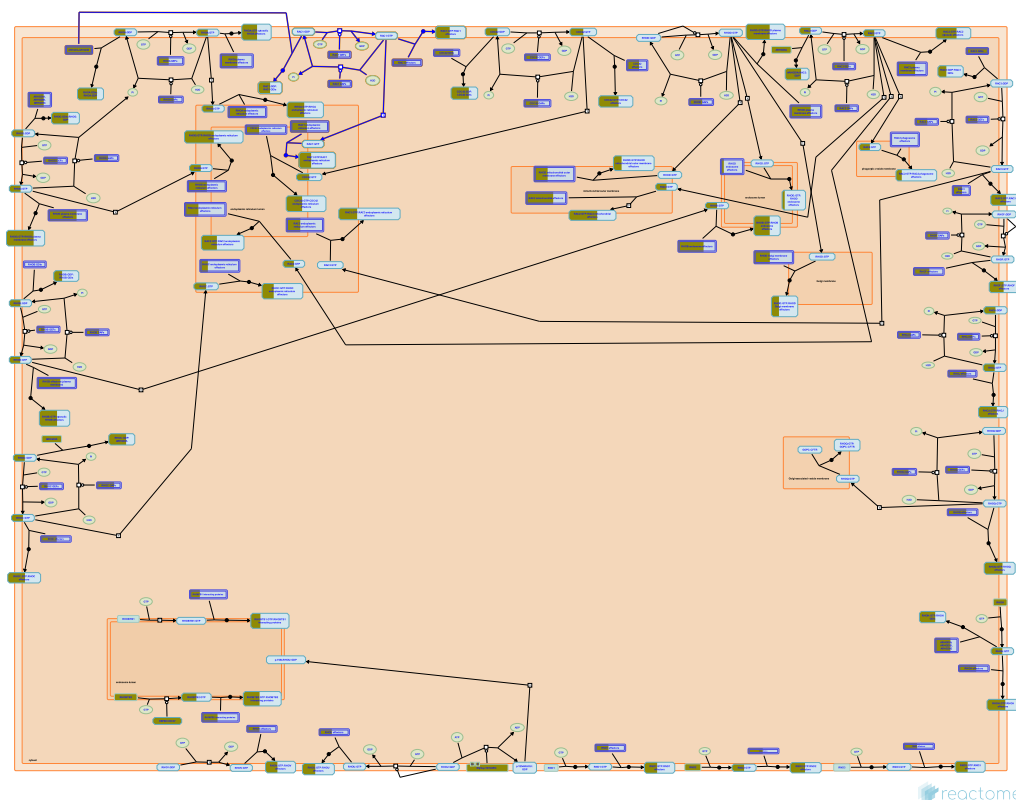
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2012-04-30	Created	Jupe S
2012-10-08	Reviewed	Kalamajski S, Raleigh S
2012-11-12	Edited	Jupe S
2012-11-19	Reviewed	Ricard-Blum S
2025-11-15	Modified	Weiser JD

10 submitted entities found in this pathway, mapping to 11 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
BMP1	P13497	COL1A1	P02452	COL1A2	P08123

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
COL4A1	P02462	COL4A2	P08572	COL4A3	Q01955
COL4A4	P53420	COL4A5	P29400	COL4A6	Q14031
COL7A1	P29400, Q02388				

22. RAC1 GTPase cycle (R-HSA-9013149)



This pathway catalogues RAC1 guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and RAC1 effectors (reviewed by Payapilli and Malliri 2018). RAC1 is one of the three best characterized RHO GTPases, the other two being RHOA and CDC42. RAC1 regulates the cytoskeleton and the production of reactive oxygen species (ROS) (Acevedo and Gonzalez-Billault 2018) and is involved in cell adhesion and cell migration (Marei and Malliri 2017). RAC1 is involved in neuronal development (de Curtis et al. 2014). In neurons, RAC1 activity is regulated by synaptic activation and RAC1-mediated changes in actin cytoskeleton are implicated in dendritic spine morphogenesis, which plays a role in memory formation and learning (Tajeda-Simon 2015; Costa et al. 2020). RAC1 is involved in metabolic regulation of pancreatic islet β -cells and in diabetes pathophysiology (Kowluru 2017; Kowluru et al. 2020). RAC1-mediated activation of NOX2 contributes to mitochondrial damage and the development of retinopathy in patients with diabetes (Sahajpal et al. 2019). RAC1 is important for exercise and contraction-stimulated glucose uptake in skeletal muscles (SyLOW et al. 2014). RAC1 plays an important role in the maintenance of intestinal barrier integrity under physiological conditions and during tissue repair after resolution of colitis. Toxins of many diarrhea-causing bacteria target RAC1 (Kotelevets and Chastre 2020). RAC1 is important for skin homeostasis and wound healing and is involved in the pathogenesis of psoriasis (Winge and Marinkovich 2019). RAC1 is essential to vascular homeostasis and chronically elevated RAC1 signaling contributes to vascular pathology (Marinkovic et al. 2015). RAC1 hyperactivation, mutation and copy-number gain are frequently observed in solid tumors (Zou et al. 2017; De et al. 2019; De et al. 2020; Cannon et al. 2020; Kotelevets and Chastre 2020).

References

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De P, Rozeboom BJ, Aske JC & Dey N (2020). Active RAC1 Promotes Tumorigenic Phenotypes and Therapy Resistance in Solid Tumors. *Cancers (Basel)*, 12. [🔗](#)

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Kotelevets L & Chastre E (2020). Rac1 Signaling: From Intestinal Homeostasis to Colorectal Cancer Metastasis. *Cancers (Basel)*, 12. [🔗](#)

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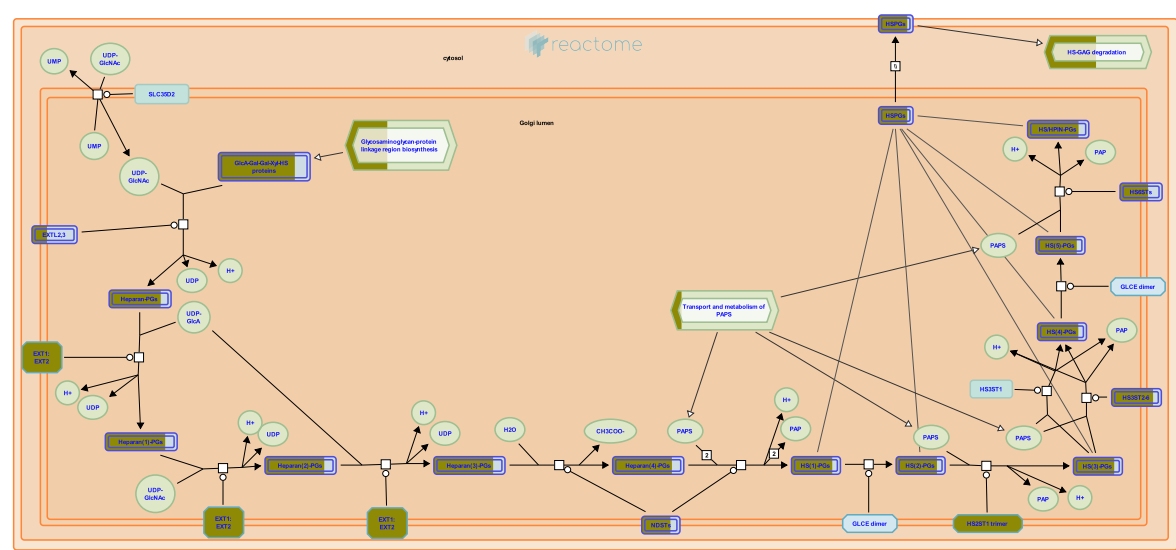
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2020-07-14	Authored	Rothfels K, Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

91 submitted entities found in this pathway, mapping to 95 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ABI1	Q8IZP0	ABL2	P42684	ABR	Q12979
AMIGO2	Q86SJ2	ARAP2	Q8WZ64	ARAP3	Q8WWN8
ARHGAP1	Q07960	ARHGAP10	A1A4S6, Q5T5U3	ARHGAP12	Q8IWW6
ARHGAP20	Q9P2F6	ARHGAP21	Q5T5U3	ARHGAP22	Q7Z5H3
ARHGAP23	Q9P227	ARHGAP24	Q8N264	ARHGAP25	P42331
ARHGAP26	Q9UNA1	ARHGAP30	Q7Z6I6	ARHGAP31	Q2M1Z3
ARHGAP35	Q9NRY4	ARHGAP42	A6NI28	ARHGAP44	Q17R89
ARHGAP9	Q9BRR9	ARHGDIA	P52565	ARHGDIB	P52566
ARHGEF15	O94989	ARHGEF25	Q86VW2	ARHGEF4	Q9NR80
ARHGEF5	Q12774	BAIAP2	Q9UQB8	CAV1	Q03135
CDC42	P60953	CDC42EP1	Q00587	CDC42EP4	Q9H3Q1
CHN2	P52757	CIT	O14578, O14578-3	CYFIP1	Q7L576
CYFIP2	Q96F07	DEF6	Q9H4E7	DEPDC1B	Q8WUY9
DIAPH3	Q9NSV4	DLC1	Q96QB1	DOCK10	Q96BY6
DOCK2	Q92608	DOCK3	Q8IZD9	DOCK4	Q8N1I0
DOCK5	Q9H7D0	DOCK8	Q8NF50	EMD	P50402
ERBIN	Q96RT1	ESYT1	Q9BSJ8	FAM13A	O94988
FAM13B	Q9NYF5	FARP1	Q9Y4F1	FGD5	Q6ZNL6
GARRE1	O15063	GIT2	Q14161	GNA13	Q14344
IQGAP2	Q13576	IQGAP3	Q86VI3	ITGB1	P05556
JAG1	P78504	LAMTOR1	Q6IAA8	MCF2	P10911
MPP7	Q5T2T1	MYO9B	Q13459	NAPA	Q9Y2A7
OPHN1	O60890	PAK1	Q13153, Q16512	PAK4	O96013
PAK5	Q9NQU5, Q9P286	PAK6	Q9NQU5	PLEKHG1	Q9ULL1
PREX2	Q70Z35	RAB7A	P51149	RACGAP1	Q9H0H5
RASGRF2	O14827	SLC1A5	Q15758	SRGAP1	Q7Z6B7

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
SWAP70	Q9UH65	SYDE2	Q5VT97	TAOK3	Q9H2K8
TRIO	O75962	VAMP3	Q15836	VANGL1	Q8TAA9
VAV1	P15498	VAV3	Q9UKW4	WASF2	Q9Y6W5
WIPF1	O43516	WIPF2	Q8TF74	WIPF3	A6NGB9
YKT6	O15498				

23. HS-GAG biosynthesis (R-HSA-2022928)



Heparan sulfate (HS) and heparin (sometimes collectively called HS-GAG) chains consist of the variably sulfated repeating disaccharide units heparan (Ido-GlcNAc) or heparosan (GlcA-GlcNAc). Heparin is exclusively made in mast cells whereas HS is made by virtually every type of cell in the body. As the chain length increases, the polysaccharides can undergo modifications such as epimerisation of glucuronic acid to iduronic acid and deacetylation and sulfation of GlcNAc to form sulfated glucosamine (Stringer & Gallagher 1997, Sasisekharan & Venkataraman 2000).

References

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Sasisekharan R & Venkataraman G (2000). Heparin and heparan sulfate: biosynthesis, structure and function. *Curr Opin Chem Biol*, 4, 626-31. [🔗](#)

Edit history

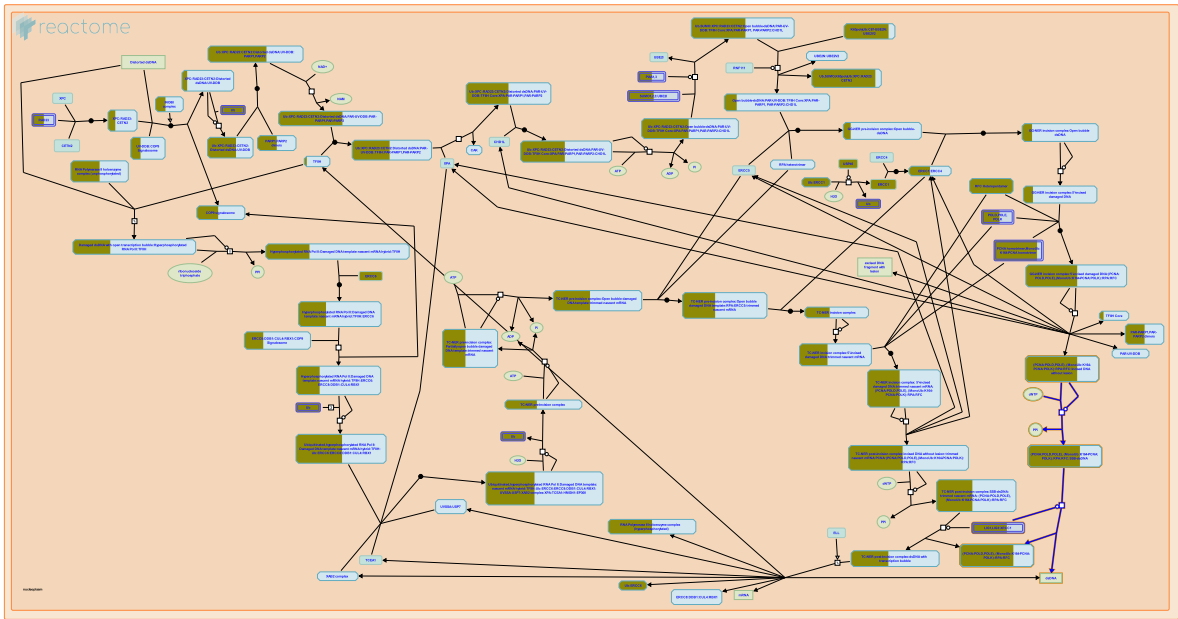
Date	Action	Author
2011-12-01	Edited	Jassal B
2011-12-01	Authored	Jassal B
2011-12-01	Created	Jassal B
2012-03-28	Reviewed	D'Eustachio P
2025-05-30	Reviewed	Matthews L
2025-06-02	Reviewed	Hill DP
2025-11-15	Modified	Weiser JD

22 submitted entities found in this pathway, mapping to 25 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
EXT1	Q16394	EXT2	Q93063	EXTL2	Q9UBQ6
GPC1	P35052	GPC2	Q8N158	GPC3	P51654
GPC4	O75487	GPC5	P78333	GPC6	Q9Y625
HS2ST1	Q7LGA3	HS3ST3A1	Q9Y663	HS3ST3B1	Q9Y662, Q9Y663

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
HS3ST4	Q9Y661	HS3ST5	Q8IZT8, Q96QI5	HS6ST2	Q96MM7
HS6ST3	Q8IZP7	HSPG2	P98160	NDST1	P52848
NDST3	O95803, Q9H3R1	SDC1	P18827	SDC2	P34741
SDC4	P31431				

24. Gap-filling DNA repair synthesis and ligation in GG-NER (R-HSA-5696397)



Cellular compartments: nucleoplasm.

Global genome nucleotide excision repair (GG-NER) is completed by DNA repair synthesis that fills the single stranded gap created after dual incision of the damaged DNA strand and excision of the ~27-30 bases long oligonucleotide that contains the lesion. DNA synthesis is performed by DNA polymerases epsilon or delta, or the Y family DNA polymerase kappa (POLK), which are loaded to the repair site after 5' incision (Staresincic et al. 2009, Ogi et al. 2010). DNA ligases LIG1 or LIG3 ligate the newly synthesized stretch of oligonucleotides to the incised DNA strand (Arakawa et al. 2012, Paul-Konietzko et al. 2015).

References

Staresincic L, Fagbemi AF, Enzlin JH, Gourdin AM, Wijgers N, Dunand-Sauthier I, ... Schärer OD (2009). Coordination of dual incision and repair synthesis in human nucleotide excision repair. EMBO J., 28, 1111-20. [🔗](#)

Ogi T, Limsirichaikul S, Overmeer RM, Volker M, Takenaka K, Cloney R, ... Lehmann AR (2010). Three DNA polymerases, recruited by different mechanisms, carry out NER repair synthesis in human cells. Mol. Cell, 37, 714-27. [🔗](#)

Arakawa H, Bednar T, Wang M, Paul K, Mladenov E, Bencsik-Theilen AA & Iliakis G (2012). Functional redundancy between DNA ligases I and III in DNA replication in vertebrate cells. Nucleic Acids Res, 40, 2599-610. [🔗](#)

Paul-Konietzko K, Thomale J, Arakawa H & Iliakis G (2015). DNA Ligases I and III Support Nucleotide Excision Repair in DT40 Cells with Similar Efficiency. Photochem Photobiol, 91, 1173-80. [🔗](#)

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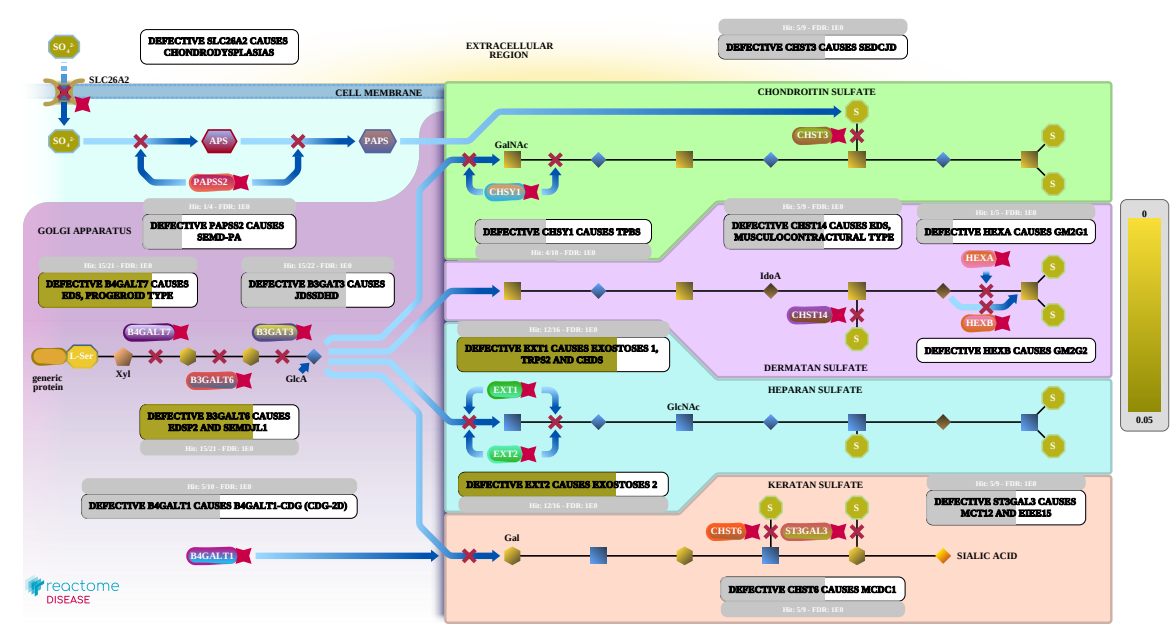
Date	Action	Author
2004-01-29	Authored	Hoeijmakers JH
2004-02-02	Authored	Gopinathrao G
2015-05-28	Created	Orlic-Milacic M

Date	Action	Author
2015-06-16	Revised	Orlic-Milacic M
2015-06-16	Edited	Orlic-Milacic M
2015-06-16	Authored	Orlic-Milacic M
2015-08-03	Reviewed	Fousteri M
2024-02-22	Edited	Orlic-Milacic M

13 submitted entities found in this pathway, mapping to 19 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CDC27	Q15054	LIG3	P49916	POLD2	P49005
POLD4	Q9HCU8	POLE2	P56282	POLE3	Q9NRF9
RFC1	P35251	RFC3	P40937, P40938	RFC4	P35249, P35250
UBA52	P62979, P62987	UBB	P0CG47, P62979, P62987	UBC	P0CG48, P62987
XRCC1	P18887				

25. Diseases associated with glycosaminoglycan metabolism (R-HSA-3560782)



Diseases: congenital disorder of glycosylation.

A number of genetic disorders are caused by mutations in the genes encoding glycosyltransferases and sulfotransferases, enzymes responsible for the synthesis of glycosaminoglycans (GAGs) as well as hexosaminidase degradation of GAGs (Mizumoto et al. 2013).

References

Mizumoto S, Ikegawa S & Sugahara K (2013). Human genetic disorders caused by mutations in genes encoding biosynthetic enzymes for sulfated glycosaminoglycans. J. Biol. Chem., 288, 10953-61. [🔗](#)

Edit history

Date	Action	Author
2013-05-21	Edited	Jassal B
2013-05-21	Authored	Jassal B
2013-05-21	Created	Jassal B
2014-07-09	Reviewed	Spillmann D
2023-10-12	Modified	Weiser JD

30 submitted entities found in this pathway, mapping to 30 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ACAN	P16112	B3GALT6	Q96L58	B3GAT3	O94766
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BGN	P21810	CHST14	Q8NCH0	CHST3	Q7LGC8
CHST6	Q9GZX3	CSPG4	Q6UVK1	DCN	P07585
EXT1	Q16394	EXT2	Q93063	FMOD	Q06828
GPC1	P35052	GPC2	Q8N158	GPC3	P51654
GPC4	O75487	GPC5	P78333	GPC6	Q9Y625

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
HEXA	P06865	HSPG2	P98160	LUM	P51884
OMD	Q99983	PAPSS2	O95340	SDC1	P18827
SDC2	P34741	SDC4	P31431	ST3GAL3	Q11203

6. Identifiers found

Below is a list of the input identifiers that have been found or mapped to an equivalent element in Reactome, classified by resource.

4014 of the submitted entities were found, mapping to 4850 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
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AASS	Q9UDR5	ABCA1	O95477	ABCB7	O75027
ABCB8	Q9NUT2	ABCC10	Q5T3U5	ABCC3	O15438
ABCC4	O15439	ABCC5	O15440	ABCC8	O60706, Q09428
ABCC9	O60706	ABCD4	O14678	ABCF2	Q9UG63
ABCG1	P45844	ABHD12	Q8N2K0	ABHD14B	Q96IU4
ABHD17B	Q5VST6	ABHD5	Q8WTS1	ABI1	Q8IZP0
ABL2	P42684	ABLIM1	O14639	ABLIM3	O94929
ABR	Q12979	ABRAXAS1	Q6UWZ7	ACAA1	P09110
ACACA	Q13085	ACAD11	Q709F0	ACAD8	Q9UKU7
ACAD9	Q9H845	ACADM	P11310	ACADVL	P49748
ACAN	P16112	ACBD7	Q8N6N7	ACD	Q96AP0
ACER2	Q5QJU3	ACKR3	P25106	ACLY	P53396
ACO2	Q99798	ACOT13	Q9NPJ3	ACOT2	P49753
ACOT7	O00154	ACOX1	O15254	ACOX2	Q99424
ACOX3	O15254	ACOXL	Q9NUZ1	ACP2	Q9BT43
ACP6	Q9NPH0	ACSL1	P33121, Q9ULC5	ACSL3	O95573
ACSL6	Q9UKU0	ACSM6	Q6P461	ACSS3	Q9H6R3
ACTA1	P68133	ACTA2	P62736	ACTB	P60709, P63261
ACTC1	P68032	ACTG1	P60709, P63261	ACTN1	P12814
ACTN2	P35609	ACTN4	O43707	ACTR10	Q9NZ32
ACTR1A	P61163	ACTR1B	P42025	ACTR2	P61160
ACTR3	P61158	ACVR1C	Q8NER5	ACVR2A	P27037
ACVR2B	Q13705	ADA2	Q9NZK5	ADAM12	O43184
ADAM15	Q13444	ADAM19	Q9H013	ADAM20	O43506
ADAM22	Q9P0K1	ADAM23	O75077	ADAM9	Q13443
ADAMTS1	Q9UHI8	ADAMTS16	Q8TE57	ADAMTS17	Q8TE56
ADAMTS19	Q8TE59	ADAMTS2	O95450	ADAMTS20	P59510
ADAMTS3	O15072	ADAMTS4	O75173	ADAMTS5	Q9UNA0
ADAMTS6	Q9UKP5	ADAMTS8	Q9UP79	ADAMTSL1	Q8N6G6
ADAMTSL4	Q6UY14	ADARB1	P78563	ADAT2	Q7Z6V5
ADCY1	Q08828	ADCY2	Q08462	ADCY3	O60266
ADCY6	O43306	ADCY7	P51828	ADCY8	P40145
ADCY9	O60503	ADCYAP1R1	P41586	ADD1	P35611
ADD2	P35612	ADGRE2	Q9UHX3	ADGRE5	P48960
ADGRG6	Q86SQ4	ADGRV1	Q8WYG9	ADH6	P00325, P28332
ADI1	Q9BV57	ADIRF	Q15847	ADM	P35318
ADM2	Q7Z4H4	ADPRS	Q9NX46	ADRA1B	P35368
ADRA1D	P25100	ADRA2A	P08913	ADRA2B	P18089
ADRB2	P07550	ADRM1	Q16186	ADSS2	P30520
AEBP2	Q6ZN18	AFDN	P55196-2	AGAP2	Q99490

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
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AGPAT5	Q9NUQ2	AGTPBP1	Q9UPW5	AGTR1	P30556
AHCYL2	Q96HN2	AHR	P35869	AHRR	A9YTD3
AIFM2	Q9BRQ8	AIM2	O14862	AIP	O00170
AK1	P00568, P54819	AK4	P27144	AK5	Q9Y6K8
AK7	Q96M32	AKAP1	Q92667	AKR1C1	Q04828
AKR1C2	P52895	AKR1E2	Q96JD6	AKR7A2	O43488, O95154
AKT1	P31749	AKT1S1	Q96B36	AKT3	Q9Y243
ALAD	P13716	ALAS1	P13196	ALCAM	Q13740
ALDH1A2	O94788	ALDH1A3	P47895	ALDH1B1	P30837
ALDH1L2	Q3SY69	ALDH3A2	P51648-1	ALDH3B1	P43353
ALDH5A1	P51649	ALDH6A1	Q02252	ALDOA	P04075
ALDOC	P09972	ALG1	Q9BT22	ALG3	Q92685
ALG6	Q9Y672	ALKAL2	Q6UX46	ALMS1	Q8TCU4
ALPK1	Q96QP1	ALPL	P05186	ALYREF	Q86V81
AMDHD2	Q9Y303	AMFR	Q9UKV5	AMIGO2	Q86SJ2
AMN	Q9BXJ7	AMPD2	Q01433	AMT	P48728
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ANAPC5	Q9UJX4	ANGPT4	Q9Y264	ANK2	Q01484
ANK3	Q12955	ANKH	Q9HCJ1	ANKLE2	Q86XL3
ANKRD26	Q9UPS8	ANLN	Q9NQW6	ANO1	Q5XXA6
ANO2	Q9NQ90	ANO9	A1A5B4	ANOS1	P23352
ANP32A	P39687	ANPEP	P15144	ANTXR1	Q9H6X2-1, Q9H6X2-2
ANTXR2	P58335-1, P58335-4	ANXA1	P04083	ANXA2	P07355
ANXA3	P12429	ANXA5	P08758	ANXA6	P08133
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AP1S1	P56377, P61966	AP1S2	P56377, Q96PC3	AP2A1	O95782
AP2A2	O94973	AP2M1	Q96CW1	AP2S1	P53680
AP4M1	O00189	APBA1	Q02410	APBA2	Q99767
APBB1IP	Q7Z5R6	APOB	P04114	APOBEC3B	Q9UH17
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APPL2	Q06481	AQP4	P55087	AR	P10275
ARAF	P10398	ARAP2	Q8WZ64	ARAP3	Q8WWN8
ARCN1	P48444	ARF1	P18085	ARF4	P18085
ARF6	P62330	ARFGAP1	Q9NP61	ARFGAP2	Q8N6H7
ARFGAP3	Q9NP61	ARFRP1	Q13795	ARHGAP1	Q07960
ARHGAP10	Q5T5U3	ARHGAP12	Q8IWW6	ARHGAP18	Q8N392
ARHGAP19	Q14CB8	ARHGAP20	Q9P2F6	ARHGAP21	Q5T5U3
ARHGAP22	Q7Z5H3	ARHGAP23	Q9P227	ARHGAP24	Q8N264
ARHGAP25	P42331	ARHGAP26	Q9UNA1	ARHGAP28	Q9P2N2
ARHGAP30	Q7Z6I6	ARHGAP31	Q2M1Z3	ARHGAP35	Q9NRY4
ARHGAP42	A6NI28	ARHGAP44	Q17R89	ARHGAP8	P85298
ARHGAP9	Q9BRR9	ARHGDIA	P52565	ARHGDIB	P52566
ARHGEF12	Q9NZN5	ARHGEF15	O94989	ARHGEF16	Q5VV41
ARHGEF17	Q96PE2	ARHGEF2	Q92974	ARHGEF25	Q86VW2, Q96PX9
ARHGEF26	Q96DR7	ARHGEF35	A5YM69	ARHGEF4	Q9NR80
ARHGEF5	Q12774	ARHGEF9	O43307	ARID2	Q68CP9
ARID4A	P29374	ARID4B	Q4LE39	ARID5B	Q14865

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ARIH1	Q9Y4X5	ARL2BP	Q9Y2Y0	ARL6IP5	O75915
ARMCX3	Q9UH62	ARPC1A	Q92747	ARPC1B	O15143
ARPC2	O15144	ARPC4	P59998	ARPC5	O15511
ARRB1	P49407	ARRB2	P32121	ARSA	O43681
ARSB	P15848	ARSI	Q5FYB1	ARSJ	Q5FYB0
ASAP1	Q9ULH1	ASB1	Q9Y576	ASB6	Q9NWX5
ASB8	Q9H765	ASCC2	Q9H1I8	ASL	P04424
ASPH	Q12797	ASRGL1	Q7L266	ASS1	P00966
ASXL1	Q8IXJ9	ATAD1	Q8NBU5	ATAD3A	Q5T9A4
ATAD3B	Q5T9A4	ATAT1	Q5SQI0	ATF3	P18847
ATF5	Q9Y2D1	ATF6B	Q99941	ATF7IP	Q6VMQ6
ATG10	Q9H0Y0	ATG101	Q9BSB4	ATG13	O75143
ATG14	Q6ZNE5	ATG16L2	Q676U5, Q8NAA4	ATG4A	Q8WYN0, Q9Y4P1
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ATP10B	O94823	ATP10D	Q9P241	ATP11C	Q8NB49
ATP12A	P54707	ATP13A1	Q9HD20	ATP1A2	P50993, Q13733
ATP1A3	P13637	ATP1B2	P14415	ATP2A3	Q93084
ATP2B4	P23634	ATP2C2	O75185	ATP5F1D	P30049
ATP5MC1	P05496, P48201, Q06055	ATP6AP1	Q15904	ATP6V0B	Q99437
ATP6V0C	P27449	ATP6V0D1	P61421	ATP6V0E1	O15342
ATP6V1E1	P36543	ATP6V1F	Q16864	ATP6V1H	Q9UI12
ATP7B	P35670	ATP8A1	Q9Y2Q0	ATP8A2	Q9NTI2
ATP8B1	O43520	ATP8B2	P98198	ATP9A	O75110
ATRIP	Q8WXE1	ATRNL1	Q9Y2I2	ATRX	P46100
ATXN3	P54252, Q9H3M9	AUP1	Q9Y679	AURKA	O14965
AXIN2	Q9Y2T1	AXL	P30530	AZIN1	O14977
B2M	P61769	B3GALT1	Q9Y5Z6	B3GALT5	Q9Y2C3
B3GALT6	Q96L58	B3GAT1	Q9P2W7	B3GAT3	O94766
B3GLCT	Q6Y288	B3GNT2	Q9NY97	B3GNT3	Q9Y2A9
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B4GALNT1	Q00973	B4GALT1	P15291	B4GALT2	O60909
B4GALT4	O60513	B4GALT6	Q9UBX8	B4GALT7	Q9UBV7
B9D2	Q9BPU9	BACE1	P56817	BAD	Q92934
BAG2	O95816	BAG3	O95817	BAIAP2	Q9UQB8
BANF1	O75531	BAP1	Q99496	BASP1	P80723
BBS4	Q96RK4	BCAN	Q96GW7	BCAP31	P51572
BCAT1	P54687	BCAT2	O15382	BCHE	P06276
BCL11A	Q9H165	BCL11B	Q9C0K0	BCL2L11	O43521
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BCL9	O00512, Q9Y3Y4	BCL9L	Q86UU0	BCS1L	Q9Y276
BDH2	Q9BUT1	BDKRB1	P46663	BDNF	P23560
BDP1	A6H8Y1	BEST1	O76090	BEST4	O76090, Q8NFO0
BET1L	Q9NYM9	BEX3	Q00994	BGN	P21810
BHLHE40	O14503	BHMT	Q93088	BICD2	Q8TD16
BIRC2	Q13489, Q13490	BIRC5	O15392	BLM	P54132
BLMH	Q13867	BLOC1S3	Q6QNY0	BLVRB	P30043
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BRCA2	P51587	BRD1	O95696	BRD7	Q9NPI1
BRF2	Q9HAW0	BRI3	O95415	BRMS1	Q9HCU9
BRPF1	P55201	BRWD1	Q9NSI6	BRWD3	Q9NSI6
BSN	Q9UPA5	BST1	Q10588	BST2	Q10589
BTBD6	Q96KE9	BTG1	P62324	BTN2A1	Q7KYR7
BUB1B	O60566	C19orf84	I3L1E1	C1GALT1	Q9NS00
C1R	P00738	C1S	P09871	C2CD5	Q86YS7
C5	P01031	C9orf78	Q9NZ63	CA12	O43570
CA13	Q8N1Q1	CA14	Q9ULX7	CA2	P00918
CABP1	Q15084	CACNA1B	Q00975	CACNA1C	Q13936
CACNA1D	Q01668	CACNA1E	Q15878	CACNA1H	O95180
CACNA2D2	Q9NY47	CACNA2D3	Q8IZS8	CACNB1	Q02641
CACNB2	Q08289	CACNB4	O00305	CACNG7	P62955
CACNG8	Q8WXS5	CADM1	Q9BY67	CADM2	Q8N3J6
CADM3	Q8N126	CALB1	P05937	CALCB	P01258, P10092
CALCR	P30988	CALD1	Q05682	CALM3	P0DP23
CALR	P27797	CALU	O43852	CAMK2D	Q13557
CAMK4	Q16566	CAMKK1	Q8N5S9	CAMKK2	Q96RR4
CAMKMT	Q7Z624	CAND1	Q86VP6	CANT1	Q8WVQ1
CAP1	Q01518	CAP2	P40123	CAPG	Q9BPX3
CAPN1	P07384	CAPN11	Q9UMQ6	CAPN12	Q6ZSI9
CAPN13	Q6MZZ7	CAPN15	O75808	CAPN2	P17655
CAPN3	P20807	CAPN6	Q9Y6Q1	CAPNS1	P04632, Q96L46
CAPZB	P47756	CARD11	Q9BXL7	CARMIL1	Q5VZK9
CARS2	Q9HA77	CASK	O14936	CASP4	P49662
CASP7	P55210	CASP8AP2	Q9UKL3	CASQ1	P31415
CAST	O15446	CAV1	Q03135	CAV2	P51636
CAVIN1	Q6NZI2	CBFB	Q13951	CBR1	P16152
CBR3	P16152	CBR4	Q8N4T8	CBS	P35520
CBX2	Q14781	CBX3	Q13185	CBX6	O95503
CC2D1B	Q13415	CCDC88A	Q3V6T2	CCDC88C	Q9P219
CCKBR	P32239	CCL2	P13500	CCN1	O00622
CCN2	P29279	CCNA2	P20248	CCNC	P24863
CCND1	P24385	CCND2	P30279	CCND3	P30281
CCNE1	P24864	CCNF	P41002	CCNG1	P51959
CCNL1	P49736	CCNT1	O60563	CCNT2	O60583
CCR9	P51686	CCS	O14618	CD101	Q93033
CD109	Q6YHK3	CD151	P48509	CD200	P41217
CD24	P25063	CD247	P20963-1	CD274	Q9NZQ7
CD2AP	Q9Y5K6	CD2BP2	O95400	CD320	Q9NPF0
CD40	P25942-1	CD44	P16070	CD53	P19397
CD55	P08174	CD59	P13987	CD63	P08962
CD72	P21854	CD74	P04233	CD81	P60033
CD99	P14209	CD99L2	Q8TCZ2	CDA	P32320
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CDC25A	P30304	CDC25B	P30304	CDC27	Q15054

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COL28A1	Q2UY09	COL2A1	P02458	COL3A1	P02461
COL4A1	P02462, Q14055	COL4A2	P08572	COL4A3	Q01955
COL4A4	P53420	COL4A5	P29400	COL4A6	Q14031
COL5A1	P20908	COL5A2	P05997	COL6A1	P12109
COL6A2	P12110	COL6A3	A8TX70, P12111	COL7A1	P29400, Q02388
COL8A1	P27658	COL8A2	P25067	COL9A1	P20849
COL9A3	Q14050	COLEC12	Q5KU26	COLGALT1	Q8NBJS
COMMD3	Q9UBI1	COMMD4	Q9H0A8	COMMD5	Q9GZQ3
COMMD9	Q9P000	COMP	P49747	COMT	P21964
COPA	P53621	COPB1	P53618	COPE	O14579
COPG1	Q9Y678	COPG2	Q9UBF2	COPS6	Q7L5N1
COPS7A	Q9H9Q2, Q9UBW8	COPZ1	P61923	COPZ2	Q9P299
COQ10A	Q96MF6	COQ10B	Q9H8M1	COQ4	Q9Y3A0
COTL1	Q14019	COX14	Q96I36	COX4I1	P13073
COX5B	P10606	COX6A1	P12074	COX6B1	P14854
COX7A1	P24310	COX7C	P15954	COX8A	P10176
CPN1	Q99829	CPNE1	Q99829	CPPED1	Q9BRF8
CPSF6	Q16630-1, Q16630-2	CPT1A	P50416	CPTP	Q5TA50
CR1L	P17927	CR2	P20023	CRABP1	P29762
CRABP2	P29373	CRACR2A	Q9BSW2	CRAT	P43155
CRB3	Q9BUF7	CREB1	P16220-1	CREB3	O43889
CREB3L1	Q96BA8	CREB3L2	Q70SY1	CREB3L4	O43889, Q8TEY5
CRHR1	P34998	CRIPTO	P13385	CRIPTO3	P51864
CRISPLD2	Q9H0B8	CRLF1	O75462	CRTAP	O75718
CRTC2	Q53ET0	CRTC3	Q6UUUV7	CRY2	Q49AN0
CRYAB	P02511	CRYM	Q14894	CSF1	P09603
CSGALNACT1	Q8TDX6	CSGALNACT2	Q8N6G5	CSK	P41240
CSNK1D	P48730	CSNK2A3	Q8NEV1	CSPG4	Q6UVK1
CSRPI	P21291	CST3	P01034	CST6	P04080
CTBP1	Q13363	CTDNEP1	O95476	CTH	P32929
CTNNA2	P26232	CTPS2	Q9NRF8	CTSA	P10619
CTSB	P07858	CTSD	P07339	CTSH	P09668
CTSK	P43235	CTSL	P07711	CTSV	P07711
CTSZ	Q9UBR2	CTTN	Q14247	CUBN	O60494
CWC22	Q9HCG8	CWC27	Q6UX04	CWF19L2	Q2TBE0
CXADR	P78310	CXCL1	P09341	CXCL12	P48061
CXCL5	P42830, P80162	CXCL8	P10145	CXCR4	P61073
CXCR5	P32302	CXXC4	Q9H2H0	CYB5A	P00167-1
CYB5R1	Q9UHQ9	CYB5R3	P00387	CYBRD1	Q53TN4
CYC1	P08574	CYFIP1	Q7L576	CYFIP2	Q96F07
CYLD	Q9NQC7	CYP1B1	Q16678	CYP24A1	Q07973
CYP26A1	O43174	CYP27B1	O15528	CYP2B6	P20813
CYP2J2	P51589	CYP2R1	Q6VVX0	CYP2S1	Q96SQ9
CYP39A1	Q9NYL5	CYRIB	Q9NUQ9	CYTH3	O43739
DAAM1	Q9Y4D1	DAB2	P98082	DAD1	P61803
DAP	P46527	DAPK1	P53355	DBN1	Q16643
DBNL	Q9UJU6	DCAF10	Q5QP82	DCAF13	Q9NV06

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
DCAF17	Q5H9S7	DCAF6	Q58WW2	DCK	P27707
DCLRE1A	Q6PJP8	DCLRE1B	Q9H816	DCN	P07585
DCP2	Q8IU60	DCTD	P32321	DCTN1	Q14203-2
DCTN2	Q13561	DCTN3	O75935	DCUN1D1	Q96GG9
DCX	O43602	DDA1	Q9BW61	DDAH1	O94760
DDHD1	Q8NEL9	DDOST	P39656	DDR1	Q08345
DDR2	Q16832	DDRGK1	Q96HY6	DDX39A	O00148
DDX41	Q7L591	DDX49	Q9Y6V7	DEF6	Q9H4E7
DEGS1	O15121	DEK	P51580	DENND1C	Q8IV53
DENND2A	Q9ULE3	DENND2B	P78524	DENND4A	Q7Z401
DENND5A	Q6IQ26	DEPDC1B	Q8WUY9	DERL1	Q9GZP9
DES	P17661	DGAT1	O75907	DGKB	Q9Y6T7
DGKI	O75912	DGKK	Q5KSL6	DHPS	P49366
DHRS3	O75911	DHRS7B	Q6IAN0	DHTKD1	Q96HY7
DHX16	O60231	DHX36	Q9H2U1	DHX38	Q92620
DIAPH3	Q9NSV4	DICER1	Q9UPY3	DIO2	Q92813
DKK1	O94907	DLC1	Q96QB1	DLG2	Q15700
DLG3	Q92796	DLG5	Q8TDM6	DLGAP1	O14490
DLGAP2	Q9P1A6	DLGAP3	O95886	DLGAP4	Q9Y2H0
DLGAP5	Q15398	DLST	P36957	DMAC2	Q7Z7M8
DMKN	Q6E0U4	DNA2	P51530	DNAJA4	Q8WW22
DNAJB1	P25685	DNAJC2	Q99543	DNAJC3	Q13217
DNAJC5	Q9H3Z4	DNASE2	O00115	DNER	Q8NFT8
DNM2	P50570	DNMBP	Q6XZF7	DNMT3A	Q9Y6K1
DNMT3B	Q9UBC3	DNTTIP1	Q9H147	DOCK10	Q96BY6
DOCK2	Q92608	DOCK3	Q8IZD9	DOCK4	Q8N1I0
DOCK5	Q9H7D0	DOCK8	Q8NF50	DOK1	Q93070
DOK4	Q8TEW6	DOK5	Q9P104	DPEP1	P16444
DPF3	Q92784	DPH1	Q9UMX0	DPH6	Q7L8W6
DPM2	O94777	DPM3	Q9P2X0	DPP4	P27487
DPPA2	Q7Z7J5	DPPA4	Q7L190	DPYD	Q12882
DPYSL3	Q14195	DPYSL5	Q9BPU6	DRAP1	Q14919
DRG2	P55039	DSC2	Q02487	DSC3	Q08554
DSCC1	Q9BVC3	DSG2	Q14126	DTNA	Q9Y4J8
DTNB	O60941	DTNBP1	Q96EV8	DTX1	Q86Y01
DTX4	Q9Y2E6	DTYMK	P23919	DUSP1	P28562
DUSP16	Q9BY84	DUSP3	P51452	DUSP6	Q16828
DUSP7	Q16829	DVL1	O14640	DYNC1H1	Q14204
DYNC1I2	Q13409	DYNC2H1	Q8NCM8	DYNC2I2	Q96EX3
DYNLRB1	Q9NP97	DYRK3	Q92630	DZIP3	Q86Y13
E2F1	O00716, Q01094	E2F4	Q16254	E2F5	Q15329
EAF2	Q96CJ1	EARS2	Q5JPH6	EBF1	Q9UH73
EBF2	Q9HAK2	EBNA1BP2	Q99848	ECE1	P42892
ECHS1	P30084	ECI1	P42126	ECM1	Q16610
EDA	Q92838, Q92838-3	EDAR	Q9UNE0	EDARADD	Q8WWZ3
EDC4	Q6P2E9	EDEM1	Q92611	EDIL3	O43854
EDNRB	P24530	EEF1D	P29692	EEF2KMT	Q96G04
EEFSEC	P57772	EEPD1	Q7L9B9	EFEMP1	Q12805
EFEMP2	O95967, Q12805	EFHD2	Q96C19	EFNA2	O43921

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EFNA5	P52803	EGF	P01133	EGFR	P00533, P04626
EGLN1	Q9GZT9	EGLN3	Q9H6Z9	EGR2	P11161
EHD1	Q9H4M9	EHD2	Q9NZN3, Q9NZN4	EHD3	Q6NVY1
EIF2AK4	Q9P2K8	EIF3B	P55884	EIF3C	Q99613
EIF3E	P60228	EIF3G	O75821	EIF3K	Q9UBQ5
EIF4A2	Q14240	EIF4B	P23588	EIF4E3	Q8N5X7
EIF4EBP1	Q13541	EIF4G1	Q04637	EIF5A	P63241, Q9GZV4
EIF5A2	Q9GZV4	EIF5B	O60841	ELAVL2	Q12926
ELF1	O43921	ELF3	P78545	ELK1	P19419
ELL2	O00472	ELMO1	Q92556	ELMO2	Q96JJ3
ELN	P15502	ELOB	Q15370	ELOVL1	Q9BW60
ELOVL6	Q9H5J4	ELOVL7	A1L3X0	ELP2	Q6IA86
EMB	Q6PCB8	EMD	P50402	EME1	A4GXA9, Q96AY2
EMILIN1	Q9Y6C2	EMILIN2	Q9BXX0	EML4	Q9HC35
EMX2	Q04743	ENDOD1	O94919	ENO1	P06733
ENO2	P09104	ENO4	A6NNW6	ENPP2	Q13822
ENPP4	Q9Y6X5	ENSA	O43768	ENTPD1	P49961
ENTPD5	O75356	ENTPD6	O75354	EPAS1	Q99814
EPB41	O43491, P11171	EPB41L2	O43491	EPB41L3	Q9Y2J2
EPB41L5	Q9HCM4	EPCAM	P16422	EPHA4	P54764
EPHA6	Q9UF33	EPHA7	Q15375	EPHA8	P29322
EPHX2	P34913	EPN1	Q9Y6I3	EPPIN	O95925
EPPIN-WFDC6	O95925	EPS8	Q12929	ERAL1	O75616
ERAP1	Q9NZ08	ERBB3	P21860-1	ERBB4	Q15303-1, Q15303-2
ERBIN	Q96RT1	ERC1	Q96FL8	ERCC1	P07992
ERCC2	P18074	ERCC6	Q03468	ERF	P50548
ERLEC1	Q96DZ1	ERLIN1	O75477	ESCO1	Q5FWF5
ESD	P10768	ESPL1	Q14674	ESR2	Q92731
ESRP1	Q6NXG1	ESRP2	Q9H6T0	ESRRG	P62508
ESYT1	Q9BSJ8	ETFB	P38117	ETHE1	O95571
ETNK1	Q9HBU6	ETNPPL	Q8TBG4	ETS1	P14921
EVA1A	Q9H8M9	EVC	P57679	EVC2	Q86UK5
EXOC3	O60645	EXOC7	Q9UPT5	EXOSC4	Q9NPD3
EXT1	Q16394	EXT2	Q93063	EXTL2	Q9UBQ6
EYA1	Q99502	EYA2	O00167	EZR	P15311
F10	P00742	F11	P03951	F11R	Q9Y624
F2RL1	P25116	F3	P13726	FAAH2	Q6GMR7
FABP3	P05413, Q0Z7S8	FABP5	Q01469	FABP6	P51161
FABP7	O15540	FADD	Q13158	FAH	P16930
FAHD1	Q6P587	FAM120B	Q96EK7	FAM135A	Q9P2D6
FAM13A	O94988	FAM13B	Q9NYF5	FAM169A	Q9Y6X4
FAM20B	O75063	FAM20C	Q8IXL6	FAM50A	Q14320
FAM83B	Q5T0W9	FANCD2	Q9BXW9	FANCL	Q9NW38
FAP	P27487	FAR2	Q96K12	FARP1	Q9Y4F1
FARSA	Q9Y285	FAS	P25445	FASTK	Q14296-4
FAU	P62861	FBLIM1	Q8WUP2	FBLN2	P98095
FBLN5	Q12805, Q9UBX5	FBN1	P35555	FBN3	Q75N90
FBP1	P09467	FBXL14	Q8N1E6	FBXL16	Q8N461
FBXL18	Q96ME1	FBXL3	Q9UKT7	FBXL4	Q9UKA2

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FBXL7	Q9UJT9	FBXO17	Q96EF6	FBXO2	Q96EF6, Q9H4M3, Q9UK22
FBXO21	O94952	FBXO22	Q8NEZ5	FBXO31	Q5XUX0
FBXO41	Q8TF61	FBXW12	Q6X9E4	FBXW5	Q969U6
FDFT1	P37268	FDX2	P10109, Q6P4F2	FEM1A	Q9BSK4
FEM1B	Q9UK73	FETUB	P04196	FGD4	Q96M96
FGD5	Q6ZNL6	FGF1	P05230	FGF11	Q92914
FGF12	P61328	FGF13	Q92913	FGF14	Q92915
FGF19	O95750	FGF2	P09038	FGF5	P12034-1
FGFBP3	Q8TAT2	FGFR2	P21802, P21802-17	FGFR4	P22455
FGFRL1	Q8N441	FGL2	Q14314	FHL2	Q14192
FHOD3	Q8IVF7	FIG4	Q92562	FILIP1	Q7Z7B0
FIS1	Q9Y3D6	FITM2	A5D6W6, Q8N6M3	FKBP1A	P62942
FKBP1B	P68106	FKBP5	Q13451	FKBP9	O95302
FKRP	Q9H9S5	FLI1	Q01543	FLNA	P21333
FLNB	O75369	FLNC	Q14315	FLOT1	O75955
FLRT2	O43155	FLT1	P17948	FLT3	P36888
FLT4	P35916	FLVCR1	Q9Y5Y0-1, Q9Y5Y0-2	FMN1	Q969G6
FMO2	Q99518	FMOD	Q06828	FN1	P02751
FN3KRP	Q9H479, Q9HA64	FNBP1	Q96RU3	FNBP1L	Q5T0N5
FNTB	P49356	FOLR3	P41439	FOSL1	P15407
FOXA3	P55318	FOXC1	Q12948	FOXF1	Q12946
FOXH1	O75593	FOXJ2	Q9P0K8	FOXK1	P85037
FOXK2	Q01167	FOXM1	Q08050	FOXO1	Q12778
FOXP1	Q9H334-8	FOXP4	Q9BZS1	FPGS	Q05932-1, Q05932-2
FRAT2	O75474	FST	P19883	FSTL1	Q12841
FSTL3	O95633	FTH1	P02794	FTL	P02792
FTSJ1	Q9UET6	FURIN	P09958	FUT1	P19526
FUT2	Q10981	FUT8	Q9BYC5	FUT9	Q9Y231
FXR1	P51114	FXYD5	Q14802	FXYD6	Q9H0Q3
FZD1	Q9UP38	FZD3	O00144, Q9NPG1	FZD5	Q13467
FZD8	Q9H461	FZR1	Q9UM11	G0S2	P27469
G3BP2	Q9UN86	G6PC3	Q9BUM1	G6PD	P11413
GAB1	Q13480	GABBR1	Q9UBS5	GABBR2	O75899
GABPB2	Q06547-1, Q06547-2, Q06547-3, Q06547-4	GABRA3	P34903	GABRB3	P28472
GABRG2	P18507	GABRQ	Q9UN88	GADD45A	P24522
GAK	O14976	GAL3ST1	Q99999	GALE	Q14376
GALNS	P34059	GALNT1	Q10472	GALNT10	Q86SR1
GALNT12	Q8IXK2	GALNT13	Q8IUC8	GALNT14	Q96FL9
GALNT15	Q7Z4T8, Q8N3T1	GALNT17	Q49A17, Q6IS24	GALNT18	Q6P9A2
GALNT2	Q10471	GALNT3	Q14435	GALNT4	Q8N4A0
GALNT5	Q7Z7M9	GALNT6	Q14435, Q8NCL4	GALNT7	Q86SF2
GALNT8	Q9NY28	GALR1	P47211	GAN	Q9H2C0
GANAB	Q14697	GAPDH	P04406	GARRE1	O15063
GART	P22102	GAS2	O43903	GAS6	Q14393
GATA2	P23769	GATA6	Q92908	GATAD2A	Q86YP4
GATM	P50440	GBA1	P04062	GBE1	Q04446
GBF1	Q92538	GBP1	P32455, Q9H0R5	GBP2	P32456, Q9H0R5

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GBP3	Q9H0R5	GCA	P28676	GCC2	Q8IWJ2
GCFC2	P16383	GCH1	P30793	GCLC	P48506
GCLM	P48507	GCN1	Q92616	GCNT1	Q02742
GCNT4	Q9P109	GCSH	P23434	GDA	P16260
GDAP1	Q8TB36	GDI1	P50395	GDPD1	Q8N9F7
GEM	P55040	GEMIN2	O14893	GET3	O43681
GFER	P55789	GFPT2	O94808	GFRA1	P56159
GFRA2	O00451	GFUS	Q13630	GGCT	O75223
GIPC1	O14908	GIT2	Q14161	GJA1	P17302
GJA5	P36382	GLB1L2	Q8IW92, Q8NCI6	GLB1L3	Q8NCI6
GLDC	P23378	GLIPR1	P48060	GLMN	Q92990
GLO1	Q04760	GLP1R	P43220	GLRX	P35754
GLS	O94925	GLS2	O94925, Q9UI32	GLTP	Q9NZD2
GLYATL1	Q969I3	GMDS	O60547	GMPPA	Q96IJ6
GMPPB	Q9Y5P6	GMPS	P49915	GNA11	P29992
GNA13	Q14344	GNA14	O95837, P29992	GNAI2	P04899
GNAL	P38405	GNAO1	P09471	GNAS	P63092, Q5JWF2
GNB1	P62873	GNB2	P62879	GNG11	P61952
GNG12	Q9UBI6	GNG4	P50150	GNG5	P63218
GNL3	Q9BVP2	GNPDA1	P46926, Q8TDQ7	GNS	P15586
GOLGA1	Q92805	GOLGA3	Q08378	GOLIM4	O00461
GORASP1	Q9BQQ3	GORASP2	Q9H8Y8	GOSR2	O14653
GP1BA	P07359	GP6	Q9HCN6	GPAA1	O43292
GPAT2	Q6NUI2	GPAT3	Q53EU6	GPAT4	Q53EU6, Q86UL3
GPC1	P35052	GPC2	Q8N158	GPC3	P51654
GPC4	O75487	GPC5	P78333	GPC6	Q9Y625
GPFR1	Q99527	GPKOW	Q92917	GPLD1	P80108
GPNMB	Q14956	GPR143	P51810	GPR176	Q14439
GPR18	Q14330	GPR183	P32249	GPR27	Q9NS67
GPR37	O15354	GPS1	Q13098	GPS2	Q13227
GPSM1	Q86YR5	GPX1	P07203	GPX4	P36969
GPX7	Q96SL4	GPX8	Q8TED1	GRB14	Q14449
GRB2	P62993-1	GRB7	Q14451	GREB1	Q4ZG55
GREM1	O60565	GREM2	Q9H772	GRHL1	Q9NZI5
GRHPR	Q9UBQ7	GRIA1	P42261	GRIA3	P42263
GRIA4	P48058	GRIK3	Q13003	GRIK4	Q16099
GRIK5	Q16478	GRIN2A	Q12879	GRIN2B	Q13224
GRIP1	Q9Y3R0	GRIP2	Q9C0E4	GRK3	P35626
GRK5	P34947	GRM3	Q14832	GRM4	Q14833
GRM8	O00222	GRN	P28799	GRPR	P30550
GSDME	O60443	GSN	P06396	GSPT2	Q8IYD1
GSS	P48637	GSTK1	Q9Y2Q3	GSTM2	P28161, P46439
GSTM3	P21266	GSTO1	P78417, Q9H4Y5	GSTO2	P78417, Q9H4Y5
GSTP1	P09211	GTF2E1	P29083	GTF2F1	P35269
GTF3C5	Q9Y5Q8	GTPBP2	Q9BX10	GTPBP3	Q969Y2
GTSE1	Q9NYZ3	GUCA1A	P43080	GUCY2C	P25092
GUCY2D	P51841, Q02846	GUK1	Q16774	GYG1	P46976
GYG2	O15488	GYS1	P13807	H2AC15	P0C0S8
H2AC16	P20671	H2AC18	Q6FI13	H2AC19	Q6FI13

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H2AC21	Q8IUE6	H2AX	P16104	H2BC10	P62807
H2BC12	O60814, Q93079, Q99877, Q99880	H2BC21	P06899, Q16778	H2BC4	P62807, Q93079
H3-3A	P84243	H3C7	P68431	HAAO	P46952
HABP4	Q5JVS0	HACD4	Q5VWC8	HACE1	Q8IYU2
HACL1	Q9UJ83	HADHA	P40939	HAGH	Q16775-2
HAND2	P61296	HARS1	P12081	HAS2	Q92819
HAS3	O00219	HAUS1	Q96CS2	HAUS6	Q7Z4H7
HAUS8	Q9BT25	HCFC1	P51610	HCFC2	Q9Y5Z7
HCN1	O60741	HCN4	Q9Y3Q4	HDAC2	Q92769
HDAC7	Q8WUI4	HDAC9	Q9UKV0	HDLBP	Q00341
HEBP1	Q9NRV9	HECTD3	Q5T447	HELLS	Q9NRZ9
HEPHL1	Q9BQS7	HERC3	Q15034	HERC4	Q15034, Q5GLZ8
HERC5	Q9UII4	HERC6	Q8IVU3, Q9UII4	HERPUD1	Q15011
HEXA	P06865	HEY2	Q9UBP5	HGS	O14964
HGSNAT	Q68CP4	HHIP	Q96QV1	HIBCH	Q6NVY1
HIC1	Q14526	HIF1A	Q16665	HIF3A	Q9Y2N7
HIPK2	Q9H2X6	HIVEP3	Q5T1R4	HK2	P52789
HLA-B	P01889	HLA-C	P10321	HLA-DOA	P06340
HLA-DQB1	P01920, P05538	HLA-E	P13747	HM13	Q8TCT9
HMBS	P08397	HMCN2	Q15109	HMGA2	P52926
HMGB1	P09429	HMGB2	P26583	HMGCR	P04035-1, P04035-2
HMGCS1	P54868	HMOX1	P09601	HMOX2	P30519
HNF4G	Q14541	HNMT	P50135	HNRNPA1	P09651
HNRNPD	Q14103	HNRNPUL1	Q9BUJ2	HOMER1	Q86YM7
HOMER2	Q9NSB8	HOMER3	Q9NSC5	HOXA11	P31270
HOXA4	Q00056	HOXB2	P14652	HOXB3	P14651
HOXC11	O43248	HOXC4	P09017	HOXC8	P31273
HOXD4	P09016	HPGDS	O60760	HPN	P05981
HPS1	Q92902	HPS4	Q9NQG7	HRAS	P01112
HRC	P23327	HRH1	P35367	HRH2	P25021
HS2ST1	Q7LGA3	HS3ST3A1	Q9Y663	HS3ST3B1	Q9Y662, Q9Y663
HS3ST4	Q9Y661	HS3ST5	Q8IZT8, Q96QI5	HS6ST2	Q96MM7
HS6ST3	Q8IZP7	HSD11B2	P80365	HSD17B10	Q99714
HSD17B4	P51659	HSD17B7	P56937	HSD17B8	Q92506
HSD3B7	Q9H2F3	HSF1	Q00613	HSP90AA1	P07900
HSP90AB1	P08238	HSPA12A	O43301	HSPA12B	Q96MM6
HSPA1A	P0DMV8	HSPA1B	P0DMV8, P0DMV9	HSPA2	P54652
HSPA4	P34932	HSPA4L	O95757	HSPA5	P11021
HSPB1	P04792	HSPB8	Q9UJY1	HSPD1	P10809
HSPE1	P61604	HSPG2	P98160	HTATSF1	O43719
HTR1D	P28221	HTR2C	P28335	HTR3A	P46098
HTR7	P34969	HTRA1	Q92743	HTRA2	O43464
HYAL2	Q12891	HYAL3	P38567	IAPP	P10997
ICAM1	P05362	ICAM3	P32942	ICE1	Q9Y2F5
ICE2	Q659A1	ICMT	O60725	ID1	P41134
ID2	Q02363	ID3	Q02535	ID4	P47928
IDH1	O75874	IDH3A	P50213	IDH3G	P51553
IDI1	Q13907	IDO1	P14902, Q6ZQW0	IER3	P46695
IFI16	Q16666	IFI27	P40305	IFI27L2	P09912

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
IFI35	P80217	IFI44	Q8TCB0-1	IFI6	P09912
IFIT1	P09914	IFIT3	O14879, P09914	IFIT5	Q13325
IFITM2	P13164, Q01629	IFITM3	Q01628	IFNGR2	P38484
IFNLR1	Q8IU57	IFT43	Q96FT9	IFT81	Q8WYA0
IGF2BP1	Q9NZI8	IGF2BP3	O00425	IGFBP2	P18065
IGFBP3	P17936	IGFBP4	P22692	IGFBP6	P24592
IGFBP7	Q16270	IGSF3	Q93033	IKBIP	Q70UQ0
IKBKE	Q14164	IKBKG	Q9Y6K9	IKZF1	Q13422-1
IL10RB	P38484	IL11	P20809	IL13RA1	P78552
IL13RA2	Q14627	IL15RA	Q13261	IL17RA	Q96F46
IL17RC	Q8NAC3	IL17RD	Q8NFM7	IL1R1	Q9HB29
IL1RAPL2	Q9NP60	IL1RL2	Q9HB29	IL27RA	Q6UWB1
IL2RB	P14784	IL33	O95760	IL4R	P24394
IL6ST	P40189	IL7R	P16871	ILK	Q13418
IMP4	Q96G21	IMPACT	Q9P2X3	IMPDH1	P20839
INCENP	Q9NQS7	ING3	Q9NXR8	INHBA	P08476
INHBB	P09529	INO80C	Q6PI98	INO80D	Q53TQ3
INPP1	P49441	INPP4B	O15327	INPP5A	Q14642
INPP5D	Q92835	INPP5F	Q9Y2H2	INPP5K	Q9BT40
INPPL1	O15357	INSR	P06213	INTS11	Q5TA45
INTS13	Q9NVM9	INTS2	Q9H0H0	INTS6	Q9UL03
INTS8	Q75QN2	INTU	Q9ULD6	IP6K1	Q92551, Q96PC2
IPMK	Q8NFU5	IQCB1	Q15051	IQCE	Q6IPM2
IQGAP2	Q13576	IQGAP3	Q86VI3	IQSEC3	Q9UPP2
IRAG1	Q9Y6F6	IRAK1	P51617	IRAK3	Q9Y616
IREB2	P48200	IRF1	P10914	IRF6	O14896
IRF9	Q00978	IRS1	P35568	IRS2	Q9Y4H2
IRS4	O14654	IRX1	P78414	ISLR	O14498
ITFG2	Q969R8	ITGA11	Q9UKX5	ITGA2	P17301
ITGA2B	P08514	ITGA3	P26006	ITGA4	P13612
ITGA5	P08648	ITGA9	Q13797	ITGB1	P05556
ITGB3	P05106	ITGB3BP	Q13352	ITGB5	P18084
ITGB8	P26012	ITGBL1	O95965	ITLN2	Q8WWA0
ITPA	Q9BY32	ITPKC	Q96DU7	ITPR2	Q14571
IZUMO2	Q6UXV1	JADE1	Q6IE81	JADE2	Q9NQC1
JADE3	Q92613	JAG1	P78504	JAG2	Q9Y219
JAK1	P23458	JAK3	P52333	JAM2	P57087
JAM3	Q9BX67	JARID2	Q92833	JMJD1C	Q15652
JMJD4	Q9H9V9	JMJD6	Q6NYC1	JMY	Q8N9B5
JOSD2	Q8TAC2	JUNB	P17275	KAT14	Q9H8E8
KAT2B	Q92831	KAT5	Q92993	KAT7	O95251
KAZN	Q674X7	KBTBD7	Q8WVZ9	KBTBD8	Q8NFY9
KCNA3	P22001	KCNA7	Q96RP8	KCNAB1	Q14722
KCNB1	Q14721	KCNC3	Q14003	KCND2	Q9NZV8
KCNE3	Q9Y6H6	KCNE4	Q8WWG9	KCNG2	Q9UJ96
KCNG3	Q8TAE7, Q8TDN1	KCNH1	O95259	KCNH2	Q12809
KCNH5	Q8NCM2	KCNH7	Q9NS40	KCNH8	Q96L42
KCNIP1	Q9NZI2	KCNIP3	Q9Y2W7	KCNJ11	Q14654
KCNJ2	P63252	KCNJ6	P48051	KCNJ8	Q15842

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
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KCNMB1	Q16558	KCNMB2	Q9NPA1, Q9Y691	KCNMB3	Q9NPA1, Q9Y691
KCNMB4	Q86W47	KCNN2	Q9H2S1	KCNN4	O15554
KCNQ2	O43526	KCNQ5	Q9NR82	KCNS1	Q96KK3
KCNS3	Q9BQ31	KCTD1	Q719H9	KCTD10	Q8WZ19
KCTD13	Q8WZ19	KCTD15	Q96SI1	KDELR1	P24390
KDELR2	P24390, P33947	KDELR3	O43731	KDM1A	O60341
KDM2B	Q8NHM5	KDM4A	O75164	KDM5B	Q9UGL1
KDM6A	O15550	KDM7A	Q6ZMT4	KDR	P35968
KEAP1	Q14145	KEL	Q9H2C0	KHDRBS2	Q5VWX1
KIAA0319	Q5VV43	KIAA1671	Q9C0C2	KIF13B	Q9NQT8
KIF18B	Q86Y91	KIF1A	Q12756	KIF1C	O43896
KIF21A	Q7Z4S6	KIF21B	O75037	KIF23	Q02241
KIF26A	Q9ULI4	KIF26B	Q2KJY2	KIF2A	O00139, Q8N4N8
KIF5A	Q12840	KIF5C	Q6PKD1	KIF6	Q6ZMV9
KIF7	Q2M1P5	KIFAP3	Q92845	KIFC2	Q96AC6
KIFC3	Q9BVG8-2	KIRREL1	Q96J84	KIRREL2	Q6UWL6
KIRREL3	Q8IZU9	KIT	P10721	KITLG	P21583-1
KLC1	Q07866	KLF15	Q9UIH9	KLF16	Q9BXX1
KLF4	O43474	KLHL12	Q53G59	KLHL13	Q9P2J3, Q9P2N7
KLHL21	Q9UJP4	KLHL22	Q53GT1	KLHL3	Q9UH77
KLK1	P06870	KLK13	Q9UKR3	KLK8	O60259
KLKB1	P03952	KLRG1	Q96E93	KMO	O15229
KMT5A	Q9NQR1	KMT5B	Q4FZB7, Q86Y97	KNTC1	P50748
KPNA5	O60684	KRBOX5	Q7Z2F6	KRT14	P08727
KRT16	P08779	KRT19	P08727	KRT33B	O76011, Q14525
KRT34	P02533	KRT7	P08729	KRT8	P13647
KRT80	P13647	KRT81	P78385, Q14533	KRTAP1-1	Q07627
KRTAP1-5	Q07627, Q9BYS1	KRTAP2-3	P0C7H8	KRTAP2-4	Q9BYR9, Q9BYU5
KSR1	Q8IVT5	KSR2	Q6VAB6, Q8WXI2	KYAT3	Q6YP21
KYNU	Q16719	L2HGDH	Q9H9P8	L3MBTL2	Q969R5
LAMA1	P25391	LAMA2	P24043	LAMA4	Q16363
LAMA5	O15230	LAMB1	P07942	LAMB2	P11047, P55268
LAMC1	P11047	LAMC3	Q9Y6N6	LAMP1	P11279
LAMP2	P13473	LAMTOR1	Q6IAA8	LAMTOR2	Q9Y2Q5
LARGE2	Q8N3Y3	LARS1	Q9P2J5	LCK	P06239
LCP1	P13796	LCP2	Q13094	LDHA	P00338
LDHB	P07195	LDLR	P01130	LDLRAP1	Q5SW96
LEAP2	Q969E1	LEFTY1	O00292, O75610	LEFTY2	O00292
LENG1	Q96BZ8	LETM1	O95202	LGALS1	P09382
LGALS3	P17931	LGI2	Q8N0V4	LGR6	Q9HBX8
LHX9	Q9NQ69	LIG3	P49916	LIMA1	Q8TDZ2
LIMS1	P48059	LIN28A	Q9H9Z2	LIN37	Q96GY3
LIN54	Q6MZIP7	LIN7B	Q9HAP6	LINGO1	Q96FE5
LIPA	P18054	LIPG	Q9Y5X9	LIPT2	A6NK58
LMAN2	Q12907	LMF2	Q9BU23	LMNA	P02545
LMO3	P25800	LMO7	Q8WWI1	LMOD1	P29536
LNPEP	Q9UIQ6	LONRF1	Q17RB8	LOX	P28300

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LOXL1	P58215, Q08397	LOXL2	Q9Y4K0	LOXL3	P58215
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LPAR4	Q99677	LPCAT2	Q7L5N7	LPGAT1	Q92604
LPL	P06858	LRAT	O95237	LRFN2	Q9ULH4
LRIG1	Q96JA1	LRP1	Q07954	LRP10	O75096
LRP12	Q9Y561	LRP2	P98164	LRP6	O75581
LRP8	Q14114	LRPPRC	P42704	LRRC1	Q9BTT6
LRRC14	Q15048	LRRC41	Q15345	LRRC8A	Q8IWT6
LRRC8B	Q6P9F7	LRRC8C	Q8TDW0	LRRC8E	Q6NSJ5
LRRFIP1	Q32MZ4	LRRN2	O43300	LRRTM2	O43300
LRRTM3	Q86VH5	LRRTM4	O43300, Q86VH4	LSM10	Q969L4
LSM6	P62312	LSS	P48449	LTA4H	P09960
LTB4R	Q15722, Q9NPC1	LTBP1	Q14766	LTBP2	Q14767
LTBP3	Q14767, Q9NS15	LTBR	P36941	LUC7L3	O95232
LUM	P51884	LY6E	Q16553	LY6K	Q17RY6
LYL1	P12980	LYN	P07948	LYPLA1	O75608
MAEA	Q7L5Y9	MAFG	O15525	MAGED1	Q9Y5V3
MAGED2	Q9UNF1	MAGI2	Q86UL8	MAGOHB	P61326, Q96A72
MALSU1	Q96EH3	MALT1	Q9UDY8	MAML2	Q8IZL2
MAML3	Q96JK9	MAMLD1	Q13495	MAN1B1	Q9UKM7
MAN2A1	Q16706	MAN2B2	Q9Y2E5	MAOA	P21397
MAP1B	P46821	MAP1LC3B	Q9GZQ8	MAP1LC3C	Q9BXW4
MAP2	P50579	MAP2K3	P46734	MAP2K6	P52564
MAP3K1	Q13233	MAP3K3	Q99759	MAP4K4	O95819
MAPK10	P53779	MAPK3	P27361, P28482	MAPK4	P31152
MAPK6	Q16659	MAPK7	Q13164	MAPKAP1	Q9BPZ7
MAPKAPK2	P49137	MAPRE3	Q9UPY8	MASP1	P48740
MASP2	O00187-1	MAST1	Q7Z460	MAT1A	Q00266
MAVS	Q7Z434	MBD1	Q9UIS9	MBD2	Q9UBB5
MBD3	O95983	MBD6	Q96DN6	MBOAT1	Q6ZNC8
MBOAT2	Q6ZWT7	MBOAT7	Q96N66	MBP	P13727
MBTPS1	Q14703	MCAT	Q8IVS2	MCCC2	Q9HCC0
MCF2	P10911	MCFD2	Q8NI22	MCM3	P25205
MCOLN1	Q9GZU1	MCOLN2	Q8IZK6	MCOLN3	Q8TDD5
MCRS1	Q96EZ8	MCUB	Q9NWR8	MDGA1	Q8NFP4
MDGA2	Q7Z553	MDH2	P40926	ME3	Q16798
MECP2	P51608-1, P51608-2	MED10	Q9BTT4	MED13L	Q71F56
MED14	O60244	MED15	Q96RN5	MED16	Q9Y2X0
MED19	A0JLT2	MED22	Q15528	MED28	Q9H204
MED29	Q9NX70	MED8	Q96G25	MEF2C	Q06413
MEF2D	Q14814	MEGF11	P07942	MERTK	Q12866
MET	P08581	METAP2	P50579	METTL21A	Q8WXB1
MEX3C	Q5U5Q3	MFAP4	P55083	MFAP5	Q13361
MFNG	O00587	MFSD4B	Q5TF39	MGA	O43451
MGAT1	P26572	MGAT2	Q10469	MGAT3	Q09327
MGAT4A	Q9UM21	MGAT4B	Q9UBM8, Q9UQ53	MGAT4C	Q9UBM8
MGLL	Q99685	MGME1	Q9BQP7	MGMT	P16455
MGRN1	O60291	MGST3	O14880	MIB1	Q86YT6
MICA	Q29983	MICAL1	Q8TDZ2	MICOS13	Q5XKP0

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
MICU1	Q9BPX6	MICU3	Q86XE3, Q8IYU8	MID1	Q9NQW6
MIGA1	Q8NAN2	MIOS	Q9NXC5	MIS18A	Q9NYP9
MIS18BP1	Q6P0N0	MLKL	Q8NB16	MLLT1	Q03111
MLX	Q9UH92	MLXIP	Q9NP71	MMP1	P03956, P08254, P22894
MMP12	P39900	MMP14	P50281	MMP16	P51512
MMP19	Q99542	MMP2	P08253	MMP25	Q9NPA2
MMP3	P08254	MMP9	P14780	MMRN2	Q9H8L6
MND1	Q9BWT6	MOCS2	O96007, O96033	MON1A	Q86VX9
MOV10	Q9HCE1	MPC2	O95563	MPDU1	O75352
MPG	P29372	MPL	P40238	MPP2	Q08050
MPP7	Q5T2T1	MPRIIP	Q6WCQ1	MRAS	O14807
MRC2	Q9UBG0	MRE11	P49959	MRI1	Q9BV20
MRPL12	P52815	MRPL14	Q6P1L8	MRPL17	Q9NRX2
MRPL19	P49406	MRPL2	Q5T653, Q9BZE1	MRPL24	Q96A35
MRPL27	Q8IXM3, Q9P0M9	MRPL38	Q96DV4	MRPL40	Q9NQ50
MRPL41	Q8IXM3	MRPL42	Q9Y6G3	MRPL43	Q8N983
MRPL50	Q8N5N7	MRPL52	Q86TS9	MRPL53	Q96EL3
MRPL54	Q6P161	MRPL55	Q7Z7F7	MRPL58	Q14197
MRPS11	P82912	MRPS12	O15235	MRPS18A	Q9NVS2
MRPS2	Q9Y399	MRPS24	Q96EL2	MRPS9	P82933
MRS2	Q9HD23	MRTFA	Q969V6	MSH2	P43246
MSH3	P20585	MSH6	P52701	MSI1	O43347
MSI2	Q96DH6	MSL3	Q8N5Y2	MSMO1	Q15800
MSN	P26038	MSRB1	Q9NZV6	MSRB3	Q8IXL7
MST1R	Q04912	MSX1	P28360	MT-ATP8	P03928
MT-ND1	P03886	MT-ND2	P03891	MT-ND4	P03905
MT1E	P04732	MT1H	P80294	MT1M	P13640, Q8N339
MT2A	P02795	MTA2	O94776	MTA3	Q9BTC8
MTARC1	Q5VT66	MTF2	Q9Y483	MTHFD2L	Q9H903
MTIF2	P46199	MTMR6	Q9Y217	MTMR7	Q9Y216
MTMR8	Q96QG7	MTREX	P42285	MTRF1	O75570
MTX1	Q13505	MUC16	Q8WXI7	MUC19	Q7Z5P9
MUC3A	Q02505	MUC4	Q99102	MUL1	O15151, Q969V5
MUS81	Q96NY9	MVP	Q14764	MX1	P20591
MX2	P20591, P20592	MXRA8	Q9BRK3	MYB	P10242
MYBBP1A	Q9BQG0	MYBL1	P10243	MYBPC2	Q14324
MYCN	P04198	MYDGF	Q969H8	MYH10	P35580
MYH11	P35749	MYH14	Q7Z406	MYH2	Q9UKX2
MYH3	P11055	MYH9	P35579	MYL12A	P19105
MYL12B	O14950	MYL5	Q02045	MYL6	P60660
MYL6B	P14649	MYL9	P24844	MYLIP	Q8WY64
MYLK	Q15746	MYO10	Q9HD67	MYO1C	O00159
MYO3A	Q8NEV4	MYO5B	Q9ULV0	MYO7A	Q13402
MYO9B	Q13459	MYRIP	Q8NFW9	MYSM1	Q5VVJ2
MZT2B	Q6NZ67	NA	P03468	NAA30	Q147X3
NAA38	Q9BRA0	NAAA	Q02083	NAALAD2	Q9Y3Q0
NAB2	Q15742	NABP1	Q96AH0	NAE1	Q13564
NAF1	Q15025	NAGK	Q9UJ70	NAMPT	P43490
NANOG	Q9H9S0	NANS	Q9NR45	NAPA	Q9Y2A7

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
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NCAM1	P13591	NCAPH2	Q6IBW4	NCEH1	Q6PIU2
NCOA6	Q14686	NCOR2	Q9Y618	NCS1	P43080
NCSTN	Q92542	NDE1	Q9NXR1	NDEL1	Q9GZM8
NDST1	P52848	NDST3	O95803, Q9H3R1	NDUFA11	Q86Y39
NDUFA13	Q9P0J0	NDUFA5	Q16718	NDUFA7	O95182
NDUFAB3	Q9BU61	NDUFAB6	Q330K2	NDUFAB7	Q7L592
NDUFB11	Q9NX14	NDUFB7	P17568	NDUFS6	O75380
NDUFS7	O75251	NDUFS8	O00217	NDUFV1	P49821
NEB	Q15058	NECAP2	Q9NVZ3	NEDD4	P46934, Q96PU5
NEFL	P07196	NEGR1	Q7Z3B1	NEK2	P51955
NEK3	Q9UHY1	NEK6	Q9HC98	NEK7	Q8TDX7
NELFE	P18615	NELL2	Q99435	NEMF	O60524
NEURL1B	A8MQ27	NF2	P35240	NFASC	O94856
NFE2L2	Q16236	NFIA	Q12857	NFIC	P08651
NFIX	Q14938	NFKB1	P19838	NFKBIB	Q15653
NFYB	P25208	NGF	P01138	NGLY1	Q96IV0
NHLRC3	Q5JS37	NHP2	Q9NX24	NID1	P14543
NID2	Q14112	NINL	Q9Y2I6	NIPAL1	Q6NVV3
NIPAL2	Q9H841	NIPAL3	Q6P499	NIPBL	Q6KC79
NKIRAS2	Q9NYR9	NLGN1	Q8N2Q7	NLGN3	Q9NZ94
NLGN4X	Q8N0W4	NLGN4Y	Q8NFX3	NLK	Q9UBE8
NLN	Q9BYT8	NLRC5	Q86WI3	NLRP1	Q9C000
NME2	P22392	NMNAT1	Q9HAN9	NMNAT3	Q96T66
NMRK2	Q9NPI5	NMT1	O60551	NMT2	O60551
NMU	P48645	NNMT	P40261	NOB1	Q9ULX3
NODAL	Q96S42	NOG	Q13253	NOL12	Q9UGY1
NOP58	Q9Y2X3	NOS1	P29475	NOS2	P35228
NOSIP	Q9Y314	NOTCH2	Q04721	NOTCH2NLA	Q7Z3S9
NOTCH2NLB	P0DPK3	NPAS2	Q99743	NPC1	O15118
NPDC1	Q9NQX5	NPFFR2	Q9Y5X5	NPHP4	O75161
NPL	Q9BXD5	NPLOC4	Q8TAT6	NPM1	P06748
NPNT	Q6UXI9	NPRL3	Q12980	NPY	P01303
NPY1R	P25929	NPY4R	P50391	NQO1	P15559
NR1D1	P20393	NR1H2	P55055	NR1I2	O75469-1, O75469-2, O75469-3, O75469-4, O75469-5, O75469-6, O75469-7
NR2C1	P13056	NR2C2AP	Q86WQ0	NR2F1	P10589
NR2F2	P24468	NR3C1	P04150-1, P04150-2, P04150-3, P04150-4, P04150-5, P04150-6, P04150-7, P04150-8, P04150-9	NR5A2	O00482-1, O00482-2, O00482-3
NR6A1	Q15406-1, Q15406-2, Q15406-3, Q15406-4, Q15406-5	NRBP1	Q9UHY1	NRCAM	Q92823
NRG1	Q02297, Q02297-10	NRG3	P56975	NRG4	Q8WWG1
NRIP1	P48552	NRN1	Q9NPD7	NRP1	O14786
NRP2	O60462	NRXN1	P58400, Q9ULB1	NRXN3	Q9HDB5, Q9Y4C0
NSD1	Q96L73	NSD2	O96028	NSFL1C	Q9UNZ2
NSL1	Q96IY1	NSMCE1	Q8WV22	NSMCE4A	Q8N140, Q9NXX6

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
NSUN6	Q8TEA1	NT5E	P21589	NT5M	Q8TCD5, Q9NPB1
NTF3	P20783	NTM	Q9P121	NTN4	Q9HB63
NTNG1	Q9Y2I2	NTS	P30990	NUBPL	Q8TB37
NUCB1	Q02818	NUDT10	Q9BW91-1	NUDT11	Q96G61
NUDT12	Q9BQG2	NUDT13	Q86X67	NUDT19	A8MXV4
NUDT2	P50583	NUDT21	O43809	NUMA1	Q14980
NUMB	P49757	NUP107	P57740	NUP133	Q8WUM0
NUP205	Q92621	NUP210	Q8TEM1	NUP54	Q7Z3B4
NUP62	P37198	NXT1	Q9UUKK6	OAS3	Q9Y6K5
OAT	P04181	OAZ1	P54368	OBSCN	Q5VST9
OCLN	Q16625	ODF2	Q5BJF6	OFD1	O75665
OGDH	Q02218	OGG1	O15527-4	OGT	O15294
OMD	Q99983	OMG	P23515	ONECUT1	Q9UBC0
OPA1	O60313	OPHN1	O60890	OPRD1	P41143
OR14A2	Q96R54	OR2D3	Q8NGH3	OR2H2	O95918
OR52A1	Q9UKL2	ORAI2	Q96SN7	ORC1	Q13415
ORC2	Q13416	ORC3	Q9UBD5	ORC4	O43929
OS9	Q13438	OSBPL10	Q9BXB5	OSBPL1A	Q9BXW6
OSBPL3	Q9H4L5	OSBPL5	Q9H0X9	OSBPL6	Q9BZF3
OSBPL9	Q96SU4	OSMR	Q99650	OST4	P0C6T2
OSTF1	Q92882	OSTM1	Q86WC4	OTOG	Q6ZRI0
OTUD3	Q5T2D3	OTX2	P32243	OVCA2	Q9BZG8
OXTR	P30559	P2RY1	P47900	P2RY13	Q9BPV8
P2RY14	Q15391	P3H1	Q32P28	P3H3	Q8IVL6
P4HA2	O15460	P4HA3	Q7Z4N8	P4HB	P07237
PACS1	Q6VY07	PACSIN1	Q9BY11	PACSIN2	Q9UNF0
PADI2	Q9Y2J8	PADI3	Q9ULW8	PAF1	P28328
PAH	P00439	PAICS	P22234	PAK1	Q13153
PAK4	O96013	PAK5	Q9NQU5, Q9P286	PAK6	Q9NQU5
PALB2	Q86YC2	PAM16	Q9Y3D7	PAN2	Q9HBH5
PANK1	Q8TE04	PANK3	Q9H999	PAPOLG	Q9BWT3
PAPPA	Q13219	PAPSS2	O95340	PARD6B	Q9BYG5
PARD6G	Q9BYG4	PARG	Q86W56	PARP1	P09874
PARP8	Q8N3A8	PARVA	Q9NVD7	PARVB	Q9HBI1
PATJ	Q8NI35	PAX3	P23760	PBX1	P40424
PCBP1	Q15365	PCBP4	P57723	PCCA	P05165
PCDH15	Q96QU1	PCGF5	Q86SE9	PCK2	Q16822
PCLO	Q9Y6V0	PCM1	Q15154	PCNT	O95613
PCOLCE	Q15113	PCSK5	Q92824	PCSK6	P29122
PCSK7	Q16549	PCYT1A	P49585	PCYT1B	Q9Y5K3
PCYT2	Q99447	PDAP1	Q13442	PDCD1LG2	Q9BQ51
PDCD6IP	Q8WUM4	PDE12	Q6L8Q7	PDE1C	Q01064
PDE2A	O00408	PDE3B	Q13370	PDE4A	P27815
PDE5A	O76074	PDE6B	P35913	PDE7A	Q13946
PDE7B	Q9NP56	PDE8A	O60658	PDE9A	O76083
PDGFB	P01127	PDGFC	Q9NRA1-1	PDGFRA	P16234
PDGFRB	P09619	PDK1	O15530	PDK2	Q15119
PDK3	Q15120	PDLIM5	Q96HC4	PDLIM7	Q9NR12
PDP1	Q9P0J1	PDPK1	O15530	PDPN	Q86YL7

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
PDPR	Q8NCN5	PDXK	O00764	PEA15	Q15121
PEAK1	Q9H792	PEAR1	P11047	PELI1	Q96FA3
PELI2	Q9HAT8	PELO	Q9BRX2	PEMT	Q9UBM1
PENK	P01210	PER3	P56645	PES1	O00541
PEX1	O43933	PEX10	O60683	PEX11B	O96011
PEX14	O75381	PEX26	Q7Z412	PEX6	Q13608
PFKFB2	O60825	PFKFB3	Q16875	PFKL	P17858
PFKP	Q01813	PFN1	P07737	PGAP1	Q75T13
PGF	P49763	PGGT1B	P53609	PGLS	O95336
PGM2L1	Q6PCE3	PGP	A6NDG6	PHB1	P35232
PHC1	P78364	PHC2	P78364, Q8IXK0	PHETA2	Q9H4I9
PHF1	O43189	PHF19	Q5T6S3	PHF2	O75151
PHIP	Q8WWQ0	PHKA1	P46020	PHKG1	Q16816
PHLPP1	O60346	PI4K2A	Q9BTU6	PI4KA	P42356
PI4KB	Q9UBF8	PIAS2	O75928-1	PIAS3	Q9Y6X2
PICALM	Q13492	PIEZO1	Q92508	PIF1	Q9H611
PIGA	P37287	PIGH	Q14442	PIGL	Q9Y2B2
PIGQ	Q9BRB3-2	PIGS	Q96S52	PIGT	Q969N2
PIGZ	Q86VD9	PIK3AP1	Q6ZUJ8	PIK3C2A	O00443
PIK3C2B	O00750	PIK3CB	P42338	PIK3R5	Q8WYR1
PIKFYVE	Q9Y2I7	PINK1	Q9BXM7	PIP4K2A	P48426
PIP5K1B	O14986	PIP5K1C	O60331	PIPOX	Q9P0Z9
PIR	O00625	PITPNM2	O00562, Q9BZ72	PITPNM3	Q9BZ71
PITRM1	Q5JRX3	PITX2	Q99697	PIWIL2	Q8TC59
PJVK	Q0ZLH3	PKD1	P98161	PKM	P14618
PKMYT1	Q99640	PKP2	Q99959	PLA2G12A	Q9BZM1
PLA2G15	Q8NCC3	PLA2G3	Q9NZ20	PLA2G4C	Q9UP65
PLA2G7	Q13093	PLAAT1	Q9HDD0	PLAAT5	Q96KN8
PLAGL1	Q9UM63	PLAT	P00750	PLAU	P00749
PLAUR	Q03405	PLBD1	Q6P4A8	PLCB3	Q01970
PLCB4	Q15147	PLCD3	Q8N3E9	PLCG2	P16885
PLCH1	Q4KWH8	PLD4	Q96BZ4	PLEC	Q15149
PLEKHA2	Q9HB19	PLEKHA5	Q9HAU0	PLEKHA6	Q9Y2H5
PLEKHA8	Q96JA3	PLEKHG1	Q9ULL1	PLEKHG4B	Q96PX9
PLEKHO2	Q8TD55	PLIN2	Q99541	PLIN3	Q8IUC6
PLK1	P53350	PLK2	Q9NYY3	PLK3	Q9H4B4
PLOD1	Q02809	PLOD3	O60568	PLPP3	O14495
PLPP4	Q5VZY2	PLPPR1	Q8TBJ4	PLPPR2	Q96GM1
PLPPR5	Q32ZL2	PLS1	Q14651	PLXNA1	Q9UIW2
PLXNA2	O75051	PLXNA3	P51805	PLXNB1	O43157
PLXNC1	O60486	PLXND1	Q9Y4D7	PMAIP1	Q13794
PMEL	P40967	PML	P29590	PMM1	Q92871
PMM2	O15305	PMP22	Q9NR77	PMPCA	Q10713
PMS1	P54278	PMVK	Q15126	PNKD	Q8N490-2
PNN	Q9H307	PNPLA2	Q96AD5	PNPLA3	Q9NST1
PNPO	Q9NVS9	POC1A	Q16829	PODXL2	Q9NZ53
POFUT1	Q9H488	POFUT2	Q9Y2G5	POLA1	P09884
POLA2	Q14181	POLB	P06746	POLD2	P49005
POLD4	Q9HCU8	POLE2	P56282	POLE3	Q9NRF9

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
POLG2	Q9UHN1	POLH	Q9Y253	POLI	Q9UNA4
POLQ	O75417	POLR1F	Q3B726	POLR2A	P24928
POLR2E	P19388	POLR2F	P61218	POLR2I	P36954
POLR2J	P52435	POLR2L	P62875	POLR3E	Q9NVU0
POLR3G	O15318, Q9BT43	POLR3H	Q9Y535	POM121	Q96HA1
POM121C	A8CG34, Q96HA1	POMGNT1	Q8WZA1	POMT2	Q9UKY4
POP7	O75817	PORCN	Q9H237	POT1	Q9NUX5
POU2F1	P14859	POU3F1	Q03052	POU5F1	Q01860
PPARA	Q07869	PPARD	Q03181	PPARG	P37231
PPARGC1A	Q9UBK2	PPARGC1B	Q86YN6	PPAT	Q06203
PPFIA2	O75334	PPFIBP2	Q8ND30	PPIB	P23284
PPIG	Q13427	PPIL2	Q13356	PPL	P23434
PPM1B	O75688	PPM1E	Q8WY54	PPM1F	P49593
PPM1L	Q5SGD2	PPME1	Q9Y570	PPP1CA	P62136
PPP1CC	P36873	PPP1R10	Q96QC0	PPP1R12B	O60237
PPP1R3C	Q9UQK1	PPP2R1A	P30153	PPP2R5B	Q15173
PPP2R5D	Q14738	PPP3CC	P48454	PPP4R2	Q9NY27
PPP5C	P53041	PPP6R1	Q9UPN7	PPRC1	Q5VV67
PPT1	Q96H96	PPWD1	Q96BP3	PRC1	O43663
PRDM14	Q9GZV8	PRDM8	Q9NQV8	PRDX4	Q13162
PRDX5	P30044-2	PREB	Q9HCU5	PRELID1	Q9Y255
PREX2	Q70Z35	PRICKLE1	Q96MT3	PRIM1	P49642
PRKAA1	Q13131	PRKAA2	P54646	PRKACA	P17612
PRKAG1	P54619	PRKAR2A	P13861	PRKAR2B	P31323
PRKCA	P17252	PRKCB	P05771	PRKCE	Q02156
PRKCG	P05129	PRKCH	P24723	PRKCI	P41743
PRKCQ	Q04759	PRKCSH	P14314	PRKD1	Q15139
PRKG1	Q13976-1	PRKG2	Q13237	PRKX	P51817
PRMT2	Q96LA8	PRMT3	O60678	PRNP	P04156
PROCR	Q9UNN8	PROK2	Q9HC23	PROKR1	Q8TCW9
PROKR2	Q8NFJ6	PROM1	O43490	PRPF31	Q8WWY3
PRPF38A	Q8NAV1	PRPF40A	O75400	PRPF6	O94906
PRPS1	P60891	PRSS23	O95084	PRSS3	P35030
PRSS8	Q16651	PRXL2B	Q8TBF2	PSAP	Q8IWL1
PSG1	P11464	PSG3	Q16557	PSG5	Q15238
PSIP1	O75475	PSMA2	P25787	PSMA3	P25788
PSMB3	P49720	PSMB6	P28072	PSMB8	P28062
PSMB9	P28065	PSMC3	P17980	PSMD2	Q13200
PSMD3	O43242	PSMD8	P48556	PSMD9	O00233
PSME1	Q06323	PSME4	Q14997	PSMF1	Q92530
PTAFR	P25105	PTCD3	Q96EY7	PTGES	O14684
PTGES2	Q9H7Z7	PTGFR	P43088	PTGIR	P43119
PTGIS	Q16647	PTGR2	Q8N8N7	PTGS1	P23219
PTGS2	P35354	PTH2R	P49190	PTHLH	P12272
PTPA	P18433	PTPN13	Q12923	PTPN14	Q15678
PTPN18	Q99952	PTPN2	P17706	PTPN20	Q4JDL3
PTPN23	Q9H3S7	PTPN3	P26045	PTPN4	P29074
PTPN5	P54829	PTPN6	P29350	PTPRD	P23468
PTPRJ	Q12913	PTPRN2	Q92932	PTPRZ1	P23471

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
PTRH2	Q9Y3E5	PTX3	P26022	PUF60	Q9UHX1
PURA	P30520	PUS7	Q96PZ0	PVALB	P20472
PVR	P16234	PXDN	Q92626	PXMP2	Q9NR77
PXMP4	Q9Y6I8	PXN	P11678	PXYLP1	Q8TE99
PYCR1	P32322	PYCR3	Q53H96	PYGB	P11216
QPRT	Q15274	QSOX1	O00391	QTRT1	Q9BXR0
RAB11B	Q15907	RAB11FIP3	O75154	RAB17	Q9H0T7
RAB19	A4D1S5	RAB1A	P62820	RAB1B	Q9H0U4
RAB20	Q9NX57	RAB22A	Q9UL26	RAB23	Q9ULC3
RAB25	P57735	RAB27A	P51159	RAB2A	P61019
RAB32	Q13637	RAB34	Q9BZG1	RAB35	Q15286
RAB36	O95755	RAB37	Q96AX2	RAB38	P57729
RAB39A	Q14964	RAB39B	Q96DA2	RAB3B	P20337
RAB3C	Q96E17	RAB3IP	Q96QF0	RAB5B	P61020
RAB5C	P51148	RAB6A	P20340	RAB7A	P51149
RAB7B	Q96AH8	RAB8A	P61006, Q92930	RAB8B	Q92930
RABGEF1	Q9UJ41	RABGGTA	Q92696	RAC2	P15153
RACGAP1	Q9H0H5	RAD23A	P54725	RAET1G	Q6H3X3
RAG1	P15918	RAI1	Q7Z5J4	RALB	P11234
RALGAPA2	Q2PPJ7	RANBP2	P49792	RANGAP1	P46060
RANGRF	Q9HD47	RAP1A	P62834	RAP1B	P61224
RAP1GAP2	Q684P5	RAP1GDS1	P52306	RAPGEF2	Q9Y4G8
RAPGEF4	Q8WZA2	RARG	P13631	RARRES2	Q99969
RASA1	P20936	RASA3	Q14644	RASA4	O43374
RASA4B	O43374	RASAL1	O95294	RASGEF1A	Q8N9B8
RASGRF2	O14827	RASGRP2	Q7LDG7	RASGRP4	Q8TDF6
RB1CC1	Q8TDY2	RBBP4	Q09028	RBBP6	Q7Z6E9
RBFOX3	A6NFN3	RBM25	P49756	RBM26	Q5T8P6
RBM27	Q9P2N5	RBM39	Q14498	RBM42	Q9BTD8
RBP1	P29374	RBPJ	Q06330	RCAN3	Q9UKA8
RCC1L	Q96I51	RCC2	Q9P258	RDH10	Q8IZV5
RDH11	Q96NR8	RDH13	Q8NBN7	RECK	O95980
RELA	Q04206	RELN	P78509	REPS2	Q8NFH8
RET	P07949	REV1	Q9UBZ9	RFC1	P35251
RFC3	P40937, P40938	RFC4	P35249, P35250	RFFL	O75771
RFK	Q969G6	RGL1	Q9NZL6	RGL3	Q3MIN7
RGMA	Q96B86	RGMB	Q6NW40	RGS10	O43665
RGS12	O14924	RGS16	O15492	RGS17	Q9UGC6
RGS19	P49795	RGS2	Q9UGC6	RGS4	P49798
RGS5	O15539	RGS6	P49758	RGS7	P49802
RGS9	O75916	RHOA	P61586	RHOB	P62745
RHOBTB2	Q9BYZ6	RHOBTB3	O94955	RHOC	P08134
RHOG	P84095	RHOH	Q15669	RHOJ	Q9H4E5
RHOT1	Q8IXI2	RHPN1	Q8TCX5	RHPN2	Q8IUC4
RIDA	P52758	RIF1	Q5UIP0	RIMKLA	Q8IXN7
RIMKLB	Q9ULI2	RIN1	Q13671	RIN2	Q8WYP3
RIN3	Q8TB24	RING1	Q06587	RIOK3	O14730
RIPOR2	Q9Y4F9	RIT2	Q99578	RLF	P51460
RLN1	P04090	RMI2	Q96E14	RMND5A	Q9H871

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
RNASEK	Q6P5S7	RND2	P52198	RNF125	Q96EQ8
RNF126	Q9BV68	RNF130	Q86XS8	RNF138	Q8WVD3
RNF14	Q9UBS8	RNF144A	P50876	RNF146	Q9NTX7
RNF181	Q9P0P0	RNF182	Q8N6D2	RNF185	Q96GF1
RNF19A	Q9NV58	RNF213	Q63HN8	RNF216	Q9NWF9
RNF25	Q96BH1	RNF31	Q96EP0	RNF40	O75150
RNH1	O60930	RNMT	O43148	ROBO1	Q9Y6N7
ROBO2	Q9HCK4	ROR1	Q01973	RORB	Q92753
RP2	O75695	RPL10	P27635, Q96L21	RPL13A	P40429
RPL15	P61313	RPL18	Q07020	RPL18A	Q02543
RPL19	P84098	RPL26L1	Q9UNX3	RPL28	P46776, P46779
RPL29	P47914	RPL35	P42766	RPL36	Q9Y3U8
RPL8	P62424, P62917	RPLP1	P05386	RPLP2	P05387
RPN1	Q13200	RPP21	Q9HCM9	RPRD1A	Q96P16
RPS11	P62280	RPS16	P62249	RPS28	P62857
RPS5	P46782	RPS6KA1	Q15418	RPS6KA2	Q15349
RPS6KA4	O75676	RPS6KA5	O75582	RPS6KA6	Q9UK32
RPS6KB2	Q9UBS0	RPTOR	Q8N122	RRAGA	Q5VZM2, Q7L523
RRAGC	Q9HB90	RRAGD	Q9NQL2	RRBP1	Q9P2E9
RRM2B	Q7LG56	RRP7A	Q9Y3A4	RRP9	O43818
RSC1A1	Q92681	RSF1	Q96T23	RSP04	Q2I0M5
RSU1	Q15404	RTEL1	Q9NZ71	RTN4	Q9NQC3
RTN4R	Q9HBX8	RTN4RL1	Q86UN2	RTN4RL2	Q86UN3
RUNX1	Q01196	RUNX2	Q13950-1, Q13950-2	RUNX3	Q13761
RUVBL2	Q9Y230	RWDD2B	Q9Y3V2	RWDD3	Q9Y3V2
RXRA	P19793	RXRB	P28702	RYR1	P21817
RYR2	Q92736	S100A10	P60903	S100A11	P31949
S1PR1	P21453	S1PR2	O95136	S1PR3	Q99500
SACM1L	Q9NTJ5	SALL4	Q9UJQ4	SAP30BP	Q9UHR5
SAR1B	Q9Y6B6	SARS1	P49591	SART1	O43290
SATB1	Q01826	SATB2	Q9UPW6	SBF1	O95248
SBSN	Q6UWP8	SBSPON	Q8IVN8	SC5D	O75845
SCAF1	O14548	SCAI	Q8N9R8	SCARB2	Q14108
SCARF1	Q14162	SCD	O00767	SCD5	Q86SK9
SCG3	Q8WXD2	SCGB3A2	Q96PL1	SCMH1	Q96GD3-2
SCN11A	Q9UI33	SCN1A	P35498	SCN4A	P35499
SCN5A	Q14524	SCNN1A	P37088	SCO2	O43819
SCUBE1	Q8IWY4	SCUBE3	Q8IX30	SDC1	P18827
SDC2	P34741	SDC4	P31431	SDCCAG8	Q86SQ7
SDHA	P31040	SDHAF4	Q5VUM1	SDK1	Q7Z5N4
SEC13	P55735	SEC16A	O15027	SEC22B	O75396
SEC23A	Q15436	SEC24C	P53992	SEC24D	O94855
SEC31A	O94979	SEC61A1	P61619	SEC61B	P60468
SECISBP2	Q96T21	SEL1L	Q9UBV2	SELENOI	Q9C0D9
SELENOP	P49908	SELENOS	Q9BQE4	SELPLG	Q14242
SEMA3E	O15041	SEMA4A	Q9H3S1	SEMA4D	Q92854
SEMA5A	Q13591	SEMA5B	Q9P283	SEMA6A	Q9H2E6
SEMA7A	O75326	SEMG1	P04279	SEPSECS	Q9HD40
SERINC1	Q9NRX5	SERINC2	Q96SA4	SERINC3	Q13530

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
SERINC5	Q86VE9	SERPINA10	Q9UK55	SERPINA5	P05154
SERPINB1	P30740	SERPINB2	P05120	SERPINB6	P35237
SERPINB8	P50452	SERPINE1	P05121	SERPINE2	P07093
SERPINH1	P50454	SESN3	P58005	SET	Q01105
SETD1B	Q9UPS6	SETD6	Q8TBK2	SETD7	Q8WTS6
SETDB1	Q15047	SF3B1	O75533	SF3B2	Q13435
SF3B4	Q15427	SF3B5	Q9BWJ5	SFRP1	Q8N474
SFRP2	Q96HF1	SGCB	Q16585	SGCD	Q92629
SGCE	O43556	SGK1	O00141	SGMS1	Q86VZ5
SGMS2	Q8NHU3	SGSH	P51688	SH2B3	Q9UQQ2
SH2D2A	Q9NP31	SH3BP4	Q9P0V3	SH3GL1	Q99961
SH3GL2	Q99962	SH3GL3	P82987	SH3KBP1	Q96B97
SH3PXD2A	Q5TCZ1	SH3RF1	Q7Z6J0	SHANK2	Q9UPX8
SHARPIN	Q9H0F6	SHB	Q15464	SHISA5	Q8N114
SHMT2	P34897	SHPRH	Q149N8	SHTN1	A0MZ66
SIGLEC10	Q96LC7	SIGMAR1	Q99720	SIPA1L2	Q96FS4
SIRPA	P78324	SIRPB1	O00241	SIRT2	Q8IXJ6
SIX1	Q15475	SKA1	Q96BD8	SKAP2	O75563
SKI	Q5VTY9	SKIC2	Q15477	SKIL	P12757
SLA	Q13239	SLC10A1	Q14973	SLC13A3	Q8WWT9
SLC13A5	Q86YT5	SLC15A1	P46059	SLC16A1	P53985
SLC16A10	Q8TF71	SLC16A2	P36021	SLC16A7	O60669
SLC19A2	O60779	SLC19A3	Q9BZV2	SLC1A3	P43003
SLC1A4	P43007	SLC1A5	Q15758	SLC1A6	P48664
SLC20A2	Q08357	SLC22A4	O76082	SLC22A5	O76082
SLC22A7	Q9Y694	SLC23A1	Q9UGH3, Q9UHI7	SLC24A2	Q9UI40
SLC24A3	Q9HC58	SLC24A5	Q71RS6	SLC25A10	Q9UBX3
SLC25A11	Q02978	SLC25A13	Q9UJS0	SLC25A16	P16260
SLC25A21	Q9BQT8	SLC25A22	Q9H936	SLC25A26	Q70HW3
SLC25A27	O95847	SLC25A28	Q96A46	SLC25A6	P12236
SLC27A2	O14975	SLC27A4	Q6P1M0	SLC27A6	Q9Y2P4
SLC29A2	Q14542	SLC2A10	O95528	SLC2A12	Q8TD20
SLC2A13	Q96QE2	SLC2A14	Q8TDB8	SLC2A3	P11169
SLC2A4	P14672	SLC30A3	Q99726	SLC34A2	O95436
SLC35A2	P78381	SLC35A4	Q96G79	SLC35B2	Q8TB61
SLC35C1	Q96A29	SLC35D1	Q9NTN3	SLC36A1	Q7Z2H8
SLC37A1	P57057	SLC37A4	O43826	SLC38A1	Q9H2H9
SLC38A4	Q969I6	SLC38A5	Q8WUX1	SLC38A9	Q8NBW4
SLC39A10	Q9ULF5	SLC39A8	Q9C0K1	SLC41A1	Q8IVJ1
SLC41A2	Q96JW4	SLC44A2	Q8IWA5	SLC44A3	Q8N4M1
SLC44A4	Q53GD3	SLC44A5	Q8NCS7	SLC4A10	Q6U841
SLC4A2	P04920	SLC4A4	Q9Y6R1	SLC4A7	Q9Y6M7
SLC52A2	Q9HAB3	SLC52A3	Q9NQ40	SLC5A10	A0PJK1
SLC5A12	Q1EHB4	SLC5A4	Q9NY91	SLC5A5	Q92911
SLC66A1	Q6ZP29	SLC6A11	P48066	SLC6A15	Q9H2J7
SLC6A6	P31641	SLC6A9	P48067	SLC7A1	P30825
SLC7A2	P52569-1, P52569-2	SLC7A3	Q8WY07	SLC7A5	Q01650
SLC7A7	Q9UM01	SLC8A1	P32418	SLC8A3	P57103
SLC9A1	P19634	SLC9B2	Q4ZJ14, Q86UD5	SLC9C1	Q4G0N8

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
SLCO1A2	P46721	SLCO2A1	Q92959	SLCO4A1	Q96BD0
SLCO4C1	Q6ZQN7	SLIT2	O94813	SLIT3	O75094
SLITRK2	Q9H156	SLITRK3	O94933	SLITRK4	Q8IW52
SLITRK5	O94991	SLITRK6	Q9H5Y7	SMAD2	Q15796
SMAD3	P84022	SMAD6	O43541	SMAD7	O15105
SMAD9	O15198	SMAP2	Q9H0E9	SMARCA2	P51531
SMARCA5	O60264	SMARCB1	Q12824	SMARCC1	Q92922
SMARCD1	Q96GM5	SMARCD3	Q6STE5	SMC3	Q9UQE7
SMC6	Q96SB8	SMG5	Q9UPR3	SMG9	Q9H0W8
SMN1	Q16637	SMO	Q9NWM0-3	SMOX	Q9NWM0-3
SMPD1	P17405	SMUG1	Q53HV7	SMURF2	Q9HAU4
SMYD1	Q8NB12	SMYD2	Q9NRG4	SNAI2	O43623
SNAP29	O95721	SNCA	P37840	SNRNP35	Q16560
SNRNP48	Q6IEG0	SNRPA1	P09661	SNRPB	P14678
SNRPD1	P62314, P62316	SNRPN	P63162	SNTA1	Q13424
SNTB2	Q13425	SNTG2	Q9NY99	SNURF	P78317
SNX1	O60749	SNX2	O60749	SNX3	O60493
SNX5	Q9Y5X3	SNX9	Q9Y5X1	SOAT1	P35610
SOCS1	O15524	SOCS5	O75159	SOCS7	Q8WXH5
SORBS1	Q9BX66	SORBS3	O60504	SORD	Q00796
SORL1	Q92673	SORT1	Q99523	SOX13	Q9UN79
SOX2	P48431	SOX3	P41225	SOX4	Q06945
SOX9	P48436	SP100	P23497	SP110	P23497
SPARC	Q14515	SPARCL1	Q14515	SPAST	Q9UBP0
SPCS1	Q9Y6A9	SPDL1	Q96EA4	SPINK1	P00995
SPINK5	Q9NQ38	SPINT1	O43278	SPINT2	O43291
SPOCD1	Q6ZMY3	SPOCK3	Q9BQ16	SPON2	Q9BUD6
SPP1	P49642	SPPL2B	Q8TCT7	SPRY1	O43609
SPSB1	Q96BD6	SPSB4	Q96A44, Q99619	SPTAN1	Q13813
SPTBN2	O15020	SPTLC2	O15270	SPTSSB	Q8NFR3
SQLE	Q14534	SQSTM1	Q13501	SRC	P12931
SRGAP1	Q7Z6B7	SRGN	P10124	SRM	P19623
SRP9	P49458	SRPK1	Q96SB4	SRPRA	P08240
SRPRB	Q9Y5M8	SRRT	Q9BXP5	SRSF10	O75494, Q8WXF0
SRSF11	Q05519	SRSF12	Q8WXF0	SRSF5	Q13243
SRXN1	Q9BYN0	SSB	P05455	SSC5D	A1L4H1
SSNA1	O43805	SSPN	Q14714	SSR2	P43308
SSR3	Q6STE5	SSR4	P51571	SSTR1	P30872
SSTR2	P30874	ST14	Q9Y5Y6	ST3GAL1	Q11201
ST3GAL2	Q16842	ST3GAL3	Q11203	ST3GAL4	Q11206
ST3GAL6	Q9Y274	ST6GAL1	P15907	ST6GAL2	Q96JF0
ST6GALNAC3	Q8NDV1	ST6GALNAC4	Q9H4F1	ST6GALNAC6	Q969X2
ST8SIA1	Q92185	ST8SIA3	O43173	ST8SIA5	O15466
ST8SIA6	P61647	STAG2	Q8N3U4	STAG3	Q9UJ98
STAM2	O75886	STAMPB	O95630	STARD13	Q9Y3M8
STARD4	Q96DR4	STAT1	P42224	STAT2	P52630
STAT6	P42226	STBD1	O95210	STC2	O76061
STEAP3	Q658P3	STIM1	Q13586	STING1	Q86WV6
STK10	O94804	STK11IP	Q8N1F8	STK26	Q9P289

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
STK38	Q15208	STOM	P27105	STON2	Q8WXE9
STRC	Q7RTU9	STRN	O43815	STS	P08842
STUB1	Q9UNE7	STX10	O60499	STX3	Q13277
STX4	Q12846	STX5	Q13190	STX6	O43752
SUCLG1	P53597	SUDS3	Q9H7L9	SULT1A1	P50225
SULT1C2	O75897	SULT1C4	O75897	SULT4A1	Q9BR01
SUMF2	Q92526	SUMO1	P63165	SUMO3	P55854
SUN2	Q9UH99	SUOX	P51687	SUPT16H	Q9Y5B9
SUPT5H	O00267	SURF4	O15260	SUV39H1	O43463
SUV39H2	Q9H5I1	SV2A	Q7L0J3	SVBP	Q8N300
SVIP	Q8NHG7	SWAP70	Q9UH65	SYDE1	Q6ZW31
SYDE2	Q5VT97	SYK	P43405	SYN2	Q9NY99
SYN3	O14994	SYNE1	Q8NF91	SYNE2	Q8WXH0
SYNGAP1	Q9UJF2	SYNJ1	O43426	SYNJ2	O15056
SYT1	P21579, Q8NBV8	SYT10	Q6XYQ8	SYT12	Q8IV01
SYT2	Q8N9I0	SYT4	Q9BT88	SYT5	O00445
SYT7	O43581	SYT9	Q86SS6	SYTL4	Q96C24
SYVN1	Q86TM6	SZT2	Q5T011	TAC1	P20366
TAC3	Q9UHF0	TACO1	Q9BSH4	TACR3	P29371
TADA2B	Q86TJ2	TAF1	P21675, Q8IZX4	TAF10	Q12962
TAF1A	Q15573	TAF1D	Q9H5J8	TAF4B	Q92750
TAF6	P49848	TAGLN2	P37802	TALDO1	P37837
TAOK3	Q9H2K8	TAP1	Q03518	TAPBP	O15533
TARBP2	Q15633	TAS2R19	P59542	TAS2R3	Q9NYW6
TAS2R30	P59541	TAS2R31	P59538	TAS2R4	Q9NYW5
TAS2R5	Q9NYW4	TASP1	Q9H6P5	TAX1BP3	O14907
TBC1D10B	Q4KMP7	TBC1D13	Q9NVG8	TBC1D15	Q8TC07
TBC1D2	Q9BYX2	TBC1D20	Q96BZ9	TBC1D24	Q9ULP9
TBC1D25	Q3MII6	TBC1D7	Q9P0N9	TBCB	P30622
TBK1	Q9UHD2	TBL1X	O60907	TBL1XR1	Q9BZK7
TBL3	Q12788	TBX2	Q13207	TBX3	O15119
TBX5	Q99593	TBXA2R	P21731, P21731-2, P21731-3	TCERG1	O14776
TCF25	Q9BQ70	TCF7	P36402	TCF7L1	Q9HCS4
TCIRG1	Q13488	TCN2	P20062	TDG	Q13569
TDRD1	Q9BXT4	TDRD12	Q587J7	TDRKH	Q9Y2W6
TEAD1	P28347	TEAD2	Q15562	TEC	Q92570-1, Q92570-2
TEK	Q02763	TERF1	P54274	TERT	O14746
TES	Q9NZU5	TESK1	Q15569	TET1	Q8NFU7
TET2	Q6N021	TET3	O43151	TEX10	Q9NXF1
TEX15	Q9BXT5	TEX2	Q8IWB9	TEX264	Q9Y6I9
TFAM	P09429	TFAP2A	P05549	TFB1M	Q8WVM0
TFDP2	Q14188	TFG	Q92734	TFPI	P10646
TFPI2	P48307	TFPT	P0C1Z6	TFR2	Q9UP52
TGFB1	P01137	TGFB1	Q15582	TGFBR2	P37173
TGFBR3	Q03167	TGIF1	Q15583	TGIF2	Q9GZN2
TGOLN2	O43493	TGS1	Q96RS0	THBD	P07204
THBS1	P07996	THBS2	P35442	THBS4	P35443
THOC1	Q96FV9	THOC2	Q8NI27	THOP1	P52888
THRA	P10827	THRB	P10828	THSD1	Q9NS62

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
THSD4	Q6ZMP0	THUMPD1	Q9NXG2	THY1	P04216
TIA1	P31483	TICAM1	Q8IUC6	TICAM2	Q86XR7
TICRR	Q7Z2Z1	TIFA	P38919	TIMM13	Q9Y5L4
TIMM22	Q9Y584	TIMM50	Q3ZCQ8	TIMP1	P01033
TIMP2	P16035	TIMP3	P35625	TJP2	Q9UDY2
TK1	P04183	TK2	O00142	TLE1	Q04724
TLN1	Q9Y490	TLR2	O60603	TLR4	O00206
TLR5	O60602	TMBIM1	Q969X1	TMC1	Q8TDI8
TMC6	Q7Z403	TMED10	P49755	TMED2	Q15363
TMED3	Q9Y3Q3	TMED5	Q9Y3A6	TMED9	Q9BVK6
TMEM11	P17152	TMEM115	Q12893	TMEM177	Q53S58
TMEM179B	Q7Z7N9	TMEM216	Q9P0N5	TMEM59	Q9BXS4
TMEM67	Q5HYA8	TMEM87B	Q8NBN3	TMOD2	Q9NZR1
TMOD3	Q9NYL9	TMPRSS2	O15393	TMSB4X	P62328
TMT1A	Q9H8H3	TNC	P24821	TNFAIP1	Q13829
TNFAIP3	P21580	TNFRSF10A	O00220	TNFRSF10C	O14798
TNFRSF11A	Q9Y6Q6	TNFRSF11B	O00300	TNFRSF12A	Q9NP84
TNFRSF19	Q9HAV5	TNFRSF1A	P19438	TNFRSF8	P28908
TNFSF11	O14788	TNFSF12	O43508	TNFSF4	P23510
TNIK	Q9UKE5	TNIP2	Q8NFBZ5	TNIP3	Q96KP6
TNKS	O95271, Q9H2K2	TNNC1	P19429, P63316	TNNI1	P19237
TNNI3	P19429	TNNT2	P45379	TNRC6A	Q8NDV7
TNRC6B	Q9UPQ9	TNRC6C	Q9HCJ0	TNS3	Q68CZ2
TOLLIP	Q9H0E2	TOM1	O60784	TOMM40	O96008
TOP1	P11387	TOP2B	Q02880	TOR1A	O14656
TOR1B	O14657	TP53I3	Q53FA7	TPCN1	Q9ULQ1
TPD52	Q9Y2Q9	TPH2	Q8IWU9	TPI1	P60174
TPM1	P09493	TPM2	P07951	TPM4	P67936
TPO	P07202	TPP1	Q96AP0	TPP2	P29144
TPR	P12270	TPST1	O60507	TPX2	Q9ULW0
TRADD	Q15628	TRAF3	Q13114	TRAK2	O60296
TRAPPC1	Q9Y5R8	TRAPPC12	Q8WVT3	TRAPPC2L	Q9UL33
TRAPPC3	O43617	TRAPPC5	Q8IUR0	TRDMT1	O14717
TRDN	Q13061	TREH	O43280	TREX1	Q9NSU2
TRIB1	Q96RU8	TRIB3	Q96RU7	TRIM11	Q96F44
TRIM24	O15164	TRIM25	Q14258	TRIM26	Q12899
TRIM3	P53365	TRIM32	Q13049	TRIM33	Q9UPN9
TRIM36	Q9NQ86	TRIM37	O94972	TRIM38	O00635
TRIM45	Q9H8W5	TRIM56	Q9BRZ2	TRIM58	Q9C037
TRIM71	Q2Q1W2	TRIM8	Q9BZR9	TRIM9	Q9C026
TRIO	O75962	TRIOBP	Q9H2D6-4, Q9H2D6-5	TRIT1	Q9H3H1
TRMT11	Q7Z4G4	TRMT12	Q53H54	TRMT61A	Q96FX7
TRMT9B	Q9P272	TRNT1	Q96Q11	TRPC4AP	Q8TEL6
TRPM3	Q9HCF6	TRPM6	Q9BX84	TRPV4	Q9HBA0
TRUB2	O95900	TSC2	P49815	TSC22D4	Q99576
TSEN34	Q9BSV6	TSPAN15	O95858	TSPAN33	Q86UF1
TSPAN5	P62079	TSPAN7	P41732	TSPO	P30536
TST	Q15477	TSTD1	Q8NFBZ3	TTBK2	Q6IQ55
TTC19	Q6DKK2	TTC21B	Q7Z4L5	TTC9	Q9UIM3

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
TTL	Q8NG68	TTLL11	Q8NHH1	TTLL4	Q14679
TTLL6	Q8N841	TTN	Q8WZ42	TTYH1	Q9BSA4, Q9H313-3
TUBA1A	Q71U36	TUBA1B	P68363	TUBA1C	P68363, Q9BQE3
TUBB	P04350, P07437	TUBB2B	Q13885, Q9BVA1	TUBB3	Q13509
TUBB4A	P04350	TUBB4B	P04350, P68371	TUBB6	Q9BUF5
TUBG1	P23258, Q9NRH3	TUBG2	Q9NRH3	TUBGCP2	Q9BSJ2
TUFM	P49411	TULP3	O75386	TUSC3	Q13454
TUT4	Q5TAX3	TUT7	Q5VYS8	TWF1	Q12792
TWF2	Q6IBS0	TWIST1	Q15672	TWIST2	Q8WVJ9
TXLNA	P40222	TXN2	Q99757	TXNDC11	Q6PKC3
TXNDC5	Q8NBS9	TXNRD1	Q16881	TYK2	P29597
TYW3	Q6IPR3	TYW5	A2RUC4	U2AF2	P26368
U2SURP	O15042	UAP1	Q16222	UBA52	P62979, P62987
UBA7	P22314, P41226	UBB	P0CG47, P62979, P62987	UBC	P0CG48, P62987
UBE2A	P49459	UBE2C	O00762	UBE2G2	P60604
UBE2J1	Q9Y385	UBE2J2	Q8N2K1-1	UBE2L3	P68036
UBE3B	Q7Z3V4	UBE3C	Q15386	UBE3D	Q7Z6J8
UBIAD1	Q9Y5Z9	UBL4A	P11441	UBTF	P17480
UBXN11	Q5T124	UBXN7	O94888	UCHL1	P09936
UCK1	Q9HA47	UCKL1	Q9NWZ5	UCMA	Q8WVF2
UCP2	P55851, P55916	UGP2	Q16851	UGT3A1	Q3SY77, Q6NUS8
UGT3A2	Q3SY77	UGT8	Q16880	UHMK1	Q8TAS1
UHRF1	Q96T88	ULBP1	Q9BZM5	ULBP2	Q9BZM5
UNC5B	Q8IZJ1	UNC5D	Q6UXZ4	UNC79	Q9P2D8
UNC80	Q8N2C7	UNG	P13051-1	UPF2	Q9HAU5
UPF3B	Q9BZI7	UPP1	Q16831	UQCC3	Q6UW78
UQCR10	Q9UDW1	UQCR11	O14957	UQCRC1	P31930
UQCRQ	O14949	URM1	Q9BTM9	UROD	P06132
UROS	P10746	USE1	Q9NZ43	USH1C	Q9Y6N9
USP18	Q9UMW8	USP25	Q9UHP3	USP28	Q96RU2
USP34	Q70CQ2	USP37	Q86T82	USP43	Q70EL4
USP44	Q70CQ1, Q9H0E7	USP45	Q70EL2	USP48	Q86UV5
USP49	Q70CQ1, Q9H0E7	USP5	P45974	USP6NL	Q92738
USP9X	Q93008	UTP25	Q68CQ4	UTRN	P46939
UTS2	O95399	UVRAG	Q9P2Y5	VAMP2	P63027
VAMP3	Q15836	VAMP8	Q9BV40	VANGL1	Q8TAA9
VANGL2	Q9ULK5	VARS1	P26640	VASH2	Q86V25
VAT1	Q99536	VAV1	P15498	VAV3	Q9UKW4
VCP	P55072	VDR	P11473	VEGFB	P49765
VEGFC	P49767	VENTX	O95231	VIM	P08670
VIPR1	P32241	VIT	O75030	VKORC1	Q9BQB6
VLDLR	P98155	VPS11	Q9H270	VPS16	Q9H269
VPS18	Q9P253	VPS28	Q9UK41	VPS37B	A5D8V6, Q9H9H4
VPS4A	Q9UN37	VPS53	Q5VIR6	VRK1	Q99986
VSIG10	Q9NYS7	WARS1	P23381	WASF2	Q9Y6W5
WDCP	Q9H6R7	WDR1	O75083	WDR19	Q8NEZ3
WDR35	Q9P2L0	WDR36	Q8NI36	WDR43	Q15061
WDR45	Q9Y484	WDR46	O15213	WDR48	Q8TAF3
WDR59	Q6PJI9	WDR75	Q8IWA0	WDR81	Q562E7

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
WDR83	Q9BRX9	WFDC2	Q14508	WHAMM	Q8TF30
WIPF1	O43516	WIPF2	Q8TF74	WIPF3	A6NGB9
WIP11	Q5MNZ9	WIP12	Q9Y4P8	WLS	Q5T9L3
WNK2	Q9Y3S1	WNK3	Q9BYP7	WNK4	Q96J92
WNT2	O00755, P09544	WNT5A	P41221	WNT5B	Q9H1J7
WRN	Q14191	WSB1	Q9Y6I7	WSB2	Q9NYS7
WWP1	Q9H0M0	WWP2	O00308	WWTR1	Q9GZV5
XAF1	Q6GPH4	XBP1	P17861-2	XK	P51811
XPO1	O14980	XRCC1	P18887	XRCC2	O43543
XYLT1	Q86Y38	YAF2	Q8IY57	YES1	P07947
YIF1A	O95070	YIPF6	Q96EC8	YJU2	Q9BW85
YKT6	O15498	YOD1	Q5VVQ6	YPEL5	P62699
YTHDC1	Q96MU7	YTHDC2	Q9H6S0	ZBTB17	Q13105
ZC3H15	Q8WU90	ZC3H8	Q8N5P1	ZCCHC7	Q8N3Z6
ZDHHC18	P31947	ZDHHC21	Q8IVQ6	ZDHHC5	Q9C0B5
ZDHHC7	Q9NXF8	ZDHHC8	Q9ULC8	ZDHHC9	Q9Y397
ZEB1	P37275	ZEB2	O60315	ZFAND6	Q6FIF0
ZFHX3	Q15911	ZFP14	Q9HCL3	ZFP36	P26651
ZFP36L2	P47974	ZFP37	P17032, Q9Y6Q3	ZFP69	Q49AA0
ZFP69B	Q9UJL9	ZFP90	Q8TF47	ZFPM2	Q8WW38
ZIC2	O95409	ZIC3	O60481	ZIK1	Q3SY52, Q9BWM5
ZKSCAN7	Q6ZNG0, Q9P0L1	ZKSCAN8	Q15776	ZMIZ1	Q92520
ZMYM2	Q9UBW7	ZNF10	P21506	ZNF100	Q8IYN0
ZNF101	Q8IZC7	ZNF114	Q8NC26	ZNF12	P17014
ZNF124	Q15973	ZNF133	P52736	ZNF14	P17017
ZNF141	Q15928	ZNF143	P52747	ZNF154	A6NNF4, Q13106
ZNF155	Q12901	ZNF157	P51786	ZNF160	Q9HCG1
ZNF18	P17022	ZNF181	Q9NR11	ZNF195	O14628
ZNF197	O14709	ZNF202	O95125	ZNF208	O43345
ZNF211	Q13398	ZNF213	O14771	ZNF214	Q9UL59
ZNF215	Q8NE65, Q9UL58	ZNF221	Q9UK13	ZNF225	P18146
ZNF226	Q9NZL3	ZNF227	Q86WZ6	ZNF233	Q9NZL3
ZNF235	Q14590, Q86WZ6	ZNF248	Q8NDW4	ZNF25	P17030
ZNF253	O75346	ZNF254	O75437	ZNF256	Q9Y2P7
ZNF260	Q8TAQ5	ZNF267	Q14586	ZNF268	Q96BR6
ZNF273	Q14593	ZNF28	P17035	ZNF285	Q96NJ3
ZNF286A	Q9HBT8	ZNF3	P17036, Q6AW86	ZNF30	P17039
ZNF300	Q96RE9	ZNF302	Q9NR11	ZNF311	Q5JNZ3
ZNF320	A2RRD8	ZNF331	Q9NQX6	ZNF334	Q9HCZ1
ZNF335	Q9H4Z2	ZNF337	Q9Y3M9	ZNF33A	Q06730
ZNF33B	Q06730	ZNF354A	O60765	ZNF391	Q8NCK3
ZNF397	Q96MU6	ZNF398	Q9BS31	ZNF415	Q09FC8
ZNF417	Q8TAU3	ZNF418	Q8TF45	ZNF419	Q96HQ0
ZNF420	Q8TAQ5	ZNF423	Q2M1K9	ZNF426	Q9BUY5
ZNF429	Q86V71	ZNF43	P17038	ZNF431	Q8TF32
ZNF436	Q9C0F3	ZNF443	Q9Y2A4	ZNF445	P59923
ZNF454	Q8N9F8	ZNF480	Q8WV37	ZNF483	Q8TF39
ZNF484	Q5JVG2, Q8N8J6	ZNF485	Q8NCK3	ZNF493	Q6ZR52
ZNF519	Q8TB69	ZNF521	Q96K83	ZNF544	Q6NX49

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ZNF549	Q6P9A3	ZNF550	Q7Z398	ZNF551	Q7Z340
ZNF552	Q9H707	ZNF561	Q8N587	ZNF563	Q8TA94
ZNF564	Q8TBZ8	ZNF567	Q8N184	ZNF569	Q5MCW4
ZNF571	Q7Z3V5	ZNF577	Q9BSK1	ZNF582	Q96NG8
ZNF587	Q96SQ5	ZNF589	Q86UQ0	ZNF595	Q8IYB9
ZNF598	Q86UK7	ZNF605	Q86T29, Q9Y462	ZNF606	Q8WXB4
ZNF607	Q96SK3	ZNF610	Q8N9Z0	ZNF613	O94892, Q6PF04
ZNF614	Q8N883	ZNF615	Q8N8J6	ZNF616	Q08AN1
ZNF620	Q6ZNG0	ZNF625	Q96I27	ZNF626	Q68DY1
ZNF627	Q7L945	ZNF638	Q14966	ZNF648	Q96GC6
ZNF649	Q9BS31	ZNF660	Q6AZW8	ZNF670	Q9BS34
ZNF675	Q8TD23	ZNF676	Q8N7Q3	ZNF678	Q5SXM1
ZNF681	Q96N22	ZNF682	O95780	ZNF697	Q5TEC3
ZNF700	Q9H0M5	ZNF703	Q9H7S9	ZNF704	Q6ZNC4
ZNF708	P17019	ZNF710	Q8N1W2	ZNF711	Q9Y462
ZNF713	Q8N859	ZNF714	Q96N38	ZNF717	Q9BY31
ZNF721	Q8TF20	ZNF724	A8MTY0	ZNF730	Q6ZMV8
ZNF732	B4DXR9	ZNF737	O75373	ZNF738	Q8NE65
ZNF74	Q16587	ZNF746	Q6NUN9	ZNF749	O43361
ZNF761	Q86XN6	ZNF765	Q7L2R6	ZNF770	Q6IQ21
ZNF776	Q68DI1	ZNF782	Q6ZMW2	ZNF790	Q6PG37
ZNF792	Q3KQV3	ZNF793	Q6ZN11	ZNF8	Q86UQ0
ZNF816	Q0VGE8	ZNF84	P51786	ZNF850	Q96PQ6
ZNF853	Q6IV72	ZNF860	A6NHJ4	ZNF875	P10072
ZNF883	P52742	ZNF92	Q03936	ZNF93	P35789
ZP3	P21754	ZSCAN10	Q96SZ4	ZSWIM8	A7E2V4
ZYX	Q8WUP2	ZZZ3	Q8IYH5		

Input	ChEBI Id
CIT	16947

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
AAMP	ENSG00000127837	ABCA1	ENSG00000165029, ENST00000374736.7	ABCC3	ENSG00000108846
ABCF2	ENSG00000033050	ABCG1	ENSG00000160179	ACACA	ENSG00000278540
ACADM	ENSG00000117054	ACADVL	ENSG00000072778	ACLY	ENSG00000131473
ACOX1	ENSG00000161533	ACSL1	ENSG00000151726	ACSS3	ENSG00000111058
ACTA2	ENST00000224784	ACTB	ENST00000331789	ADD1	ENSG00000087274
AGPAT2	ENSG00000169692	AIFM2	ENSG00000042286	AIP	ENSG00000110711
AKT1	ENSG00000142208	ALAD	ENST00000409155	ALAS1	ENSG00000023330
ALDH1A3	ENST00000329841	ANPEP	ENST00000300060	ANXA1	ENST00000257497
ANXA2	ENST00000451270	ANXA3	ENST00000264908	ANXA4	ENST00000394295
APOC1	ENSG00000130208	APOE	ENSG00000130203	APP	ENSG00000142192
ARF1	ENSG00000143761	ARFGAP1	ENSG00000101199	ATF3	ENSG00000162772
ATF5	ENSG00000169136, ENST00000423777	ATM	ENSG00000149311, ENST00000278616	ATP6V0B	ENSG00000117410
ATP6V0C	ENSG00000185883	ATP6V0D1	ENSG00000159720	ATP6V0E1	ENSG00000113732
ATP6V1E1	ENSG00000131100	ATP6V1F	ENSG00000128524	ATP6V1H	ENSG00000047249
AXIN2	ENSG00000168646, ENST00000307078	B2M	ENSG00000166710	BCL2L11	ENSG00000153094

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
BCL2L14	ENSG00000121380	BDNF	ENSG00000176697	BDP1	ENSG00000072135
BHLHE40	ENSG00000134107	BIRC5	ENST00000301633	BMAL1	ENSG00000133794
BMP4	ENSG00000125378	BOC	ENSG00000144857	BOLA2B	ENSG00000183336
BRWD1	ENSG00000185658	BST2	ENSG00000130303	BTG1	ENSG00000133639
CA2	ENST00000285379	CALR	ENSG00000179218	CAV1	ENSG00000105974
CCL2	ENSG00000108691	CCN1	ENSG00000145386	CCN2	ENSG00000118523
CCNA2	ENSG00000145386	CCND1	ENSG00000110092	CCNE1	ENSG00000105173
CCNG1	ENSG00000113328	CCNL1	ENSG00000073111	CD24	ENST00000606017
CD274	ENST00000381573.3	CD44	ENST00000428726	CD74	ENST00000009530
CDC25A	ENSG00000164045	CDC25B	ENSG00000101224	CDC42	ENSG00000070831
CDC45	ENSG00000093009	CDC7	ENSG00000097046	CDH1	ENST00000261769
CDH11	ENST00000268603, ENST00000394156	CDH2	ENSG00000170558	CDK2	ENSG00000123374
CDKN2A	ENSG00000147889, ENST00000304494, ENST00000579755	CDKN2B	ENSG00000147883	CER1	ENST00000380911
CFL1	ENSG00000172757	CGB3	ENSG00000104827	CGB5	ENSG00000104827
CGB8	ENSG00000104827	CHD1	ENSG00000153922, ENST00000284049.7	CHEK1	ENSG00000149554, ENST00000438015
CITED2	ENSG00000164442	CKLF	ENSG00000102554	CLDN10	ENST00000299339
CLDN7	ENSG00000181885	CNN2	ENSG00000064666	COL1A1	ENSG00000108821
COL1A2	ENSG00000164692	COL4A6	ENSG00000197565	COX7A1	ENSG00000161281
CPT1A	ENSG00000110090	CREB1	ENSG00000118260	CRIPTO	ENSG00000241186
CRY2	ENSG00000121671	CSF1	ENSG00000184371	CSPG4	ENST00000308508
CSRP1	ENSG00000159176	CTSD	ENSG00000117984	CXCL1	ENSG00000163739
CXCL12	ENSG00000107562	CXCL8	ENSG00000169429	CXCR4	ENST00000241393
DCTN1	ENSG00000204843	DICER1	ENSG00000100697	DIO2	ENSG00000211448, ENST00000438257
DKK1	ENSG00000107984	DLGAP5	ENSG00000126787	DNAJB1	ENSG00000132002
DNAJC3	ENSG00000102580	DPPA4	ENSG00000121570	E2F1	ENSG00000101412
EBF1	ENSG00000164330	EDEM1	ENSG00000134109	EDIL3	ENSG00000164176
EEPDI	ENSG00000122547	EGF	ENSG00000138798	EGFR	ENST00000275493
EGR2	ENSG00000122877	ELL2	ENST00000237853	ELOVL6	ENSG00000170522
EMX2	ENSG00000170370	ENPP2	ENST00000427067	EPAS1	ENSG00000116016
EPCAM	ENST00000263735	EPHA4	ENSG00000116106	ERBB4	ENSG00000178568
EXTL2	ENSG00000162694	EYA1	ENSG00000104313	F3	ENSG00000117525
FABP6	ENSG00000170231	FABP7	ENSG00000164434	FANCD2	ENSG00000144554, ENST00000419585
FAS	ENSG00000026103	FDFT1	ENSG00000079459	FGF2	ENSG00000138685
FHL2	ENSG00000115641	FKBP5	ENSG00000096060	FLT4	ENSG00000037280
FN1	ENSG00000115414	FOXA3	ENST00000302177	FOXF1	ENSG00000103241
FOXO1	ENSG00000150907	FTH1	ENST00000273550	FTL	ENST00000331825
FURIN	ENSG00000140564	G0S2	ENSG00000123689	G6PD	ENSG00000160211
GADD45A	ENSG00000116717	GATA6	ENST00000269216	GBP1	ENSG00000117228
GBP2	ENSG00000162645	GBP3	ENSG00000117226	GCLC	ENSG00000001084
GCLM	ENSG00000023909	GEM	ENSG00000164949	GLIPR1	ENSG00000139278
GLS2	ENSG00000135423	GOSR2	ENSG00000108433	GP1BA	ENSG00000185245
GPR143	ENSG00000101850	GPRC5A	ENST00000014914	GREB1	ENSG00000196208, ENST00000381486.6
GREM1	ENSG00000166923	GRHL1	ENSG00000134317	GRIN2A	ENSG00000183454
GRIN2B	ENSG00000273079	GSDME	ENSG00000105928	GSTO1	ENSG00000148834

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
HAND2	ENSG00000164107	HERPUD1	ENSG000000051108	HEY2	ENSG00000135547
HHIP	ENSG00000164161	HIF1A	ENSG00000100644	HLA-B	ENSG00000234745
HLA-C	ENSG00000204525	HLA-DQB1	ENSG00000179344	HLA-E	ENSG00000204592
HMGCR	ENSG00000113161	HMGCS1	ENSG00000112972	HMOX1	ENSG00000100292
HNF4G	ENSG00000164749	HOXA4	ENSG00000197576, ENST00000360046	HOXB2	ENSG00000173917, ENST00000330070
HOXB3	ENSG00000120093, ENST00000311626	HOXC4	ENSG00000198353, ENST00000303406	HOXD4	ENSG00000170166, ENST00000306324
HSP90AA1	ENSG00000080824	HSPA1A	ENSG00000204389, ENSG00000215328, ENSG00000234475, ENSG00000235941, ENSG00000237724	HSPA1B	ENSG00000204388, ENSG00000212866, ENSG00000224501, ENSG00000231555, ENSG00000232804
HSPA5	ENSG00000044574	HSPB1	ENSG00000106211	HSPB2	ENSG00000170276
HSPD1	ENSG00000144381	HSPE1	ENSG00000115541	HTRA2	ENSG00000115317
IAPP	ENSG00000121351	ICAM1	ENSG00000090339	ID1	ENSG00000125968
ID2	ENSG00000115738	ID3	ENSG00000117318	ID4	ENSG00000172201
IDH1	ENSG00000138413	IDI1	ENSG00000067064	IFI27	ENSG00000165949
IFI35	ENSG00000068079	IFI6	ENSG00000126709	IFIT1	ENSG00000185745
IFIT3	ENSG00000119917	IFIT5	ENSG00000152778	IFITM2	ENSG00000185201
IFITM3	ENSG00000142089	IGFBP3	ENSG00000146674	IGFBP7	ENSG00000163453
IKBIP	ENSG00000166130	IL1R1	ENSG00000115594	IL4R	ENSG00000077238
IQSEC3	ENSG00000120645	IRAK1	ENSG00000184216	IRF1	ENSG00000125347
IRF6	ENSG00000117595	IRF9	ENSG00000213928	IRX1	ENSG00000170549
ITGA2	ENSG00000164171	ITGA2B	ENSG00000005961	ITGA4	ENSG00000115232
ITGA5	ENSG00000161638	ITGB1	ENST00000302278	ITGBL1	ENSG00000198542
JAG1	ENSG00000101384	JUNB	ENSG00000171223	KDEL3	ENSG00000100196
KDM5B	ENST00000235790	KEAP1	ENSG00000079999	KIT	ENST00000288135
KLF15	ENSG00000163884	KLF4	ENSG00000136826	KLK1	ENST00000301420
KRT14	ENST00000167586	KRT19	ENST00000361566	KRT7	ENST00000331817
KRT8	ENST00000293308	LAMA5	ENSG00000130702	LCP1	ENSG00000136167
LDHB	ENST00000350669	LEFTY1	ENST00000272134	LGALS3	ENSG00000131981
LINC01139	ENST00000400946	LMNA	ENSG00000160789	LPL	ENSG00000175445
LRIG1	ENST00000273261	LSS	ENSG00000160285	MAOA	ENSG00000189221
MAP4K4	ENSG00000071054	MAPK4	ENST00000400384	MAPK6	ENSG00000069956
MECP2	ENST00000303391, ENST00000453960	MED12L	ENST00000273432	MEF2C	ENSG00000081189
MET	ENSG00000105976	MGLL	ENSG00000074416	MID1	ENSG00000101871
MMP1	ENSG00000196611	MMP2	ENSG00000087245	MMP3	ENSG00000149968
MMP9	ENSG00000100985	MSH2	ENSG00000095002	MSN	ENSG00000147065
MSX1	ENSG00000163132	MT-ATP8	ENST00000361851	MT-ND1	ENST00000361390
MT-ND2	ENST00000361453	MT-ND4	ENST00000361381	MT2A	ENSG00000125148
MX1	ENSG00000157601	MX2	ENSG00000183486	MYB	ENSG00000118513
MYCN	ENSG00000134323	MYDGF	ENSG00000074842	MYL9	ENSG00000101335
MYLIP	ENSG00000007944	NAB2	ENSG00000166886	NAMPT	ENSG00000105835
NANOG	ENSG00000111704	NCAM1	ENSG00000149294	NFE2L2	ENSG00000116044
NLK	ENST00000407008	NOS2	ENSG00000007171	NOTCH2	ENSG00000134250, ENST00000256646
NOTCH2NLA	ENSG00000264343, ENST00000362074	NOTCH2NLB	ENSG00000286019, ENST00000593495	NPAS2	ENSG00000170485
NPNT	ENSG00000168743	NPY	ENSG00000122585	NQO1	ENSG00000181019
NR1D1	ENSG00000126368	NR5A2	ENSG00000116833	OAS3	ENSG00000111331

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
OCLN	ENSG00000197822	ONECUT1	ENST00000305901	OPRD1	ENSG00000116329
OR14A2	ENSG00000241128	OR2D3	ENSG00000178358	OR2H2	ENSG00000204657
OR52A1	ENSG00000182070	ORC1	ENSG00000085840	OTX2	ENST00000339475
PAIP2	ENST00000265192	PAX3	ENSG00000135903	PBX1	ENSG00000185630
PCBP4	ENSG00000090097	PDPN	ENSG00000162493	PER3	ENSG00000049246
PINK1	ENSG00000158828	PITX2	ENSG00000164093	PLAGL1	ENSG00000118495
PLIN2	ENSG00000147872	PLK1	ENSG00000166851	PLK2	ENSG00000145632
PLK3	ENSG00000173846	PLXND1	ENSG00000004399	PMAIP1	ENSG00000141682
PMEL	ENSG00000185664	PML	ENSG00000140464	PMP22	ENSG00000109099
PMVK	ENSG00000163344	PNPLA2	ENSG00000177666	POLA1	ENSG00000101868
POLG2	ENSG00000256525	POU2F1	ENSG00000143190	POU3F1	ENSG00000185668
POU5F1	ENSG00000204531	PPARA	ENSG00000186951	PPARG	ENSG00000132170
PPARGC1A	ENSG00000109819	PPP2R5B	ENSG00000068971	PREB	ENSG00000138073
PRKCB	ENSG00000166501	PRKCQ	ENSG00000065675	PROM1	ENST00000447510
PRSS3	ENST00000379405	PSAP	ENSG00000122852, ENSG00000185303	PSMB8	ENSG00000204264
PTAFR	ENSG00000169403	PTGS2	ENSG00000073756	PTPN13	ENSG00000163629
PTPN14	ENSG00000152104	PTPN18	ENSG00000072135	PTPN2	ENSG00000175354
PTPN20	ENSG00000204179	PTPN23	ENSG00000076201	PTPN4	ENSG00000088179
PTPN5	ENSG00000110786	PTPN6	ENSG00000111679	PVALB	ENSG00000100362
PXDN	ENSG00000130508	PXN	ENSG00000089159	RAB27A	ENSG00000069974
RABGGTA	ENSG00000100949	RAG1	ENSG00000166349	RAP1B	ENSG00000127314
RBBP4	ENSG00000162521	RBFOX3	ENSG00000167281	RET	ENSG00000165731
RGL1	ENSG00000143344	RLN1	ENSG00000107018	ROBO1	ENSG00000169855
ROBO2	ENSG00000185008	RRM2B	ENSG00000048392	RUNX1	ENSG00000159216, ENST00000344691
RUNX2	ENSG00000124813	S1PR1	ENSG00000170989	SALL4	ENSG00000101115
SC5D	ENSG00000109929	SCD	ENSG00000099194	SCD5	ENSG00000145284
SCO2	ENSG00000130489	SDC4	ENST00000372733	SEC31A	ENSG00000138674
SERPINB2	ENSG00000197632	SERPINE1	ENSG00000106366	SERPINH1	ENSG00000149257
SESN3	ENSG00000149212	SGK1	ENST00000237305	SIX1	ENSG00000126778
SLC2A3	ENSG00000059804	SLC2A4	ENSG00000181856	SLC4A4	ENST00000264485
SLIT2	ENSG00000145147	SMAD6	ENSG00000137834	SMAD7	ENSG00000101665
SMYD1	ENSG00000115593	SNAI2	ENSG00000019549	SNRPA1	ENSG00000131876
SOCS1	ENSG00000185338	SOCS7	ENSG00000180008	SOX2	ENSG00000181449
SOX9	ENST00000245479	SP100	ENSG00000067066	SPARC	ENSG00000113140
SPINK1	ENST00000296695	SPP1	ENSG00000118785	SQLE	ENSG00000104549
SQSTM1	ENSG00000161011	SRPRA	ENSG00000182934	SRPRB	ENSG00000144867
SRXN1	ENSG00000271303	STAT1	ENSG00000115415	STEAP3	ENSG00000115107
SYT10	ENSG00000110975	SYVN1	ENSG00000162298	TALDO1	ENSG00000177156
TBX2	ENSG00000121068	TBX5	ENSG00000089225	TEC	ENSG00000119508
TERT	ENSG00000164362	TET2	ENST00000265149	TFAM	ENSG00000108064
TFAP2A	ENST00000379613	TFB1M	ENSG00000029639	TFPI2	ENST00000222543
TGFB1	ENSG00000105329	TGFBR3	ENSG00000069702, ENST00000212355	THBS1	ENSG00000137801
TIMP1	ENSG00000102265	TK1	ENSG00000167900	TLN1	ENSG00000137076
TMEM97	ENST00000226230	TNFRSF10A	ENSG00000104689	TNFRSF10C	ENSG00000173535
TNFRSF1A	ENSG00000067182	TNIN	ENSG00000154310	TP53I3	ENSG00000115129
TPP1	ENSG00000166340	TRIB1	ENSG00000173334	TRIB3	ENSG00000101255
TRIM25	ENSG00000121060	TRIM26	ENSG00000234127	TRIM3	ENSG00000110171

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
TRIM38	ENSG00000112343	TRIM45	ENSG00000134253	TRIM8	ENSG00000171206
TUBB4B	ENST00000340384	TUT4	ENSG00000134744	TUT7	ENSG00000083223
TWIST1	ENSG00000122691	TXNRD1	ENSG00000198431	UBB	ENSG00000170315
UCMA	ENSG00000165623	VIM	ENST00000544301	WFDC2	ENSG00000101443
WIPI1	ENSG00000070540	XAF1	ENSG00000132530	XBP1	ENSG00000100219
YIF1A	ENSG00000174851	ZBTB17	ENSG00000116809	ZEB1	ENSG00000148516
ZEB2	ENSG00000169554	ZIC2	ENSG00000043355	ZIC3	ENSG00000156925
ZNF225	ENSG00000120738	ZNF521	ENSG00000198795		

Input	miRBase Id	Input	miRBase Id	Input	miRBase Id
MIR145	MI0000461	MIR302A	MI0000738	MIR302B	MI0000772

Interactors (4557)

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A1CF	Q9NQ94	Q14192	A2ML1	A8K2U0	P01160
AAK1	Q2M2I8	Q96D71	AAMP	C9JG97	O14503
AAR2	Q9Y312	Q15185	AARSD1	Q9BTE6-2	P63172
AASS	Q9UDR5	P0DTC5	ABCA1	O95477	P02647
ABCC4	O15439	P17931	ABCD4	O14678	P23945
ABCF3	Q9NUQ8	P14373	ABHD11	Q8NFV4-4	Q15915
ABHD14A	Q9BUJ0	Q9UNA3	ABHD15	Q6UXT9	P21673
ABHD17B	Q5VST6	Q13304	ABHD18	Q0P651	Q15116
ABHD5	Q8WTS1	O75563	ABI1	Q8IZP0	Q07889
ABL2	P42684	P19174	ABLM1	O14639-4	Q14192
ABLM3	O94929-2	Q13573	ABRAXAS1	Q6UWZ7	P53611
ACAA1	P09110	O00628	ACACA	Q13085	Q13085
ACAD11	Q709F0	Q14493	ACAD9	Q9H845	P21453
ACADM	P11310	P0DTC5	ACADVL	P49748	Q53T94
ACAN	P16112	P05067	ACBD7	Q8N6N7	Q9NWQ8
ACD	Q96AP0	O43768	ACKR3	P25106	P49407
ACLY	P53396	P13569	ACO2	Q99798	Q8IWL3
ACOT13	Q9NPJ3	Q969H8	ACOT7	O00154	Q9HCK8
ACOX2	Q99424	P63172	ACP2	P11117	P21453
ACRV1	P26436	P50897	ACSL3	O95573	P27824
ACSS3	Q9H6R3	Q96CV9	ACTA1	P68133	Q16665
ACTA2	P62736	P15498	ACTB	EBI-5276484	P26358
ACTC1	P68032	P23528	ACTG1	P63261	P23528
ACTN1	P12814	P12931	ACTN2	P35609	Q92934
ACTN4	O43707	Q07157	ACTR10	Q9NZ32	Q8N5V2
ACTR1A	P61163	Q14203	ACTR1B	P42025	Q8N5V2
ACTR2	P61160	Q14247	ACTR3	P61157	O00401
ACVR1	Q04771	Q9NPF0	ACVR2A	P27037	P43115
ACVR2B	Q13705	P27037	ADAM12	O43184-2	O95633
ADAM15	Q13444	P12931	ADAM22	Q9P0K1	Q14160
ADAM23	Q9UKQ2	P00749	ADAM28	Q9UKQ2	P00749
ADAM9	Q13443	Q99962	ADAMTS1	Q9UHI8	P08833
ADAMTS4	O75173	O00462	ADAMTSL4	Q6UY14-3	Q96PM5
ADARB1	P78563-4	P19525	ADAT2	Q7Z6V5	Q96GM5
ADCY3	O60266	P17931	ADCY6	O43306	Q9Y4Z2

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ADCY9	O60503	P15514	ADCYAP1R1	P41586-3	P18509
ADD1	P35611	Q13153	ADD2	P35612	P35611
ADGRA2	Q96PE1	Q12959	ADGRA3	Q8IWK6	P34925
ADGRB3	O60242	O60883	ADGRE2	Q9UHX3	A1L3X0
ADGRE5	P48960	P08174	ADGRG5	Q8IZF4	P17302
ADGRL2	O95490	P21926	ADGRL3	Q9HAR2	Q9NZU0
ADI1	Q9BV57	P50281	ADIRF	Q15847	Q14116
ADISSP	Q9GZN8	P36873	ADNP	Q9H2P0	Q13257
ADNP2	Q6IQ32	Q13185	ADRA1B	P35368	P61073
ADRA1D	P25100	Q14160	ADRB2	P07550	P04156
ADRM1	Q16186	P60896	ADSS2	P30520	Q8NB12
AEBP2	Q6ZN18-2	Q9NRD5	AFDN	P55196	P12931
AGAP2	Q8CGU4	Q8IZJ1	AGFG2	O95081	Q9NQZ5
AGO1	Q9UL18	P04156	AGPAT3	Q9NRZ7	Q9H2J7
AGPAT5	Q9NUQ2	Q96BA8	AGTR1	P30556	Q03135
AHCYL2	Q96HN2	P31152	AHNAK	Q09666	P18827
AHNAK2	Q8IVF2-3	P24593	AHR	P35869	Q9Y618
AIF1	P55008	P06396	AIM2	O14862	P29466
AIP	O00170	Q92985	AK1	P00568	P10636-8
AK4	P27144	P13501	AKAP1	Q92667	P13861
AKAP7	Q9P0M2	P17612	AKIP1	Q9NQ31	P17612
AKR1C1	Q04828	P42330	AKR1C2	P52895	P42330
AKR7A2	O43488	P54252	AKT1	P31749	P54278
AKT1S1	Q96B36	P63104	AKT3	Q9Y243	P07900
ALAS1	P13196	Q16822	ALCAM	Q13740	P17931
ALDH1B1	P30837	Q9UMX1	ALDH1L2	Q3SY69	P42858
ALDH3A2	P51648	P27824	ALDH3B1	P43353	Q13231
ALDH6A1	Q02252	Q9H2A2	ALDOA	P04075-2	P42858
ALDOC	P09972	P13569	ALG1	Q9BT22	P13569
ALG3	Q92685	Q9H015	ALG6	Q9Y672	Q9BVK2
ALKAL2	Q6UX46	Q9UM73	ALMS1	Q8TCU4	O75923
ALX1	Q15699	P10276	ALYREF	Q86V81	P06493
AMDHD2	Q9Y303-2	P55212	AMFR	Q9UKV5	P51668
AMIGO1	Q86WK6	Q9H2H9	AMMECR1	Q9Y4X0	Q9Y2T1
AMN	Q9BXJ7	O60494	AMPD2	Q01433	P18827
ANAPC1	Q9H1A4	Q9UM11, Q12834	ANAPC11	Q9NYG5-2	Q9H2F3
ANAPC4	Q9UJX5	P51955	ANAPC5	Q9UJX4	P51955
ANGEL1	Q9UNK9	P06730	ANGPTL2	Q9UKU9	P11021
ANK2	Q01484	Q9P2E9	ANK3	Q12955	Q15796
ANKEF1	Q9NU02	P07237	ANKLE2	Q86XL3	P67775
ANKRD10	Q9NXR5-2	Q15915	ANKRD13A	Q8IZ07	P51684
ANKRD36	A1A5B0	Q08379	ANKRD36B	Q8N2N9-4	O00560
ANKRD45	Q5TZF3-1	Q96CV9	ANKRD46	Q86W74-2	Q969X2
ANKRD52	Q8NB46	P53350	ANKRD53	Q8N9V6-2	Q8IVS8
ANKRD54	Q6NXT1	Q06187	ANLN	Q9NQW6	P07948
ANO1	Q5XXA6	P13569	ANO9	A1A5B4	Q12959
ANOS1	P23352	P11362	ANP32A	P39687	O15169
ANTXR1	Q9H6X2	P11021	ANTXR2	P58335	P13423
ANXA1	P04083	Q13546	ANXA2	EBI-8785901, P07355	Q8NBP7
ANXA3	P12429	P13569	ANXA4	P09525	P55212

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ANXA5	P08758	Q96CV9	ANXA6	P08133	P10636-8
ANXA8	P13928	Q00403	AP1B1	Q10567	P42566
AP1M1	Q9BXS5	P18848	AP1S1	P61966-2	P37173
AP1S2	P56377	P56539	AP2A1	O95782	P29353
AP2A2	O94973	P29353	AP2M1	Q96CW1	P29353
AP2S1	P53680	Q96D71, Q2M2I8, P42566	AP4M1	O00189	P49354
APBA1	Q02410	O14936	APBA2	O35431	P05067-4
APEX2	Q9UBZ4	Q13257	APMAP	Q9HDC9	P13569
APOB	P04114	P01130	APOBEC3B	Q9UH17	P11802
APOC1	P02654	Q8N5I4	APOE	P02649	P09238
APOL6	Q9BWW8	Q9UHA4	APP	P05067	P09958
APPL2	Q8NEU8	O14908	AR	P10275	Q15648
ARAF	P10398	Q92985	ARAP3	Q8R5G7	P12931
ARCN1	P48444	P42858	ARF1	P84080	Q9NZ52
ARF4	P18085	Q96PM5	ARF6	P62330	Q9NZ52
ARFGAP1	Q9NP61	P50454	ARFGAP2	Q8N6H7	P05155
ARFGAP3	Q9NP61	P50454	ARFGEF1	Q9Y6D6	Q9NRM0
ARFGEF3	Q5TH69	P15907	ARGLU1	Q9NWB6	P52298
ARHGAP1	Q07960	P61586	ARHGAP10	Q5T5U3	P29692
ARHGAP12	Q8IWW6	P60709	ARHGAP19	Q14CB8	P54252
ARHGAP21	Q5T5U3	P29692	ARHGAP22	Q7Z5H3	P48730
ARHGAP23	Q9P227	P61981	ARHGAP24	Q8N264	P21333
ARHGAP26	Q9UNA1	A1A4S6	ARHGAP31	Q2M1Z3	Q9Y490
ARHGAP35	Q9NRY4	P05120	ARHGAP42	A6NI28	P46108
ARHGAP44	Q17R89	P49662	ARHGAP8	P85298-4	Q13526
ARHGAP9	Q9BRR9	Q14192	ARHGDIA	P52565	P61586
ARHGDIB	P52566	P01100	ARHGEF12	Q9NZN5	Q9UM54
ARHGEF15	O94989	Q13526	ARHGEF16	Q5VV41	Q14160
ARHGEF17	Q96PE2	P61981	ARHGEF2	Q92974	O43524
ARHGEF26	Q96DR7	O14936	ARHGEF35	A5YM69	P13716
ARHGEF4	Q9NR80-3, Q9NR80	P25054, P63104	ARHGEF5	Q12774	P29353
ARHGEF9	O43307	Q92993	ARID2	Q68CP9	P52292
ARID3B	Q8IVW6	P48431	ARID4A	P29374	Q13547
ARID4B	Q4LE39	P14921	ARID5B	Q14865	Q13547
ARIH1	Q9Y4X5	Q9BYZ6	ARK2C	Q6ZSG1	P12755
ARL10	Q8N8L6	P21453	ARL11	Q969Q4	Q96EF0
ARL15	Q9NXU5	O43772	ARL2BP	Q9Y2Y0	P36404
ARL4D	P49703	P13284	ARL5B	Q96KC2	Q9Y5M8
ARL6IP1	Q15041	Q00059	ARL6IP5	O75915	P25103
ARMC12	Q5T9G4-2	Q05329	ARMC6	Q6NXXE6	O14975
ARMC9	Q7Z3E5	P21709	ARMCX3	Q9UH62	P04004
ARPC1A	Q92747	Q9UJU6	ARPC1B	O15143	O00401
ARPC2	O15144	Q16236	ARPC4	P59998	O00401
ARPC5	O15511	P00533	ARPC5L	Q9BPX5	P47756
ARPIN	Q7Z6K5	Q96NR8	ARRB1	P49407	P12931
ARRB2	P32121	Q13233	ARRDC1	Q8N5I2	Q96PU5
ARSA	O43681	P10451	ARSB	P15848	P01185
ARSI	Q5FYB1	P11021	ARSJ	Q8N5N6	Q13077
ART5	Q96L15	O15173	ASAP1	Q9ULH1	P12931

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ASAP2	O43150	P19174	ASB1	Q9Y576	P01222
ASB6	Q9NWX5	O60551	ASB8	Q9H765	Q15369, Q93034
ASCC2	Q9H1I8	Q99633	ASF1B	Q9NVP2	P49736
ASPH	Q12797-6	Q9HBE4	ASPHD2	Q6ICH7	O75063
ASRGL1	Q7L266	Q13287	ASS1	P00966	Q9Y4Z2
ASTN1	O14525-2	Q14145	ASXL1	P59598	Q15788
ATAD1	Q8NBU5	P04201	ATAD3A	Q9NVI7	Q14197
ATAD3B	Q5T9A4	P21802	ATAD5	Q96QE3	P40937, P35249
ATAT1	Q5SQI0	P63104	ATCAY	Q86WG3	Q9UNE7
ATF3	P18847	P18847	ATF5	Q9Y2D1	Q13227
ATF6B	Q99941	Q8TC07	ATF7IP	Q6VMQ6-2	Q96KQ7
ATG10	Q9H0Y0	Q9H1Y0, O94817	ATG101	Q9BSB4	O75385
ATG13	O75143	O75385	ATG14	Q6ZNE5	P41208
ATG16L2	Q8NAA4	Q9H1Y0	ATG2A	Q2TAZ0	Q9GZQ8
ATG4A	Q8WYN0	Q9GZQ8	ATG7	O95352	P29692
ATG9A	Q7Z3C6	Q96QE2	ATL3	Q6DD88	Q92911
ATM	Q13315	Q9NZQ7	ATN1	Q86V38	Q13371
ATOSB	Q7L5A3	Q9NRD5	ATOX1	O00244	P35670
ATP13A1	Q9HD20	Q96FT7	ATP1A3	P13637	P21453
ATP2A3	Q93084	P0DTC5	ATP2B4	P23634	P01258
ATP5F1D	P30049	P23786	ATP5MC1	P05496	P21217
ATP6AP1	Q15904	Q9NRD5	ATP6V0B	Q99437	P22732
ATP6V0C	P27449	Q15904	ATP6V0D1	P61421	Q15904
ATP6V0E1	O15342	P05090	ATP6V1E1	P36543	Q9UI12
ATP6V1F	Q16864	Q9UI12	ATP6V1H	Q9UI12	P42772
ATP7B	P35670	O00244	ATP8A1	Q9Y2Q0	Q9NP91
ATP9A	O75110	Q15363	ATRIP	Q8WXE1	P50750
ATRX	P46100	P84243, Q9UER7	ATXN1	P54253	Q6VY07
ATXN3	P54252	Q969S8	ATXN7L1	Q9ULK2-2	P06396
AUP1	Q9Y679	Q9UBV2	AURKA	O14965	Q9ULW0
AURKC	Q9UQB9	O15392	AVPI1	Q5T686	Q9NPB3
AXIN2	Q9Y2T1	Q99728	AXL	P30530	P19174
B2M	P61769	P59596	B3GALT5	Q9Y2C3	P81534
B3GALT6	Q96L58	P40855	B3GAT3	O94766	Q9H5K3
B3GLCT	Q6Y288	P13671	B3GNT3	Q9Y2A9	P27824
B4GALT4	O60513	P11021	B4GALT6	Q9UBX8	P01185
B4GALT7	Q9UBV7	P53611	B9D2	Q9BPU9	P47897
BACE1	P56817	P10997	BACE2	Q9Y5Z0	P10997
BACH2	Q9BYV9	O15525	BAD	Q92934	O43521
BAG2	O95816	Q9ULW0	BAG3	O95817	Q9BRX2
BAHCC1	Q9P281	Q13547	BAIAP2	Q9UQB8	Q9UQB8
BANF1	O75531	O75531	BANK1	Q8NDB2	P51451
BAP1	Q99496	P08183	BASP1	P80723	P10636-8
BATF2	Q8N1L9-1	P05412	BAZ2B	Q9UIF8	Q9UBB9
BBS4	Q96RK4	P07237	BCAN	Q96GW7	P07948
BCAP31	P51572	Q14790	BCAR3	O75815, O75815-1	P56945
BCAT1	P54687	Q8TBF2	BCAT2	O15382	P10809
BCHE	P06276	P06276	BCL11A	Q9H165	Q13547
BCL11B	Q9C0K0	P46531	BCL2L11	O43521	P07237
BCL2L14	Q9BZR8	O14757	BCL2L2	Q92843	Q13794

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
BCL3	P20749	O95999	BCL6B	A8KA13	O95429
BCL7B	Q9BQE9	Q969G3, Q12824	BCL7C	Q8WUZ0	Q9H8M2
BCL9	O00512	P35222	BCL9L	Q67FY2	P35222
BCS1L	Q9Y276	O14786	BDNF	P23560-2	P18428
BDP1	P46678	Q9UL45	BEND3	Q5T5X7	Q13547
BEND5	Q7L4P6	Q8N8N7	BEND7	Q8N7W2-2	P11309
BET1L	Q9NYM9	Q9P2W9	BEX1	Q9HBH7	Q9NRD5
BEX2	Q9BXY8	P19883	BEX3	Q00994	Q00994
BEX5	Q5H9J7	O14561	BFAR	Q9NZS9	P51668
BFSP1	Q12934-2	Q13895	BHLHE40	O14503	O94955
BICD2	Q8TD16	P53350	BIN3	Q9NQY0	Q9BRT9
BIRC2	Q13490	Q14790	BIRC5	O15392	P06493
BIVM	Q86UB2	Q13114	BLM	P54132	P39748
BLMH	Q13867	O95336	BLOC1S3	Q6QNY0	Q9UL45, Q96EV8, P78537
BLVRB	P30043	Q8WVK2	BMAL1	O00327-8	O14977
BMERB1	Q96MC5	Q8WZA2	BMF	Q96LC9	Q07817
BMI1	P35226	P60484	BMP1	P13497	O60216
BMP4	P12644	Q9UMX1	BMP7	P18075	Q9BUH8
BMPR1A	P36894	P12644	BMS1	Q14692	Q9H9Z2
BNIP1	Q12981	Q9NZ43, Q9P2W9, O75396	BOC	Q9BWV1	Q14623
BOD1	Q96IK1-2	Q9UI12	BOD1L1	Q8NFC6	P52292
BOK	Q9UMX3	P16284	BOLA2B	Q9H3K6	P19320
BOLL	Q8N9W6-4	Q08426	BOP1	Q14137	P06748
BORCS8	Q96FH0	P78537	BPTF	Q12830-4	P62805
BRCA2	P51587	Q86YC2	BRD1	O95696	Q9NRD5
BRD7	Q9NPI1	Q09472	BRF2	Q9HAW0	P20226
BRI3	O95415	P26572	BRINP1	O60477	P09651
BRMS1	Q9HCU9	P56524	BRPF1	P55201	P62805
BRSK2	Q8IWQ3-3	Q9Y4K3	BRWD1	Q9NSI6-4	P07550
BSPRY	Q6P6S3	P63104	BST2	Q10589	Q10589
BTAF1	O14981	Q9Y639	BTBD6	Q96KE9	Q9UER7
BTF3L4	Q96K17	Q9NRG4	BTG1	P62324	Q13158
BTN2A1	Q7KYR7-4	Q8IY26	BUB1B	O60566, Q9Z1S0	Q12834
C11orf24	Q96F05	Q96BA8	C11orf54	Q9H0W9-3	Q13153
C11orf65	Q8NCR3	P12830	C11orf68	Q9H3H3	Q15056
C12orf4	Q9NQ89	P55212	C12orf57	Q99622	O14907
C14orf39	Q8N1H7	P80297	C15orf39	Q6ZRI6	Q13257
C16orf54	Q6UWD8	P54252	C16orf87	Q6PH81	Q9UM11
C19orf47	Q8N9M1-2	P60900	C19orf53	Q9UNZ5	P10636-8
C1GALT1	Q9NS00	P78423	C1QTNF1	Q9BXJ1	P02751
C1QTNF3	Q9BXJ4	P42858	C1QTNF5	Q9BXJ0	P43080
C1R	P00736	P05155	C1S	P09871	P05155
C1orf122	Q6ZSJ8	Q16576	C1orf226	A1L170	P53350
C1orf54	Q8WWF1	Q14703	C1orf94	Q6P1W5	Q9NWB1
C2CD5	Q86YS7	P43115	C2orf15	Q8WU43	Q15109
C2orf68	Q2NKX9	O75496	C2orf72	A6NCS6	O15431
C5	P01031	P13671, P02748, P10643	C5orf24	Q7Z6I8	Q15831
C6orf141	Q5SZD1	Q15149	C6orf47	O95873	P29371

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
C7orf50	Q9BRJ6	Q9H9Z2	C9orf72	Q96LT7, Q96LT7-2, Q96LT7-1	O75385
C9orf78	Q9NZ63	O75152	CA11	Q9NS71	Q8IY26
CA14	Q9ULX7	Q96BI1	CA8	P35219	P02768
CABP5	Q9NP86	O00560	CACNA1C	Q13936	Q06413
CACNB1	Q02641	P55040	CACNB2	Q08289	O14744
CACNB4	O00305-2	P21579	CACNG5	Q9UF02	Q8N9R8
CADM1	Q9BY67	Q9BY67	CADPS	A2RRN7	Q96CV9
CALB1	P05937	Q9Y6K9	CALCOCO1	Q9P1Z2	Q8TC07
CALCOCO2	Q13137	P30626	CALD1	Q05682	Q13813
CALHM5	Q8N5C1	Q00534	CALM3	P62158	O15554
CALN1	Q9BXU9	Q9NZ43	CALR	P27797	Q9ULW0
CALU	O43852	Q05513	CAMK1D	Q8IU85	Q14012
CAMK2D	Q13557	Q96PM5	CAMK4	Q16566	Q9Y2X7
CAMKK1	Q8N5S9-2	P42858	CAMKK2	Q96RR4	Q16539
CAMKMT	Q7Z624	P08123	CAMKV	Q8NCB2	P98160
CAMSAP3	Q9P1Y5	P61981	CAMTA2	O94983-5	Q9Y316
CAND1	Q86VP6	Q93034, Q9UBF6	CANT1	Q8WVQ1	Q96JA1
CAP1	Q01518	P0DTC3	CAP2	P40123	P23528
CAPG	P40121	P11142	CAPN1	P07384	Q92934
CAPN13	Q6MZZ7	Q9UPT8	CAPN15	O75808	Q9NWF9
CAPN2	P17655	P07237	CAPN3	P20807-4	Q969Y2
CAPN6	Q9Y6Q1	Q6PJG9	CAPNS1	P04632	P20936
CAPZB	P47756	Q96P48	CARD11	Q9BXL7	Q9UDY8
CARD19	Q96LW7	O95999	CARMIL1	Q5VZK9	Q00613
CASK	O14936	Q13009	CASP4	P49662	P51878
CASP7	P55210	Q13177	CASP8AP2	Q9UKL3	Q14790
CASS4	Q9NQ75-2	Q08379	CAST	O15446	O15160, O95602, P19388, P61218
CAV1	P33724	P59632	CAV2	P51636	Q15392
CAVIN1	Q6NZI2	P18827	CAVIN2	O95810	Q6NZI2
CAVIN3	Q969G5	Q6NZI2	CAVIN4	Q5BKX8	Q13895
CBFB	Q13951	Q13761	CBLC	Q9ULV8	Q9BQP7
CBR1	P16152	P00533	CBR3	O75828	P16152
CBS	P35520	P17612	CBX2	Q14781-2	Q00403
CBX3	Q13185	Q9UI95	CBX6	O95503	O00425, Q9Y6M1
CCDC102B	Q68D86	P01189	CCDC107	Q8WV48	Q12981, Q9P2W9
CCDC112	Q8NEF3-2	Q9Y2T1	CCDC113	Q9H0I3	P36941
CCDC124	Q96CT7	P06748	CCDC136	Q96JN2-2	Q96HC4
CCDC137	Q6PK04	P78563	CCDC138	Q96M89-2	Q16512
CCDC14	Q49A88-3	Q96T60	CCDC141	Q6ZP82-1	P06396
CCDC150	Q8NCX0-3	Q99633	CCDC152	Q4G0S7	O95363
CCDC167	Q9P0B6	P25103	CCDC17	Q96LX7-5	P54252
CCDC172	P0C7W6	P42566	CCDC186	Q7Z3E2	P07947
CCDC25	Q86WR0	Q9H2D1	CCDC27	Q2M243	Q9BYJ1
CCDC32	Q9BV29	Q9NUX5	CCDC71	Q8IV32	Q13547
CCDC71L	Q8N9Z2	Q9Y4Z2	CCDC85A	Q96PX6	P51955
CCDC86	Q9H6F5	P06748	CCDC88A	Q3V6T2	Q13257
CCDC88C	CCDC88C	O76024	CCDC89	Q8N998	P43897
CCDC92	Q53HC0	Q9UPV0	CCDC93	Q567U6	P36941

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CCIN	Q13939	P42858	CCKBR	P32239	Q06124
CCL2	P13500	P13501	CCM2	Q9BSQ5	O60216
CCM2L	Q9NUG4	Q9NZ43	CCN1	P20248	Q13415
CCN2	P29279	P02751	CCN3	P48745	O43559
CCN4	O95388-2	P06396	CCN5	O76076	P49639
CCNA2	P20248	Q13415	CCNB1IP1	Q9NPC3	Q93034
CCNC	P24863	Q15648	CCND1	P24385	Q13761
CCND2	P30279	P11802	CCND3	P30281	P06493
CCNDBP1	O95273	Q96T88	CCNE1	P24864	P61024
CCNF	P41002	O43303	CCNG1	P51959	P42858
CCNJ	Q5T5M9	P24941	CCNJL	Q8IV13	O00443
CCNL1	P49736	P03433	CCNT1	O60563	P04608
CCNT2	O60583-1, O60583-2, O60583	P50750	CCNY	Q8ND76	P61981
CCS	O14618	P98170	CCSER2	Q9H7U1	P61981
CD109	Q6YHK3	P23946	CD151	P48509	P05556
CD164	Q04900	P61073	CD200	P41217	Q8TD46
CD24	P25063	P07948	CD247	P20963	P43403, Q9Y2R2
CD248	Q9HCU0	P08887	CD274	Q9NZQ7	Q13315
CD2AP	Q9Y5K6	Q8WUM4	CD2BP2	O95400	P36873
CD320	Q9NPF0	P20062	CD40	P25942	O15397
CD44	P16070	P10451	CD53	P19397	P78382
CD55	P08174	P48960	CD59	P13987	Q15363
CD63	P08962	P01033	CD72	P21854	Q96BA8
CD74	P04233	P16070	CD81	P60033	P48509
CD83	Q01151	P40818	CD99L2	A0A0S2Z5R8	Q96BA8
CDA	P32320	Q96QG7	CDC14A	Q9UNH5	P61981
CDC14B	O60729	P43246	CDC20	Q12834	Q09472
CDC25A	P30304	O96017	CDC25B	P30305	Q9Y297, P63104
CDC27	P30260	P51955	CDC37	Q16543	P12931
CDC42	P60953	P52565	CDC42BPB	Q9Y5S2	Q07157
CDC42BPG	Q6DT37	Q9UHD2	CDC42EP1	Q00587-2	P21673
CDC42EP2	O14613	Q9NRD5	CDC42EP3	Q9UKI2	Q5S007
CDC42EP4	Q9H3Q1	P42858	CDC45	O75419	Q9HAW4
CDC7	O00311	P42858	CDCA3	Q99618	Q9BY43
CDCA4	Q9BXL8	Q96LA8	CDCA7	Q9BWT1	P42858
CDCA7L	Q96GN5	P46087	CDCA8	Q53HL2	P21579
CDH1	P12830	Q9ULW0	CDH13	P55290	P05556
CDH2	P19022	O14907	CDH3	P22223	P35222
CDH8	P55286	O60716	CDIPT	O14735	Q8N9R8
CDK15	Q96Q40	Q13451	CDK18	Q07002	Q9Y6D9
CDK19	Q9BWU1	Q15648	CDK2	P24941	Q09472
CDK2AP1	O14519	P52294	CDK5RAP3	Q96JB5	Q9NZQ7
CDK6	Q00534	P42772	CDKAL1	Q5VV42	P06239
CDKL1	Q00532	Q13451	CDKN2A	Q8N726	P08047, P04198
CDKN2B	P42772	Q9UI12	CDKN2C	P42773	P01185
CDKN2D	P55273	P22736	CDO1	Q16878	Q14012
CDON	O35158	Q13635	CDR2L	Q86X02	Q969Z0
CDS1	O96017	P30304	CDYL	Q9Y232	Q13257
CDYL2	Q8N8U2	O14744	CEBPZ	Q03701	P49841

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CELF1	Q92879	P07910	CELF5	Q8N6W0	O15234
CELSR2	Q9HCU4	Q92838	CENPB	P07199	P67870
CENPC	Q03188	P62736	CENPH	Q9H3R5	P20700
CENPJ	Q9HC77	P61981	CENPO	Q9BU64	Q13352
CENPT	Q96BT3	P56539	CENPU	Q71F23-1, Q71F23	P53350
CENPV	Q7Z7K6	P42858	CEP162	Q5TB80	Q9NPJ6
CEP164	Q9UPV0	Q6IQ55	CEP170B	Q9Y4F5	P63104
CEP295	Q9C0D2	Q9NV70	CEP44	Q9C0F1	Q9UL45
CEP55	Q53EZ4	O00267	CEP57	Q86XR8	P01106
CEP70	Q8NHQ1	P43897	CEP83	Q9Y592-2	P18848
CEP89	Q96ST8, Q96ST8-3	Q9NRD5	CEP95	Q96GE4	P36941
CER1	O95813	P35813	CERCAM	Q5T4B2	O95231
CERK	Q8TCT0	O75496	CERS4	Q9HA82	Q9UKG4
CERS6	Q6ZMG9	Q9H244	CERT1	Q9Y5P4	O00186
CETN3	O15182	Q01831	CFAP410	O43822	O00244
CFAP91	Q7Z4T9	Q9C005	CFL1	P23528	P53667
CFL2	Q549N0, Q9Y281	Q9NUX5	CFLAR	O15519, O15519-1	Q14790
CGB3	P0DN86	P01215	CGB5	P0DN86	P01215
CGB8	P0DN86	P01215	CGN	Q9P2M7	Q07157
CGNL1	Q0VF96	P61981	CHCHD2	Q9Y6H1	P07237
CHCHD5	Q9BSY4	P53701	CHD1	O14646	Q96KQ7
CHD3	Q12873	Q99728	CHD7	Q9P2D1	P48431
CHD9	Q3L8U1-3	Q07869	CHDH	Q8NE62	O95861
CHEK1	O14757	Q06609	CHEK2	O96017	P30304
CHFR	Q96EP1-2, Q96EP1	P51668	CHGA	P26339	P00441
CHGB	P05060	Q5S007	CHI3L1	P36222	P17931
CHIC2	Q9UKJ5	O96015	CHID1	Q9BWS9	Q8IV16
CHML	P26374	Q92696	CHMP1A	Q9HD42	P40818
CHMP1B	Q7LBR1	P40818	CHMP3	Q9Y3E7	Q9NPC8
CHMP4B	Q9H444	Q96FZ7, Q8WUX9, Q9Y3E7	CHMP4C	Q96CF2	P40818
CHMP6	Q96FZ7	Q9H444	CHN2	P52757	P63000
CHODL	Q9H9P2	P24593	CHORDC1	Q9UHD1	P53041
CHP1	Q99653	Q86UL3	CHPF2	Q9P2E5	P0DTC8
CHRM3	P20309	Q92538	CHRNA3	P32297	Q8NC56
CHRNA5	P30532	P59665	CHRNA7	P36544	Q13573
CHRNA9	Q9UGM1	O14757	CHRNBI	P11230	Q5T9L3
CHRNA9	Q9UGM1	O14757	CHST14	Q8NCH0	P27824
CHRNA9	Q9UGM1	O14757	CHST6	Q9GZX3	Q8NCG5
CHST3	Q7LGC8	P01185	CIAO2B	Q9Y3D0	P08670, P06396
CHST9	Q7L1S5	O75197	CIC	Q9U1H0	P28482
CIB1	Q99828	O60422	CIMIP2A	Q6J272	O14964
CIMAP3	Q8TCI5	Q9BT73	CIPC	Q9C0C6	O15516
CINP	Q9BW66	Q9H3L0	CIT	O14578	O14907
CISD3	P0C7P0	Q6PJG9	CIZ1	Q9ULV3	Q9P0V3
CITED2	Q99967	Q09472	CKAP4	Q07065	O75326
CKAP2	Q8WWK9	Q13257	CKMT1A	P12532	O76024
CKLF	Q13887	P0CG48	CLCN4	P51793	Q8WTS1
CLC	Q9UBD9, Q9UBD9-PRO_0000015616	P26992, O75462	CLDN1	Q9NY35	Q9ULC8
CLCN7	P51798	Q53GS7			

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CLDN10	P78369	Q9H902	CLDN11	O75508	P26715
CLDN19	Q8N6F1-2	P49682	CLDN6	P56747	P45983
CLDN7	O95471	P21854	CLEC11A	Q9Y240	Q14703
CLEC16A	Q2KHT3	Q9H4P4	CLEC2B	Q92478	P42858
CLEC2D	Q9UHP7, Q9UHP7-3	Q86TM6, O76024	CLIC1	O00299	O75832
CLIC4	Q9Y696	O15247	CLIP1	P30622, Q9JK25, P30622-1	Q14203
CLIP3	Q96DZ5	O76024	CLIP4	Q8N3C7	Q9NPC8
CLK2	P49760	Q9C0J8, Q7Z6E9	CLK4	Q9HAZ1	P52294
CLMP	Q9H6B4	Q15058	CLN6	Q9NWW5	Q96BA8
CLPB	Q9H078	P46199	CLPP	Q16740	P54252
CLPTM1	O96005	P31213	CLPTM1L	Q96KA5	Q9H1C4
CLPX	O76031	Q9Y297	CLSTN1	O94985	Q8IV16
CLTA	P09496	Q13627	CLTB	P09497	O00443, P42566
CLU	P10909	P30101	CLUH	O75153	Q96PE7
CMPK1	P30085	Q6DKK2	CMTM4	Q8IZR5	Q9NZQ7
CMTM8	Q8IZV2	P00533	CNFN	Q9BYD5	P49639
CNGA3	Q16281	Q86XL3	CNKSR1	Q969H4	P51587
CNKSR2	Q8WXI2	P00533	CNMD	O75829-2	Q8WXH5
CNN1	P51911	Q8N0S2	CNN2	Q99439	P11142
CNOT1	A5YKK6	Q8NDV7	CNOT6	Q9ULM6	P47756
CNOT7	Q9UIV1	P09237	CNP	P09543	Q12824
CNPPD1	Q9BV87	O14975	CNPY4	Q8N129	Q9BRQ8
CNRIP1	Q96F85	O95070	CNST	Q6PJW8	Q9NUX5
CNTN1	Q12860	Q92823	CNTN2	Q02246	P05067
CNTNAP1	P78357	Q9H2A9	CNTNAP2	Q9UHC6	Q12860
CNTNAP3	Q9BZ76	P01222	CNTROB	Q8N137	Q96T60
COBL	O75128	P13196	COCH	O43405	P08185
COG1	Q8WTW3	Q9NZQ7	COG4	Q9H9E3	Q9Y639
COG7	P83436	Q9NZQ7	COIL	P38432	Q96HC4
COL10A1	Q03692	Q8N9N5	COL12A1	Q99715	P08253
COL13A1	Q5TAT6	Q6ISS4	COL1A1	P02452	P30101
COL1A2	P08123	Q7Z624	COL25A1	Q8NE08	P50454
COL26A1	Q96A83-2	Q8WVX9	COL2A1	P02458-1	P06396
COL3A1	P02461-1	P09486	COL4A1	P02462	P14780
COL4A2	P08572	P14780	COL4A5	P29400-2	Q8TB40
COL4A6	Q14031	P05067	COL5A1	P20908	P14780
COL5A2	P05997	Q6ISS4	COL6A1	P12109	P14780
COL6A2	P12110	P14780	COL6A3	P12111	Q6ISS4
COL7A1	Q02388	Q5JRA6	COL8A1	P27658	O43559
COL8A2	Q4VAQ0	Q00059	COL9A3	Q14050	P00533
COLEC12	Q5KU26	P08253	COLGALT1	Q8NBJS	P02452
COMMD3	Q9UBI1	Q96PC5	COMMD4	Q9H0A8	Q14186, O75461
COMP	P49747	O00244	COMT	P21964	P30518
COPA	P53621	Q8WUJ3	COPB1	P53618	P56945
COPE	O14579	Q9NTJ5	COPG1	Q9Y678	Q9NTJ5
COPG2	Q9UBF2	P27824, P04233	COPS6	Q7L5N1	O94844
COPS7A	Q9UBW8	O94844	COPS9	Q8WXC6	Q92905
CORO1B	Q9BR76	Q9NWB7	CORO1C	Q9ULV4	P19320
CORO2A	Q92828	Q8N695	COTL1	Q14019	P09917

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COX4I1	P13073	P11182	COX5B	P10606	P11182
COX6A1	P12074	P42858	COX6B1	P14854	P00403
COX7C	P15954	P00403	COX8A	P10176	P06748
CPA4	Q9UI42	P01160	CPLX2	Q6PUV4	P43897
CPN1	P15169	P17252	CPNE1	Q99829	P13569
CPNE2	Q96FN4	Q8TBC4	CPSF6	Q16630	O95639
CPVL	Q9H3G5	P47756	CR2	P20023	P01024
CRABP1	P29762	P13569	CRABP2	P29373	Q9HAW8
CRACD	Q6ZU35	P47756	CRACR2A	Q9BSW2	P42858
CRB3	Q9BUF7-2	Q13190	CREB1	P16220	P18846
CREB3	O43889	O15554	CREB3L1	Q96BA8	Q96AA3
CREB3L2	Q70SY1	Q96BA8	CREB3L4	Q8TEY5	O43889
CREBZF	Q9NS37	O14867	CRELD2	Q6UXH1-2	P12830
CRHR1	P34998	P06850	CRIP1	P50238	P04183
CRIP2	P52943	P82979	CRIPTO	P13385	Q14116
CRLF1	O75462-PRO_0000011039, O75462	P26992, Q9UBD9	CRTAP	O75718	P02452
CRTC2	Q53ET0	Q9UMX1	CRTC3	Q6UUV7	P63104
CRY2	Q49AN0	Q9H0D6	CRYAB	P02511	P12830
CRYBB3	P26998	P05814	CRYM	Q14894	Q96CV9
CSE1L	P55060	P00533	CSF1	P09603	P07333
CSGALNACT1	Q8TDX6	Q9Y6M5	CSGALNACT2	Q8N6G5	Q8NBL1
CSK	P41240	Q68CZ2	CSMD1	Q96PZ7	P84022
CSNK1D	P48730	O75330	CSNK1G1	Q9HCP0	P78368
CSNK2A3	Q8NEV1	P67870	CSPG4	Q6UVK1	P17931
CSRP1	P21291	P00533	CSRP2	Q16527	Q7LC44
CST1	P01037	O94844	CST3	P01034	P05067
CST6	Q15828	P07711	CSTPP1	Q9H6J7-2	P02649
CTBP1	Q13363	P48552	CTDNBP1	O95476	Q9Y5M8
CTDSP1	Q9GZU7	O00560	CTH	Q96IQ7	Q15058
CTNNA2	P26232	O14907	CTNNA3	Q9UI47-2	O75582
CTNND2	Q9UQB3	O14907	CTPS2	Q9NRF8	Q8IWZ8
CTSA	P10619	Q9UM11	CTSB	P07858	P40692
CTSD	P07339	Q8NBJ7	CTSH	P09668	P07711
CTSK	P43235	P07711	CTSL	P07711	P43235
CTSV	O60911	P07711	CTSZ	Q9UBR2	Q7Z3S9, P0DPK4
CTTN	Q14247	P32121	CTTNBP2	Q8WZ74	Q9GZV8
CUBN	O60494	P27352	CUEDC1	Q9NWM3	Q9UM54
CUTA	O60888	P22736	CWC22	Q9HCG8	P21741
CWC27	Q6UX04	P0DTC4	CWF19L2	Q2TBE0	Q9Y3R4
CXADR	P97792-1	P19793	CXCL1	P09341	Q12933
CXCL12	P48061	P78556, P13501, O15444, Q99616	CXCL5	Q6I9S7	Q7Z3S9
CXCL8	P10145	P25025	CXCR4	P61073	O15554
CXXC4	Q9EQC9	O14640	CYB561	P49447	P09601
CYB561A3	Q8NBI2	Q9NUQ2	CYB561D1	Q8N8Q1	Q05329
CYB561D2	O14569	Q96BA8	CYB5A	P00167	O76024
CYB5D2	Q8WUJ1	Q9H5K3	CYB5R1	Q9UHQ9	P05919
CYB5R3	P00387	Q12893	CYBC1	Q9BQA9	O15155
CYBRD1	Q53TN4	P09601	CYC1	P08574	P40855

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CYFIP1	Q7L576	Q9UQB8	CYFIP2	Q96F07	Q9UQB8
CYLD	Q9NQC7	Q9Y6K9	CYTH3	O43739	Q13227
DAAM1	Q9Y4D1	Q9Y4D1	DAB2	P98078, P98082	Q9UM54
DAD1	P61803	Q14165	DAPK1	P53355	P14618-1
DAZAP2	Q15038	Q9HAU4	DAZL	Q64368	P84103
DBN1	Q16643	P60484	DBNDD1	Q9H9R9	P48730
DBNL	Q9UJU6	P06396	DCAF10	Q5QP82	Q13356
DCAF13	Q9NV06	P03372	DCBLD1	Q8N8Z6	P32971
DCBLD2	Q96PD2	P46109	DCLK1	O15075	O43602
DCLK2	Q8N568	P62258	DCLRE1B	Q9H816	P02649
DCN	P07585	O75063	DCP2	Q8IU60	Q92900
DCTD	P32321	Q9NRD5	DCTN1	Q14203	P30622
DCTN2	Q13561	P42858	DCTN3	O75935, Q9Z0Y1	Q14203
DCUN1D1	Q96GG9	P61081	DCX	O43602	Q9BSE5
DDA1	Q9BW61	Q9C0C7	DDAH1	O94760	O75832
DDI2	Q5TDH0-2	P14136	DDIT4L	Q96D03	Q9Y316
DDOST	P39656	Q14165	DDR1	Q08345-1	P29353
DDR2	Q16832	Q16288	DDRKG1	Q96HY6	P55017
DDX18	Q9NVP1	Q9H9Z2	DDX25	Q9UHL0	Q9UBB9
DDX39A	O00148	P21673	DDX41	Q9UJV9	Q13573
DDX49	Q9Y6V7	P04591	DDX51	Q8N8A6	P06748
DDX55	Q8NHQ9	P52294	DDX56	Q9NY93	Q9H9Z2
DEF6	Q9H4E7	Q8WZA2	DEF8	Q6ZN54-2	Q14957
DEGS1	O15121	Q68D85	DEK	P35659	Q9UER7
DENND11	A4D1U4	Q9Y337	DENND2A	Q9ULE3-2	Q96B97
DENND2B	P78524	P42858	DENND4A	Q7Z401	P18433
DEPDC1B	Q8WUY9	P13284	DERL1	Q9BUN8	Q8TB40
DES	P31001, P17661	Q13496	DESI1	Q6ICB0	P0CG47, P62987, P62979, P0CG48
DGAT1	O75907	Q9H5K3	DGCR2	P98153	Q8TC07
DHPS	P49366	P28482	DHRS13	Q6UX07	P32971
DHRS3	O75911	O60602	DHRS7B	Q6IAN0	P13569
DHX16	O60231	Q92620	DHX36	Q9H2U1	Q9H9Z2
DHX38	Q92620	Q8WUA2, Q92917, O60231	DIAPH3	Q9NSV4	P06276
DICER1	Q9UPY3, Q9UPY3-1	Q9UKV8	DIPK1B	Q5VUD6	P21673
DIRAS1	O95057	P52306	DIRAS3	O95661	P46199
DISC1	Q9NRI5	Q8WY54	DIXDC1	Q155Q3	O14641
DKK1	O94907	O75197	DKK3	Q9UBP4	P12236
DLC1	P63167	O75330	DLG2	Q15700	Q13224
DLG3	Q62936, Q92796	Q13224	DLG5	Q8TDM6	P61981
DLGAP1	O14490, P97836-5	P78352	DLGAP2	Q9P1A6-3	P54252
DLGAP3	O95886	Q08379	DLGAP4	Q9Y2H0-1	Q8N3R9
DLGAP5	Q15398	P61769	DLST	P36957	Q9ULW0
DMKN	Q6E0U4	P11021	DMRTA1	Q5VZB9	O76024
DMWD	G5E9A7	Q8WUX9	DMXL1	Q9Y485	P21281
DNA2	P51530	Q15185	DNAAF2	Q9NVR5	Q8NAV1
DNAJA3	Q96EY1-2, Q96EY1	P08581	DNAJA4	Q8WW22-2	Q8N9N5
DNAJB1	P25685	P11021	DNAJB12	Q9NXW2	P13569
DNAJB4	Q9UDY4	Q9Y266	DNAJC18	Q9H819	P43119

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DNAJC3	Q91YW3	P11021	DNAJC30	Q96LL9	P21802
DNAJC5	Q9H3Z4	Q92692	DNASE2	O00115	P27824
DNLZ	Q5SXM8	Q9BYZ2	DNM2	P50570, P39052	Q9P0V3
DNMBP	Q6XZF7, Q6XZF7-1	P50570, O00401	DNMT3A	Q9Y6K1	Q9Y6K1
DNMT3B	Q9UBC3	Q9Y6K1	DNTTIP1	Q9H147	P20248
DNTTIP2	Q5QJE6	Q6NZI2	DOCK10	Q96BY6	P42357
DOCK2	Q92608-2	P10636	DOCK3	Q8IZD9	Q92556
DOCK4	Q8N1I0	P62993	DOCK5	Q9H7D0	P46109
DOCK8	Q8NF50	Q99836	DOK1	Q99704	P46109
DOK4	Q8TEW6	Q05397	DPEP1	P16444	O43913
DPEP3	Q9H4B8	P11021	DPF3	Q92784-1, Q92784, Q92784-2	Q8NFD5, P51531, O96019
DPH1	Q9BZG8	P11142	DPH6	Q7L8W6	P43119
DPM3	Q9P2X0-2	O75396	DPP4	P27487	P27487
DPP9	Q86TI2-2	Q11203	DPPA2	Q7Z7J5	P06396
DPPA4	Q7L190	Q9BYJ1	DPY19L3	Q6ZPD9	P13671
DPYSL5	Q9BPU6	P06396	DRAM1	Q8N682	Q9NPL8
DRAP1	Q14919	Q00403	DRAXIN	Q8NBI3	O95631
DRG2	P55039	Q15742	DSC2	Q02487	Q14126
DSC3	Q02487	Q14126	DSCC1	Q9BVC3	P0CG13, P40938, P40937, P35249
DSG2	Q14126	P13569	DSTN	P60981	P60709
DSTYK	Q6XUX3	Q9Y639	DTNA	Q9Y4J8-9	P40763
DTNB	O60941-5	Q9BUH8	DTNBP1	Q96EV8	A1L4H1
DTWD2	Q8NBA8	P07437	DTX1	Q86Y01	Q96J02
DUSP1	P28562	Q16539	DUSP14	O95147	Q9BYZ6
DUSP16	Q9BY84	Q16539	DUSP6	Q16828	P28482
DUSP7	Q16829	Q16539	DVL1	O14640	Q9Y4D1
DYDC2	Q96IM9	O95602	DYNC1H1	Q14204	P43034
DYNC1I2	Q13409	P43034	DYNC2I2	Q96EX3	P84243
DYNLRB1	Q9NP97	P10636-8	DYRK4	Q9NR20	P07900
DZIP3	Q86Y13	P51668	E2F1	Q01094	P28749, P06400
E2F4	Q16254	O75496	E2F5	Q15329	Q08999, P06400
EAF2	Q96CJ1	Q16665	EARS2	Q5JPH6	Q9Y639
EBF2	Q9HAK2	Q13077	EBF3	Q9H4W6-2	Q9HAK2
EBNA1BP2	Q99848	P19525	ECE1	P42892	P17931
ECHS1	P30084	P11182	ECM1	Q16610	Q9NQ94
ECSCR	Q19T08	O14818	EDA	Q92838	P17480
EDAR	Q9UNE0	Q92838	EDARADD	Q8WWZ3	Q9Y4K3
EDC4	Q6P2E9	Q8IZH2	EDEM1	Q92611	Q9UMX1
EDF1	O60869	P37231	EEF1D	P29692-2, P29692	P26641, P29692
EEF2KMT	Q96G04	Q9NRD5	EEIG1	Q5T9C2-3	P56945
EEPD1	Q7L9B9	P18827	EFEMP1	Q12805	P24592
EFEMP2	Q95967	Q9UBX5, P15502	EFHC2	Q5JST6	Q13882
EFNA2	O43921	P54764	EFNA5	P52803	Q9UBD6
EGF	P01133	P00533	EGFL6	Q8IUX8	Q99962
EGFR	P00533	P01135, P01133	EGLN1	Q9GZT9	Q16665, Q99814
EGLN3	Q9H6Z9	P14618-1	EGR2	A0A0S2Z3U4	Q08379
EHBP1L1	Q8N3D4	Q96ES7	EHD1	Q9H4M9	Q9H1K0
EHD2	Q9NZN4	Q86UL3	EHD3	Q9NZN3	Q9H1K0

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EHD4	Q9H223	Q9H1K0	EIF2AK4	Q9P2K8	P36873
EIF3B	P55884	Q04637, P06730	EIF3C	Q99613	P47813
EIF3E	P60228	Q99814	EIF3G	O75821	Q14694
EIF3K	Q9UBQ5	P30281	EIF4A2	Q14240	Q04637, P06730
EIF4B	P23588	P53350	EIF4E3	Q8N5X7-2	Q16613
EIF4EBP1	Q13541	Q9UKV8	EIF4G1	Q04637-9	Q9NQ94
EIF5A	P63241	Q9NRD5	EIF5A2	Q9GZV4	O00560
EIF5B	O60841	P14921	EIF6	P56537	P19525
EIPR1	Q53HC9	O43323	ELAVL2	Q12926	O00425, Q9NZI8
ELAVL4	P26378-2	Q00994	ELF1	P32519	P08047
ELF3	P78545	Q16236	ELK1	P19419	P63279
ELK3	P41970	O00712, Q12857	ELL2	O00472	P07550
ELMO1	Q92556	Q8IZD9	ELMO2	Q96JJ3	Q8IZD9
ELN	P15502	Q9UBX5	ELOB	Q15370	O14543
ELOVL6	Q9H5J4	P09601	ELOVL7	A1L3X0	Q96EV8
EMB	Q6PCB8	P17931	EMC10	Q5UCC4	Q8N4V1
EMC2	Q15006	O43868	EMC8	O43402	P27824
EMC9	Q9Y3B6	Q8N4V1	EMD	P50402	Q99962
EME1	Q96AY2	Q96NY9	EMILIN1	Q9Y6C2	Q8NBN7
EMILIN2	Q9BXX0	Q99435	EML1	O00423	Q9Y266
EML3	Q32P44	P63104	EML4	Q9HC35	Q96PC5
EMP1	P54849	Q9Y5Q5	EMP3	P54852	A1L3X0
EMSY	Q7Z589	Q13547	EMX2	Q04743	P06730
EN1	Q05925	O00712, Q12857	ENC1	O14682	Q13501
ENDOD1	O94919	Q9NRZ7	ENDOG	Q14249	P11142
ENG	P17813	P08648	ENO1	P06733	P03496
ENO2	P09104	P06733	ENOX1	Q8TC92	O00560
ENPP4	Q9Y6X5	P17931	ENTPD1	P49961	Q96S59
ENTPD5	O75356	P06276	ENTPD6	O75354	P27824
ENTREP1	Q15884	Q9UM54	ENTREP3	P81408	Q96PU5
EPAS1	Q99814	P40818	EPB41	P11171-2, P11171-7	P05067
EPB41L2	O43491	Q8IX03	EPB41L3	Q9Y2J2	P49841
EPB41L5	Q9HCM4	Q8N3R9	EPCAM	P16422	P12830
EPHA4	P54764	P05177	EPHA6	Q9UF33	P04070
EPHA7	Q15375	Q16288	EPHA8	P29322	P11021
EPN1	Q9Y6I3	P42566	EPPK1	P58107	P08670
EPS8	Q08509, Q12929	Q9UQB8	EPS8L1	Q8TE68	Q9UQB8
ERAL1	O75616	P08123	ERAP1	Q9NZ08	Q15392
ERBB3	P21860	Q14451	ERBB4	Q15303	P23528
ERBIN	Q96RT1	P36873	ERC1	Q8IUD2	P01222
ERC2	O15083	P10242	ERCC1	P07992-3, P07992	P23025
ERCC2	P18074	P00533	ERCC6	Q03468	P00519
ERF	P50548	P28482	ERGIC1	Q969X5	P04601
ERGIC3	Q9Y282	Q96AA3	ERICH2	A1L162	P42858
ERLEC1	Q96DZ1	Q9UBV2	ERLIN1	O75477	Q86TM6
ERMP1	Q7Z2K6	Q9UBD6	ESF1	Q9H501	P06748
ESPL1	P60330, Q14674	O95997	ESR2	Q92731	Q15648
ESRP1	Q6NXG1-3	Q96CV9	ESRP2	Q9H6T0	P11142
ESRRG	P62508, P62508-3	Q9UBK2	ESYT1	Q9BSJ8	P22309
ETHE1	O95571	Q99757	ETS1	P14921-3	Q12778

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ETV1	P50549	Q8NHY2	ETV5	P41161	Q8NHY2
EVA1B	Q9NVM1	O43826	EXOC3	O60645	Q96NL6
EXOC3L2	Q2M3D2	Q9UNG2	EXOC7	Q9UPT5	Q9Y639
EXOSC4	Q9NPD3	O00560	EXT1	Q16394	Q03405
EXTL2	Q9UBQ6	P0DTC5	EYA1	Q99502	Q15475, Q9NPC8
EYA2	O00167-2, O00167, O00167-1	Q15475, Q9NPC8	EZR	P15311	P52565
F10	Q9BQ89	P49674	F11R	Q9Y624	Q05513
F2RL1	P55085	P04156	F3	P13726	P08709
F8A2	P23610	P42858	F8A3	P23610	P42858
FABP3	P05413	P05556	FABP5	Q01469	P13569
FABP7	O15540	O76024	FADD	Q13158	Q14790
FADS3	Q9Y5Q0	Q8N5I4	FAH	P16930	P00533
FAHD1	Q6P587-2	O76024	FAHD2B	Q6P2I3	P10809
FAM110C	Q1W6H9	P49674	FAM111A	Q96PZ2	P04637
FAM111B	Q6SJ93	P42858	FAM114A1	Q8IWE2	Q9UBN6
FAM117A	Q9C073	Q9NRD5	FAM117B	Q6P1L5	P78333
FAM118B	Q9BPY3	Q9Y639	FAM120A	Q9NZB2	Q92900
FAM120B	Q96EK7	Q12933	FAM124A	Q86V42	Q96QF0
FAM124B	Q9H5Z6-2	Q8TDD2	FAM133A	Q8N9E0	Q9NRZ7
FAM13A	O94988	P63104	FAM13C	Q8NE31	Q9UBB9
FAM161A	Q3B820	Q9BUH8	FAM161B	Q96MY7	Q9NRD5
FAM162A	Q96A26	P08238	FAM167A	Q96KS9	P42858
FAM168A	Q92567-2	Q96T21	FAM168B	A1KXE4-2	O75593
FAM169A	Q9Y6X4	Q96QG7	FAM177A1	Q8N128	O14975
FAM184A	Q8NB25	P06396	FAM20B	O75063	P07585
FAM20C	Q8IXL6	P10451	FAM210B	Q96KR6	P09601
FAM217B	Q9NTX9	Q8WY64	FAM219A	Q8IW50	O43924
FAM221A	A4D161	P55854	FAM222B	Q8WU58	Q9UBE8
FAM234B	A2RU67	O43913	FAM241A	Q8N8J7	Q96FE5
FAM241B	Q96D05	O43914	FAM3A	Q9BU27	Q9BXN2
FAM3B	P58499	P21217	FAM76B	Q5HYJ3-3	P42858
FAM83B	Q5T0W9	P49674	FAM83F	Q8NEG4	P48729
FAM83G	A6ND36	P48729	FAM89B	Q9QUI1	P12755
FANCD2	Q9BXW9	O00255	FANCD2OS	Q96PS1	Q13627
FANCL	Q9NW38	Q13485	FAR2	Q96K12	P40855
FARP1	Q9Y4F1	P18827	FARSA	Q9Y285	P10636-8
FAS	P25445-1, P25445	Q14790	FASTKD1	Q53R41	P07237
FAU	P62861	P06748	FAXC	Q5TGI0	Q9H668, Q2NKJ3
FBH1	Q8NFZ0	P42858	FBLIM1	Q8WUP2, Q71FD7	P52952
FBLN2	P98095	P14543	FBLN5	Q9UBX5	Q8N2S1
FBN1	P35555	P15502	FBP1	P09467	Q16665, Q99814
FBRS	Q9HAH7	P68400, P67870	FBXL14	Q8N1E6	Q9UPY3
FBXL16	Q8N461	Q9Y4D1	FBXL18	Q96D16	P61978
FBXL2	Q9UKC9	P53611	FBXL3	Q9UKT7	Q9UJ68
FBXL4	Q9UKA2	Q9Y3R4	FBXO16	Q8IX29	P41208
FBXO17	Q96EF6	O60551	FBXO2	Q9UK22	Q8N2S1
FBXO21	O94952	P62877, P63208, Q13616	FBXO22	Q8NEZ5	O14867
FBXO25	Q8TCJ0	P60709	FBXO31	Q5XUX0	Q13315

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
FBXW5	Q969U6	P14373	FCHSD2	O94868	P55072
FDF1	P37268	P35414	FEM1A	Q9BSK4	P09110
FEM1B	Q9UK73	P49593	FETUB	Q9UGM5	Q9H2K0
FEZ2	Q9UHY8	P54252	FGD4	Q96M96	P17252
FGD5	Q6ZNL6	O43463	FGD6	Q6ZV73	P63104
FGF1	P05230	P98160	FGF11	Q92914	Q96HN2
FGF13	Q92913	Q96EB6	FGF14	Q92915-2	P49638
FGF19	O95750	P31749	FGF2	P09038	P21802
FGFR2	P21802	O15520	FGFR4	P22455	Q03167
FGFRL1	Q8N441	Q15363	FGL2	Q14314	Q9BYC5
FHIT	P49789	P35222	FHL1	Q13642	P04156
FHL2	Q14192	Q9UBZ9	FHOD3	Q2V2M9, Q2V2M9-4	Q13501
FIBP	O43427	P80370	FIGN	Q5HY92	O14964
FIS1	Q9Y3D6	Q14790	FITM2	Q8N6M3	Q9H244
FKBP10	Q96AY3	P02452	FKBP15	Q5T1M5	O95816
FKBP1A	Q0VDC6	Q8WUX9	FKBP2	P26885	P42858
FKBP3	Q00688	Q00987	FKBP5	Q13451	O43318
FKBP7	Q9Y680	P49638	FKBP9	O95302	Q14118
FKRP	Q9H9S5	Q9Y2B1	FLI1	Q01543	P84022
FLII	Q13045	P13569	FLNA	P21333	P05556
FLNB	O75369	Q13233	FLNC	Q14315	Q92629
FLOT1	O75955	P42858	FLT1	P17948	Q12913
FLT3	P36888	Q06124	FLT4	P35916	P35968
FLVCR1	Q9Y5Y0	P31785	FMN2	Q9NZ56	Q12974
FMOD	P13605	P08603	FN1	P02751	P08648
FN3KRP	Q9HA64	P02788	FNBP1	Q8R511	P52198
FNBP1L	Q5T0N5	Q9Y4D1	FNDC3B	Q53EP0	Q9NPD5
FNTB	P49356	P01185	FOSL1	P15407	P26640
FOSL2	P15408	Q8NFP9	FOXA3	P55318	Q5VWX1
FOXB1	Q99853	P14373	FOXC1	Q12948	P08651
FOXF1	Q12946	Q9UPW6	FOXF2	Q12947	Q07954
FOXH1	P70056	Q13485	FOXJ2	Q9P0K8	Q9P0K8
FOXK1	P85037	P53611	FOXK2	Q01167	O43278
FOXN1	Q08050	Q13416	FOXN3	O00409	Q13573
FOXO1	Q12778	P06493	FOXP1	Q9H334	Q9BZS1
FOXP4	Q8IVH2	P48431	FRAS1	Q86XX4	P08311
FRAT2	O75474	P49841	FRMD6	Q96NE9-2	P17252
FRRS1L	Q9P0K9	Q969Z0	FSBP	O95073-2	P16885
FST	P19883	P03950	FSTL1	Q12841	P08571
FTH1	P02794	P02753	FTL	P02792	Q96L34, O43303
FTSJ1	Q9UET6	Q9Y5P4-2	FURIN	P09958	P50281
FUT1	P19526	Q92521	FUT2	Q10981	P36957
FUT8	Q9BYC5	P26572	FUT9	Q9Y231	Q9NWM8
FXR1	P51114	Q13416	FXR2	P51116	P11309
FXYD6	Q9H0Q3	Q9UBD6	FZD8	Q9H461	P56704
FZR1	Q9UM11	Q9HAW4	G0S2	P27469	Q07817
G2E3	Q7L622	P42858	G3BP2	Q9UN86-2, Q9UN86	Q14694
G6PD	P11413	Q8IXJ6	GAB1	Q13480	Q12913
GABBR1	Q9UBS5	P46459	GABBR2	O75899	P46459
GABPB2	Q06547-2, Q06547	P51610	GABRG2	P18507	P27824

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
GADD45A	P24522	P28715	GADD45B	O75293	O14733
GAK	O14976	Q9NQ11	GAL3ST1	Q99999	Q96JA1
GALNS	P34059	Q8NBK3	GALNT10	Q86SR1	Q07817
GALNT12	Q8IXK2	Q9H2A9	GALNT15	Q8N3T1	O15552
GALNT18	Q6P9A2	P01185	GALNT2	Q10471	P40855
GALNT4	Q8N4A0	Q13790	GALNT6	Q8NCL4	O75063
GALNT7	Q86SF2	Q8IV16	GALR1	P47211	P08908
GAN	Q9H2C0	Q99426	GANAB	Q14697	P01106
GAPDH	P04406	P35228	GARRE1	O15063	P0DTC5
GART	P22102	P13569	GAS2L1	Q99501	Q96B97
GAS2L3	Q86XJ1	Q96GD4	GATA2	P23769	O00712, P08651, Q12857
GATAD1	Q8WUU5	Q13547	GATAD2A	Q86YP4	P52292
GATD1	Q8WUU5	Q13547	GATM	P50440	Q9P1W8
GBA1	P04062	P51159	GBF1	Q92538	O43914
GBP1	P32455	P32455	GBP2	P32456	P32455
GBP3	Q8WXF7	P04156	GCA	P28676	P42858
GCC2	Q8IWJ2	P28482	GCFC2	P16383	P68400
GCH1	P30793	Q93009	GCLC	P48506	Q6IPR3
GCLM	P48507	P10746	GCN1	Q92616	Q13418
GCNT1	Q02742	P62987	GCSH	P23434, Q53XL7	Q13287
GDA	P16260	Q15116	GDAP1	Q8TB36	P54252
GDF3	Q9NR23	P35813	GDF9	O60383	P04183
GDI1	P31150	P51149	GEM	P55040	Q9BUH8
GEMIN2	O14893	P26367	GET3	O43681	P10451
GFER	P55789	P11802	GFPT2	O94808	Q9NUX5
GFRA1	P56159-2, Q62997	Q92673	GIPC1	O14908	Q03167
GIT2	Q14161	P53611	GJA1	P17302, P08050	Q07157
GJA5	P36382	Q9UKG4	GKAP1	Q5VSY0	O00560
GLB1L2	Q8IW92	P21217	GLCCI1	Q86VQ1	P61981
GLDC	P23378	Q9BQB4	GLIPR1	P48060	Q9BYJ1
GLIS2	Q9BZE0	O75496	GLIS3	Q8NEA6	Q9UMX1
GLMN	Q92990	P15907	GLO1	Q04760	P13569
GLP1R	P43220	Q9HD20	GLRX2	Q9NS18	Q7Z4W1
GLS	O94925	Q9NXA8	GLS2	Q9UI32	Q6GPH4
GLTP	Q9NZD2	Q9NQZ5	GLYR1	Q49A26-4	Q96CV9
GMCL1	Q96IK5	O43148	GMDS	O60547	O60547
GMEB2	Q9UKD1	P18827	GMPPA	Q96IJ6	Q92917
GMPPB	Q9Y5P6	Q8IVS8	GMPS	P49915	Q93009
GNA11	P29992	P31749	GNA13	Q14344	Q92888
GNA14	O95837	O96015	GNAI2	P04899-1	P08173
GNAO1	P09471	P55072	GNAS	P04896	P07550
GNB1	P62873	Q13371	GNB2	P62879	Q13371
GNG12	Q9UBI6	Q14192	GNG4	P50150	Q13371
GNG5	P63218	Q9UBC2	GNL3	Q9BVP2	Q9UKV8
GNPDA1	P46926	Q9Y303	GNS	P15586	Q8NBK3
GOLGA1	Q92805	P18848	GOLGA2P5	Q9HBQ8	O76024
GOLGA6A	Q9NYA3	Q9HAQ2	GOLGA6L9	A6NEM1	Q13415
GOLIM4	O00461	Q9Y2C2	GOLPH3L	Q9H4A5	P21453
GOLT1A	Q6ZVE7	Q9NRZ7	GORASP1	Q9BQQ3	P30622

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
GORASP2	Q9H8Y8	P11926	GOSR2	O14653	Q13190
GP1BA	P07359	P04275	GP6	Q9HCN6, Q9HCN6-1	P06241
GPAA1	O43292	Q12974	GPAT3	Q53EU6	Q9H2J7
GPAT4	Q86UL3	O60443	GPATCH2L	Q9NWQ4-1, Q9NWQ4	Q9GZV8
GPBP1	Q86WP2	P63244	GPBP1L1	Q9HC44	P01100
GPC1	P35052	O15530	GPC3	Q8IYG2	P42858
GPC4	O75487	O15530	GPC5	P78333	Q13485
GPC6	Q9Y625	P32249	GPKOW	Q92917	P23193
GPM6A	P51674	Q09328	GPM6B	P35803	P31645
GPN2	Q9H9Y4	P62875	GNPMB	Q14956-1, Q14956	P00533
GPR141	Q7Z602	O00186, Q12846	GPR143	P51810	Q16655, P14679
GPR156	Q8NFN8	Q9ULK5	GPR160	Q9UJ42	P17252
GPR18	Q14330	P42357	GPR182	O15218	Q9BVG9
GPR183	P32249	O14786	GPR21	Q99679	P17302
GPR37	O15354	P29274	GPR50	Q13585	P04731, P02795
GPRASP2	Q96D09	P08311	GPRASP3	Q6PI77	P48552
GPRC5A	Q8NFIJ5	P13569	GPRC5B	Q9NZH0	P48960
GPRIN2	O60269	P31947	GPS1	Q13098	Q9BYZ6
GPS2	Q13227	P18848	GPSM1	Q86YR5-3	O14503
GPX1	P07203	P13569	GPX7	Q96SL4	P19525
GPX8	Q8TED1	O43916	GRAMD1B	Q3KR37	P68400
GRAMD2B	Q96HH9	Q12981	GRB14	O88900	P11362
GRB2	P62993	P98082	GRB7	Q14451	Q92730
GREM1	O60565	P35968	GRHL1	Q9NZI5	Q9NZI5
GRHL2	Q6ISB3	Q02548	GRIA1	P42261	Q9UIH9
GRIA3	P42263	Q9UBS3	GRIA4	P48058-2	Q8WTS1
GRID1	Q9ULK0	P68871	GRIN2A	Q12879	P78352
GRIN2B	Q13224	P78352	GRIP1	Q9Y3R0	Q14653
GRIP2	Q9C0E4	Q15223	GRK3	P35626	P25098
GRK5	P34947	P25963	GRM3	Q14832	O43791
GRN	P28799	P98160	GRPR	P30550	Q7L8W6
GSDME	O60443	Q86UL3	GSE1	P38247	Q9UNE7
GSN	P06396	Q8N5V2	GSPT2	Q8IYD1	Q92900
GSTK1	Q9Y2Q3	O00142	GSTM3	P21266	Q9ULW0
GSTO1	P78417	P04792	GSTO2	Q9H4Y5	Q92922
GSTP1	P09211	Q12933	GTF2E1	P29083	Q9Y6K9
GTF2F1	P35269	Q9Y5B0	GTF2I	P78347	Q06187
GTF3C5	Q9Y5Q8	P53350	GTPBP2	Q9BX10	P08833
GTPBP3	Q969Y2	Q969Y2	GTSE1	Q9NYZ3	P53350
GUCD1	Q96NT3-2	Q9BUH8	GUK1	Q16774	Q6DKK2
GYS1	P13807	Q04446, Q9UQK1	H1-6	P22492	Q15361
H2AC15	P0C0S8	O95760	H2AC16	P0C0S8	O95760
H2AC18	Q6FI13	Q12824	H2AC19	Q6FI13	Q12824
H2AC21	Q8IUE6	P17096	H2AX	P16104	Q14191
H2BC10	P62807	Q96CW1	H2BC12	O60814	Q12824
H2BC21	Q16778	Q15843	H2BC4	P62807	Q96CW1
H3-3A	P84243	Q9Y6K1	H3C7	P68431	Q01831
HAAO	P46952	Q9NUX5	HABP4	Q5JVS0	Q99873
HACE1	Q8IYU2	Q96CV9	HADHA	P40939	O00468

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
HAND2	P61296-2	Q96CV9	HARS1	P12081	P12081
HAUS1	Q96CS2	Q12981	HAUS6	Q7Z4H7	O60341
HAUS8	Q9BT25	Q6VN20	HAX1	O00165	P01583
HCFC1	P51610	P08047	HCFC2	Q9Y5Z7	Q92560
HCN1	O60741	P01024	HDAC2	Q92769	P53350
HDAC7	Q9UKV0	Q06413	HDAC9	Q9UKV0	Q06413
HDLBP	Q00341	Q06609	HDX	Q7Z353	Q9UMX1
HECTD3	Q5T447	Q6NZI2	HEG1	Q9ULI3	P17931
HELB	Q8NG08	P20248	HELLS	Q9NRZ9	P01106
HEPHL1	Q6MZM0	Q9P2M4	HERC3	Q15034	Q9UNU6
HERC5	Q9UII4	P15880	HERPUD1	Q15011	Q86TM6
HESX1	Q9UBX0	P40763	HEY2	Q9UBP5	Q9Y5J3
HGH1	Q9BTY7	P08238	HGS	O14964	Q9Y5J6
HHIP	Q96QV1	Q15465	HIC1	Q14526	P35222
HID1	Q8IV36	O15357	HIF1A	Q16665	P40818
HIF3A	Q9Y2N7-4	Q16665	HIPK2	Q9H2X6	Q09472
HIRIP3	Q9BW71	P42858	HIVEP2	P31629	Q9Y297, P63104
HIVEP3	Q5T1R4	P61981	HK2	P52789	P19367
HLA-B	P01889	P61769	HLA-C	P10321	P01222
HLA-E	P13747	Q09328	HLX	Q14774	O75593
HM13	Q8TCT9	P04578	HMBOX1	Q6NT76	Q02548
HMBS	P08397	Q9NRD5	HMG20A	Q9NP66	P52292
HMGA2	P52927	P06400	HMGB1	P09429	P43246
HMGB2	P26583	P48729	HMGB3	O15347	Q02548
HMGCR	P04035	Q9Y5Z9	HMGCS1	Q01581	P18827
HMGN5	P82970	Q8WVK2	HMOX1	P09601	Q9NUX5
HMOX2	P30519	P0DTC3	HNRNPA1	P09651	Q8IZH2
HNRNPD	Q14103-4, Q14103	Q9NZI8, Q9Y6M1	HNRNPUL1	Q9BUJ2	Q9NPI1
HOMER1	Q86YM7	O75928-2	HOMER2	Q9NSB8	Q9H4P4
HOMER3	Q9NSC5	P50458	HOOK1	Q9UJC3	P53611
HOOK2	Q96ED9-2	Q9NPB3	HOOK3	Q86VS8	Q9NYA4
HOXA10	P31260	Q9Y4X5	HOXA7	P31268	O00560
HOXA9	EBI-1801917	Q03164	HOXB5	P09067	Q9Y4D7
HOXB7	P09629	P78527	HOXB9	P17482	P04004
HOXC11	O43248	P42858	HOXC4	P09017	Q9Y3P9
HOXC5	Q00444	P40424	HOXC8	P31273	Q5SW96
HOXC9	P31274	O75496	HOXD4	P09016	Q9NW38
HOXD8	P13378	Q5SW96	HPCAL4	Q9UM19	P13726
HPGDS	O60760	Q13370	HPN	P05981	P26572
HPS1	Q92902	Q9C0C7	HRAS	P01112	Q8N5V2
HS2ST1	Q7LGA3	P53794	HS6ST2	Q96MM7	P27824
HSBP1L1	C9JCN9	Q9UBN6	HSD17B10	Q99714	P13639
HSD17B4	P51660	P42858	HSD17B7	P56937	Q05940
HSD17B8	Q92506	Q96EN8	HSD3B7	Q9H2F3	Q96FE5
HSDL1	Q3SXM5	Q00169	HSF1	Q00613	P49137
HSP90AA1	P07900	P12931	HSP90AB1	P08238	P12931
HSPA12A	O43301	Q9Y639	HSPA12B	Q96MM6	O00418
HSPA1A	P08107	P27695	HSPA1B	P08107	P27695
HSPA2	P54652	Q8WUX9	HSPA4	P34932	Q53H54
HSPA4L	O95757	P11021	HSPA5	P11021	P01222

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
HSPB1	P04792	Q8N5V2	HSPB2	Q16082	Q9NQ94
HSPB6	O14558	P04792	HSPB7	Q9UBY9	Q9GZV8
HSPB8	Q9UJY1	P49247	HSPBP1	Q9NZL4	P32519
HSPD1	P10809	Q16822	HSPE1	P61604	P19320
HSPG2	P98160-PRO_0000391621, P98160-PRO_0000391622, P98160	P35968	HTATSF1	O43719	O75533
HTR2C	P28335	P08173	HTR3A	P46098	Q16585
HTRA1	Q92743	P01185	HTRA2	O43464	Q8WXH5
HTRA3	P83110-2, P83110	P60709	HYAL2	Q12891	P21217
HYAL3	O43820	O75553	HYI	Q5T013	Q05519
HYLS1	Q96M11	Q08379	IAPP	P10997	P10997
ICA1	Q05084	Q9NRD5	ICA1L	Q8NDH6	Q9NRD5
ICAM1	P05362	P15514	ICAM3	P32942	P11686
ICE1	Q9Y2F5	P55199	ICE2	Q659A1	P55199
ICMT	O60725	O00186	ID1	P41134	O14936
ID2	Q02363	Q96HC4	ID3	Q02535	P51149
ID4	P47928	Q9Y4Z2	IDH1	O75874	P0DP23
IDH3A	P50213	P13569	IDH3G	P51553	P10809
IER3	P46695	Q9UKG4	IER5	Q5VY09	Q00613
IFFO1	Q0D2I5-5	P02545	IFI16	Q16666	P38398
IFI27	P40305-2	P22736	IFI35	P80217-2	Q8N2W9
IFIT1	P09914	Q86WV6	IFIT3	O14879	P05067
IFIT5	Q13325	Q9H4P4	IFITM2	Q6IBB0	Q14973
IFITM3	Q01628	Q9NXB9	IFNGR2	P38484	P13861
IFNLR1	Q8IU57	P23458	IFRD2	Q12894	Q9Y6K5
IFT43	Q96FT9-2	P17252	IFT81	Q8WYA0	P36941
IGDCC4	Q8TDY8	P02788	IGF2BP1	Q9NZI8	O00425, Q9Y6M1
IGF2BP3	O00425	Q9NZI8	IGFBP2	P18065	P59665
IGFBP3	P17936	Q86XT9	IGFBP4	P22692	P05019
IGFBP6	P24592	P53611	IGFBPL1	Q8WX77	P43146
IGSF3	O75054	P21926	IGSF9	Q9P2J2-2	O43169
IKBIP	Q70UQ0	O14920	IKBKE	Q14164	Q14653, Q9UHD2
IKBKG	Q9Y6K9	P12931	IKZF1	Q13422-7	Q99728
IKZF3	Q9UKT9	Q13882	IL10RB	Q08334	P22301
IL13RA1	P78552	Q8TDR0	IL13RA2	Q14627	Q14118
IL15RA	Q13261	P31785	IL17RA	Q96F46	Q9NRM6
IL17RD	Q8NFM7	Q9NYV6	IL1R1	P14778	Q9Y4K3
IL1RAPL2	Q9NP60	Q13332	IL27RA	Q6UWB1	O75843
IL2RB	P14784	P60568	IL4R	P24394	O94955
IL6ST	P40189	P05231	IL7R	P16871	O43791
ILK	Q13418	P49327, Q13085	ILKAP	Q9H0C8	Q13418
IMP4	Q8TCT7	O75063	IMPACT	Q9P2X3	P20618
IMPDH1	P20839	Q8WVC6	INA	Q16352	P08670
INAVA	Q3KP66-1	Q07157	INCENP	Q9NQS7-1, Q9NQS7	O14965
INF2	Q27J81	P50548	ING3	Q9NXR8	P52294
INHBA	P08476	P31749	INKA1	Q96EL1	Q9HAQ2
INO80C	Q6PI98	P50454	INO80D	Q53TQ3	Q9Y265, Q9Y230
INPP1	A0A0S2Z3X1	P20393	INPP5A	Q14642	P53611
INPP5D	Q92835, Q92835-2	O75525	INPP5F	Q9Y2H2	P63104

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INPP5K	Q9BT40	P22102	INPPL1	O15357	P35813
INSR	P06213	P29353	INTS11	Q5TA45	P52294
INTS13	Q9NVM9	Q13951	INTS2	Q9H0H0	P21462
INTS6	Q9UL03	P22891	INTS8	Q75QN2	P67775
IP6K1	Q92551	P21802	IPO13	O94829	P47813
IPO4	Q8TEX9	P01106	IQCB1	Q15051	P36941
IQCE	Q6IPM2	P25788	IQGAP2	Q13576	P01106
IQGAP3	Q86VI3	P40855	IQUB	Q8NA54	Q15669
IRAG1	Q9Y6F6-3	O76024	IRAK1	P51617	Q92985
IRAK3	Q9Y616	Q9Y4K3	IREB2	P48200	P43353
IRF1	P15314	Q13114	IRF9	Q00978	P52630
IRS1	P35570	Q06124	IRS2	Q9Y4H2	P27986
IRS4	O14654	Q15185	ISLR	O14498	Q99741
ISOC1	Q96CN7	P04004	ISY1-RAB43	Q9ULR0-1	O43559
ITFG2	Q969R8	Q5T2D3	ITGA2	P17301	P05556
ITGA2B	P08514	P08514	ITGA3	P26006	P48509
ITGA4	P13612	P02751	ITGA5	P08648	P02751
ITGB1	P05556	P48509	ITGB1BP1	O14713	Q15669
ITGB1BP2	Q9UKP3-2	Q8N2W9	ITGB3	P05106	P02751
ITGB3BP	Q13352	Q969Y2	ITGB5	P18084	P04004
ITGB8	P26012	O43914	ITLN2	Q8WWU7	P36776
ITM2A	O43736	P05067-4	ITPA	Q9BY32	P56705
ITPKC	Q96DU7	O14965	ITPR2	Q14571	Q9H244
ITPRID2	P28290	P22681	ITPRIP	Q8IWB1	Q96FT7
ITPRIPL1	Q6GPH6-2	Q12933	ITPRIPL2	Q3MIP1	Q9BRI3
IVNS1ABP	Q9Y6Y0	P53350	JADE1	Q6IE81-3	Q96CV9
JADE2	Q9NQC1-2	P01100	JADE3	Q92613	Q00403
JAG1	P78504	P46531	JAK1	P23458	O60674
JAK3	P52333	Q13485	JAKMIP1	Q96N16	P51451
JAKMIP2	Q96AA8	Q969G3	JAM2	P57087	Q9BX67
JAM3	Q9BX67	Q9BX67	JARID2	Q92833-1, Q92833	Q15022
JMJD1C	Q15652	P10275	JMJD6	Q6NYC1	O60885
JMY	Q9QXM1	Q09472	JOSD2	Q8TAC2	Q2T9J0
JPH1	Q9HDC5	Q5SY16	JPH3	Q8WXH2	Q9NR63
JPH4	Q96JJ6	Q9ULK5	JUNB	P17275	P18847
KANK2	Q63ZY3	Q15669	KAT14	Q9H8E8	Q7L190
KAT2B	Q92831	Q09472	KAT5	Q92993	O94844
KAT7	O95251	P53350	KATNAL1	Q9BW62	Q14061
KATNAL2	Q8IYT4	P03372	KAZN	Q674X7-2	O75604
KBTBD7	Q8WVZ9	Q86UK7	KBTBD8	Q8NFY9	Q9UF56
KCNA3	P15384	P78352	KCND2	Q9NZV8	P11142
KCNE3	Q9Y6H6	P40818	KCNH1	O95259	Q7L8L6
KCNIP1	Q9NZI2-2	P40189	KCNIP3	Q9Y2W7	P40189
KCNJ11	Q14654	Q8WTR4	KCNJ2	P63252	P63252
KCNJ6	P48051	O75063	KCNK2	Q13303	P42858
KCNK5	O95279	P24593	KCNMA1	Q08460	P46109
KCNMB2	Q9Y691	P42858	KCNN4	O15554	P61073
KCNS3	Q9BQ31	Q13370	KCNT2	Q6UVM3	Q14160
KCTD1	Q719H9	Q9NRD5	KCTD12	Q96CX2	P0CG13
KCTD13	Q8WZ19	P46108	KCTD14	Q9BQ13	Q4G176

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
KCTD15	Q96SI1-2	P42858	KCTD21	Q4G0X4	Q9NWW6
KCTD5	Q9NXV2	Q9UBU9	KCTD9	Q7L273	O14786
KDELR1	P24390	Q92796	KDM1A	O60341	P46531
KDM2B	Q8NHM5	P63208	KDM4A	O75164	Q15831
KDM6A	O15550	O43474	KDR	P35968	P98160
KEAP1	Q14145	Q86YC2	KEL	P23276	P50454
KHDRBS2	Q5VWX1	Q13882	KIAA0040	A0A384DVV8	P42858
KIAA0319L	Q8IZA0	Q8IUQ4	KIAA0408	Q6ZU52	P08670
KIAA0825	Q8IV33	Q96CV9	KIAA1671	Q9BY89	Q9Y3R4
KIAA1958	Q8N8K9	O14543	KIAA2013	Q8IYS2	Q15904
KIF13B	Q9NQT8	P63104	KIF18B	Q86Y91	Q99435
KIF1A	Q12756	O76024	KIF1C	O43896	P63104
KIF21A	Q7Z4S6	Q14678	KIF21B	O75037	P61981
KIF23	E9Q5G3, Q02241-2, Q02241	Q9H0H5	KIF2A	O00139	Q5S007
KIF5A	Q12840	P63104	KIF5C	O60282	P68400, P67870
KIF7	Q2M1P5	P10070	KIFAP3	Q92845	Q13177
KIFBP	Q96EK5	Q15185	KIFC3	Q9BVG8-5	Q9BUH8
KIRREL1	Q96J84	Q07157	KIRREL2	Q6UWL6	P25788
KIRREL3	Q8IZU9	P15311	KIT	P10721	Q14451
KITLG	P21583	P10721	KIZ	Q2M2Z5	Q96NL6
KLC1	Q07866-2	P05067	KLF10	Q13118	P08047
KLF13	Q9Y2Y9	P43694	KLF15	Q9UIH9	P12931
KLF16	Q9BXX1	Q12857	KLF2	Q9Y5W3	Q9HCE7
KLF4	O43474	O14964	KLF6	Q99612	P02545
KLF8	O95600	Q12857	KLHDC8A	Q8IYD2	Q13371
KLHDC9	Q8NEP7	O76003	KLHL12	Q53G59	O95050
KLHL13	Q9P2N7	P01282	KLHL21	Q9UJP4	P23025
KLHL22	Q53GT1	Q9BPX6	KLHL23	Q8NBE8	P29279
KLHL24	Q6TFL4	O43639	KLHL3	Q9UH77	P06396
KLHL7	Q8IXQ5	Q8N4J0	KLK10	O43240	Q9HAW8
KLK8	O60259	P50454	KLKB1	P03952	Q8N0U8
KLRG1	Q96E93	P37173	KMO	O15229-2	P42858
KMT5A	Q9NQR1	P62805	KMT5B	Q4FZB7	Q9H2G4
KNOP1	Q1ED39	Q96CW1	KNTC1	P50748	Q9Y639
KPNA5	O15131	O43148	KRT14	P02533	Q15149
KRT16	P08779	Q969G3	KRT19	P08727	Q9P0N9
KRT33B	Q14525	P06396	KRT34	O76011	Q9NQ94
KRT7	P08729	P09429	KRT8	P05787	Q9Y6K9
KRT80	Q6KB66-2	Q9BRX5	KRT81	Q14533	P49639
KRTAP1-1	Q07627	P43897	KRTAP1-5	Q9BYS1	P06850
KRTAP2-3	P0C7H8	Q8WUK0	KRTAP2-4	Q9BYR9	P49638
KRTAP4-12	Q9BQ66	O15169	KSR1	Q8IVT5	P04049
KSR2	Q6VAB6	Q93008	L1TD1	Q5T7N2	P52292
L2HGDH	Q9H9P8	O15431	L3MBTL2	Q969R5	Q13185
L3MBTL3	Q96JM7	P26358	L3MBTL4	Q8NA19	Q9Y2I6
LAMA1	P25391	P17931	LAMA4	Q16363-3	P05067
LAMA5	O15230	Q6ISS4	LAMB1	P07942	P17931
LAMB2	P55268	Q03405	LAMC1	P11047	P17931
LAMP1	P11279	Q15836	LAMP2	P13473-2	Q8WUX9
LAMTOR1	Q6IAA8	P08670, P06396	LAMTOR2	Q9Y2Q5	P61421

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
LAPTM4B	Q86VI4	P08581	LARGE2	Q8N3Y3	Q03405
LARP6	Q9BRS8	Q9Y3F4	LARP7	Q4G0J3	P23193
LARS1	Q9P2J5	O00560	LASP1	Q14847	Q9H2D1
LAYN	Q6UX15-2	P24593	LCK	P06239	P08575
LCP1	P13796	Q8NG68	LCP2	Q13094	O15117
LDHA	P00338	P00338	LDHB	P07195	P00338
LDLR	P01130	Q8NBP7	LDLRAP1	D3ZAR1	P01130
LDOC1	O95751	P48552	LEMD1	Q68G75	Q15125
LENG1	Q96BZ8	Q9BUH8	LEPROT	O15243	Q9UBD6
LETM1	O95202	P04798	LGALS1	P09382	P48509
LGALS14	Q8TCE9	Q9H668	LGALS3	P17931	P08648
LGALSL	Q3ZCW2	Q9Y6D9	LHFPL1	Q86WI0	O75396
LHX8	Q68G74	Q9BRT9	LHX9	Q9NQ69	P68400
LIG3	P49916	Q9NTX7	LIMA1	Q9UHB6	P63104
LIMCH1	Q9UPQ0-1	Q96JA1	LIMS1	P48059	Q9Y3P9
LIN28A	Q9H9Z2	P19338	LIN28B	Q6ZN17	Q8IZH2
LIN37	Q96GY3	P10243	LIN54	Q6MZIP7	P10243
LIN7B	Q9HAP6	O14936	LINC-PINT	A0A455ZAR2	Q08050
LINGO1	Q96FE5	P08138	LINGO2	Q7L985	Q9HAU4
LIPG	Q9Y5X9	Q9Y3A6	LLGL2	Q6P1M3	Q05513
LMAN2	Q12907	P13569	LMNA	P02545-1, P02545-2, P02545	P20700
LMNB2	Q03252	P20700	LMO3	Q8TAP4-4	O94844
LMO4	P61968	Q9BRX2	LMO7	Q8WWI1-3	Q96JA1
LNPEP	Q9UIQ6	P17931	LNPK	Q9C0E8-2	P04083
LNx2	Q8N448	O95050	LONRF1	Q17RB8	Q9UQK1
LONRF2	Q1L5Z9	P99999	LOX	P28300	Q9UBX5, P15502
LOXL1	Q08397	P06276	LOXL2	Q9Y4K0	P02751
LPAL2	Q16609	P06307	LPAR1	P61793	Q9NZN5
LPAR2	Q9HBW0	P17931	LPAR3	Q9UBY5	Q96BA8
LPAR4	Q99677	Q13315	LPCAT2	Q7L5N7	Q15125
LPGAT1	Q92604	P22732	LPL	P06858	Q8N0U8
LPP	Q93052	P05120	LPXN	O60711	Q13177
LRAT	O95237	Q7Z5P4	LRATD2	Q96KN1	O60551
LRBA	P50851	P16410	LRCH2	Q5VUJ6	Q13309
LRFN5	Q96NI6	P23468, Q13332	LRIG1	Q96JA1	P22681
LRIG2	O94898	Q96JA1	LRP1	Q07954	P12236
LRP10	O75096	Q9BQB4	LRP11	Q86VZ4	Q9Y5M8
LRP12	Q9Y561	Q9NPI5	LRP2	P98164	P29279
LRP6	O75581	P98082	LRP8	Q14114	P35968
LRPAP1	P30533	P98155	LRPPRC	P42704	Q9UBK2
LRRC1	Q9BTT6	P36873	LRRC14	Q15048	Q92905
LRRC32	Q14392	P11021	LRRC40	Q9H9A6	Q96JA1
LRRC41	Q15345	Q93034, Q9UBF6	LRRC4C	Q9HCJ2	Q9NT99
LRRC59	Q96AG4	P04049	LRRC61	Q9BV99	O14964
LRRC8A	Q8IWT6	O43914	LRRC8B	Q6P9F7	Q13370
LRRC8C	Q8TDW0	P25090	LRRC8E	Q6NSJ5	P43119
LRRFIP1	Q32MZ4	P42858	LRRFIP2	Q9Y608	P13569
LRRN1	O75427	A1L3X0	LRRN2	O75325	P98170
LRRTM2	O43300	P01375	LRRTM3	Q86VH5	O95470

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LRRTM4	Q86VH4	O14786	LRWD1	Q9UFC0	Q9Y5N6
LSM10	Q969L4	P62306	LSM6	P62312	O00560
LSMEM2	Q8N112	P21854	LSS	P48449	P09917
LTBP1	Q14766	P09238	LTBP2	Q14767	Q9UBX5
LTBP3	Q14767	Q9UBX5	LTBR	P36941	Q9NPF0
LTO1	Q8WV07	P61221	LUC7L	Q9NQ29-3	Q13247, Q16629
LUC7L3	O95232-2	P40692	LUM	P51884	P50281
LURAP1L	Q8IV03	O00560	LYN	P07948	P12931
LYPLA1	Q6IAQ1, O75608	Q08379	LZTR1	Q8N653	Q13485
M1AP	Q8TC57	P20700	MAB21L2	Q9Y586	Q9UBB9
MAEA	Q7L5Y9	P01275	MAFG	O15525	O15525
MAGEA4	P43358	Q9NUX5	MAGED1	Q9Y5V3	P52292
MAGED2	Q9UNF1	Q5S007	MAGEL2	Q9UJ55	Q9UMX1
MAGI1	Q6P9H4	P42772	MAGI2	Q86UL8	P01135
MAGOHB	Q96A72	Q13573	MAK	P20794	Q9UM11
MAL2	Q969L2	Q05329, Q99259	MALL	Q13021	Q13324-2
MALSU1	Q96EH3	Q14197	MALT1	Q9UDY8	Q14790
MAML3	Q96JK9	P46531	MAN2A1	Q16706	Q03405
MAN2B2	Q9Y2E5	Q96BA8	MANBAL	Q9NQG1	Q12893
MAP1A	P78559	Q9GZQ8	MAP1B	P46821	Q9GZQ8
MAP1LC3B	Q9GZQ8	O75385	MAP1LC3C	Q9BXW4	O75385
MAP2	P11137-4	P02649	MAP2K3	P46734	Q16539
MAP2K6	P52564	Q16539	MAP3K1	Q13233	P32121
MAP3K20	Q9NYL2	O75582	MAP3K21	Q5TCX8	O15111
MAP3K3	Q99759	Q13163	MAP3K6	O95382	P61981
MAP3K9	P80192	P17252	MAP4	P27816	P10636-8
MAP4K1	P70218	Q16584	MAP4K3	Q924I2	Q99962
MAP4K4	O95819	Q9HOR5	MAP7	Q14244	Q9Y3M2
MAP7D1	Q3KQU3	P01106	MAPK10	P53779	P49407
MAPK3	P27361	P19419	MAPK4	P31152	Q96HN2
MAPK6	Q16659	P02647	MAPK7	Q13164	Q13163
MAPKAP1	Q9BPZ7	P31749	MAPKAPK2	P49137	Q16539
MAPRE3	Q9UPY8	P05412	MARCHF3	Q86UD3	Q12959
MARCKSL1	P49006	O60551	MARK1	Q9P0L2	P05067
MARVELD2	Q8N4S9	Q8NG68	MASP1	P48740-2	P17927
MASP2	O00187, O00187-PRO_0000027598	29108, P05155	MAST1	Q9Y2H9	O14734
MAST2	Q6P0Q8	P27037	MAT1A	Q00266	P42858
MATN2	O00339	Q96ES7	MATR3	P43243	Q13573
MAVS	Q7Z434	Q13568	MAZ	P56270	P43146
MBD1	Q9UIS9	P45973	MBD2	Q9UBB5	P26358
MBD3	O95983-2	Q9Y5J6	MBD6	Q96DN6	O43639
MBNL1	Q86VM6, Q9NR56	O75553	MBNL3	Q9NUK0	O75553
MBOAT7	Q96N66	Q86WV6	MBP	P13727	Q12946
MBTD1	Q05BQ5	P62805	MBTPS1	Q14703	Q8IV16
MCAT	Q8IVS2	Q00839	MCCC2	Q9HCC0	P13569
MCFD2	Q8NI22	P78382	MCM3	P25205	Q13416
MCOLN1	Q9GZU1	P0DTC3	MCOLN2	Q8IZK6	Q9GZU1
MCOLN3	Q8TDD5	Q86XT9	MCRS1	Q96EZ8	Q9Y5K5
MCUB	Q9NWR8	Q15047	MDFI	Q99750	Q9Y6K1

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MDH2	P40926	P19320	MDN1	Q9NU22	Q9Y639
MEA1	Q16626	Q96CW1	MEAK7	Q6P9B6	O60551
MECP2	Q00566	P46100	MED10	Q9BTT4	Q15648
MED12L	Q86YW9	P61024	MED13L	Q71F56	Q15648
MED14	O60244	Q15648	MED15	Q96RN5	Q15648
MED16	Q9Y2X0	Q15648	MED19	A0JLT2	Q15648
MED22	Q15528-2	P42858	MED28	Q9H204	Q15648
MED29	Q9NX70	Q15648	MED8	Q96G25	Q15648
MEDAG	Q5VYS4	Q9H0E2	MEF2C	Q06413	P56524
MEF2D	Q14814	Q16539	MEIOC	A2AG06	Q9H6S0
MEIS2	O14770-4	Q13882	MEPCE	Q7L2J0	Q9Y3E7
MEST	Q5EB52	Q05329	MET	P08581	P12931
METAP2	P50579	Q93034	METRNL	Q641Q3	Q01523
METTL21A	Q8WXB1	P11142	METTL21C	Q5VZV1	Q15369
MEX3C	Q5U5Q3	Q93009	MFAP4	P55083	Q16625
MFHAS1	Q9Y4C4	Q9Y496	MFSD10	Q14728	P45973
MFSD13A	Q14CX5	P0DPK4	MFSD14B	Q5SR56	Q9NRZ7
MFSD4B	Q5TF39	Q7L8W6	MFSD5	Q6N075	P25103
MGA	Q8IWI9	Q13185	MGAT1	P26572	Q9BYC5
MGAT2	Q10469	P40855	MGAT4B	Q9UQ53	Q96CV9
MGAT4C	Q9UBM8	O75197	MGLL	Q99685	P07550
MGME1	Q9BQP7	Q02790	MGRN1	O60291	P14138
MGST3	O14880	O60551	MIB1	Q86YT6	Q9UHD2
MICA	Q29983	P17931	MICAL2	O94851	P54252
MICAL3	Q7RTP6, Q7RTP6-1	Q02241	MICALL1	Q8N3F8	Q96B97
MICOS13	Q5XKP0	P55056, P02654	MICU1	Q9BPX6	Q9H4I9, Q9BPX6, Q8NE86
MID1	P29372	P04156	MID2	Q9UJV3-2	P12236
MIER3	Q7Z3K6	Q13547	MILR1	Q7Z6M3	Q92538
MIOS	Q9NXC5	P55735	MIPOL1	Q8TD10	Q8WY64
MIS18A	Q9NYP9	P23434	MIS18BP1	Q6P0N0	P56539
MKX	Q8IYA7	Q9BZK7	MLF1	P58340	P51587
MLF2	Q15773	P04049	MLKL	Q8NB16	Q9Y572
MLLT1	Q03111	P04608	MLLT10	P55197-2	P35680
MLLT6	P55198	P26640	MLX	Q9UH92-3	O75928-2
MLXIP	Q9HAP2	P61981	MMD	Q15546	P01350
MMP1	EBI-9996975	P17676	MMP14	P50281	P09958
MMP2	P08253	P39060	MMP25	Q9H239	P09958
MMP3	P08254	P09238	MMP9	P14780	P14780
MND1	Q9BWT6	P09917	MNS1	Q8NEH6	Q00613
MOB3B	Q86TA1	Q13188	MOCS2	O96007	O96033
MORC4	Q8TE76	P40763	MOV10	Q9HCE1	Q92900
MOXD1	Q6UVY6	O76003	MPC2	O95563	P02654
MPG	P29372	P04156	MPL	P40238	O60674
MPP2	Q08050	Q13416	MPP3	Q13368	Q9HAP6
MPP4	Q96JB8	Q8N3R9	MPP7	Q5T2T1	Q9HAP6
MPPED2	Q15777	Q9Y244	MPRIP	Q6WCQ1	P53350
MPZL1	O95297	Q06124	MRC2	Q9UBG0	P17931
MRE11	P49959	P04637	MRI1	Q9BV20	Q9BV20
MRPL12	P52815	Q9NWR8	MRPL17	Q9NRX2	P67809

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
MRPL19	P49406	P40429	MRPL2	Q9BZE1, Q5T653	Q14197
MRPL24	Q96A35	P62753	MRPL27	Q9P0M9, Q8IXM3	Q14197
MRPL38	Q96DV4	Q9NPA3	MRPL40	Q9NQ50	P67809
MRPL41	Q8IXM3	Q14197	MRPL42	Q9Y6G3	Q14197
MRPL43	Q8N983-3	P17252	MRPL50	Q8N5N7	Q14197
MRPL52	Q86TS9	Q14197	MRPL53	Q96EL3	Q15370
MRPL58	Q14197	Q00059	MRPS11	P82912	Q00839
MRPS12	O15235	Q04721	MRPS2	Q9Y399	Q00839
MRPS24	Q96EL2	P10809	MRPS9	P82933	P04618
MRS2	Q9HD23	Q8N5V2	MRTFA	Q969V6	P11831
MRTFB	Q9ULH7	P11831	MSANTD3	Q96H12	P06396
MSH2	P43246	P52701, P20585	MSH3	P20585	P43246
MSH6	P52701	P43246	MSI1	O43347	O43347
MSI2	Q96DH6	Q9Y5J6	MSL3	Q8N5Y2	O76024
MSMO1	Q15800	P50281	MSN	P26038	P16070
MSRB1	Q9NZV6	P14373	MSRB3	Q8IXL7	Q9UKN5
MST1R	Q04912	P26927	MT-ATP8	P03928	P25090
MT-ND1	P03886	Q9P0J0	MT-ND2	P03891	O75380
MT-ND4	P03905	Q9P0J0	MT2A	P02795	P09429
MTA2	O94776	Q96KQ7	MTA3	Q9BTC8	Q96KQ7
MTCH1	Q9NZJ7	Q92611	MTCL2	O94964-4	O14503
MTF2	Q9Y483-4	P54252	MTFR2	Q6P444	P13196
MTG2	Q9H4K7	O95861	MTHFD2L	Q9H903	Q5T8I9
MTIF2	P46199	P43897	MTMR6	Q9Y217	Q9Y216, Q96QG7
MTMR7	Q9Y216	Q96QG7	MTMR8	Q96EF0-1, Q96EF0, Q96QG7	Q9Y216, Q96QG7
MTREX	P42285	P52298, Q09161	MTSS2	Q765P7	P60709
MTUS1	Q9ULD2-3	Q13190	MTX1	Q13505	Q05940
MUL1	Q969V5	O60443	MUS81	Q96NY9	P39748
MVP	Q14764	Q9H0E2	MX1	P20591	Q13507
MX2	P20592	Q96KQ7	MXD3	Q9BW11	Q9UBB9
MXRA8	Q9BRK3	Q9NPB3	MYADM	Q96S97	Q9Y624
MYB	P10242	P63165, P61956	MYBBP1A	Q9BQG0	P01106
MYCL	P12524-2	P54646	MYCN	P04198	Q8N726
MYDGF	Q969H8	Q14554	MYEF2	Q9P2K5-2	Q9UGI0
MYG1	Q9HB07	Q9BRX2	MYH10	P35580	Q14164
MYH11	P35749	P13569	MYH14	Q7Z406	O15111
MYH2	Q9UKX2	Q16665	MYH3	P11055	Q16665
MYH9	P35579	O00255	MYL12A	P19105	P35579
MYL12B	O14950	P35579	MYL6	P60660	O43809
MYL6B	P14649	P35579	MYL9	P24844	P35579
MYLIP	Q8WY64	P54252	MYO10	P79114	P05556
MYO1C	Q12965	P50570	MYO5B	Q9ULV0	P40855
MYO5C	Q9NQX4	O75496	MYOZ3	Q8TDC0	Q13882
MYRIP	Q8NFW9	P51159	MZT2B	Q6NZ67	Q9Y6L6
N4BP3	O15049	P40937	NAA10	P41227	Q96FF9
NAA16	Q6N069-4	P14136	NAA60	Q9H7X0	P55212
NAA1AD2	Q9Y3Q0	O76024	NAB2	Q15742	P40937
NABP1	Q96AH0	Q5VWX1	NAE1	Q13564	Q15843
NAF1	Q15025	P19838	NAGK	Q9UJ70	P33992

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NAMPT	P43490	P43490	NANOG	NANOG	Q01860
NAP1L1	P55209	P04004	NAP1L2	Q9ULW6	P42858
NAP1L3	Q99457	P14136	NAPA	P54920	P51809
NAPSA	O96009	P11021	NARS2	Q96I59	P00480
NASP	P49321	P62805	NAT8L	Q8N9F0	P40855
NAV1	Q8NEY1	P61981	NAV2	Q8IVL1	P63104
NAXD	Q8IW45	Q7Z3S9	NBEA	Q8NFP9	Q9BYJ1
NBL1	P41271-2	P05067	NBPF1	Q3BBV0	O60341
NCAM1	P13591	P25098	NCAPH2	Q6IBW4	P61073
NCCRP1	Q6ZVX7	Q15915	NCDN	Q9UBB6	Q15185
NCEH1	Q6PIU2	P27824	NCLN	Q969V3	P42858
NCOA4	Q13772	O95714	NCOA6	Q14686	P06400
NCOR2	Q9Y618	Q07869	NCS1	P62166	P10451
NCSTN	Q92542	P35613	NDE1	Q9NXR1, Q9CZA6	P43034
NDEL1	Q9GZM8	P23025	NDRG2	Q9UN36	P03372
NDRG4	Q9ULP0-2	O60906	NDST1	P52848	Q03405
NDUFA11	Q86Y39	O75489	NDUFA13	Q9P0J0	P23786
NDUFA5	Q16718	Q9P0J0	NDUFA7	O95182	P49638
NDUFAF3	Q9BU61	Q9NPB3	NDUFAF7	Q7L592	Q5S007
NDUFB11	Q9NX14	P08034	NDUFB7	P17568	P19404, O75380, Q9UI09, O75489, O43920
NDUFS6	O75380	P25103	NDUFS7	O75251	Q8NE62
NDUFS8	O00217	Q8IWL3	NDUFV1	P49821	P17252
NEB	P20929	Q07889	NEBL	O76041	Q9Y6D9
NECAB2	Q7Z6G3-2	Q96QG7	NECAP2	Q9NVZ3	P42566
NEDD4	P46934-4	Q9HAU4	NEFL	P07196	Q13569
NEFM	P07197	P06400	NEGR1	Q7Z3B1	P61916
NEK2	P51955	Q8NG66	NEK3	P51956	P32971
NEK6	Q9HC98-4	P12830	NEK7	Q8TDX7	Q9ULW0
NELFE	P18615	Q8WX92	NELL2	Q99435	Q6IPR3
NENF	Q9UMX5	Q14554	NEPRO	Q6NW34	P06748
NF2	P35240	Q16584	NFE2L1	Q14494	O15525
NFE2L2	Q16236	O15525	NFE2L3	Q9Y4A8	O15525
NFIA	Q12857	Q9BXX1	NFIC	P08651	O00712
NFIX	Q14938	O00712	NFKB1	P19838-PRO_0000030311	O00255
NFKBIB	Q15653	P25963	NFXL1	Q6ZNB6	Q96FT7
NFYB	P25208	P04637	NGF	P01138-PRO_0000019600, P01138	P08138
NGLY1	Q96IV0	Q96FE5	NHERF2	Q15599	Q9H4P4
NHLH2	Q02577	Q99081	NHLRC3	Q5JS37	O00253
NHP2	Q9NX24	P42858	NHSL2	Q5HYW2	P53350
NIBAN1	Q9BZQ8	P31749	NIBAN2	Q96TA1	Q14145
NID2	Q8IV28	P10745	NINL	Q9Y2I6	Q7L590
NIPAL1	Q6NVV3	Q09328	NIPAL3	Q6P499	Q92838
NIPBL	Q6KC79	P45973	NIPSNAP1	Q9BPW8	Q13501
NIPSNAP3A	Q9UFN0	P10809	NKD1	Q969G9	P67775
NKTR	P30414	P68400	NKX3-1	Q99801	Q96BI1
NLGN3	Q9NZ94-2	P00403	NLK	Q9UBE8	Q96PM5

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NLN	Q9BYT8	Q5SW96	NLRC5	Q86WI3	O15111, O14920
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NME2	P19804	Q05193	NMNAT1	Q9HAN9	P52292
NMT1	P30419	P04601	NMT2	O60551	Q9BY76
NMU	P48645	P12644	NOB1	Q9ULX3	P51530
NOC3L	Q8WTT2	P19525	NOL12	Q9UGY1	P40429
NOL3	O60936	Q14790	NOL4	O94818-2	P07237
NOP16	Q9Y3C1	Q8WYQ5	NOP58	Q9Y2X3	P52294
NOPCHAP1	Q8N5I9	P52294	NOS1	P29476	P23634
NOS1AP	O75052	P53350	NOS2	P60321	Q14192
NOSIP	Q9Y314	Q9Y3R4	NOTCH2	Q04721	Q8N2S1
NOTCH2NLA	Q7Z3S9	Q96PM5	NPAS2	Q99743	O00327
NPAS3	Q8IXF0	Q9HBZ2	NPAT	Q14207	Q16576
NPC1	O15118	P61916, 16113	NPDC1	Q9NQX5	Q9BW85
NPHP4	O75161	O95816	NPLOC4	Q8TAT6	P55072
NPM1	P06748	Q14694	NPRL3	Q12980	Q07954
NQO1	P15559	P15559	NR1D1	P20393	P06727
NR1H2	F1D8P7	P06396	NR1I2	O75469	Q15788
NR2C1	P13056-2	P10276	NR2C2AP	Q86WQ0	P42858
NR2F1	P10589	Q00403	NR2F2	P24468	P43119
NR3C1	P06536	Q15788	NR5A2	O00482	P0DTC5
NRBP1	Q9UHY1	P51530	NRCAM	Q92823	Q12860
NRG1	Q02297-10	P21854	NRIP1	P48552	Q9Y6K1
NRIP2	Q9BQI9	Q8N9N5	NRIP3	Q9NQ35	Q9BZL1
NRM	Q8IXM6	P04000	NRN1	Q9NPD7	Q86XT9
NRP1	P97333	P08648	NRP2	Q99435	Q6IPR3
NRROS	Q86YC3	P11021	NRSN2	Q9GZP1	Q8N6M3
NRXN1	Q63373	O14936	NSD1	Q96L73	P17096
NSD2	O96028	P17096	NSFL1C	Q9UNZ2	P55072
NSG1	P42857	Q96BA8	NSL1	Q96IY1	Q13185
NSMCE1	Q8WV22	Q9H668	NSMCE4A	Q9NXX6	P51587
NT5E	P21589	P36404	NTF3	P20783	P23560
NTM	Q9P121-3	P01229	NTMT1	Q9BV86	P18754
NTN4	Q9HB63	P16284	NTPCR	Q9BSD7	Q8NEZ3, Q96RY7
NTS	P20068-PRO_0000019532	P63211	NUBPL	Q8TB37	Q9Y5Y2
NUCB1	Q02818	P08254	NUCB2	P80303	P17275
NUDCD3	Q8IVD9	P43034	NUDT16L1	Q9BRJ7	Q9NPB3
NUDT19	A8MXV4	O95861	NUDT2	P50583	Q14566
NUDT21	O43809	O43809, Q16630-1	NUDT22	Q9BRQ3	P52747
NUDT6	P53370	P46199	NUFIP2	Q7Z417	Q9UBX5
NUMA1	Q14980	P03372	NUMB	Q9QZS3-2	P08151
NUMBL	Q9Y6R0	Q9NWB1	NUP107	P57740	Q9UBP0
NUP133	Q8WUM0	Q9UBP0	NUP205	Q92621	P19320
NUP210	Q8TEM1	Q9UK39	NUP54	Q7Z3B4	O14964
NUP62	P37198	Q6W0C5	NUP62CL	Q9H1M0	O14641
NUTM1	Q86Y26	P09917	NXN	Q6DKJ4	O14641
NXPH2	O95156	O75197	NXT1	Q9UKK6	Q9Y3Y2
OAF	Q86UD1	P56559	OARD1	Q9Y530	P09874
OAS3	Q9Y6K5	Q7Z434	OAT	P04181	P05067
OAZ1	P54368	O14977	OBSCN	Q5VST9-3	P16157

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OCLN	Q16625	P18827	ODAD2	Q5T2S8	P43354
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OFD1	O75665	Q4AC94	OGDH	Q02218	P42858
OGDHL	Q9ULD0	P25963	OGFOD3	Q6PK18	P27824
OGT	O15294-3	Q8N9R8	OLFM3	Q96PB7, Q96PB7-3	Q9UNE7
OLFML2A	Q68BL7	Q15848	OLFML2B	Q68BL8	Q13217
OPA1	O60313	P01106	OPRD1	P41143	Q9UKG4
ORC1	Q13415	O43929, Q9UBD5, Q13416, O43913	ORC2	Q13416	Q13415
ORC3	Q9UBD5	Q13415	ORC4	O43929	Q13415
OS9	Q13438	O00468	OSBPL1A	Q9BXW6	P51149
OSBPL3	Q9H4L5	P63104	OSBPL5	Q9H0X9	P54802
OSBPL6	Q9BZF3	P53350	OSBPL9	Q96SU4	Q92685
OSGIN1	Q9UJX0	P48507	OSMR	Q99650	P23458
OST4	P0C6T2	A6NI15	OSTF1	Q92882	O75695
OSTM1	Q86WC4	P17931	OTULINL	Q9NUU6	Q8NBQ5
OTX2	P32243-2	Q96FT7	P3H1	Q32P28	Q70UQ0
P3H3	Q8IVL6	Q03405	P3H4	Q92791	Q9H2A9
P4HA2	O15460	P39060	P4HA3	Q7Z4N8	P21359
P4HB	P07237	O43521	PABPC3	Q9H361	P67809
PABPC4L	P0CB38	P16104	PACS1	Q6VY07	Q9NZ52
PACSIN1	Q9BY11	Q9NUX5	PACSIN2	Q9UNF0	P50570, O00401
PADI3	Q9ULW8	Q9Y496	PAF1	Q8N7H5	P23193
PAICS	P22234	Q9UBW7	PAIP2	Q9BPZ3	P11940
PAIP2B	Q9ULR5	O15234	PAK1	Q13153	P53667
PAK4	O96013	O00401	PAK5	Q9P286	P53350
PAK6	Q9NQU5	Q5S007	PALB2	Q86YC2	Q9Y253
PALD1	Q9ULE6	Q92985	PALLD	Q8WX93	P13569
PALM2AKAP2	Q9Y2D5	P17612	PALM3	A6NDB9	Q6IA17
PALS2	Q9NZW5	Q9HAP6	PAM	P19021	P08185
PAN2	Q9HBH5	P25103	PAPSS2	O95340	Q96CV9
PAQR6	Q6TCH4	P50876	PAQR7	Q86WK9	P40763
PARD6B	Q9BYG5	Q05513	PARD6G	Q9BYG4	Q05513
PARP1	P09874	Q9NTX7	PARP12	Q9H0J9	P48507
PARP8	Q8N3A8	P61981	PARVA	Q9NVD7	Q96EN8
PARVB	Q9HBI1	Q15404, Q9NVD7, P48059	PARVG	Q9HBI0	Q15404, P48059
PASK	Q9UEW8	P43119	PATJ	Q8NI35	Q8N3R9
PAWR	Q96IZ0, Q62627	P11021	PAX3	P24610	Q08999, P28749
PAXBP1	Q9Y5B6	O75533	PAXX	Q9BUH6	Q14191
PBK	Q96KB5	O94955	PBX1	P40424	Q12948
PCBP1	Q15365	P08581	PCBP4	P57723	P11142
PCDH10	Q9P2E7	P43119	PCDHA3	Q9Y5H8	Q16850
PCDHAC2	Q9Y5I4	A0PJW6	PCDHB11	Q9Y5F2	Q14118
PCDHB13	Q9Y5F0	Q9UMR7	PCDHB14	Q9Y5E9	P61978
PCDHB5	Q9Y5E4	Q12974	PCDHGA9	Q9Y5G4	Q14118
PCDHGB5	Q9Y5G0	Q16850	PCDHGC3	Q9UN70	P12830
PCGF5	Q86SE9	Q96LA8	PCK2	Q16822	P35558
PCM1	Q15154	Q8IW35, O43303	PCMTD1	Q96MG8	Q93034, Q9UBF6
PCMTD2	Q9NV79	Q93034, Q9UBF6	PCNT	O95613	P67809

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PCNX2	A6NKB5-5	Q9BRI3	PCOLCE	Q15113	P07996
PCSK5	Q92824	Q03405	PCSK6	P29122	P0DTC8
PCSK7	Q16549	P50454	PCYT1A	P49585	Q8N0U8
PCYT1B	Q9Y5K3-3	Q8N0U8	PCYT2	Q99447	O15357
PDAP1	Q13442	P10636-8	PDCD1LG2	Q9BQ51	Q9NZQ7
PDCD2L	Q9BRP1	P51813	PDCD6IP	Q8WUM4	P12931
PDE2A	O00408	O00170	PDE3B	Q13370	O95398
PDE4A	P27815	O75344	PDE7A	Q13946	P17612
PDE9A	O76083-2	O00560	PDGFB	P01127	P16234
PDGFC	Q9NRA1	P16234	PDGFRA	P16234	P19174
PDGFRB	P09619	Q14451	PDIK1L	Q8N165	Q13451
PDK1	O15530	O00468	PDK2	Q15119	P42858
PDK3	Q15120	P10515	PDLIM4	P50479	P12814
PDLIM5	Q96HC4	P42858	PDLIM7	Q9NR12	Q96QF0
PDP1	Q9P0J1	P12644	PDPK1	O15530	O00468
PDPR	Q8NCN5	O95861	PDZD2	O15018	P59637
PDZD4	Q17RL8	Q9UJW3	PDZRN3	Q9UPQ7	P31040
PEA15	Q15121	P28482	PEAK1	Q9H792	P46109
PEF1	Q9UBV8	O14964	PEL1	Q96FA3	O14653
PELI2	Q9HAT8	Q12891	PELO	Q9BRX2	P17252
PEMT	P15941-PRO_0000317447	P17676	PER3	P56645	O96017
PES1	O00541	P42858	PEX1	O43933	P19429
PEX11B	O96011	P40855	PEX14	O75381	P40855
PEX26	Q7Z412	Q969S8	PEX6	Q13608	P17980
PFKFB2	O60825	P63104	PFKFB3	Q16875	P63104
PFKL	P17858	Q969Y2	PFKP	Q01813	Q5S007
PFN1	P07737	P60709	PGAP4	Q9BRR3	Q6ZMH5
PGBD5	Q8N414	P53611	PGF	P49763	P17948
PGGT1B	P53609	Q9NQ87	PHB1	P35232	P40763
PHC1	P78364	P11142	PHC2	Q8IXK0	P08047
PHF1	O43189	Q15742	PHF14	O94880	P16104
PHF19	Q5T6S3	Q14012	PHF2	O75151	Q96RI1
PHF21B	Q96EK2	Q13547	PHF24	Q9UPV7	P07204
PHF3	Q92576	P28482	PHLDB1	Q86UU1	P63104
PHLPP1	O60346	Q8TAF3, O94782	PHOX2A	O14813	O43464
PHPT1	Q9NRX4	P0DTC3	PI16	Q6UXB8	O75509
PI4K2A	Q9BTU6	Q96J02	PI4KA	P42356	P21462
PI4KB	Q9UBF8, A0A0B4J1S8	Q9H3P7	PIAS2	O75928-2	Q13114
PIAS3	Q9Y6X2	Q13485	PICALM	Q13492	O00443, P42566
PID1	Q7Z2X4	Q8N9N5	PIEZO1	Q92508	P61073
PIGH	Q14442	O95256	PIGQ	Q9BRB3	Q9P0P0
PIGS	Q96S52	Q7Z3S9, P0DPK4	PIGT	Q969N2	Q8NBL1
PIK3AP1	Q6ZUJ8-3, Q6ZUJ8	O43639	PIK3C2A	O00443	O95630, Q96FJ0
PIK3C2B	O00750	P00533	PIK3CB	P42338	Q13485
PINK1	Q9BXM7	Q8WXH5	PIP4K2A	P48426	P15907
PIP5K1C	O60331	P53611	PIPOX	Q9P0Z9	Q9NUX5
PIR	O00625	P02675	PITRM1	Q5JRX3	P0DTC5
PITX1	P78337	Q13882	PITX2	Q99697-3	Q12948
PKD1	Q15139	P12830	PKIA	P61925	O43663
PKIB	Q9C010-2	P17612	PKIG	Q9Y2B9	P17612

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
PKM	P14618	P32121, P49407	PKMYT1	Q99640	P06493
PKP2	Q99959	P03431	PLA2G12A	Q9BZM1	O76024
PLA2G4C	Q9UP65	P42858	PLAAT1	Q9HDD0	O14964
PLAAT5	Q96KN8	Q96JH7	PLAGL2	Q9UPG8	O94955
PLAT	P00750	Q04721	PLAU	P00749	P05121
PLAUR	Q03405	Q09328	PLBD1	Q6P4A8	P15498
PLCD3	Q8N3E9	Q9Y4X5	PLCG2	P16885	O95997
PLCH1	Q4KWH8	P00533	PLEC	Q15149	P42858
PLEKHA2	Q9HB19	O14908	PLEKHA5	Q9HAU0	Q9NWB1
PLEKHA7	Q6IQ23-2	Q15669	PLEKHB1	Q9UF11	P52306
PLEKHB2	Q96CS7	Q86T03	PLEKHG4B	Q96PX9	O76003
PLEKHH1	Q9ULM0	Q7Z5R6	PLEKHM1	Q9Y4G2	P51149
PLEKHO1	Q53GL0	Q9HCE7	PLEKHO2	Q8TD55	P63104
PLIN2	Q99541	O76024	PLIN3	O60664	Q8TB40
PLK1	P53350	Q15648	PLK2	Q9NYY3	Q13177
PLK3	Q9H4B4	P04156	PLLP	Q9Y342	Q86WV6
PLOD1	Q02809	P55317	PLOD3	O60568	P39060
PLP1	P60201-2	Q12846	PLP2	Q04941	Q96QE2
PLPP3	O14495	O60716	PLPP4	Q5VZY2	P17302
PLPPR1	Q8TBJ4	P50454	PLPPR2	Q96GM1	Q9BVC3
PLS1	Q14651	P25963	PLSCR3	Q9NRY6	Q15915
PLXDC2	Q6UX71	P08833	PLXNA1	Q9UIW2	Q01523
PLXNA2	O75051	P01375	PLXNB1	O43157	P52198
PLXNB2	O15031	P52198	PLXNC1	O60486	O75326
PLXND1	Q9Y4D7	P22736	PM20D2	Q8IYS1	O75695
PMAIP1	Q13794	P10415	PML	P29590	Q13164
PMM1	Q92871	P20340	PMP22	A0A6Q8PF08	Q969S8
PMPCA	Q10713	P01160	PMS1	P54277	P40692
PMVK	Q15126	P01160	PNKD	Q8N490-3	Q13153
PNN	Q9H307	Q9UJV9	PNPLA2	Q96AD5	P09917
PNPO	Q9NVS9	Q99551	POC1A	Q8NBT0	Q13371
PODXL2	Q9NZ53	P19320	POFUT1	Q9H488	P0DTC8
POGLUT3	Q7Z4H8	P0DTC8	POGZ	Q7Z3K3	P14859, P52747
POLA1	P09884	Q9H668	POLA2	Q14181	Q9H668
POLB	P06746	Q01831	POLD2	P49005	O60673
POLD4	Q9HCU8	P12004	POLDIP2	Q9Y2S7	P13196
POLE2	P56282	P52594	POLE3	Q9NRF9	O60264
POLG2	Q9UHN1	Q02252	POLH	Q9Y253	Q86YC2
POLI	Q9UNA4	Q15843	POLQ	O75417-1	Q06609
POLR1F	Q3B726	Q9NYV6	POLR2A	P24928	Q9Y5B0
POLR2E	Q2T9T3, P19388	O00267	POLR2F	P61218	Q9Y5B0
POLR2I	P36954	O00267	POLR2J	P52435	Q15648
POLR2J3	Q9H1A7	Q9UPW6	POLR2L	P62875	O00267
POLR3E	Q9NVU0	Q9H063	POLR3G	O15318	P52292
POLR3H	Q9Y535	P61081	POM121	Q96HA1	P36941
POM121C	A8CG34	Q14974	POMGNT1	Q8WZA1	Q96FT7
POMT2	Q9UKY4	P37268	POP7	O75817	Q9BQ90
PORCN	Q9H237-2	P55056	POT1	Q9NUX5	P50747
POTEF	A5A3E0	Q9UNA3	POTEI	P0CG38	Q9UM54
POU2F1	P14859	P14859	POU5F1	Q01860	P48431

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
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PPARD	Q03181	P31749	PPARG	P37231, P37231-2	Q9UBK2
PPARGC1A	Q9UBK2	P37231	PPAT	Q06203	P61769
PPFIBP2	Q8ND30	P46108	PPIB	P23869	P37659
PPIC	P45877	Q8N9N5	PPIG	Q13427	P21741
PPL	O60437	Q96QF0	PPM1B	O75688	P05161
PPM1E	Q8WY54	P61289	PPM1H	Q9ULR3	P63104
PPM1J	Q5JR12	Q96GM5	PPM1L	Q8BHN0	Q9H3P7
PPME1	Q9Y570	P67775, P63151	PPP1CA	P62136	Q9UQK1
PPP1CC	P36873-1	P51955	PPP1R10	Q96QC0	P36873
PPP1R11	O60927	P36873	PPP1R12B	O60237-2	P06396
PPP1R12C	Q9BZL4	Q9UI12	PPP1R16B	Q96T49	P40937
PPP1R18	Q6NYC8	P28482	PPP1R1A	Q13522	Q96LA8
PPP1R3B	Q86XI6	P36873	PPP1R3C	Q9UQK1	O95278
PPP1R3F	Q6ZSY5	Q13627	PPP1R7	Q15435	Q6VY07
PPP1R9A	Q9ULJ8	Q9UM54	PPP1R9B	Q96SB3	P38398
PPP2R1A	P30153	P56211, O43768	PPP2R2B	Q00005	O15530
PPP2R3A	Q06190	Q9NYZ3	PPP2R5B	Q15173	O96017
PPP2R5D	Q14738	P30307	PPP3CC	P48454	P24592
PPP4R2	Q9NY27	O75533	PPP4R3A	Q6IN85	P18509
PPP5C	P53041	Q15185	PPP6R1	Q9UPN7	Q14694
PPP6R2	O75170	P53350	PPT1	P50897	O75925
PPWD1	Q96BP3	P55735	PRADC1	Q9BSG0	P59665
PRAF2	O60831	P80370	PRC1	O43663	P36941
PRDM11	Q9NQV5	Q14192	PRDM14	Q9GZV8	O43439
PRDM5	Q9NQX1-2	Q9NX46	PRDX4	Q13162	P21731
PRDX5	P30044	P00441	PREB	Q9HCU5	Q16625
PRELID1	Q9Y255	O43715	PREX2	Q70Z35	P15056
PRICKLE1	Q96MT3	Q9HAQ2	PRICKLE3	O43900	Q9UJK0
PRIM1	P49642	P42858	PRKAA1	Q13131	O75385
PRKAA2	P54646	O75385	PRKACA	P17612	P13861
PRKAG1	P54619	O75385	PRKAR2A	P13861	P17612
PRKAR2B	P31323	P13861	PRKCA	P17252	O94844
PRKCB	P05771	P31749	PRKCE	Q02156	P17252
PRKCG	P63319	P53667	PRKCH	P24723	P17252
PRKCI	P41743	Q8N5V2	PRKCQ	Q04759	Q06187
PRKCSH	P14314	O14975	PRKD1	Q15139	P12830
PRKX	P51817	Q13451	PRMT2	P55345	P15121
PRMT3	O60678	P15880	PRNP	P04156	P29372
PROCR	Q9UNN8	P0DTC3	PROKR2	Q8NFJ6	P24593
PROM1	O43490	Q9UBN7	PROSER2	Q86WR7	Q9Y316
PROSER3	Q2NL68	Q96ES7	PRPF31	Q8WWY3	P05455
PRPF38A	Q8NAV1	Q96CW1	PRPF38B	Q5VTL8	P61981
PRPF39	Q86UA1	P0DTC5	PRPF40A	O75400-2	Q9BV20
PRPF6	O94906	P01106	PRPS1	P60891	O60551
PRPSAP2	O60256	P14635	PRR11	Q96HE9	Q15361
PRR14L	Q5THK1	Q15185	PRR16	Q569H4	P78352
PRR5-ARHGAP8	B1AHC3	P62993	PRR5L	Q6MZQ0	Q9UI95
PRRC2C	Q9Y520	Q9H9Z2	PRSS23	O95084	P60709

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PRXL2C	Q7RTV5	Q9NRD5	PSAP	P07602	P07948
PSD4	Q8NDX1-2	Q5SW96	PSG1	P11464	P01137
PSIP1	O75475	Q03164	PSKH1	P11801	P08238
PSMA2	P25787	Q9Y5K5	PSMA3	P25788	Q9BUF7
PSMB3	P49720	O00487	PSMB6	P28072	Q9Y5K5
PSMB8	P28062-2	Q13114	PSMB9	P28065	Q15788
PSMC3	P17980	O00487	PSMD2	Q13200	P50454
PSMD3	O43242	P25963	PSMD8	P48556	P0DTC5
PSMD9	O00233	P03952	PSME1	Q06323	P60896
PSME4	Q14997	P60896	PSMF1	Q92530	Q9NWB1
PTAFR	P25105	P04156	PTAR1	Q7Z6K3	P53611
PTBP2	Q9UKA9	P0DTD2	PTBP3	O95758	Q92900
PTCD1	O75127	O00425	PTCD3	Q96EY7	Q9H9Z2
PTGES	O14684	O15155	PTGES2	Q9H7Z7	Q9UG63
PTGIR	P43119	Q7L8W6	PTGR2	Q8N8N7	P49327
PTGR3	Q8N4Q0	P11182	PTGS1	P23219	P48645
PTH2R	P49190	P13591	PTPA	P18433	P12931
PTPN13	Q12923	P59637	PTPN14	Q15678	P46937
PTPN18	Q99952	P00533	PTPN2	P17706	Q9Y5K5
PTPN21	Q16825	P51813	PTPN23	Q9H3S7	P12931
PTPN3	P26045	Q96QG7	PTPN4	P29074	Q9BY67
PTPN5	P54829	P28482	PTPN6	P29350	P35968
PTPRD	P23468	O14965	PTPRG	P23470	Q6VY07
PTPRJ	Q62884	P12931	PTPRM	P28827	O60469
PTPRN	Q16849	Q13332	PTPRT	O14522	O60469
PTPRZ1	P23471	Q12860	PTRH1	Q86Y79	Q9NRD5
PTRH2	Q9Y3E5	Q9H2J7	PTTG1IP	P53801	Q16585
PTX3	P26022	P09038	PUF60	Q9UHX1	P05455
PUM3	Q15397	P40429	PURA	Q00577	P06400
PURB	Q96QR8	Q9H9Z2	PURG	Q9UJV8	Q9H6S0
PUSL1	Q8N0Z8	P10809	PVR	P15151	O43929
PWWP2A	Q96N64	Q13547	PXDN	Q92626	P01282
PXMP2	Q9NR77	P46059	PXN	P49023	P12931
PYCR1	P32322	Q99497	PYCR3	Q53H96	P42772
PYGB	P11216	Q13451	QPRT	Q15274	O00443
QTRT1	Q9BXR0	P10809	RAB11B	Q15907	P12931
RAB11FIP1	Q6WKZ4	P61981	RAB11FIP3	O75154	Q9H0H5
RAB11FIP4	Q86YS3	Q8NFX5	RAB11FIP5	Q9BXF6	P63104
RAB17	Q9H0T7	P24386	RAB1A	P62820	Q92696
RAB1B	Q9H0U4	Q92696	RAB20	Q9NX57	P52306
RAB25	P57735	P15056	RAB27A	P51159	Q92696
RAB2A	P61019	P51149	RAB32	Q13637	Q92696
RAB35	Q15286	P42858	RAB37	Q96AX2	P53611
RAB38	P57729	Q5S007	RAB39A	Q14964	Q08379
RAB39B	Q96DA2	P46059	RAB3B	P20337	Q8N695
RAB3C	Q96E17	P36957	RAB3IP	Q96QF0	Q96T51
RAB5B	P61020	P42858	RAB5C	P51148	P10636-8
RAB6A	P20340-2	P05067	RAB7A	P51149	P21796

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RAB8B	Q92930	Q9UK39	RABAC1	Q9UI14	P36941
RABGAP1L	Q5R372-2	P06400	RABGEF1	Q9UJ41	Q9H0F6
RABGGTA	Q92696	P53611	RABIF	P47224	Q9NZU5
RAC2	P15153	P52306	RACGAP1	Q9H0H5	O14757
RAD23A	P54725	Q15843	RAG1	P15918	P52294
RAI1	Q7Z5J4	Q13526	RALB	P11234	P11182
RALY	Q53GL6	Q969Y2	RANBP2	P49792	P61769
RANBP3	Q9H6Z4	P18754	RANGAP1	P46060	P61769
RANGRF	Q9HD47	Q8N307	RAP1A	P62834	P52306
RAP1B	P61224	Q12967	RAP1GDS1	P52306	P17480
RAPGEF2	Q9Y4G8	P46937	RAPGEF4	Q8WZA2	P05455
RAPGEF5	Q92565	Q12955	RAPH1	Q70E73	Q99962
RARG	P13631	Q08999	RASA1	P20936, P50904	Q13480
RASD2	Q96D21	O75197	RASEF	Q8IZ41	P13569
RASGRP4	Q8TDF6-2	Q12982	RASL10B	Q96S79	Q9H3L0
RASL12	Q9NYN1	Q14192	RASSF1	Q9NS23	P31749
RASSF2	P50749	P36957	RAVER1	Q8IY67	O75791
RB1CC1	Q8TDY2	O75385	RBAK	Q9NYW8	Q13263
RBBP4	Q09028	P07237	RBBP6	Q7Z6E9-3	O76024
RBFOX3	A6NFN3	P22732	RBM11	P57052	Q93009
RBM24	Q9BX46	P05120	RBM25	P49756	O75533
RBM26	Q5T8P6	Q9Y4Z2	RBM39	Q14498-3	Q9UH92
RBM41	Q96IZ5	Q9Y2I6	RBM42	Q9BTD8	O00425
RBM47	A0AV96	P0DTC9	RBM6	P78332	P09651
RBMS2	Q15434	O00425	RBMXL1	Q96E39	P23246
RBP1	P09455	Q9UBN6	RBPJ	Q06330-6, Q06330	P46531
RBPMS	Q93062	O94955	RBPMS2	Q6ZRY4	O75593
RCAN1	P53805	P16298	RCAN3	Q9UKA8	P11802
RCC1L	Q96I51	O00425	RCN2	Q14257	Q9Y4K3
RCN3	Q96D15	Q9UMX1	RCOR2	Q8IZ40	Q13547
RDH10	Q8IZV5	Q05329	RDH13	Q8NBN7	Q01523
RDM1	Q8NG50	P26441	RECK	Q6P9E2	Q96LC9
REEP4	Q9H6H4	Q9Y584	RELA	Q04206	Q13568
RELL1	Q8IUW5	P26572	RELN	Q60841	P98155
RELT	Q969Z4	Q8NFM7	REPS2	Q8NFH8	Q96D71
RESF1	Q9HCM1	Q9NV70	RET	P07949	P22455
RETREG2	Q8NC44	Q9GZQ8	REV1	Q9UBZ9	Q9UI95
RFC1	P35251	P35250, P40938, P40937, P35249	RFC3	P40938	P40937, Q9BVC3, P35249
RFC4	P35249	P35250, P0CG13, P40938, P40937, Q9BVC3, Q8WVB6	RFFL	Q8WZ73	P29353
RFTN1	Q14699	P18827	RFTN2	Q52LD8	Q70CQ3
RFX2	P48378	O75928-2	RFX3	P48380	Q14192
RFX5	P48382	Q92769	RFX7	Q2KHR2	P61981
RFXANK	O14593	P04049	RGPD2	P0DJD1	P25103
RGS12	O14924	P56559	RGS17	Q9UGC6	Q9H2F3
RGS19	P49795	P25786	RGS2	P41220	Q9Y2I6
RGS4	P49798	P00533	RGS6	P49758	P08238
RHBDD2	Q6NTF9-3	P23434	RHOA	P61586	P52565

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RHOB	P62745	Q92888	RHOBTB2	Q9BYZ6	Q9Y4X5
RHOBTB3	O94955	P53611	RHOC	P08134	P52565
RHOG	P84095	P52565	RHOH	Q15669	Q92673
RHOJ	Q9H4E5	Q13177	RHOT1	Q8IXI2	O60260
RHOXF2	Q9BQY4	P63165	RHPN1	Q8TCX5	Q9NZ53
RHPN2	Q8IUC4	P61981	RIC3	Q7Z5B4-5	O14653
RIC8A	Q9NPQ8-4	Q96QE2	RIDA	P52758	P0CG47
RIF1	Q5T3J3	Q96A70	RILPL1	Q5EBL4-3	Q86T03
RIMKLA	Q8IXN7	Q96EN8	RIMS3	Q9UJD0	P08670
RIN1	Q13671	P00519	RIN3	Q8TB24	P19174
RING1	Q06587	P51668	RIOK3	O14730	Q14653, Q9UHD2
RIPOR2	Q9Y4F9, Q9Y4F9-2	P61586	RIPPLY3	P57055	Q13416
RIT2	Q5BJQ5, Q99578	Q01959	RLF	Q13129	Q13185
RMI2	Q96E14-1, Q96E14	P54132	RMND5A	Q9H871	P12755
RNASEH2A	O75792	Q99943	RNASEH2C	Q8TDP1	P04792
RNASEK	Q6P5S7	Q5QGT7	RND2	P52198	O43157
RNF11	Q9Y3C5	Q8N5V2	RNF122	Q9H9V4	O95674
RNF125	Q96EQ8	P55072	RNF126	Q9BV68	P51668
RNF130	Q86XS8	Q6UX41	RNF138	Q8WVD3	P07550
RNF14	Q9UBS8	P04004	RNF144A	P50876	Q96AA3
RNF146	Q9NTX7	Q86WJ1	RNF150	Q9ULK6-3	P17252
RNF181	Q9P0P0	P51668	RNF182	Q8N6D2	P51668
RNF185	Q96GF1	P51668	RNF187	Q5TA31	Q93034
RNF19A	Q9NV58	P00441	RNF213	Q63HN8	P21462
RNF216	Q9NWF9	P68036	RNF24	Q9Y225-2	O95406
RNF25	Q96BH1	P51668	RNF31	Q96EP0	P32455
RNF40	O75150	P06400	RNF44	Q7L0R7	Q8TDS5
RNH1	P13489	Q12834	RNMT	O43148	P06730
RO60	P10155	P05455	ROBO1	Q9Y6N7	O75044
ROR1	Q01973	P12931	RORB	Q92753-1	Q96GM5
RP2	O75695	P49674	RPAP3	Q9H6T3	P36873
RPGR	Q92834-2, Q92834	O43924	RPGRIPI	Q96KN7	Q9P0N9
RPL10	P27635	P12931	RPL13A	P40429	P06748
RPL15	P61313	Q9H9Z2	RPL18	Q07020	P40429
RPL18A	Q02543	Q15361	RPL19	P84098	P40429
RPL26L1	Q9UNX3	P42858	RPL28	P46779	P43146
RPL29	P47914	P09429	RPL35	P42766	P05455
RPL36	Q9Y3U8	P62753	RPL8	P62917	Q15843
RPLP1	P05386	O43809	RPLP2	P05387	Q9H9Z2
RPN1	P04843	Q14165	RPRD1A	Q96P16-3	O14964
RPS11	P62280	P19338	RPS16	P62249	Q8IZH2
RPS28	P62857	P06748	RPS5	P46782	P42858
RPS6KA1	Q15418	Q13085	RPS6KA2	Q15349	P28482
RPS6KA4	O75676	Q16539	RPS6KA5	O75582	Q16539
RPS6KA6	Q9UK32	P27361	RPS6KB2	Q9UBS0	P67870
RPTOR	Q8N122	O75385	RRAGA	Q7L523	P61916
RRAGC	Q9HB90	P17931	RRAGD	Q9NQL2	Q9Y2Q5, Q0VGL1, Q6IAA8, O43504, Q8NBW4, Q9UHA4
RRBP1	Q9P2E9	P13569	RREB1	Q92766	Q13547

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RRP9	O43818	Q9Y2X3	RSF1	Q96T23	P09429
RSKR	Q96LW2	O95231	RSPH3	Q86UC2	P40692
RSPH9	Q9H1X1	Q9UPT8	RSPO4	Q2I0M5	O14964
RSRC2	Q7L4I2-2	Q08379	RSU1	Q15404	Q9NVD7, Q9HBI1, P48059
RTEL1	Q9NZ71	Q15185	RTKN2	Q8IZC4	Q96B97
RTL6	Q6ICC9	Q9P2W9	RTL8B	Q17RB0	O00560
RTL8C	A6ZKI3	P42772	RTL9	Q8NET4	Q00613
RTN1	Q16799-3	P42858	RTN4	Q9NQC3	P0DTC5
RTN4IP1	Q8WWV3	Q9HAU4	RTN4R	Q99M75	Q96FE5
RTN4RL1	Q86UN2	Q7Z3S9, P0DPK4	RTN4RL2	Q86UN3	P17931
RUBCN	Q80U62, Q92622	P00533	RUNDC3A	Q59EK9	Q96CW1
RUNX1	Q01196	Q9H2X6	RUNX2	Q13950	P17480
RUNX3	Q13761	Q09472	RUSC1	Q9BVN2	P42772
RUSC2	Q8N2Y8	P00533	RUSF1	Q96GQ5	Q6ZPD8
RUVBL2	Q9Y230	O96019, O94805	RWDD2B	P57060	Q96HC4
RWDD3	Q9Y3V2	P63279	RXRA	P19793	Q15648
RXRB	P28702	P37231	RYR1	P11716	P62942
RYR2	Q92736	Q00987	S100A10	P60903	P03950
S100A11	P31949	P04637	S100A13	Q99584	Q92845
S100A16	Q96FQ6	Q15915	S100A2	P29034	P52292
S100A4	P26447	P15514	S100A6	P06703	P52292
S1PR1	P21453	Q7Z2K8	S1PR2	O95136	Q9NWM8
S1PR3	Q99500	P21964	SACM1L	Q9NTJ5	Q9H244
SALL2	Q9Y467	P08138	SAMD11	Q96NU1	Q96RU7
SAMD3	Q8N6K7-2	P04156	SAMD4A	G3V2R1	P42858
SAMD4B	Q5PRF9	P62736	SAP30BP	Q9UHR5	P68400
SAR1B	Q9Y6B6	Q9Y5M8	SARS1	P49591	P38936
SART1	O43290	P55854, P63165	SASS6	Q6UVJ0	P42858
SATB1	Q01826	P63279	SATB2	Q9UPW6	Q12946
SBDS	Q9Y3A5	Q15149	SBF1	O95248	Q13614
SBSN	Q6UWP8	P15498	SC5D	O75845	Q8TDU9
SCAF1	Q9H7N4	O43159	SCAI	Q8N9R8	P45973
SCAMP2	O15127	P06576	SCAMP3	O14828	P80370
SCAMP4	Q969E2	Q8WTR4	SCAMP5	Q8TAC9	P49638
SCARA3	Q6AZY7-2	P05067	SCARB2	Q14108	Q9UBD6
SCARF1	Q14162	Q96BA8	SCARF2	Q96GP6	P06241
SCD	O00767	Q9H2K0	SCFD2	Q8WU76	Q12981, Q9NZ43, Q9P2W9, O75396
SCG3	Q8WXD2	O00560	SCMH1	Q96GD3	P35227
SCML1	Q9UN30-2	O00560	SCML2	Q9UQR0-1	P38936
SCN5A	Q14524	Q99873	SCNM1	Q9BWG6	Q15669
SCNN1A	P37088	P46934	SCUBE3	Q8IX30	P37173
SCYL1	Q96KG9-4	O15169	SDC1	P18827	P09038
SDC2	P34741	P09038	SDC4	P31431	O15530
SDCCAG8	Q86SQ7	P61088	SDF2L1	Q9HCN8	Q9H5K3
SDF4	Q9BRK5	P50548	SDHA	P31040	P31749
SDHAF4	Q5VUM1	Q9NRP4, P31040	SEC13	P55735	P49674
SEC14L5	O43304	P62333	SEC16A	O15027	Q9BY84

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SEC22B	O75396	O95249, Q13190	SEC23A	Q15436	P50570
SEC24C	P53992	Q15436	SEC24D	O94855, O94855-2	Q15436
SEC31A	O94979	Q15436	SEC61A1	P61619	Q16288
SEC61B	P60468	P43307	SECISBP2	Q96T21	P42858
SEL1L	Q9UBV2	Q96FT7	SELENOM	Q8WWX9	P61604
SELENOP	P49908	P00533	SELENOS	Q9BQE4	P55072
SELENOV	P59797	Q13526	SEMA3C	Q99985	P18827
SEMA3E	O15041, P70275	Q9Y4D7	SEMA4A	Q9H3S1	Q9Y4D7
SEMA4D	Q92854	Q9ULL4	SEMA7A	O75326	O60486
SEMG1	P04279	Q6N021	SEPHS1	P49903	Q99611
SEPSECS	Q9HD40	P04591	SEPTIN11	Q9NVA2	Q15019
SEPTIN3	Q9UH03	Q15019	SEPTIN8	Q92599	Q15019
SERGEF	Q9UGK8	O75593	SERINC1	Q9NRX5	Q06710
SERINC3	Q13530	O76024	SERP2	Q8N6R1	Q9H2H9
SERPINA10	Q9UK55	P00742	SERPINA5	P05154	P09958
SERPINB1	P30740	Q00403	SERPINB2	P05120	Q9NRY4
SERPINB4	P48594	Q14116	SERPINB5	P36952	Q15831
SERPINB7	O75635	Q9NYY3	SERPINB8	P50452	Q15696
SERPINB9	P50453	Q11203	SERPINE1	P05121	P04004
SERPINF1	P36955	P11021	SERPINH1	P50454	P12830
SERTAD1	Q9UHV2	P61024	SERTAD2	Q14140	O60733
SERTAD3	Q9UJW9	Q6W0C5	SERTM1	A2A2V5	Q9H2K0
SET	Q01105	P17480	SETD1B	Q9UPS6	Q9UER7
SETD7	Q8WTS6	P26358	SETDB1	Q15047	Q9Y6K1
SETMAR	Q53H47	P11142	SF3B1	O75533	Q9UJV9
SF3B2	Q13435	Q9H9Z2	SF3B4	Q15427	P00519, Q9UQB8
SF3B5	Q9BWJ5	Q96FJ0	SFMBT1	Q9UHH3	P62805
SFRP1	Q8N474	Q8N661	SFT2D2	O95562	Q8NBQ5
SFXN3	Q9BWM7	P46059	SGCB	Q16585	P01375
SGK1	O00141	P49815	SGSH	P51688	Q8NBK3
SGSM1	Q2NKQ1-4, Q2NKQ1	P41208	SGTB	Q96EQ0	P10451
SH2D2A	Q9NP31	O14543	SH2D3A	Q9BRG2	P00533
SH2D4A	Q9H788, Q9H788-2	O14936, Q9UJU6	SH3BGRL3	Q9H299	P00533
SH3BP4	Q9P0V3	O14908	SH3BP5	O60239	Q07889
SH3BP5L	Q7L8J4	Q9NWX6	SH3GL1	Q99961	Q99962
SH3GL2	Q99962	Q99962	SH3GL3	Q99963	Q99962
SH3GLB1	Q9Y371	P02647	SH3KBP1	Q96B97	P98082
SH3PXD2A	Q5TCZ1	P63104	SH3PXD2B	A1X283	O43639
SH3RF1	Q7Z6J0	P63000	SH3RF3	Q8TEJ3	Q13177
SH3YL1	Q96HL8	Q5HYK7	SHANK2	Q9UPX8	P19174
SHARPIN	Q9H0F6	P08648	SHB	Q15464	P00519
SHCBP1	Q8NEM2	P05771	SHISA4	Q96DD7	O95406
SHISA5	Q8N114	Q9NZC7	SHISA6	Q6ZSJ9	P38936
SHMT2	P34897	P46736, Q9NWW8, Q9NXR7	SHOX2	O60902-3	O76024
SHPRH	Q149N8-1	P61088	SHROOM2	Q13796	P61981
SHTN1	A0MZ66	P63104	SIGMAR1	Q99720	P14416
SIL1	Q9H173	P11021	SIM1	P81133	Q9HBZ2
SINHCAF	Q9NP50	Q9Y6N5	SIPA1L2	Q9P2F8	P63104
SIPA1L3	O60292	P63104	SIRPA	P78324	Q08722

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SIRPB1	O00241	O43914	SIRT2	Q8IXJ6, Q8VDQ8, Q8IXJ6-2	Q9UM11, Q12834
SIRT7	Q9NRC8	Q16656	SIVA1	O15304	P26842
SIX1	Q15475	Q99502	SKA1	Q96BD8	A1L4H1
SKAP2	O75563	Q9Y2R2	SKI	P12755	P36941
SKIC2	Q15477	P06748	SKIDA1	Q1XH10	Q15022
SKIL	P12757	P23025	SLA	Q13239-3	Q13882
SLAIN1	Q8ND83	P09917	SLAIN2	Q9P270	P61981
SLC10A1	Q14973	Q9NRZ7	SLC13A3	Q8WWT9	Q96BA8
SLC15A1	P46059	Q8NE86	SLC15A2	Q16348	Q92922
SLC16A1	P53985	P35613	SLC16A2	P36021	P21926
SLC16A6	O15403	O76024	SLC16A7	O60669	P09601
SLC19A2	O60779	Q92911	SLC19A3	Q9BZV2	O43561-2
SLC1A3	P43003	Q8WVV5	SLC1A4	P43007	Q15758
SLC1A5	Q15758	P0DTC5	SLC20A2	Q08357	P43088
SLC22A4	Q9H015	O76082	SLC22A5	O76082	Q9H015
SLC23A1	Q9UHI7	Q9UBQ7	SLC24A3	Q9HC58	Q86TM6
SLC25A10	Q9UBX3	Q7Z3S9	SLC25A11	Q02978	P12830
SLC25A13	Q9UJS0	Q9Y5J6	SLC25A16	P16260	Q15116
SLC25A21	Q9BQT8	P0DTC5	SLC25A22	Q9H936	Q5S007
SLC25A28	Q96A46	P46059	SLC25A30	Q5SVS4	Q9UBS4
SLC25A43	Q8WUT9	P10809	SLC25A48	Q6ZT89	Q13077
SLC25A6	P12236	P12235	SLC27A2	O14975	Q8NF37
SLC27A4	Q6P1M0	P40855	SLC27A6	Q9Y2P4	Q9Y2C2
SLC2A12	Q8TD20	Q7Z7B1	SLC2A13	Q96QE2	Q96CV9
SLC2A14	Q8TDB8	P11169	SLC2A3	P11169	P17931
SLC2A4	P14672	Q9UER7	SLC30A3	Q99726	Q8IWU4
SLC30A4	O14863	P55010	SLC34A2	O95436	Q15904
SLC35A2	P78381	P16885	SLC35A4	Q96G79	Q9UKG4
SLC35B2	Q8TB61	Q96BA8	SLC35C1	Q96A29	P05090
SLC35E1	Q96K37	Q9Y366	SLC35E4	Q6ICL7	Q9UKG4
SLC35F1	Q5T1Q4	P21964	SLC35F6	Q8N357	Q9NPL8
SLC36A1	Q7Z2H8	P05067	SLC38A1	Q9H2H9	Q9UKG4
SLC38A10	Q9HBR0	Q30201	SLC38A7	Q9NVC3	P62495
SLC38A9	Q8NBW4	Q9Y2Q5, Q0VGL1, Q6IAA8, O43504, Q9UHA4	SLC39A10	Q9ULF5	Q6ZMH5
SLC39A13	Q96H72	O15173	SLC39A8	Q9C0K1	Q09328
SLC41A1	Q8IVJ1	P30518	SLC41A2	Q96JW4	P21217
SLC43A3	Q8NBI5	P55072	SLC44A2	Q8IWA5	P0DTC4
SLC44A3	Q8N4M1-3	Q8WTS1	SLC44A5	Q8NCS7	P50454
SLC4A7	Q9Y6M7	P17931	SLC52A3	Q9NQ40	P54252
SLC5A5	Q92911	P0DPK4	SLC66A2	Q8N2U9	Q9NR28
SLC6A15	Q9H2J7	Q9NRZ7	SLC7A1	P30825	Q09328
SLC7A2	P52569	P30825	SLC7A3	Q8WY07	P22732
SLC9A1	P19634	O75326	SLC9B2	Q86UD5	O15173
SLCO4C1	Q6ZQN7	Q14527	SLF1	Q9BQI6	P19838
SLIT2	O94813	O94813	SLITRK3	O94933	O76024
SLITRK4	Q8IW52	Q9BT09	SLITRK5	O94991	Q6PJG9
SLX4IP	Q5VYV7	P53350	SLX9	Q9NSI2	Q9NRD5
SMAD2	Q15796	P98082	SMAD3	P84022	P98082

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SMAD6	O43541-2, O35182	O95630	SMAD7	O15105	P62942
SMAD9	O15198, O15198-2	Q13485	SMAGP	Q0VAQ4	Q8IY26
SMAP2	Q9H0E9	O96019	SMARCA2	P51531	Q8NFD5
SMARCA5	O60264	P17096	SMARCA1	Q9H4L7	P43246
SMARCB1	Q12824	Q92925, Q8NFD5, P51531, O96019, O75177, Q15532	SMARCC1	Q92922	Q92925, Q8NFD5, O96019, Q15532
SMARCD1	Q96GM5	Q9H8M2	SMARCD3	Q6STE5-2, Q6STE5-1	P37231
SMC3	Q9CW03, Q9UQE7	Q7Z5K2	SMC6	Q96SB8	Q00613
SMCHD1	A6NHR9	Q96GD4	SMCO4	Q9NRQ5	P25025
SMG5	Q9UPR3	Q92540	SMG9	Q9H0W8	Q9BRV3
SMIM3	Q9BZL3	Q16585	SMN1	Q16637	Q96FZ7
SMTN	P53814-5	O76024	SMTNL2	Q2TAL5	P20248
SMUG1	Q53HV7-2	Q8IVS8	SMURF2	Q9HAU4	Q9HAU4
SMYD1	Q8NB12	O14503	SMYD2	Q9NRG4	P01160
SNAI2	O43623	Q9NPB3	SNAP29	O95721	Q13190
SNCA	P37840	Q969S8	SNRNP35	Q16560-2	P50454
SNRNP48	Q6IEG0	O75533	SNRPA1	P09661	P55199
SNRPB	P14678-2	P52747	SNRPD1	P62316	Q09161
SNRPN	P63162	O75563	SNTA1	Q13424	P60484
SNTB2	Q13425	O14936	SNTG2	Q9NY99-2	Q8N3R9
SNURF	P78317	P13500	SNX1	Q13596	Q8IVP5
SNX10	Q9Y5X0	P17252	SNX13	Q9Y5W8	P61073
SNX17	Q15036	Q13496	SNX2	O60749	Q5T8I9
SNX21	Q969T3	P17707	SNX22	Q96L94	P21802
SNX27	Q96L92	P27037	SNX29	Q8TEQ0	Q13077
SNX3	O60493	P56539	SNX30	Q8WV41	Q05193
SNX32	Q86XE0	Q06609	SNX33	Q8WV41	Q05193
SNX5	Q9Y5X3	Q9Y639	SNX8	Q9Y5X2	P16435
SNX9	Q9Y5X1	Q07889	SOAT1	P35610	O75326
SOBP	A7XYQ1	Q13363	SOCS1	O15524	P46109
SOCS5	O75159	Q93034, Q9UBF6	SOCS7	Q8WXH5	Q8N3R9
SORBS1	Q9BX66	P00519	SORBS2	O94875	Q13177
SORBS3	O60504-2	P00519	SORL1	Q92673	O95395
SORT1	Q99523	P01138	SOX13	Q9UN79	Q12933
SOX2	P48431	P30740	SOX3	P41225	Q12933
SOX4	Q06831	O00560	SOX5	P35711	O00712, P08651, Q12857
SOX9	P48436	Q12857	SP100	P23497	Q9Y6K1
SP2	Q9UKF7-2	O60906	SP4	Q02446	P14859
SPAST	Q9UBP0, Q9UBP0-1	Q9UF56	SPATA12	Q7Z6I5	P06396
SPATA17	Q96L03	P36957	SPATA20	Q8TB22	P47985
SPATA46	Q5T0L3	Q99685	SPATA6	Q9NWH7-2	P55040
SPDL1	Q96EA4	P25963	SPEG	Q15772-4	Q9NRD5
SPIB	Q01892	P42858	SPICE1	Q8N0Z3	P05120
SPINT1	O43278-2	P02787	SPINT2	O43291	Q99640
SPIRE1	Q08AE8	P60709	SPOCK1	Q08629	Q96BA8
SPOUT1	Q5T280	P0DTC9	SPP1	P10451	P16070
SPPL2B	Q8TCT7	O75063	SPRY1	O43609	O43597
SPRY3	Q8NCJ5	P22692	SPRY4	Q9C004	P22681
SPRYD3	Q8NCJ5	P22692	SPRYD7	Q5W111	Q86UT5

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SPSB1	Q96BD6	Q13153	SPSB4	Q96A44	P21926
SPTAN1	Q13813	P40692	SPTLC2	O15270	Q9H1C4
SQLE	Q14534	Q96BA8	SQSTM1	Q13501	P04629
SRA1	Q9HD15	Q92769	SRBD1	Q8N5C6	Q9H9Z2
SRC	P12931	Q68CZ2	SRCAP	Q6ZRS2	Q13257
SRFBP1	Q8NEF9	Q13895	SRGAP1	Q7Z6B7	P63104
SRGN	P10124	P10144	SRM	P19623	P19623
SRP9	P49458	O75533	SRPK1	Q96SB4	Q05397
SRPRA	P08240	Q9Y5M8	SRPRB	Q9Y5M8	P09601
SRPX	P78539	Q9BRX2	SRRM4	A7MD48	P61964
SRRT	Q9BXP5	P52292	SRSF10	O75494	Q9H9Z2
SRSF11	Q05519	Q92844	SRSF12	Q8WXF0	P61981
SRSF5	Q13243	Q9H9Z2	SRXN1	Q9BYN0	Q06830
SSB	P05455	Q8WZA2	SSBP4	Q9BWG4	Q9UBH0
SSC5D	A1L4H1	O00560	SSH1	Q8WYL5	P53667
SSNA1	O43805	Q9P0N9	SSR2	P43308	Q86WV6
SSR3	Q9UNL2	P27824	SSR4	P51571	Q5S007
SSTR2	P30875, P30874	P27986	ST14	Q9Y5Y6	Q04721
ST3GAL1	Q11201	Q9ULF5	ST3GAL2	Q16842	P11021
ST6GAL2	Q96JF0-2	Q9NPL8	ST8SIA1	Q92185	Q96CV9
ST8SIA3	O43173	Q9UBS3	STAG2	Q8N3U4	Q7Z5K2
STAM2	O75886	P23458	STAMPB	O95630	Q9Y3E7
STARD4	Q96DR4	Q9H2K0	STAT1	P42224	Q05397
STAT2	P52630	O94782	STAT6	P42226	Q9UHD2
STAU2	Q9NUL3	Q9NZI8	STBD1	O95210	O95278, Q04446, Q9UQK1
STC2	O76061	P0DTC8	STIM1	Q13586	P30101
STING1	Q86WV6	Q96N66	STK17B	O94768	P08173
STK19	B7ZL18	Q969Z0	STK25	O00506	Q9BRV8
STK26	Q9P289	Q9BRV8	STK38	Q15208	Q9Y5B0
STOM	P27105	O43916	STON2	Q8WXE9	Q15843
STOX1	Q6ZVD7	Q01974	STOX2	Q9P2F5	P49674
STRBP	Q96SI9	P07910	STRC	A1L378	P42858
STRIP2	Q9ULQ0	Q9BRV8	STRN	O55106, O43815	Q9BRV8
STS	P08842	Q8NBK3	STUB1	Q9UNE7	P13591
STX10	O60499-2	Q12846	STX12	Q86Y82	P60880
STX2	P32856-2	Q13190	STX3	Q13277	P51809
STX4	Q12846	P25103	STX5	Q13190	O15498
STX6	O43752	P51809	STXBP5	Q5T5C0	Q13277
STXBP6	Q8NFX7	Q13277	SUCLG1	P53597	Q8IWL3
SUDS3	Q9H7L9	P51610	SULF1	Q8IWU6	P34741
SULT4A1	Q9BR01	Q9NUX5	SUMF2	Q8NBK7	Q8NBK3
SUMO1	P63165	P19419	SUMO3	P55854	P63279
SUN2	Q9UH99	Q96PU5	SUOX	P51687	Q15669
SUPT16H	Q9Y5B9	Q9NTX7	SUPT5H	O00267	Q9Y6D9
SURF4	O15260	Q15363	SUSD6	Q92537	Q96BN8
SUV39H1	O43463	P50454	SUV39H2	Q9H5I1	P45973
SV2A	Q02563	P0DPI0	SWAP70	Q9UH65	Q8WZA2
SYDE1	Q6ZW31	P62258	SYDE2	Q5VT97	P63104
SYK	P43405	Q9ULZ3	SYN3	Q17R54	O75928-2

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SYNE1	Q8NF91-1, Q8NF91	Q9UH99, O94901	SYNE2	Q8WXH0-1	P02545
SYNE3	Q6ZMZ3	Q9UH99	SYNE4	Q8N205	Q08722
SYNGAP1	Q96PV0	Q6VMQ6	SYNGR3	O43761	Q05329
SYNJ1	Q62910, O43426	Q99962	SYNJ2	O15056	Q96B97
SYNPO	Q8N3V7	P04156	SYNPO2	Q9UMS6	Q13418
SYNPR	Q8TBG9	Q8IY26	SYT1	P21579	P98155
SYT12	Q8IV01	Q15119	SYT15	Q9BQS2-2	P42858
SYT16	Q17RD7	P55056, P02654	SYT17	Q9BSW7	P02787
SYT2	Q8N9I0	P98155	SYT3	Q9BQG1	P42858
SYT4	Q9H2B2	Q14596	SYT5	O00445	P21579
SYT6	Q5T7P8-2	Q9Y6D9	SYT7	O43581	Q8IUQ4
SYTL3	Q4VX76	P51159	SYTL4	Q96C24	P51159
SYVN1	Q86TM6	Q9UBV2	SZRD1	Q7Z422	Q92900
TACC1	O75410, O75410-7	Q9Y316	TACO1	Q9BSH4	P21673
TADA2B	Q86TJ2	P52294	TAF1	P21675	P05067
TAF10	Q12962	Q92831	TAF1A	Q15573	Q01105
TAF1D	Q9H5J8	P06746	TAF6	P49848	P20226
TAFA4	Q96LR4	P08648	TAGLN2	P37802	P42858
TALDO1	P37837	P42858	TANC2	Q9HCD6	O14907
TANGO2	Q6ICL3	Q99962	TAOK2	Q9UL54	P36941
TAOK3	Q9H2K8	P42858	TAP1	Q03518	P30101
TAPBP	O15533	Q30201	TARBP1	Q13395	O75496
TARBP2	Q15633	P03496	TAS2R19	P59542	P15151
TAS2R5	Q9NYW4	Q5QGT7	TASP1	Q9H6P5	Q9H6P5
TAX1BP3	O14907	Q13177	TBC1D13	Q9NVG8	O60664
TBC1D15	Q8TC07	Q99941	TBC1D2	Q9BYX2-5	Q8NC69
TBC1D20	Q96BZ9	P55056	TBC1D22A	Q8WUA7	Q9H3P7
TBC1D25	Q3MII6	P18848	TBC1D2B	Q9UPU7	P56539
TBC1D30	Q9Y2I9-2	Q9UGI0	TBC1D7	Q9P0N9	P23025
TBC1D8	O95759	Q13077	TBC1D9	Q6ZT07	P21709
TBC1D9B	Q66K14	P21462	TBCB	Q99426	Q93009
TBK1	Q9UHD2	P59596	TBL1X	O60907	P50548
TBL1XR1	Q9BZK7	P36404	TBL2	Q9Y4P3	O15530
TBL3	Q12788	P06748	TBX15	Q96SF7	O94955
TBX18	O95935	Q7LC44	TBX2	Q13207	Q6DKK2
TBX3	O15119	Q08117	TBX5	Q99593	P46937
TBXA2R	Q0VAB0	Q969S0	TC2N	Q8N9U0	Q96JC9
TCEA3	O75764	Q8WVC0	TCEAL1	Q15170	Q96T88
TCEAL5	Q5H9L2	P43354	TCEAL8	Q8IYN2	O76024
TCERG1	O14776	Q99728	TCF19	Q9Y242	Q15669
TCF25	Q9BQ70	P08311	TCF7	P36402	P35222
TCHP	Q9BT92	P36941	TCIRG1	Q13488	Q15904
TCL1B	O95988	Q15648	TCN2	P20062	Q9NPF0
TCP11L1	Q9NUJ3	Q8TBF2	TCTA	P57738	P78382
TDG	Q13569	Q9GZX7	TDRD7	Q8NHU6	P42771
TEAD1	P28347	Q06609	TEAD2	Q15562-2, Q15562	P46937
TEC	P42680	P78325	TECPR2	O15040	P35813
TEDC1	Q86SX3	P35638	TEF	Q10587	Q16236
TEK	Q02763	P29353	TELO2	Q9Y4R8	Q9Y639
TENM2	Q9NT68	P21926	TENM3	Q9P273	Q13217

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
TENM4	Q6N022	Q92608	TENT5A	Q96IP4	P43405
TENT5B	Q96A09	P06396	TERF1	P54274-2	Q15004
TERT	EBI-21348901	P19419	TES	Q9UGI8	P13569
TESK1	Q15569	Q7Z6M1	TESK2	Q96S53	P63104
TESMIN	Q9Y4I5	P08235	TET1	Q8NFU7	O15294
TET2	Q6N021	P19544	TET3	Q8BG87, O43151	O15294
TEX10	Q9NXF1	Q9Y4Z2	TEX15	Q9BXT5	P01008
TEX2	Q8IWB9	Q9NP91	TEX264	Q9Y6I9	Q13370
TEX9	Q8N6V9	O60341	TFAM	Q00059	Q14197
TFAP2A	P05549	Q09472	TFB1M	Q8WVM0	Q00059
TFDP2	Q14188	Q08999, P06400	TFG	Q92734	O75593
TFPT	P0C1Z6	P55199	TGFB1	P01137	O60551
TGFB1I1	O43294	Q14289	TGFB2	P37173	Q99741
TGFBR3	Q03167	P22455	TGIF1	Q15583	P53350
TGIF2	Q9GZN2	Q96ST3	TGM2	P21980	P02751
TGOLN2	O43493-5	P01019	TGS1	Q96RS0	P52294
THAP3	Q8WTV1	P36941	THBD	P07204	P55056
THBS1	P07996	P02751	THBS2	P35442	Q6Y288
THEM6	Q8WUY1	Q8TC07	THNSL1	Q8IYQ7	Q9BQP7
THOC1	Q96FV9	P82979, Q9Y3Y2	THOC2	Q8NI27	Q9Y3Y2
THOP1	P52888	Q96CV9	THRA	P10827	Q15648
THRB	P10828	P60709	THSD4	Q6ZMP0	P11142
THUMPD1	Q9NXG2	Q96PZ0	THUMPD2	Q9BTF0	O76024
THY1	P04216	P05067	THYN1	Q9P016	Q9UJH6
TIA1	Q80ZW7	Q05397	TICAM1	Q8IUC6	Q9UHD2
TICAM2	Q86XR7	O00206	TIFA	Q96CG3	Q9Y4K3
TIGD1	Q96MW7	P42858	TIGD4	Q8IY51	Q12933
TIGIT	Q495A1	P15151, Q9NQS3	TIMM13	Q9Y5L4	Q99558
TIMM50	Q3ZCQ8	Q9NSC2	TIMP1	P01033	P14780, P09238
TIMP2	P16035	P50281, P08253	TIMP3	P35625	P16035
TJP2	Q9UDY2	Q07157	TK1	P04183	Q9UM11
TLE1	Q04724	Q13761	TLK1	Q9UKI8	P50458
TLN1	Q9Y490, P26039	P05556	TLN2	Q9Y4G6	P00519
TLR2	O60603	O60603	TLR4	O00206	Q99836
TLR5	O60602	O75911	TM4SF18	Q96CE8	Q9NPF0
TM6SF1	Q9BZW5-2	P06396	TM9SF4	Q92544	P13726
TMBIM1	Q969X1	Q8NCR0	TMBIM4	Q7Z782	O95231
TMC6	Q7Z403	P08670	TMCC1	O94876	Q9BRI3
TMED1	Q13445	Q15363	TMED10	P49755	Q9Y3A6
TMED2	Q15363	Q9Y3A6	TMED3	Q9Y3Q3	Q15363
TMED5	Q9Y3A6	Q15363	TMED9	Q9BVK6	Q9Y3A6
TMEFF1	Q8IYR6	Q13627	TMEM106A	Q96A25	Q16585
TMEM108	Q6UXF1	Q9NYV6	TMEM109	Q9BVC6	Q9H2K0
TMEM11	P17152	Q16585	TMEM115	Q12893	Q86U10
TMEM120A	Q9BXJ8	P61073	TMEM131	Q92545	P0DTC3
TMEM132D	Q14C87	P62136	TMEM139	Q8IV31	P48509
TMEM144	Q7Z5S9	Q9NPL8	TMEM14A	Q9Y6G1	P49682
TMEM167B	Q9NRX6	Q8N130	TMEM171	Q8WVE6	O43913
TMEM178A	Q8NBL3	Q12959	TMEM179B	Q7Z7N9	P43119
TMEM182	Q6ZP80	P17931	TMEM184B	Q9Y519	O15357

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
TMEM200A	Q86VY9	P16298	TMEM201	Q5SNT2-2	Q8N183
TMEM208	Q9BTX3	Q96BA8	TMEM214	Q6NUQ4	O75326
TMEM222	Q9H0R3	P21964	TMEM230	Q96A57	P27824
TMEM248	Q9NWD8	O95070	TMEM255B	Q8WV15	Q6ZPD8
TMEM266	Q2M3C6	O60341	TMEM30B	Q3MIR4	P21802
TMEM35A	Q53FP2	O15155, O14653	TMEM38A	Q9H6F2	O60496
TMEM39B	Q9GZU3	O75326	TMEM43	Q9BTV4	Q6Q788
TMEM50B	P56557	Q07817	TMEM54	Q969K7	P15151
TMEM59	Q9BXS4	Q13332	TMEM67	Q5HYA8	P26842
TMEM74	Q96NL1	Q8NAT1	TMEM80	Q96HE8	O75063
TMEM92	Q6UXU6	Q96J02	TMEM97	Q5BJF2	Q9UBD6
TMEM9B	Q9NQ34	P51790	TMOD2	Q9NZR1	Q9UM54
TMOD3	Q9NYL9	P60709	TMPRSS2	O15393-2	P78382
TMSB10	P63313	Q9NUX5	TMSB4X	P20065	Q8WWQ8
TMT1A	Q9H8H3	P17252	TMTC4	Q5T4D3	Q9Y4Z2
TMUB2	Q71RG4-4	P06396	TMX1	Q9H3N1	Q9NPF0
TMX4	Q9H1E5	O43914	TNC	P24821	P0DTC8
TNFAIP1	Q13829	P12931	TNFAIP3	P21580	Q14790
TNFRSF10A	O00220	Q14790	TNFRSF10C	O14798	P50591
TNFRSF11A	Q9Y6Q6-2	P00533	TNFRSF11B	O00300	Q9Y6Q6
TNFRSF12A	Q9NP84	Q12933	TNFRSF19	Q9NS68	P52565
TNFRSF1A	P19438	Q13546	TNFRSF8	P28908	Q99741
TNFSF11	O14788-PRO_0000034515, O14788	Q9Y6Q6	TNFSF12	O43508	P06493
TNFSF4	P23510	P43489	TNIK	Q9UKE5	O43318
TNIP2	Q8NFBZ5	P41279, P19838	TNIP3	Q96KP6	P42858
TNKS	O95271	Q9NTX7	TNMD	Q9H2S6-2	P36957
TNNC1	P63316	P19429	TNNI1	P19237	Q9NPB3
TNNI3	P19429	O76024	TNNT2	P45379-11	O60341
TNPO2	O14787	Q9Y639	TNRC18	O15417	P21741
TNRC6A	Q8NDV7	Q9UKV8	TNRC6B	Q9UPQ9, Q9UPQ9-2	Q9UKV8
TNRC6C	Q9HCJ0	Q9UKV8	TNS1	Q9HBL0	P11532
TNS2	Q63HR2	Q9Y5Y2	TNS3	Q68CZ2	P12931
TOB2	Q14106	P11940	TOLLIP	Q9H0E2	Q9UM54
TOM1	O60784	Q9UM54	TOM1L1	O75674	P56539
TOM1L2	Q6ZVM7	Q9UM54	TOMM40	O96008	P56705
TOP1	P11387	P00519	TOP2B	Q02880	P46531
TOR1A	O14656-2	Q96FZ7	TOR1B	O14657	P32971
TP53I3	EBI-1550792	P04637	TPD52	P55327	Q86T03
TPD52L2	O43399	P13569	TPI1	P60174	P42858
TPM1	P09493-10, P09493	Q9Y6D9	TPM2	P07951-2	P00441
TPM4	P67936	Q9NRD5	TPP1	Q96AP0	O43768
TPP2	P29144	P06746	TPR	P12270	Q9Y6D9
TPST1	O60507	P61769	TPX2	Q9ULW0	O75330
TRADD	Q15628	Q14790	TRAF3	Q13114	Q13546
TRAK2	O60296	P53350	TRAPPC1	Q9Y5R8	Q96QF0
TRAPPC12	Q8WVT3	Q9HAP6	TRAPPC2L	Q9UL33	Q96QF0
TRAPPC3	O43617	Q96QF0	TRAPPC5	Q8IUR0	Q9P2M4
TRDN	Q13061-3	O14543	TREX1	Q9NSU2-1	Q15125
TRHDE	Q9UKU6	Q5SRI9	TRIB1	Q96RU8	Q8NHY2

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TRIB2	Q92519	Q8NHY2	TRIB3	Q96RU7	Q6W0C5
TRIL	Q7L0X0	O15455	TRIM11	Q96F44	P61088
TRIM16	O95361	P08670	TRIM24	O15164	Q9UPN9
TRIM25	Q14258	P03496	TRIM26	Q12899	Q6NZI2
TRIM3	O75382	P51668	TRIM32	Q13049	Q06124
TRIM33	Q9UPN9	P53611	TRIM36	Q9NQ86	Q14451
TRIM37	O94972	Q7L590	TRIM38	O00635	Q13546
TRIM45	Q9H8W5	Q9BQP7	TRIM56	Q9BRZ2	P05388
TRIM7	Q9C029	Q9UI95	TRIM71	Q2Q1W2	Q9H9Z2
TRIM8	Q9BZR9	P51668	TRIM9	Q9C026	P50552
TRIML2	Q8N7C3	O75044	TRIO	O75962	Q9NV70
TRIOBP	Q9H2D6	P04004	TRIP13	Q15645	Q9UBX5
TRIP6	Q15654	Q9Y316	TRMT11	Q7Z4G4	P10809
TRMT12	Q53H54	P11142	TRMT1L	Q7Z2T5	Q00403
TRMT61A	Q96FX7	Q14192	TRNT1	Q96Q11-3	O76024
TRO	Q12816	Q05655	TROAP	Q12815	Q8N9N5
TRPC4AP	Q8TEL6	P01160	TRPV4	Q9EPK8	Q96J02
TRUB2	O95900	O00425	TSC2	P49815	Q9P0N9
TSC22D4	Q9Y3Q8	Q8NG68	TSGA10	Q9BZW7	Q9Y6D9
TSPAN11	A1L157	P01189	TSPAN12	O95859	Q96AA3
TSPAN15	O95858	Q9H244	TSPAN17	Q96FV3	Q16585
TSPAN18	Q8WUV1	P48509	TSPAN33	Q86UF1	O14672
TSPAN4	O14817	Q92570	TSPAN5	P62079	P48509
TSPAN6	O43657	Q9NY25	TSPAN7	P41732	Q8TDQ0
TSPO	P30536	P29762	TSSC4	Q9Y5U2	O76024
TTBK2	Q6IQ55	Q9UPV0	TTC12	Q9H892-2	Q13153
TTC17	Q96AE7	Q8IV16	TTC19	Q6DKK2	Q16774
TTC21B	Q7Z4L5	P18206	TTC23L	Q6PF05	P11021
TTC28	Q96AY4	O43639	TTC3	P53804	P31751
TTC9C	Q8N5M4	Q06187	TTF2	O00358	P19419
TTL	Q8NG68	P01189	TTN	Q8WZ42	O75923
TTYH1	Q9H313	O15354	TUBA1A	Q71U36	Q9UBX3
TUBA1B	P68363	Q15398	TUBA1C	Q9BQE3	P36507
TUBB	P07437	P36404	TUBB2B	Q9BVA1	P36404
TUBB3	Q13509	P01222	TUBB4A	P04350	P22736
TUBB4B	P68371	Q9BTW9, P36404	TUBB6	Q9BUF5	P42858
TUBG1	P23258	Q96L34	TUBG2	Q9NRH3	Q13371
TUBGCP2	Q9BSJ2	P08173	TUFM	P49411, A0A384ME17	Q8IVP5
TUFT1	Q9NNX1	Q92993	TULP3	O75386	P98160
TUSC1	Q2TAM9	O95749	TUSC3	Q13454	Q14165
TUT4	Q5TAX3	Q9H9Z2	TUT7	Q5VYS8-5	Q96CV9
TWF1	Q12792	P67870	TWF2	Q6IBS0	P60709
TWIST1	Q15672	O60885	TWIST2	Q8WVJ9	P15036
TXLNA	P40222	Q9UHD2	TXLNG	Q8BHN1	Q9UHD2
TXN2	Q99757	P49247	TXNDC11	Q6PKC3	P42858
TXNDC5	Q8NBS9	P0DTC3	TXNRD1	Q16881	O75695
TYK2	P29597	O75496	TYW3	Q6IPR3	P01130
TYW5	A2RUC4	Q9NZ43	U2AF2	P26368	Q09161
U2SURP	O15042	Q9H9Z2	UBA52	P62990	P51668

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UBA7	P41226	P05161	UBASH3B	Q8TF42	Q96PM5
UBB	P0CG47	Q9NZ52	UBC	P0CG48	Q9Y253
UBE2A	P49459	Q8WZA2	UBE2C	Q5TZN3, O00762	Q13164
UBE2G2	P60604	P0CG48	UBE2J1	Q9Y385	Q9UBV2
UBE2J2	Q8N2K1	A1L3X0	UBE2L3	P68036	P50876
UBE3B	Q7Z3V4	P80370	UBE3C	Q15386	P60896
UBE3D	Q7Z6J8	P35813	UBIAD1	Q9Y5Z9	P21579
UBL4A	P11441	O15055	UBL7	Q96S82	P55072
UBN2	Q6ZU65	P54198	UBQLN4	Q9NRR5	P10451
UBR5	O95071	P00740	UBTF	P17480	P52306
UBXN10	Q96LJ8	P55072	UBXN11	Q5T124	O94955
UBXN2B	Q14CS0	P55072	UBXN6	Q9BZV1, Q9BZV1-1, Q9BZV1-2	P55072
UBXN7	O94888	Q15843	UCHL1	P09936	Q15843
UCK1	Q9HA47-2	P06396	UCKL1	Q9NWZ5	Q9Y639
UCP2	P55851	P20930	UGP2	Q16851	O76003
UGT3A2	Q3SY77	P19526	UGT8	Q16880	P19526
UHMK1	Q8TAS1-2	Q9BU61	UHRF1	Q96T88	Q9Y6K1
ULBP1	Q9BZM6	Q6ZMH5	ULBP2	Q9BZM5	P32971
UNC5CL	Q8IV45	Q96HA1	UNG	P13051-2	Q14790
UPF2	Q9HAU5	Q92900	UPF3B	Q9BZI7, Q9BZI7-2	Q92900
UPK3BL1	B0FP48	P23945	UQCR10	Q9UDW1	P31930, O14949
UQCRC1	P31930	P26641	UQCRQ	O14949	P31930, Q9UDW1, P14927, P22695
UROD	P06132	P00533	UROS	P10746	P48507
USE1	Q9NZ43	O95249	USH1C	Q9Y6N9	P63000
USHBP1	Q8N6Y0	P09683	USP18	Q9UMW8	P05161
USP25	Q9UHP3	Q9UKG9	USP28	Q96RU2	Q12888
USP34	Q70CQ2	P21709	USP37	Q86T82	Q9UM11
USP43	Q70EL4	P61981	USP44	Q9H0E7	P41208
USP46	P62068	Q96B01	USP49	Q70CQ1	P41208
USP5	P45974-2	O14773	USP53	Q70EK8	P46109
USP54	Q70EL1-9	O94955	USP9X	Q93008	P29353
UTP25	Q68CQ4	P32121	UTRN	P46939	Q14118
UVRAG	Q9P2Y5	P0DTC3	VAMP2	P63027	Q12846
VAMP3	Q15836	Q9Y6L6	VAMP5	O95183	Q13190, Q9NYM9
VAMP8	Q9BV40	Q9P2W9	VANGL1	Q8TAA9	P18827
VANGL2	Q9ULK5	O14641	VARS1	P26640	P68104
VASH2	Q86V25	P61981	VASN	Q6EMK4	P78504
VAT1	P54219	Q7L8W6	VAV1	P15498	P00519
VAV3	Q9UKW4	Q9UGI0	VAX2	Q9UIW0	Q9Y3E7
VCP	P55072	P60880	VDR	P11473	Q15648
VEGFB	P49765	P15692	VEGFC	P97953-1	O60462
VENTX	O95231	O14503	VGLL3	A8MV65-2	O14964
VIL1	P09327	P19174	VIM	P08670	P11021
VIPR1	P32241	Q9BVG9	VIT	Q6UXI7	P11021
VKORC1	Q9BQB6	Q96BA8	VLDLR	P98155	P21579
VPS11	Q9H270	P0DTC3	VPS16	Q9H269	P0DTC3
VPS18	Q9P253	P0DTC3	VPS26B	Q4G0F5	O60493
VPS28	Q9UK41	P42858	VPS37B	Q9H9H4	O14964

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VPS4A	Q9UN37	P53990, O43633	VPS53	Q5VIR6	Q96EV8
VRK1	Q99986	P48431	VRTN	Q9H8Y1	P38919
VSIG1	Q86XK7	O43808	VSIR	Q9H7M9	Q969S0
VSNL1	P62760	P04155	VSTM4	Q8IW00	P22735
VWA2	Q5GFL6-3	P26045	VWC2	Q2TAL6	P25490
VXN	Q8TAG6	Q9NPB3	WARS1	P23381	P13569
WASF2	Q9Y6W5	P00519, Q9UQB8	WBP1L	Q9NX94	P06396
WBP2NL	Q6ICG8	Q9BYE7	WDCP	Q9H6R7	P53350
WDHD1	O75717	Q9BQP7	WDR1	O75083	P60709
WDR13	Q9H1Z4	P21709	WDR19	Q8NEZ3	O75386
WDR27	A2RRH5	P08243	WDR35	Q9P2L0	O75386
WDR36	Q8NI36	P03372	WDR43	Q15061	Q13901, Q9H8H0
WDR46	O15213	P06748	WDR48	Q8TAF3	Q96B01
WDR55	Q9H6Y2	Q9BQ90	WDR59	Q6PJI9	Q9HD47
WDR62	O43379	P45983	WDR75	Q8IWA0	Q13901, Q9NQT4
WDR81	Q562E7	Q14508	WDR83	Q9BRX9	O43166
WDR90	Q96KV7	Q13451	WFDC2	Q14508	P08253
WIPF1	O43516	Q9GZQ8	WIPF2	Q8TF74	O00401
WIPF3	A6NGB9	O00401	WIP11	Q5MNZ9	P63244
WIP12	Q9Y4P8-4	Q96E11	WIZ	O95785	P08311
WLS	Q5T9L3	P41221	WNK4	Q80UE6	O00141
WNT2	P09544	P11021	WNT5A	P41221	Q5T9L3
WRN	Q14191	P39748	WSB1	Q9Y6I7	Q93034, Q9UBF6
WSB2	Q9NYS7	Q07817	WWP1	Q9H0M0	Q96J02
WWP2	O00308	Q01860	WWTR1	Q9EPK5	Q07157
XAF1	Q6GPH4	P40937	XBP1	P17861-1	Q8TCT9
XPO1	O14980	Q9UKV8	XPO4	Q9C0E2	Q9NZQ7
XPO6	Q96QU8	Q9NZQ7	XRCC1	P18887	Q9NTX7
XRCC2	O43543	O75771, O43502, O15315	YAF2	Q8IY57-5	Q6NZI2
YBX2	Q9Y2T7	Q14197	YES1	P07947	O14543
YIF1A	O95070	Q9NQ11	YIF1B	Q5BJH7	Q9H5K3
YIPF3	Q9GZM5	Q9NRZ7	YIPF5	Q969M3	O95070
YIPF6	Q96EC8	Q9UBD6	YJU2	Q9BW85	P52747
YJU2B	P13994	Q92989	YKT6	O15498	Q92696, P53611
YOD1	Q5VVQ6	Q9Y4K3	YPEL5	P62699	P01275
YTHDC1	Q96MU7	Q9H9Z2	YTHDC2	Q9H6S0	P40429
ZACN	Q401N2	P17302	ZBTB10	Q96DT7	O15397
ZBTB12	Q9Y330	O43639	ZBTB17	Q13105	P04198
ZBTB22	O15209	Q9Y316	ZBTB24	O43167	Q9NRD5
ZBTB26	Q9HCK0	O15169	ZBTB32	A0A0C4DGF1	Q9UH92
ZBTB34	Q8NCN2	Q9Y2W6	ZBTB38	Q8NAP3	O14773
ZBTB39	O15060	P55854, P61956	ZBTB4	Q9P1Z0	Q08379
ZBTB44	Q8NCP5	P60900	ZBTB45	Q96K62	O95429
ZBTB47	Q9UFB7	Q9UGI0	ZBTB7A	O95365	Q13547
ZBTB8B	Q8NAP8	O00311	ZC3H13	Q5T200	P61981
ZC3H7B	Q9UGR2	Q04637	ZC3H8	Q8N5P1	Q9H9Z2
ZC4H2	Q9NQZ6	P55040	ZCCHC3	Q9NUD5	Q9H9Z2
ZCCHC7	Q8N3Z6	P63279	ZDBF2	Q9HCK1	P48431
ZDHHHC15	Q96MV8	Q9Y5Y5	ZDHHHC17	Q8IUH5	Q8IVB4

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ZDHHC18	Q9NUE0	Q9H8X2	ZDHHC21	Q8IVQ6	Q16581
ZDHHC22	Q8N966	P15151	ZDHHC5	Q9C0B5	Q8TDQ0
ZDHHC7	Q9NXF8-2	O43561-2	ZDHHC9	Q9Y397	Q9H8X2
ZEB1	P37275	Q13363	ZEB2	O60315	Q96KQ7
ZER1	Q7Z7L7	P01350	ZFAND2A	Q8N6M9	Q9NWF9
ZFAND2B	Q8WV99	Q9Y5K5	ZFAND6	Q6FIF0	Q12933
ZFHX2	Q9C0A1	P10636	ZFHX3	Q15911-2	Q9NRD5
ZFHX4	Q86UP3	P48431	ZFP36	P26651	Q9NPI6
ZFP36L2	P47974	P63104	ZFP42	Q96MM3	P50458
ZFP62	Q8NB50	O43159	ZFP90	Q8TF47	Q9Y2W6
ZFP91	Q96JP5	P04608	ZFPL1	O95159	Q12846
ZFPM2	Q8WW38	P02545	ZFYVE19	Q96K21-3	O15169
ZGRF1	Q86YA3	Q06609	ZHX1	Q9UKY1	P42858
ZHX3	Q9H4I2, Q9H4I2-2	Q9UBE8	ZIC2	Q62520	P10071
ZKSCAN8	Q15776	P42858	ZMAT4	Q9H898-2	Q15633
ZMIZ2	Q8NF64-3, Q8NF64-2, Q8NF64	Q13526	ZMYM2	Q9UBW7	Q16822
ZMYM3	Q14202	Q13547	ZMYM4	Q5VZL5-4	Q96CV9
ZMYND8	Q9ULU4	Q13547	ZNF10	P21506	Q13263
ZNF100	Q8IYN0	Q9Y4Z2	ZNF101	Q8IZC7	Q99435
ZNF114	Q8NC26	Q9UMX1	ZNF12	P17014	Q13263
ZNF121	P58317	P01106	ZNF124	Q15973	P26367
ZNF133	P52736	P05067	ZNF134	P52741	P55081
ZNF14	P17017	Q13263	ZNF143	P52747	Q14192
ZNF165	P49910	O14641	ZNF177	Q13360-2	Q15126
ZNF18	P17022	Q96T88	ZNF181	Q2M3W8	P05067
ZNF195	O14628	Q13263	ZNF197	O14709	P05067
ZNF202	O95125	Q9H2K0	ZNF211	Q13398	Q13263
ZNF213	O14771	Q9UMX1	ZNF214	Q9UL59	Q9NS56
ZNF225	P18146	Q8N726	ZNF227	Q86WZ6	P05067
ZNF232	Q9UNY5	P12931	ZNF233	Q9NZL3	P98164
ZNF235	Q14590	Q13263	ZNF24	P17028	P31749
ZNF248	Q05CR2	P17252	ZNF267	Q14586	Q13263
ZNF268	Q14587	O15431	ZNF276	Q8N554	Q9NRD5
ZNF277	G5E9M4, Q8WWA6	P63211	ZNF28	P17035	Q16659
ZNF280C	Q8ND82	Q13185	ZNF281	Q9Y2X9	P40763
ZNF286A	Q9HBT8	Q9NRD5	ZNF292	O60281	P83916
ZNF3	P17036	Q08379	ZNF302	Q9NR11-2	Q96CV9
ZNF320	A2RRD8	Q13263	ZNF326	Q5BKZ1	O75533
ZNF331	Q9NQX6	Q9UBX5	ZNF334	Q9HCZ1	P05067
ZNF335	Q9H4Z2	P51610	ZNF354A	O60765	Q13263
ZNF397	Q8NF99	Q8NHY2	ZNF398	Q8TD17	Q96CV9
ZNF410	Q86VK4-3	Q9Y314	ZNF417	Q8TAU3	P19883
ZNF419	Q96HQ0	Q96CV9	ZNF420	Q8TAQ5	Q8N726
ZNF423	Q2M1K9	Q96K83	ZNF426	Q9BUY5	P57764
ZNF431	Q8TF32	P17643	ZNF436	Q9C0F3	Q9HAQ2
ZNF445	P59923	P46087	ZNF449	Q6P9G9	Q13526
ZNF462	Q96JM2	Q8N465	ZNF48	Q96MX3	O43159
ZNF480	Q8WV37	Q13263	ZNF483	Q8TF39	P04004
ZNF484	Q5JVG2	Q13263	ZNF485	Q8NCK3	O00560

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ZNF488	Q96MN9-2	Q00403	ZNF503	Q96F45	O75593
ZNF507	Q8TCN5	O94901	ZNF512	Q96ME7	Q9H9Z2
ZNF516	Q92618	Q13547	ZNF518A	Q6AHZ1	Q13185
ZNF521	Q96K83	Q2M1K9	ZNF526	Q8TF50	Q04721
ZNF536	O15090	P48431	ZNF544	Q6NX49	O15228
ZNF550	Q7Z398	Q9UBX5	ZNF552	Q9H707	Q13263
ZNF561	Q8N587	Q13263	ZNF563	Q8TA94	Q92673
ZNF564	Q8TBZ8	P02545	ZNF567	Q8N184	Q13263
ZNF57	Q68EA5	Q9NPB3	ZNF577	Q9BSK1	O95373
ZNF580	Q9UK33	P14373	ZNF587	Q96SQ5	P19883
ZNF592	Q92610	Q13547	ZNF593	O00488	Q9Y4Z2
ZNF598	Q86UK7	P63244	ZNF605	Q86T29	P05067
ZNF606	Q8WXB4	P05067	ZNF608	Q9ULD9	Q13547
ZNF614	Q8N883	Q14938	ZNF620	Q6ZNG0	Q13163
ZNF625	Q96I27-2	O76024	ZNF627	Q7L945	Q9UBX5
ZNF629	Q9UEG4	O15397	ZNF638	Q14966	Q9Y4Z2
ZNF644	Q9H582	P48431	ZNF648	Q5T619	Q9BUH8
ZNF649	Q9BS31	P02545	ZNF660	Q6AZW8	P05067
ZNF670	Q9BS34	Q96CV9	ZNF675	Q8TD23	Q9Y4K3
ZNF69	Q9UC07-2	P07237	ZNF695	Q8IW36	Q13263
ZNF697	Q5TEC3	P42858	ZNF703	Q9H7S9	Q8N695
ZNF704	Q6ZNC4	Q96ST3, Q9H160	ZNF711	Q9Y462	Q9UPP1
ZNF724	A8MTY0	P36941	ZNF738	Q8NE65	Q13263
ZNF74	Q16587	Q8N2S1	ZNF746	Q6NUN9	O60260
ZNF761	Q86XN6	Q13263	ZNF763	Q0D2J5	Q9UMX1
ZNF765	Q7L2R6	Q13263	ZNF792	Q3KQV3	O95995
ZNF793	Q6ZN11	Q96EB6	ZNF8	P17098	Q13185
ZNF800	Q2TB10	Q13547	ZNF808	Q8N4W9	Q13263
ZNF813	Q6ZN06	Q16659	ZNF816	Q0VGE8	Q9NS56
ZNF836	Q6ZNA1	P05067	ZNF84	P51523	Q13263
ZNF844	Q08AG5	Q13077	ZNF845	Q96IR2	Q13263
ZNF85	Q49A12	Q14192	ZNF852	Q6ZMS4	Q9Y4X5
ZNF92	Q03936	P05067	ZNFX1	Q9P2E3	Q9Y3F4
ZNG1A	Q9BRT8	P01100	ZNG1B	Q8IUF1	Q9Y337
ZNHIT1	O43257	Q14451	ZNRD2	O60232	Q13459
ZRANB2	O95218	P32121, P49407	ZRANB3	Q5FWF4	P12004
ZSCAN1	Q8NBB4-2	Q00403	ZSCAN12	O43309	Q9BYJ1
ZSCAN18	Q8TBC5	Q9UMX1	ZSCAN21	Q9Y5A6	Q04721
ZSCAN23	Q3MJ62	Q9NRD5	ZSCAN31	Q96LW9	Q9NYA4
ZSWIM8	A7E2V4	P49674	ZUP1	Q96AP4	P50281
ZYX	Q15942	P02751	ZZEF1	O43149	Q8NBL1
ZZZ3	Q8IYH5	Q9NRD5			
Input	ChEBI Id	Interacts with	Input	ChEBI Id	Interacts with
APP	P05067-PRO_0000000092	28815	BMP1	P13497	29108
CDC42	P60953	15996	CHP1	Q99653	29108
CHP2	O43745	29108	CIB1	Q99828-1	29108
CSE1L	P55060	15996	FGF1	P05230	29036
GCC2	Q8IWJ2	15996	GLRX	P35754	16856
GRB7	Q14451	16152	HADHA	P40939	28494
IDH1	O75874	16810	IL11	P20809	29108

Input	ChEBI Id	Interacts with	Input	ChEBI Id	Interacts with
LRP1	Q07954	29108	MASP2	O00187-PRO_0000027598	29108
NMT2	O60551	15530	NPC1	O15118	16113
PAK1	Q13153	15996	PLEKHM1	Q9Y4G2	15996
RAB1A	P62820	15996	RAB32	Q13637	15996
RAB38	P57729	15996	RAB5B	P61020	15996
RAB8A	P61006	17552	RHOG	P84095	16152
RILPL2	Q969X0	15996	RRAGD	Q9NQL2	15996
S100A13	P97352	29036	SNCA	P37840	17552
SRC	P12931	15530	SYT1	P21707	29036
THBS1	P07996	28815			

7. Identifiers not found

These 2391 identifiers were not found neither mapped to any entity in Reactome.

AACSP1	AASDH	ABCA11P	ABCF2-H2BK1	ABHD12B	ABHD2	ABI3BP	ACAP3
ACOT6	ACOXL-AS1	ACTG1P25	ACVR2B-AS1	ADAMTS16-DT	ADARB2	ADGRD1	ADGRG2
ADGRL4	AEBP1	AFAP1	AFAP1L2	AGA-DT	AGAP2-AS1	AHSA2P	AIF1L
AKIRIN1	ALDH1A3-AS1	ALKBH6	ALOX12P2	ALPK2	ALPK3	AMZ1	AMZ2P1
ANK2-AS1	ANKLE1	ANKMY1	ANKRD18A	ANKRD18A.1	ANKRD18B	ANKRD24	ANKRD31
ANKRD34B	ANKRD65	ANO5.1	ANXA11	ANXA2R-OT1	ANXA7	AOPEP	AP1AR
AP1G2-AS1	AP3B2	AP5S1	APCDD1L	APCDD1L-DT	APELA	APOBEC3D	APOLD1
APRG1	ARHGAP26-AS1	ARHGAP29.1	ARL5A	ARNT2-DT	ARPC4-TTLL3	ASB16-AS1	ASTN2
ASTN2-AS1	ASXL3	ATAD2B	ATF7IP2	ATOH8	ATP2C2-AS1	B4GALNT3	BAALC
BDNF-AS	BEND4	BEND6	BEX4	BICC1	BICDL1	BLACAT1	BLTP1
BMP6	BNC1	BOLA2B.1	BOLA3-DT	BRD3OS	BTBD19	BVES	C10orf105
C10orf95	C10orf95-AS1	C11orf42	C11orf91	C12orf60	C12orf75	C12orf76	C13orf42
C14orf132	C16orf96	C17orf113	C17orf58	C17orf75	C1QTNF6	C1orf115	C1orf116
C1orf159	C1orf174	C1orf210	C1orf43	C1orf53	C21orf91	C21orf91-OT1	C22orf23
C2orf50	C5orf63	C6orf163	C6orf89	CAAP1	CACHD1	CACNA2D3-AS1	CADPS2
CALHM2	CALML4	CAMK2N1	CARD6	CARMIL3	CASC9	CASD1	CBLN3
CBR1-AS1	CBR3-AS1	CBS.1	CCBE1	CCDC110	CCDC144NL-AS1	CCDC149	CCDC160
CCDC169	CCDC169-SOHLH2	CCDC171	CCDC177	CCDC18	CCDC180	CCDC181	CCDC39
CCDC66	CCDC80	CCDC92B	CCDC9B	CCND2-AS1	CCNQ	CCNYL6	CCSER1
CCT6P3	CDH26	CDHR1	CDKL2	CDRT4	CDYL-AS1	CEBPA-DT	CECR2
CELSR1	CELSR3	CENPVL1	CENPVL2	CENPVL3	CERS6-AS1	CFAP144	CFAP157
CFAP20DC	CFAP58-DT	CFAP69	CFAP95	CFDP1	CFLAR-AS1	CGB7	CHD2
CHMP1B2P	CHRFAM7A	CHURC1	CHURC1-FNTB	CIMAP2	CKAP2L	CKLF-CMTM1	CKMT1A.1
CLIC6	CLMAT3	CLMN	CLSTN2	CLYBL	CLYBL-AS2	CMKLR2	COA8
COL18A1-AS1	COLCA1	COMMD3-BMI1	COPB2-DT	CORO2B	CORO7-PAM16	CPAMD8	CPEB2
CPED1	CPNE4	CPNE9	CPQ	CPXM2	CRIM1	CRIP3	CRISPLD1
CROCCP3	CRYBG2	CRYGD	CSRNP3	CTAGE15	CTAGE4	CTAGE6	CTAGE8
CTB-1I21.1	CTCFL	CTHRC1	CTIF	CUX2	CUZD1	CYP4X1	CYRIA
CYTL1	CYTOR	CYYR1	CYYR1-AS1	D21S2088E	DAAM2	DACH2	DANCR
DANT2	DAPK1-IT1	DBNDD2	DCHS2	DDX11-AS1	DDX60L	DEPP1	DGCR5
DHRS12	DHRS4L2	DHX33	DIAPH2-AS1	DICER1-AS1	DIP2C	DIPK2A	DLEU7.1
DLX1	DLX2	DMTF1	DNAAF3	DNAAF9	DNAH10	DNAH11	DNAH14
DNAH3	DNAH9	DNASE1L3	DOCK4-AS1	DOP1B	DPP10	DPP6	DPT
DUS4L	DUSP18	DYNAP	EBLN2	EBPL	ECHDC3	EDIL3-DT	EFCAB13
EGFR-AS1	EIF3J-DT	ELAPOR2	ELF4	ELMO1-AS1	ELMOD3	ELN-AS1	ELOF1
EMBP1	EML6	EMP2	EMX1	EMX2OS	ENDOV	ENPP5	ENTREP2
EPB41L4A	EPB41L4B	EPG5	EPHA1-AS1	EPHX3	EQTN	ERICH1	ERV3-1
ERVH48-1	ERVMER34-1	ESRG	ETAA1	ETS1-AS1	EXPH5	EZR-AS1	FAM133B
FAM153A	FAM174C	FAM180A	FAM184B	FAM227A	FAM47E-STBD1	FAM66D	FAM86B3P
FAM86DP	FANCC.1	FAT3	FBXO8	FCRL5	FCRLA	FCRLB	FER1L5

FGF13-AS1	FGGY	FHDC1	FHIP1A	FHIP2B	FIBIN	FILIP1L	FIRRE
FJX1	FKBP11	FKBP9P1	FLJ20021	FLJ42393	FLNB-AS1	FLNC-AS1	FLRT2-AS1
FLVCR1-DT	FLVCR2	FMO4	FNDC1	FNDC7	FOXD1	FOXD3-AS1	FOXI3
FOXN3-AS2	FREM1	FREM2	FRMD4B	FRMD7	FRMD8	FSIP1	FSIP2
FSTL4	FTX	FXYD6-FXYD2	GABRP	GAL3ST3	GALNT7-DT	GANC	GARIN1A
GARNL3	GAS5	GAS6-DT	GASK1A	GAU1	GBA1LP	GCNT2	GDF6
GDPD2	GIRGL	GJC3	GLDN	GLIPR2	GLT1D1	GNAS-AS1	GNG12-AS1
GNPTG	GOLGA6B	GOLGA6C	GOLGA6D	GOLGA8A	GOT1-DT	GPC6-AS1	GPR137
GPR171	GPR19	GPR22	GPR34	GPR63	GPR82	GPR84-AS1	GPR85
GPR89A.1	GPRIN3	GREM1-AS1	GRHL2-DT	GRINA	G RTP1	GRWD1	GSAP
GSDMB	GTDC1	GTF2I-AS1	GTF2IP1	GTF2IP4	GTF2IRD1P1	GTPBP6	GTSF1
GULP1	GUSBP2	GVQW3	H6PD	HAGLR	HAND2-AS1	HAPLN4	HCP5
HDAC2-AS2	HDGFL3	HDHD2	HECW1-IT1	HEIH	HEY2-AS1	HFM1	HHLA1
HLA-DPB2	HMGNI	HMGNI	HMSD	HMX2	HNRNPA1P33	HOXA-AS3	HOXA10-AS
HOXB-AS1	HOXB-AS3	HOXB6	HOXC-AS3	HOXC10	HOXC6	HOXD9	HPAT5
HPS6	HRCT1	HRK	HROB	HSD17B7P2	HSPE1-MOB4	HYDIN	IAH1
IDI2-AS1	IER3-AS1	IER5L	IFITM10	IFNK	IGDCC3	IGF2BP2-AS1	IGFBP7-AS1
IGIP	IGLON5	IGSF1	IGSF10	IGSF11	IGSF21	IGSF22	IGSF5
IGSF9B	ILDR2	IMMP2L	INAFM1	INKA2	INSYN2B	INTS6L	IPCEF1
IPO9-AS1	IQCA1	IRX3	ISOC2	ITIH5	JAKMIP2-AS1	JAKMIP3	JCAD
JHY	JMJD8	KANSL1L	KANTR	KAZALD1	KC6	KCNJ13	KCNQ1OT1
KCTD11	KCTD8	KDF1	KIAA0513	KIAA1549L	KLF7	KLHDC7A	KLHDC8B
KLHL23.1	KLHL29	KLHL4	KRBA2	KRTCAP3	LAD1	LARGE-AS1	LBH
LCMT1-AS2	LDLRAD2	LHFPL6	LIFR-AS1	LIN28B-AS1	LINC-ROR	LINC00173	LINC00174
LINC00240	LINC00309	LINC00398.1	LINC00458	LINC00471	LINC00511	LINC00517	LINC00540
LINC00562	LINC00595	LINC00598	LINC00598.1	LINC00641	LINC00648	LINC00649	LINC00664
LINC00665	LINC00668	LINC00680	LINC00698	LINC00839	LINC00842	LINC00869	LINC00877
LINC00923	LINC00968	LINC01012	LINC01108	LINC01119	LINC01145	LINC01152	LINC01162
LINC01204	LINC01224	LINC01234	LINC01237.1	LINC01315	LINC01356	LINC01405	LINC01416
LINC01487	LINC01572	LINC01600	LINC01638	LINC01694	LINC01748	LINC01801	LINC01806
LINC01876	LINC01918	LINC01950	LINC01980	LINC01981	LINC02210	LINC02249	LINC02253
LINC02301	LINC02367	LINC02381	LINC02387	LINC02440	LINC02523	LINC02540	LINC02570
LINC02576	LINC02582	LINC02593	LINC02603	LINC02615	LINC02617	LINC02649	LINC02751
LINC02802	LINC02893	LINC02912	LINC02994	LINC02997	LINC03040	LINC03069	LINC03104
LINC03105	LIX1L	LNCOC1	LNCOG	LNCPRESS1	LOXHD1	LOXL1-AS1	LOXL2-AS1
LRATD1	LRGUK	LRP1-AS	LRRC15	LRRC2	LRRC3B	LRRC69	LRRC70
LRRCC1	LSP1P4	LSP1P4.1	LSP1P5	LY75	LYPLAL1	MACC1	MACC1-AS1
MAGI2-AS3	MAL2-AS1	MAN1B1-DT	MANEAL	MAP3K9-DT	MAP6	MAP6D1	MAP9
MAPKAPK5-AS1	MARCHF1	MARCHF4	MARVELD1	MARVELD3	MBNL1-AS1	MBNL2	MCF2L2
MCTP1	MDFIC	MDH1B	MEGF10	MEGF9	MESTIT1	METRNI	METTTL18
METTTL24	METTTL5	METTTL8	MEX3A	MFRP	MFSI1	MGC32805	MGP
MIAT	MIR100HG	MIR1184-2	MIR1184-3	MIR1204	MIR124-1HG	MIR1251	MIR1272
MIR1279	MIR1282	MIR1304	MIR133A1HG	MIR17HG	MIR181A2	MIR198	MIR205HG
MIR22HG	MIR302C	MIR302CHG	MIR302D	MIR31HG	MIR3606	MIR367	MIR3671
MIR3679	MIR4435-2HG	MIR4639	MIR4647	MIR4751	MIR4761	MIR5004	MIR5047
MIR571	MIR573	MIR600HG	MIR671	MIR6805	MIR6852	MIR9-1HG	MIR924HG
MIX23	MMP24OS	MNI	MOB3A	MOK	MPV17L	MPV17L2	MREG
MRGPRF	MRO	MRPS31P5	MSANTD2	MSC-AS1	MSH2-OT1	MTCL3	MTFR1L
MTLN	MTX3	MXRA5	MXRA7	MYCL-AS1	MYL12-AS1	MYMX	MYOF
MYOSLID	MYPN	MYRFL	N4BP2	N4BP2L1	N4BP2L2-IT2	NAA25	NACC2

NAGA	NAGPA	NAIP	NALF2	NANOGP8	NAP1L6P	NAPSB	NAV3
NBL1.1	NBPF10	NBPF14	NBPF15	NBPF20	NBPF25P	NBPF8	NBPF9
NDNF	NDUFB2-AS1	NEAT1	NEK5	NEXMIF	NEXN-AS1	NKAIN1	NKAIN3
NKAPP1	NKX1-2	NKX6-2	NLRP7	NME1-NME2	NNAT	NOMO2	NOVA1
NOVA2	NOXRED1	NPAP1	NPIP11	NPIP6	NPR3	NPTX2	NPY4R2
NR2F1-AS1	NRARP	NRBP2	NRK	NSUN5	NSUN7	NT5DC1	NT5DC2
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None.2086	None.2087	None.2089	None.2098	None.2103	None.2110	None.2129	None.2134
None.2137	None.2156	None.2169	None.2180	None.2181	None.2183	None.2187	None.2190
None.2216	None.2226	None.2263	None.2274	None.2283	None.2294	None.2328	None.2371
None.2379	None.2380	None.2382	None.240	None.2406	None.243	None.244	None.2444
None.2456	None.2460	None.2466	None.2467	None.2482	None.2484	None.2488	None.2489
None.2494	None.2495	None.2497	None.2498	None.2499	None.2500	None.2508	None.2534
None.2538	None.2547	None.2558	None.2565	None.2575	None.2592	None.2593	None.2594
None.2603	None.2624	None.2626	None.263	None.2635	None.2653	None.2654	None.2656
None.2703	None.2704	None.2709	None.271	None.273	None.2750	None.2751	None.2752
None.2754	None.2775	None.2776	None.2777	None.2795	None.2796	None.2797	None.2806
None.2826	None.2828	None.2843	None.2855	None.2859	None.287	None.2888	None.2889
None.2898	None.2910	None.2939	None.295	None.2965	None.2978	None.3	None.3007
None.3008	None.3016	None.302	None.3026	None.3041	None.3055	None.3056	None.3071
None.3073	None.3074	None.3075	None.3083	None.3103	None.3106	None.3118	None.3127
None.3129	None.3137	None.3152	None.3155	None.317	None.319	None.3195	None.3200
None.3243	None.3244	None.3261	None.3276	None.3286	None.329	None.3340	None.3345
None.335	None.3364	None.337	None.3374	None.3394	None.3396	None.3439	None.3446
None.3447	None.3454	None.3457	None.3465	None.3469	None.3482	None.3483	None.3491
None.3495	None.3499	None.3502	None.3503	None.3512	None.3542	None.3544	None.3553
None.3560	None.3587	None.359	None.3617	None.3626	None.3648	None.366	None.3685
None.3693	None.3695	None.3698	None.3699	None.37	None.371	None.372	None.3745
None.3788	None.3838	None.3868	None.3882	None.389	None.3892	None.3897	None.3917
None.393	None.394	None.3944	None.3950	None.3987	None.3992	None.4014	None.4022
None.4079	None.4086	None.4093	None.4096	None.4097	None.4099	None.4101	None.4105
None.4110	None.4114	None.4125	None.4149	None.4152	None.4161	None.4167	None.4168
None.4176	None.419	None.4191	None.4201	None.4202	None.4212	None.424	None.4243
None.4244	None.4262	None.4264	None.4272	None.4279	None.4288	None.432	None.4322
None.4354	None.4369	None.4370	None.4375	None.4379	None.4380	None.4381	None.4393
None.4394	None.4399	None.4439	None.4447	None.4460	None.4470	None.4472	None.4497
None.4509	None.4510	None.4511	None.4568	None.4569	None.4591	None.4593	None.4597
None.4603	None.4604	None.4618	None.4620	None.463	None.4632	None.4635	None.465
None.4658	None.466	None.4677	None.4684	None.4685	None.4686	None.4687	None.4688
None.4689	None.4690	None.4691	None.4709	None.4727	None.4728	None.4729	None.4738
None.4740	None.475	None.477	None.4777	None.4814	None.4857	None.4862	None.487
None.488	None.4904	None.4918	None.4942	None.4943	None.4955	None.4981	None.5000
None.5021	None.5040	None.5064	None.5104	None.5105	None.5106	None.5134	None.5136
None.5150	None.5164	None.5168	None.5169	None.5176	None.5177	None.5199	None.5211
None.5219	None.5221	None.5236	None.5246	None.5248	None.5265	None.5272	None.5335
None.5349	None.5358	None.5360	None.5371	None.5386	None.5415	None.5445	None.5446
None.5447	None.5458	None.5459	None.5530	None.5558	None.5582	None.5584	None.5602
None.5603	None.5604	None.5607	None.5609	None.5610	None.5615	None.5621	None.5639
None.5644	None.5645	None.5653	None.5660	None.5666	None.5667	None.5668	None.5672
None.570	None.5832	None.5839	None.5843	None.5849	None.585	None.5852	None.5880
None.5887	None.5888	None.5910	None.593	None.5956	None.5978	None.5979	None.5984
None.5986	None.5991	None.601	None.602	None.6028	None.6076	None.6119	None.615
None.6203	None.6212	None.6214	None.6222	None.6223	None.6224	None.6225	None.6226
None.6232	None.6236	None.6260	None.6264	None.6321	None.6336	None.6357	None.6362
None.6363	None.6373	None.6374	None.6381	None.6389	None.6391	None.6392	None.6395
None.6415	None.6424	None.6485	None.649	None.6498	None.6499	None.6508	None.6516

None.654	None.6540	None.6563	None.6624	None.6639	None.664	None.6653	None.6663
None.6671	None.6695	None.6730	None.6754	None.6767	None.6796	None.6797	None.6799
None.6800	None.6823	None.683	None.6841	None.6849	None.6850	None.6876	None.688
None.6885	None.6886	None.690	None.6902	None.6941	None.6949	None.6969	None.6972
None.6974	None.698	None.6983	None.699	None.701	None.7010	None.7024	None.7025
None.7036	None.7037	None.7038	None.7040	None.706	None.7061	None.7068	None.7084
None.709	None.7122	None.715	None.718	None.719	None.7196	None.7218	None.7219
None.7227	None.7228	None.7256	None.7257	None.7259	None.7290	None.7291	None.730
None.7320	None.7339	None.734	None.735	None.7351	None.736	None.7363	None.7365
None.7366	None.7367	None.7368	None.7369	None.7370	None.7383	None.7404	None.7409
None.7410	None.7427	None.7452	None.7454	None.7460	None.7462	None.7463	None.7464
None.7465	None.748	None.7484	None.7495	None.7497	None.7501	None.7515	None.7516
None.752	None.7520	None.7522	None.7528	None.7529	None.7536	None.754	None.7540
None.755	None.7571	None.7598	None.7601	None.7602	None.7619	None.764	None.7649
None.7669	None.7703	None.7706	None.7720	None.7742	None.7751	None.7768	None.780
None.7812	None.7813	None.7871	None.7872	None.788	None.7882	None.789	None.7902
None.7908	None.792	None.7927	None.7933	None.7951	None.7978	None.8001	None.8014
None.8035	None.8041	None.8047	None.8076	None.8077	None.8078	None.8081	None.8091
None.81	None.8186	None.8212	None.8213	None.8224	None.8254	None.8255	None.8257
None.8278	None.8279	None.8282	None.8283	None.8284	None.8294	None.8296	None.8300
None.8323	None.8333	None.8344	None.8349	None.8356	None.8379	None.838	None.8396
None.8397	None.84	None.8404	None.8405	None.8411	None.8426	None.8459	None.8460
None.8461	None.8490	None.8491	None.8492	None.8498	None.850	None.8505	None.851
None.8520	None.8528	None.8547	None.8555	None.8588	None.8601	None.8602	None.8624
None.8625	None.8626	None.8639	None.8666	None.8707	None.8742	None.8770	None.8778
None.8799	None.8800	None.8804	None.8807	None.882	None.883	None.8837	None.884
None.8847	None.8849	None.885	None.8850	None.8856	None.8857	None.8858	None.8866
None.8867	None.8868	None.8871	None.8874	None.8876	None.8878	None.8879	None.889
None.8892	None.8898	None.8907	None.8908	None.8910	None.8924	None.8939	None.8946
None.895	None.8956	None.8965	None.8973	None.8975	None.8989	None.9005	None.9006
None.9008	None.9014	None.9026	None.9044	None.9050	None.9061	None.9069	None.9084
None.914	None.9146	None.9151	None.9160	None.9161	None.9206	None.9213	None.9217
None.9238	None.9248	None.9249	None.9254	None.9268	None.9313	None.9314	None.9321
None.9330	None.9331	None.9356	None.9357	None.9388	None.9440	None.9441	None.9455
None.9460	None.9461	None.9470	None.9539	None.9573	None.9585	None.9599	None.9618
None.9680	None.9681	None.97	None.9731	None.9754	None.9758	None.9760	None.9767
None.9768	None.977	None.9770	None.978	None.9799	None.980	None.9800	None.9814
None.9840	None.9843	None.9844	None.9845	None.989	OCIAD2	OGA	OIP5-AS1
OLFML3	OLMALINC	OMG.1	ONECUT2	OPA3	OR2A20P	OSBPL10-AS1	OTUD6B
OVCH2	OVOL2	P4HA2-AS1	P4HTM	PAK6.1	PAMR1	PANK1-AS1	PAPLN
PAPPA-AS1	PARAIL	PARM1	PARM1-AS1	PARP3	PART1	PAXBP1-AS1	PBX1-AS1
PCA3	PCAT1	PCAT1.1	PCAT7	PCBP1-AS1	PCBP2-OT1	PCDH1	PCDH11Y
PCDH11Y.1	PCDH12	PCDH17	PCDH9	PCDHA1	PCDHA10	PCDHA11	PCDHA12
PCDHA13	PCDHA2	PCDHA5	PCDHA6	PCDHA7	PCDHA8	PCDHA9	PCDHAC1
PCDHB12	PCDHB15	PCDHB18P	PCDHB2	PCDHB3	PCDHGA1	PCDHGA12	PCDHGA2
PCDHGA3	PCDHGA6	PCDHGA7	PCDHGA8	PCDHGB3	PCDHGC4	PCDHGC5	PCNX3
PCOLCE-AS1	PCP4	PDE9A-AS1	PDLIM2	PEDS1	PEPD	PEX5L	PGAM1P5
PGLS-DT	PHACTR1	PHF6	PHLDA2	PHYHIPL	PI15	PIEZO2	PIM2
PINK1-AS	PINLYP	PITRM1-AS1	PKD1L2	PKDREJ	PKHD1	PKIA-AS1	PLAC9
PLAG1	PLBD2	PLEK2	PLEKHA8P1	PLEKHH2	PLEKHM2	PMFBP1	PNISR
PNMA3	PNMA8A	PODN	PODNL1	POLR1HASP	POLR2J2	POLR2J4	POMZP3

POPDC3	POSTN	POU2AF3	POU5F1B	POU5F1P3	POU5F1P4	PPDPF	PPM1M
PPM1N	PPP1CB-DT	PPP1R14B	PPP1R14B-AS1	PPP1R17	PPP1R1C	PPP4R4	PRC1-AS1
PRICKLE2	PRICKLE2-AS1	PRIMA1	PRKCQ-AS1	PRNCR1	PROB1	PROSER1	PRPF4B
PRR36	PRRG3	PRRX1	PRRX2	PRSS12	PRSS16	PRSS35	PRSS48
PRSS54	PRTG	PSG1.1	PTCHD4	PTMS	PTPRG-AS1	PURPL	PWAR1
PWRN1	PWRN1.1	PWWP3B	PXDC1	PXK	PZP	QRICH2	QSER1
R3HCC1	R3HCC1L	RAB17-DT	RABGAP1L-AS1	RAD51AP2	RAD54B	RALGPS1	RAMP2-AS1
RANBP17	RAPGEFL1	RASGEF1B	RASGRF2-AS1	RASIP1	RASL11B	RAVER2	RBM12B
RBM12B-AS1	RBMS3	REXO6P	RFESD	RFLNB	RFX8	RGPD1	RHBDL3
RHOQ-AS1	RHOXF1-AS1	RIMS2	RIPOR1	RMST	RN7SK	RN7SL1	RN7SL2
RN7SL3	RNF103-CHMP3	RNF139-DT	RNF157	RNF175	RNF180	RNF227	RNFT2
RNPEP	RNPEPL1	RNU6-8	ROM1	ROR1-AS1	RP9P	RPL23AP7	RPP21.1
RPUSD1	RTL8A	S100A3	SAC3D1	SALL3	SAMD12	SAMD9	SAMD9L
SBDSP1	SBK1	SBK2	SCAND3	SCG5	SCIRT	SCNM1.1	SCPPPQ1
SDR42E1	SEC14L6	SEC1P	SECTM1	SEMA3G	SEMA6A-AS2	SEMA6B	SERF2
SERPIN1	SERTM2	SETD5	SEZ6	SFMBT2	SH3BGR	SH3BGRL2	SH3BP5-AS1
SH3TC2	SHANK2-AS3	SHISA2	SHISA9	SHISAL2B	SIAE	SIDT2	SIX4
SLC10A3	SLC16A12	SLC16A14	SLC16A9	SLC18B1	SLC25A36	SLC25A5-AS1	SLC25A53
SLC2A4RG	SLC31A2	SLC45A4	SLC49A3	SLC6A17	SLC9A3-OT1	SLFN11	SLFN13
SLFN5	SLTM	SMIM10L2A	SMIM12	SMIM19	SMIM24	SMIM8	SMLR1
SMN1.1	SMOC2	SMPDL3B	SNED1	SNHG1	SNHG21	SNHG26	SNHG4
SNHG7	SNN	SNORA13	SNORA26	SNORA48	SNORA6	SNORA65	SNORA66
SNORA74C-1	SNORA74D	SNORD100	SNORD108	SNORD109A	SNORD11	SNORD116-30	SNORD12
SNORD15B	SNORD21	SNORD26	SNORD28	SNORD45A	SNORD45B	SNORD53B	SNORD56
SNORD57	SNORD58A	SNORD58B	SNORD60	SNORD64	SNORD69	SNORD96A	SNX15
SNX19	SNX25	SOHLH2	SORCS1	SORD2P	SORL1-AS1	SOX11	SOX2-OT
SP140L	SP8	SPATA21	SPATA3-AS1	SPATA9	SPDYE18	SPESP1	SPINK8
SPMIP5	SRPX2	SSBP3-AS1	STAM-DT	STEAP2	STK33	STK38L	STOML1
STPG1	STRCP1	STX18-AS1	STXBP5L	SUGT1P4-STRAGLP	SUGT1P4-STRAGLP-CCDC180	SUMF1.1	SUMO4
SUSD1	SUSD4	SVEP1	SVIL-AS1	SVOPL	SYCP2L	SYNDIG1L	SYNPR-AS1
SYT13	SYT14	SYT15B	TAF4A5	TAGLN	TBC1D23	TBC1D32	TBC1D7.1
TBX5-AS1	TCEAL2	TCEAL3	TCEAL7	TCERG1L	TCF15	TCF7L1-IT1	TDRD3
TDRD5	TDRP	TEX14	TEX261	TEX50	TFAP2A-AS2	TFCP2L1	TFIP11-DT
TH2LCRR	THAP9	THBS4-AS1	THNSL2	THOC7-AS1	THUMPD3-AS1	TIPARP	TLCD2
TLCD5	TM4SF1	TM9SF1	TMC5	TMCC3	TMED7-TICAM2	TMEM104	TMEM117
TMEM119	TMEM121B	TMEM125	TMEM127	TMEM131L	TMEM132B	TMEM135	TMEM147-AS1
TMEM151B	TMEM158	TMEM167A	TMEM170A	TMEM170B	TMEM175	TMEM178B	TMEM18
TMEM185B	TMEM200C	TMEM204	TMEM221	TMEM232	TMEM250	TMEM253	TMEM263
TMEM269	TMEM37	TMEM62	TMEM63C	TMEM74B	TMEM8B	TMSB15A	TOR3A
TOX-DT	TOX3	TP53INP2	TPBG	TPRA1	TPRG1L	TPTEP1	TRABD2A.1
TRAM2	TRG-AS1	TRIM61	TRNP1	TSBP1	TSBP1-AS1	TSC22D2	TSHZ1
TSKU	TSSK4	TSTD3	TTC14	TTC22	TTC23L-AS1	TTC28-AS1	TTC36
TTC39A	TTC7A	TTC9-DT	TTN-AS1	TTN-AS1.1	TUBD1	TUG1	TUNAR
TVP23C-CDRT4	TWSG1	TXNDC15	TXNDC16	UBALD2	UBE2Q2P2	UBL3	ULK4P1

ULK4P2	ULK4P3	UPK3BL2	URGCP	URGCP-MRPS24	USF3	USH2A	USP40
USP51	UXT-AS1	VIM-AS1	VLDLR-AS1	VN1R1	VPS13A	VSIG10L	VWA5A
VWDE	WDR17	WDR38	WDR47	WDR49	WDR72	WDR83OS	WDR86-AS1
WDR87BP	WDR89	WFDC10B	WSCD1	XG	XKR4	XKR5	XKR6
XKR7	YPEL4	ZBBX	ZBED3	ZBED5	ZBTB20	ZBTB37	ZC3H12B
ZCCHC24	ZDHHC23	ZFAND1	ZFAND3	ZFAND4	ZFP82	ZFR2	ZHX2
ZIC5	ZKSCAN2	ZMAT1	ZMYM1	ZNF107	ZNF117	ZNF117.1	ZNF142
ZNF146	ZNF177.1	ZNF280B	ZNF286B	ZNF322	ZNF385B	ZNF385D-AS1	ZNF396
ZNF469	ZNF525	ZNF532	ZNF578	ZNF582-DT	ZNF587B	ZNF594	ZNF606-AS1
ZNF630	ZNF649-AS1	ZNF652	ZNF66	ZNF747-DT	ZNF761.1	ZNF766	ZNF780B
ZNF789	ZNF8-DT	ZNF814	ZNF876P	ZNF878	ZNF888	ZNF891	ZNF90
ZNF91	ZNG1E	ZNRF3-AS1	ZSCAN2	ZSCAN29	ZSWIM5	ZSWIM8-AS1	